Reference textbook: Operating System Concepts, 9th Edition
Abraham Silberschatz, Peter B. Galvin, Greg Gagne

Operating Systems

Full course handout

A4 · Two slides per page · Lin Cing

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Introduction

How can an OS provide convenience, efficiency, and isolation between untrusted programs and finite hardware? Prerequisites are program execution, CPU and memory roles, and interrupts. The objectives are to distinguish privilege modes, interrupts, exceptions, and traps; connect programs, processes, files, and storage; and assess kernel structures and computing environments.

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Managed abstractions

Resource	Program abstraction	Required enforcement
CPU and timer	Process or thread	Who runs, when control returns, who changes privilege
RAM and MMU	Address space	Ordinary programs cannot change others' mappings or permissions
Disk and controller	File and block service	Names, permissions, ordering, and error reporting
NIC	Socket and message	Endpoints, buffer ownership, packet and connection state

Convenience, utilization, response time, fairness, isolation, and reliability are separate goals. A long batch job can keep the CPU busy while interactive requests wait. Specify the workload and objective before claiming efficiency. The kernel is the trusted privileged core. Shells, GUIs, libraries, and daemons usually run in user space. A library call need not enter the kernel, and changing a shell need not change the kernel.

Policy and mechanism

Alice and Bob share a machine. Bob's program is untrusted. Policy denies Bob access to Alice's private pages and files and prevents permanent CPU monopolization. Mechanisms include protected page tables and memory-access checks, kernel file-authority checks, and a protected timer and scheduler entry.

An application's check is insufficient because Bob can bypass it. User and kernel modes are insufficient if the kernel reads sensitive files for Bob without validating authority. Enforcement must mediate every relevant operation; untrusted programs must not modify its policy or metadata. The boundary includes CPU control, memory, object rights, and mutable shared state. A hidden GUI button enforces nothing. Revoking an already authorized reference has a separate contract in Chapters 11 and 14.

Controller, driver, and DMA

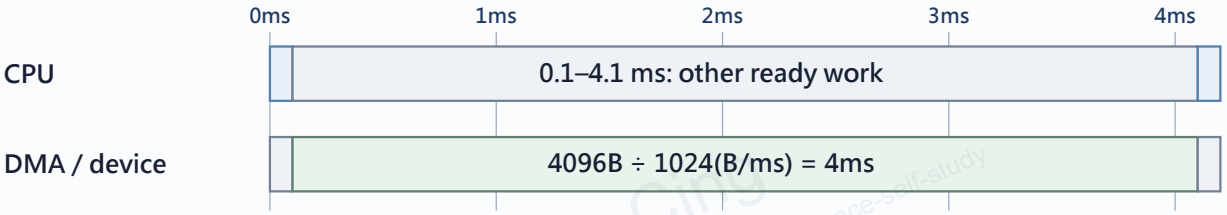
The controller implements device control in hardware or firmware. The driver is OS software that understands its registers, queues, and completion status. CPU submission, device transfer, and CPU completion handling are distinct events. Polling or an interrupt can discover completion.

The fixed model gives setup 0–0.1 ms, DMA 0.1–4.1 ms, and completion handling 4.1–4.2 ms. During the middle 4 ms, the CPU can run another ready job. DMA reduces per-byte CPU copying but does not eliminate setup, cache synchronization, bandwidth, or completion costs. A 4096 B block at 1024 B/ms takes 4 ms. CPU control costs 0.2 ms, whereas request latency is 4.2 ms. CPU service time is not application waiting time. Required data visibility and DMA completion precede buffer return.

Controller, driver, and DMA (continued)

DMA: latency 4.2 ms; CPU control 0.2 ms

Blue: CPU control. Green: DMA. Gray: no transfer. Time is to scale.



CPU setup 0–0.1 ms → DMA done 4.1 ms → CPU completion 4.1–4.2 ms

DMA avoids per-byte CPU copying; setup, ordering, completion, and buffer lifetime remain.

Interrupt, exception, and system call

Event	Relation to current instruction	Typical continuation
Device or timer interrupt	External, asynchronous	Save state, handle event, possibly schedule
Page fault or divide exception	Synchronous consequence	Repair and restart, report error, or terminate
System call	Deliberate synchronous request	Controlled entry, validation, service, return

ISA terminology and return points differ; not every exception resumes at the next instruction. An interrupt in kernel mode may cause no user-to-kernel transition, but still needs state preservation and reentry synchronization. An interrupt can return to the same thread, and a system call need not block. Neither implies a process switch. DMA can use polling or batched notification rather than one interrupt per block. State the completion rule.

Architecture follows sharing

Architecture	Typical sharing	Additional issue
One general-purpose CPU	One general execution resource	Concurrency with DMA and interrupts remains
SMP	Peer CPUs and common RAM	Synchronization and cache coherence
NUMA shared memory	Common addressable RAM, unequal locality	Location changes latency and bandwidth
Cluster	Independent networked nodes	Messages, partial failure, retry and consensus assumptions

Core, socket/package, and hardware thread count different things. SMP peer control and NUMA access asymmetry are separate dimensions; shared memory need not be UMA. Multiple CPUs alone do not establish fault tolerance. Power, RAM, kernel, or storage may fail jointly. Failover also needs detection, recovery, and exclusion of two simultaneous owners of an exclusive resource.

NUMA mean and parallel speedup

Assume sequential nonoverlapping requests without cache effects. Local memory costs 100 ns, remote 180 ns. At 80% local access, mean cost is 0.8×100+0.2×180=116 ns. At 20% local it is 164 ns. Their ratio is not automatically whole-program speedup.

A 100 s baseline has 20 s serial and 80 s perfectly parallel work with no overhead. Four processors give T=20+80/4=40 s and speedup 2.5. Generally, $T(P) = 100[s + (1 - s)/P]$, so speedup is at most $1/s$, before synchronization and data-movement costs. Operation-count fractions cannot replace baseline-time fractions. Hardware does not automatically discover the parallel work.

Multiprogramming overlaps CPU and I/O

Assume one CPU, one I/O device, free switching, A=CPU 2→I/O6→CPU 2, B=CPU 4, both ready at t=0.

Policy	CPU intervals	I/O	All complete	CPU utilization
Finish A before B	A0–2, idle2–8, A8–10, B10–14	A2–8	14	8/14=4/7
Run B while A blocks	A0–2, B2–6, idle6–8, A8–10	A2–8	10	8/10=0.8

CPU work remains 8. One CPU does not execute both CPU bursts simultaneously. The saving of 4 is exactly the overlap of B with A's I/O. Without other ready work, or when all I/O waits on one bottleneck, this saving need not exist.

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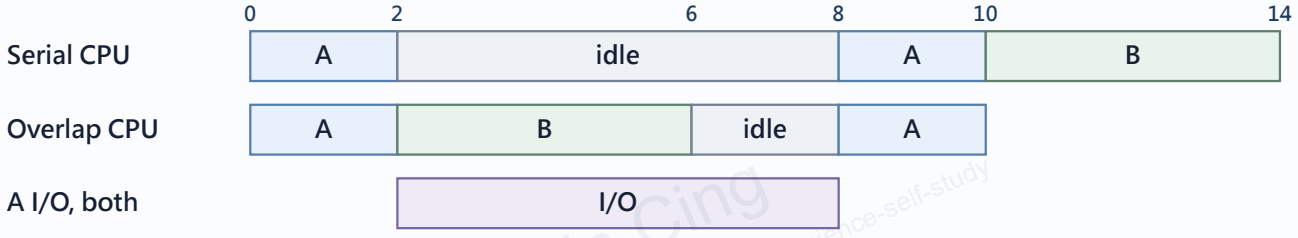
Multiprogramming overlaps CPU and I/O (continued)

Same workload: A=CPU 2→IO6→CPU 2, B=CPU 4

One CPU, independent I/O, free switches. Blue: A. Green: B. Gray: idle. Purple: I/O.



CPU work8 in both. Serial: 14, utilization4/7. Overlap: 10, utilization0.8. Saving4 is B/I/O overlap.

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Time sharing requires recoverable control

Switching only on blocking cannot protect other jobs from an infinite CPU loop. Time sharing normally adds timer preemption and a ready-queue rule. For n continuously ready jobs, quantum q, switching cost c, and fixed RR, a job that just missed its turn waits at most $(n-1)q+nc$, conservatively including switch costs. The bound fails with unbounded n, unbounded nonpreemptible kernel execution, or indefinitely inserted higher-priority work.

Smaller q reduces rotation delays but raises $c/(q+c)$. With q=4 ms and c=0.1 ms, this CPU-only model gives 0.1/4.1≈2.44%. This is a specified model, not measured universal overhead.

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Privilege and execution context

Protected privilege state prevents ordinary code changing the timer, page-table root, or interrupt entry. A system call saves ISA-required return state and enters controlled kernel code. Validation of arguments and authority precedes service.

t	One thread's execution	Mode	Process switch required?
0	Prepare read arguments	User	No
1	Entry and validation	Kernel	No
2a	Cache hit, copy, return	Kernel→user	No
2b	Cache miss, wait for I/O	Blocked in kernel	Another ready thread may run

A mode switch changes privilege. A context switch saves and loads an execution context. Another thread in the same address space need not require an address-space switch.

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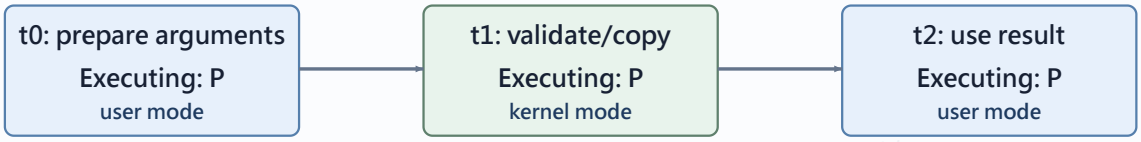
Privilege and execution context (continued)

Cache-hit syscall: 2 mode transitions,0 thread switches



Blue: user. Green: kernel. Entry/return change mode; P and its AS stay the same.

No other event here. Blocking/preemption may schedule Q. Count each switch type separately.

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A timer does not prove fairness

The OS sets a protected timer before handing control to untrusted user code. Allowing that code to disable or modify the timer defeats control recovery. Progress also requires serviceable interrupts, bounded nonpreemptible kernel sections, terminating handlers and scheduler actions, and selection without permanent starvation.

Interrupts can be delayed, batched, or prioritized. Setting q does not imply a precise switch at each q boundary. Kernel faults and malicious arguments remain possible. Mode enforcement is a foundation, not a proof of all kernel behavior.

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Programs and process state

A program is passive bytes, perhaps in a file. A process has an address space, credentials, open references, and at least one thread. A thread has a PC, registers, stack, and execution context. Threads share some process resources, not one PC.

Event	Transition	Actor
Wait for I/O	Running→waiting	Current thread and kernel register waiting
I/O completion	Waiting→ready	Completion path makes it schedulable
Selection	Ready→running	Scheduler and dispatcher
Quantum expiration	Running→ready	Controlled preemption

Completion does not immediately make a thread running; it must obtain the CPU. Blocked time is not ready-queue waiting. Process management also includes creation, termination, synchronization, IPC, and cleanup.

A non-atomic increment loses updates

Initially counter=0. Each thread loads, increments, and stores.

Step	T1	T2	Counter
1	Load r1=0		0
2		Load r2=0	0
3	Store r1+1=1		1
4		Store r2+1=1	1

The result is 1 rather than 2. A single CPU can switch between these steps; multiple CPUs add interleavings. A suitable mutex or atomic RMW changes the contract to an indivisible update. `volatile` alone does not provide mutual exclusion. Chapter 5 supplies the ordering conditions.

Equal virtual addresses, different frames

Pages are 4096 B. P maps VPN 1→PFN 7 read/write; Q maps VPN 1→PFN 9 read-only. VA0x1234 has VPN 1 and offset0x234=564. P's PA is 0x7234, Q's 0x9234.

Operation	P	Q
Read VA0x1234	Frame 7+564	Frame 9+564
Write VA0x1234	Permission allows	Permission fault
Change page table to frame 7	Ordinary user cannot modify protected tables	Same restriction

Mapping existence, residency, and authorization are distinct. The OS may repair a not-present mapping; adding a page need not repair forbidden writing. Sharing requires explicit mappings to the same frame and defined permissions. Equal VAs do not create sharing.

Frame allocation and reclamation

Allocated and free sets cannot both claim exclusive ownership of one frame. Legitimate sharing needs consistent references and permissions. Reclamation requires old mappings to be unusable, including TLB entries and outstanding I/O holders.

Eight RAM frames with kernel 2, P3, Q2 leave free 1. Releasing one unshared Q frame gives free 2. If P still references it, it cannot enter the free pool. Remaining reachability matters more than one process declaring release. Virtual memory need not keep all virtual pages resident, but RAM remains finite. Working sets, backing I/O, and commit limits constrain performance and allocation guarantees.

Define miss penalty before averaging

L1 lookup costs 1 cycle, hit probability is 0.9, and a miss adds 50 cycles. Without overlap or prefetch, mean cost is 1+0.1×50=6, equivalently 0.9×1+0.1×51. If 50 denotes total miss latency, the mean is 0.9×1+0.1×50=5.9. The inputs differ. A mean of 6 does not make every access 6 or specify tail latency.

Caching introduces copy coherence and invalidation. CPU coherence does not establish filesystem durability; data may remain volatile. Write completion, flush completion, and power-loss survival need separate definitions.

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Allocation and storage failures

Applications use file names and offsets. Filesystems map ranges to blocks, and block layers and drivers submit device requests. Allocation prevents duplicate exclusive owners; free-space indexes must agree with object pointers. A memory lock handles concurrency but cannot repair a crash that persisted half an update.

For 2^{20} blocks at one bit per block, bitmap capacity is 2^{20} bits=2^{17} B=128 KiB. This is logical capacity. An uncached query still reads its containing storage block, not one physical bit. Corruption, hardware failure, accidental deletion, and hostile access need different responses. RAID, backup, and protection do not replace one another.

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Protection and security

Protection restricts known subjects' use of objects. Security also concerns authentication, intrusion, malware, and threats to the system. A simple internal/external split cannot classify every attack.

Question	Mechanism	Still insufficient
Who requests?	Authentication and trusted credentials	Identity does not grant authority
What object and operation?	ACL or capability and access check	UI visibility cannot enforce an API
Can messages be intercepted?	Threat-appropriate encryption and key management	Cannot repair compromised endpoints
Can mappings exceed authority?	MMU rights and protected tables	Kernel bugs may bypass policy

Confidentiality, integrity, and availability are separate. Denying every request may prevent disclosure while destroying availability.

Least privilege and authorization time

An image converter needs input-file read and specified-output write rights, not write authority over a whole home directory. Bounded authorized references limit damage better than unrestricted paths or root identity. An ordinary user still has authority through inherited descriptors, shared memory, and sockets.

Removing a name's permission need not invalidate existing handles. Distinguish reference-creation checks from later-operation checks. A complete decision identifies credentials, object identity, requested rights, policy, and check time. File names, displayed identities, and privilege icons alone are insufficient.

Kernel structures and their premises

Structure	Operations	Conditions
Linked list	O(1) insertion/deletion with required node or predecessor	Worst-case key search O(n), pointer/cache costs
Stack or FIFO queue	O(1) endpoints with suitable representation	FIFO is not priority order
Balanced search tree	Ordered lookup/ranges, O(log n)	Maintain balancing invariant
Hash table	Average O(1) in specified model	Worst-case collision O(n), resizing costs
Bitmap	Membership in finite integer domain	Space follows domain size; claims need atomicity

With distinct BST keys, all left-subtree keys are below the node and all right-subtree keys above. Immediate-child comparisons are insufficient. Balanced height is O(log n), not an exact log2 n level count for any tree.

Collisions, subtree bounds, and lifetime

For h(k)=k mod4, tail-insert 5,9,13 into bucket1 as [5,9,13]. Searching 13 takes three comparisons, and absent17 also takes three. n congruent keys give linear worst-case search. Constant average cost requires distribution/load premises or a stronger design.

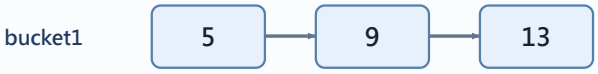
A tree with root 10, left5, right20, and 12 as 5's right child satisfies 12>5 but violates root 10's left-subtree bound. Searching 12 follows right20 and then an empty left link. Full subtree ranges must hold.

A reader may retain an old node after unlinking. Immediate free/reuse permits use-after-free. Locks, reference counting, or an RCU-style grace period have different reclamation contracts. Stale values do not authorize invalid memory.

Collisions, subtree bounds, and lifetime (continued)

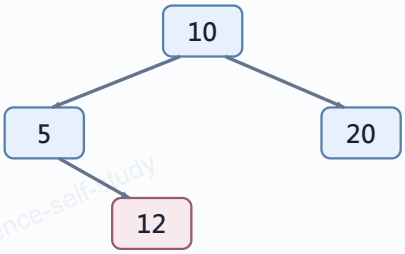
Hash collision cost and complete BST subtree bounds

h(k)=k mod4; chaining



Find 13: 3 comparisons. Reject 17: 3 comparisons.

n congruent keys: worst case O(n).



Red 12 is left of root 10 but12>10: subtree-bound violation.

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Computing environments

Environment	Boundary	Unsupported inference
Client/server	Request and service roles	A server cannot also be a client
Peer-to-peer	Nodes request and provide	No central discovery/authentication exists
Cluster/distributed	Independent nodes and network	Timeout proves remote failure
VM	Guest virtual hardware through VMM	Guest privilege equals host privilege
Conventional container	Views normally share host kernel	Each has an independent kernel
Embedded/real-time	Device or timing constraints	Every embedded system is hard real-time

IaaS, PaaS, and SaaS divide provision and management responsibilities. They are not CPU architectures. Virtualization can support them, but every cloud service is not one VM.

Deadlines are part of correctness

Assume one CPU, no blocking/overhead, A(C=1, T=D=4), B(C=2, T=D=6). Demand is 1/4+2/6=7/12. Utilization below one alone does not prove deadlines without a scheduler.

A feasible 12-unit EDF schedule is A0–1, B1–3, idle3–4, A4–5, idle5–6, B6–8, A8–9, idle9–12. A completes at1,5,9 before4,8,12; B at3,8 before6,12. It repeats because lcm=12 and no work remains at the boundary. Nonpreemptible sections, jitter, or I/O change the premises and require fresh blocking/response analysis. Low average latency does not replace a worst-case deadline check. Hard/soft real-time differ in consequences and permitted violations.

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Source visibility and open-source rights

The Open Source Definition requires more than readable source, including specified redistribution and derived-work rights. Source-available, freeware, public domain, and open source are different. Commercial charging need not conflict with open source.

A distribution combines a Linux kernel with tools, libraries, services, and packages. BSD is another family and licensing tradition. One open component does not give all files one license. Comparing implementations requires fixed versions, configuration, and dependencies. Linux 3.2 and Windows 7 retain historical contracts here; modern supplementary interfaces are identified separately.

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Match evidence to the question

Question	Evidence	Insufficient substitute
Which API outcomes are allowed?	Versioned interface/standard	One successful run
How does this version implement it?	Matching source/configuration/architecture	Same function name in another version
What is observed latency?	Workload time/resource breakdown	Operation proportions treated as time
Is an update durable?	Storage/flush contract and failure order	Write returned success

Source access permits inspection, not a proof of no races, optimal speed, or security. Preserve premises, error outcomes, and general invariants. One example does not establish all inputs.

One read connects the course

P reads4096 B through an authorized fd. Controlled entry validates the descriptor/OFD and buffer. A cache miss creates I/O and blocks P; the scheduler selects Q. DMA completion makes P ready. P later obtains the CPU, observes valid data, and copies/returns actual bytes.

Part	State	Chapters
Call/protection	Mode, credentials, requested right	2,14,15
Process/thread	Ready/waiting/running, saved registers	3–7
Mapping/data	VA/PTE/frame, cache, buffer lifetime	8,9,13
Name/storage	OFD offset, inode, block, completion/durability	10–12

These states do not complete simultaneously. Waiting→ready is not immediately running. Completing one request does not flush unrelated files.

Quantities that must remain distinct

Mode transitions, context switches, and address-space switches are separate counts. One thread's cache-hit system call may require only the first. CPU busy time, request latency, throughput, and deadlines use different denominators; less CPU work need not shorten a request. Names, object identities, and open references differ; renaming a path need not alter an fd. Average cost, worst-case cost, and general correctness differ, as collisions, timer delay, and races demonstrate.

Fix the version and abstraction, enumerate observable states/results, then use invariants and counterexamples to assess the mechanism.

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References

- Silberschatz, Galvin, and Gagne, Operating System Concepts, 9e, Chapter 1, printed pp. 3–52. Relevant figures are1.3 (p.8), interrupt timeline, and1.9 (p.19), multiprogramming. The explanations, numerical cases, counterexamples, exercises, and SVGs are original.
- BST/hash statements retain complete subtree bounds and collision premises. Core/socket/hardware-thread, SMP/NUMA, and mode/context are distinct.
- Open Source Initiative. Each component's actual license governs.
- Time, probability, and memory examples state model premises. DMA/scheduling times are assigned values, not hardware measurements. Historical details belong to the corresponding system cases.

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References and attribution

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Operating-system structures

How can an OS expose maintainable services while preserving isolation and performance? Prerequisites are privilege modes, interrupts/traps, and call stacks. Trace library-to-kernel calls, compare kernel structures, and explain debugging, system generation, and boot dependencies.

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Services and completion contracts

User interfaces, execution, I/O, file operations, communication, and error detection support applications. Allocation, accounting, protection, and security support coexistence. A service is not merely a syscall name. Kernels, libraries, daemons, and tools can cooperate to provide one interface. Actual privileges and object access determine protection boundaries.

Copy service	Provider in this model	Observable completion
Interpret source/destination	Shell and copy tool	Two pathnames, not kernel interpretation of shell syntax
Find file and check rights	VFS/filesystem	Successful open descriptor or explicit error
Transfer	Library API→kernel I/O	Actual completed bytes, possibly below request
Allocate and protect	Kernel/filesystem	Valid descriptors, cache/storage capacity, rights
Store result	Filesystem/device	File synchronization distinct from directory-entry persistence

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Services and completion contracts (continued)

User and system programs normally obey the same protection rules. The name system tool grants no kernel privilege. A privileged user-space daemon still runs in user mode; credentials and CPU mode are separate.

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A 6000-byte copy with short writes

Source is a fixed6000 B, destination a new empty regular file, buffer 4096 B, blocking I/O, no competing writer.

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A 6000-byte copy with short writes (continued)

Event	Returned bytes	Source offset	Destination offset/length	Buffer pending
Open source/create destination		0	0	0
read 4096	4096	4096	0	4096
write 4096	1000	4096	1000	3096
write buffer[1000:]	3096	4096	4096	0
read 4096	1904	6000	4096	1904
write 1904	1024	6000	5120	880
write buffer[1024:]	880	6000	6000	0
read 4096	0, EOF	6000	6000	0

A 6000-byte copy with short writes (continued)

Excluding directory synchronization, the path has2 opens,3 reads,4 writes,1 fsync,2 closes, totaling12 API operations. The essential rule is progress from actual returned bytes. One write after each read would lose3976 B.

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Copy-loop invariant and failure boundary

Let R be bytes read, W bytes written, and [j, n) the pending buffer interval. Maintain $W \leq R$ and buffer[j: n] equal to the next contiguous source fragment not yet written. A positive return $k \leq n-j$ increments both W and j by k without changing sequence. Only j=n permits overwriting the buffer with another read. EOF with no pending bytes proves W=R=6000.

Handle zero-progress EINTR according to the interface and retry the same remaining range. A positive count advances progress first; rewriting the whole chunk duplicates data. Unexpected write=0 must report no progress or otherwise avoid an infinite loop. ENOSPC/EIO leaves partial output and cannot return success.

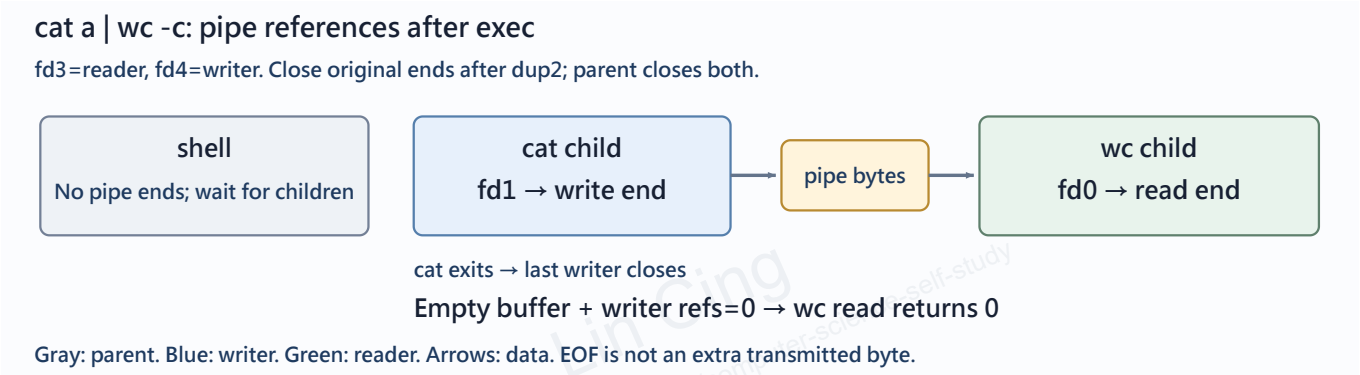
A successful write is not crash durability. Under a compliant local filesystem/device flush contract, successful destination fsync confirms its data and required metadata. A new name also requires synchronization of its containing directory. This copy still exposes a partial destination on failure. Atomic replacement needs a same-directory temporary file, data synchronization, rename, and directory synchronization protocol; close alone does not supply it.

Shell, GUI, and kernel

A command interpreter translates text/scripts into processes and I/O connections; a GUI translates input events into application actions. Both can use the same kernel services. A child-only `cd` cannot change the parent shell's cwd, which explains its usual builtin implementation. External `cat` can execute in a child. A builtin is not necessarily kernel code. Privileged frontends also expose attack surfaces such as command injection and accidental overwriting.

For POSIX-style `cat a | wc -c`, assume no alias, job control, or extra subshell. The shell creates a pipe, forks two children, connects cat's stdout to its write end and wc's stdin to its read end, then waits. Two execs replace child images without creating two more processes. The parent retains its image.

Shell, GUI, and kernel (continued)



Endpoint references determine EOF

pipe returns fd3 read and fd4 write. Initially all three processes inherit both. Cat uses dup2(4,1), wc uses dup2(3,0), then closes redundant descriptors.

Process	Remaining pipe reference	Close	Reason
Shell	None	3,4	Do not keep a writer alive
cat	fd1→write end	3 and original4	Do not hide loss of readers with own read reference
wc	fd0→read end	Original3 and4	Do not retain a writer that prevents EOF

Cat writes6000 B and exits. Wc can read arbitrary chunks and accumulate6000. Only an empty pipe with every write-end reference closed makes the next read return 0. If the shell retains fd4, wc may block after cat exits while the shell waits for wc. Process exit alone does not establish the reference condition.

Output, status, and permissions

If cat cannot open a and returns ENOENT, it can exit nonzero without writing bytes. Wc can correctly count the empty stream, output0, and exit0. Collect both wait statuses. Pipeline-wide failure follows the shell's specified rule, independently of data correctness.

A GUI cannot claim copy success merely because a progress bar ends. It must await the I/O/error and synchronization contract it promises. CLI versus GUI does not change access checks, capacity, or crash consistency. `cat 'a b'` supplies one pathname; `cat a b` supplies two. The shell parses quotes and pipes. The kernel receives an exec argument array and does not reapply shell syntax.

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API, ABI, and implementation

An API defines function names, types, and observable behavior. An ABI defines machine calling conventions, syscall numbers, registers/stack layout, and returns. Kernel implementation operates on protected objects. Source compatibility does not make a binary portable across ISAs/ABIs.

`strlen` can run entirely in user space. Buffered `printf` may retain output and combine several calls into one write. A copy API can combine many opens/reads/writes. API-call counts therefore differ from trap counts. If libc converts kernel errors into−1/errno, distinguish the two return formats.

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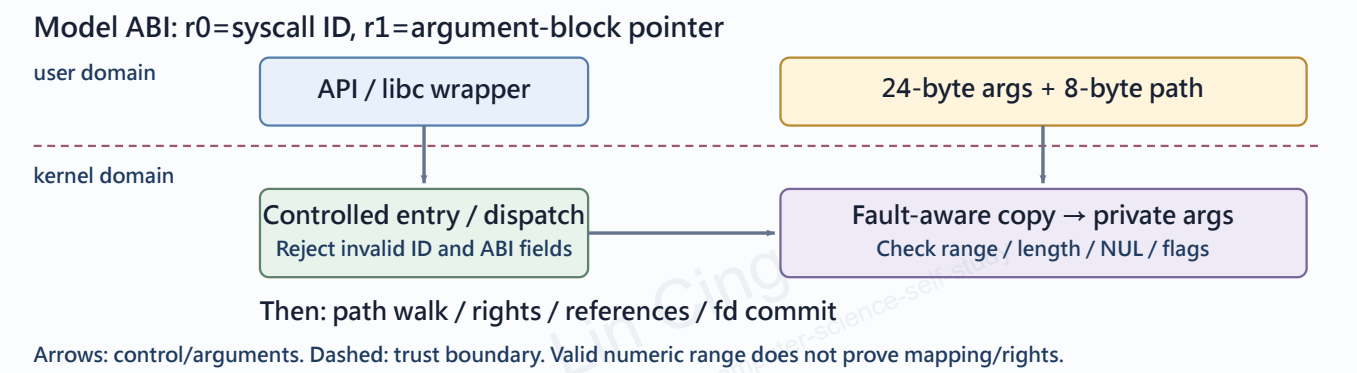
API, ABI, and implementation (continued)



The teaching ABI puts syscall ID in r0 and a pointer to a24 B argument block in r1. Its three64-bit fields are pathname pointer, flags, and mode. This is a specified model, not a claim that Linux open always uses such a block; real platforms may use registers and vary by version.

Copy arguments before validating their meaning

`/data/x` plus NUL occupies8 B. A fault-aware user copy obtains the24 B fixed block, then bounded string copying obtains at most PATH_LIMIT bytes,8 here. Total copy is32 B. Parse and use the same kernel-owned copy so later user modifications cannot substitute another pathname.

Step	Check	Failure
Decode ID/ABI	Known ID, valid fields/flags	Reject, never jump to arbitrary handler
Copy block	User range and actual fault-safe access	EFAULT or model equivalent, no partial arguments used
Copy pathname	Accessible, NUL-terminated, bounded	EFAULT/ENAMETOOLONG, no published fd
Path walk	Directory search and final-object rights	EACCES/ENOENT or other error
Allocate/publish fd	Stable reference, rollback incomplete allocation	Valid fd or no leaked reference

Copy arguments before validating their meaning (continued)

A readable-looking VMA is insufficient: another thread can unmap it and the actual access can fault. Use fault-aware copying, check that the copy completed, and validate and consume the same kernel-owned bytes. **Bytes stable after copying are not an atomic snapshot of the source during copying.**

For example, user fields begin at (length=4, capacity=4). The kernel reads length=4; user code writes length=8 and then capacity=8; the kernel reads capacity=8. The private result is (4,8), although the source visited only (4,4), (8,4), and (8,8). Treat the result as arbitrary untrusted input and revalidate cross-field relations, lengths, and authority. One copy does not guarantee coherent source fields. An API requiring a coherent snapshot needs an additional enforceable synchronization or version protocol; an adversarial caller cannot be trusted to honor a user-space lock. Filesystem namespace stability during path walk separately requires kernel object references and synchronization.

Prove arithmetic before accessing memory

For32-bit unsigned words, nonempty ranges in user addresses[0, U), U=0xC0000000, use `p < U && n <= U-p`. Do not first calculate wrapping p+n. Arithmetic success still needs mapping, permission, and fault-safe copy checks.

p	n	Mathematical exclusive end	Result
0xBFFFFF00	0x100	0xC0000000	Valid, final byte0xBFFFFFFF
0xBFFFFF00	0x101	0xC0000001	Crosses boundary
0xFFFFFFF0	0x40	0x100000030	Invalid,32-bit addition disguises it as0x30

For count elements of4 B, first require count≤SIZE_MAX/4. count 0x40000001 has product0x100000004, truncated to4 in32 bits. Allocating4 B and iterating the original count overruns memory. Zero-length pointer rules belong to each API; the present condition explicitly concerns nonempty ranges.

Syscall categories connect objects to state

Category	Requests	Protected state	Success does not imply
Process control	fork, exec, wait	Task, image, exit status	exec creates another process
Files	open/read/write/lseek/close	fd, OFD, inode	Write durability
Devices	read/write/control	Driver queue, device state	Queueing is completion
Information	Attributes, time	Credentials, metadata	Object ID grants rights
Communication	Pipe, messages, shared memory	Endpoint, message, mapping	Stream read is one record
Protection	Permissions, credentials	Policy, principal	Authentication is every authorization

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Syscall categories connect objects to state (continued)

Categories overlap. Socket reads involve I/O, communication, and protection. An fd need not name a regular file. Files, pipes, and terminals share read-like interfaces with different blocking, EOF, and seek rules. Parameters may use registers, blocks, or a stack. Persistent channels and bounded message units have different correctness obligations.

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Returns are state-machine outputs

For a regular-file read without concurrent modification, count 4096 from offset0 may return 3000 and advance to3000, then1096 and advance to4096, then0 at that moment's EOF. A short read is not permanent EOF; on a pipe it may merely reflect available bytes.

Model kill(pid100, sig9) as two32-bit arguments,8 B of representation, not necessarily8 B copied. The kernel resolves the PID namespace and validates authority/signal. Missing target gives ESRCH; insufficient authority EPERM. Success means delivery/queueing under signal rules, not observed complete termination and reclamation. Wait supplies lifecycle synchronization. Multiple successful syscalls are not automatically one atomic transaction. Identify the commit point; protection, blocking, visibility, durability, and termination remain separate.

Source to running process

For hello.c with main and libc output, the path is preprocess→compile→assemble→link→image. Dynamic execution additionally needs a compatible interpreter/loader and libraries. Compiler/linker are user programs using kernel file, memory, and process services; generating machine code does not give kernel mode.

Stage	Input/output	Independent failure
Preprocess	Source/headers→tokens	Missing header, wrong macro branch
Compile/assemble	Tokens→assembly→object	Syntax/type error, target ISA mismatch
Link	Objects/libraries→image	Unresolved symbol, ABI mismatch
Exec/load	Image/interpreter→process image	Permission, format, missing loader
Run	Entry→APIs/syscalls	Fault, I/O error

Source to running process (continued)

File tools, status tools, programming support, loading, communication, and background services have different purposes. Daemon lifetime changes neither CPU-mode enforcement nor kernel-interface rules.

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Build errors must propagate

For parser, compiler, linker, success is0 and failure nonzero. `parser && compiler && linker` runs a stage only after success. Semicolons continue regardless, and the last status may reflect only the linker. Prevent stale outputs through explicit versions and dependencies, not mere existence.

Case	Parser	Compiler	Linker	Conclusion
All succeed	0	0	0	Image exists, execution still unproved
Parser fails with &&	2	Not run	Not run	Build failed, do not publish stale image
Compiler fails with ;	0	1	0 using stale object	Last0 hides earlier failure

Source API compatibility, binary ABI compatibility, and complete runtime dependencies are separate. Link success or a matching digest alone proves neither a bootable target nor error-free execution.

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Mechanism and policy

Timer interrupts, context save/restore, and controlled ready-queue operations are mechanisms. Quantum and selection order are policy. Timer resolution, nonpreemptible code, and switch cost constrain achievable policy latency.

One CPU has A/B/C bursts5/3/1 ms, all ready at0, initial RR order A, B, C, no I/O, zero switch cost.

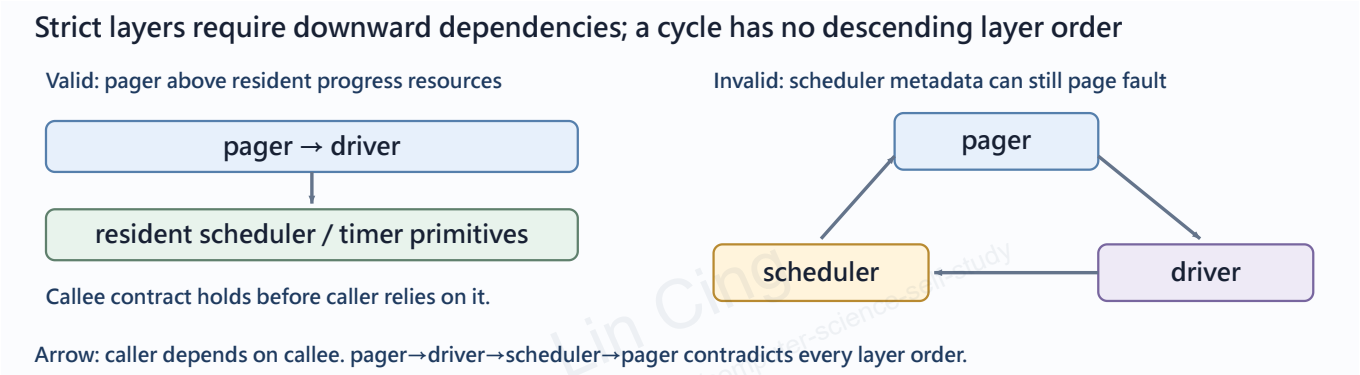
q	CPU timeline	Completion A, B, C	First response A, B, C
1	A0–1, B1–2, C2–3, A3–4, B4–5, A5–6, B6–7, A7–8, A8–9	9,7,3	0,1,2
4	A0–4, B4–7, C7–8, A8–9	9,7,8	0,4,7

Both do9 ms useful work. Mean response is1 versus11/3 ms. The q 1 trace has7 actual task switches; q 4 has3. Final A→A is no task switch. Nonzero switch cost changes wall times and requires recalculation.

Layered proof requires acyclic dependencies

Strict layers use only lower-layer contracts. If layer0 satisfies C0 and each layer i establishes Ci assuming C0…C(i−1), induction establishes upper properties. Tests do not make lower layers infallible; false contracts, premises, or coverage defeat the inference.

Layered proof requires acyclic dependencies (continued)



Pager→disk driver→scheduler→pager cannot have strictly descending layer numbers. Keep necessary scheduler metadata and fault/I/O-progress resources resident, and separate lower scheduling primitives that do not depend on paging. Hiding an arrow does not remove the cycle.

Implementation language and measurable goals

Higher-level languages can improve readability, portability, and optimization. They do not automatically provide memory safety, atomicity, or hardware ordering; C pointer errors can corrupt a kernel. Assembly remains at machine-specific entry, context, and privileged-instruction boundaries. One library API does not erase ABI differences.

With queue capacity1024, identical pregenerated operations, and identical machine resources, candidate A loses no elements. Candidate B loses an enqueued element in one recorded interleaving. B violates correctness even if faster. This is logical exclusion, independent of a throughput threshold.

CPU time per request, end-to-end latency, throughput, and p99 have different denominators. Define workload and quantile rules. Zero-cost schedule tables describe only that model. Real comparison needs switching, cache, and wait measurements, without inventing throughput or p99.

A fixed cold read in two structures

Use read(fd, user_buf,4096) from `/srv/data.bin`, offset0, cold page cache. ABI arguments use registers, with no additional parameter copy. The monolithic VFS, filesystem, and block driver share kernel execution. In the microkernel model, file server and driver have separate user address spaces, kernel mapping is in the current user AS, and each IPC uses sender→kernel→receiver two-copy delivery.

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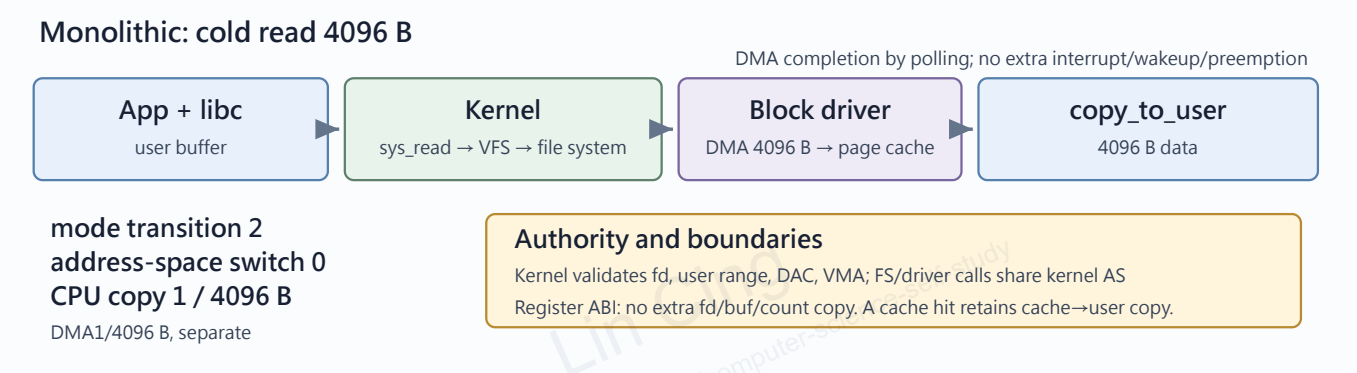
A fixed cold read in two structures (continued)



The application uses libc, enters kernel mode, validates fd/rights/buffer/offset, calls VFS→filesystem→driver, transfers4096 B by DMA into page cache, copies4096 B to user, and returns.

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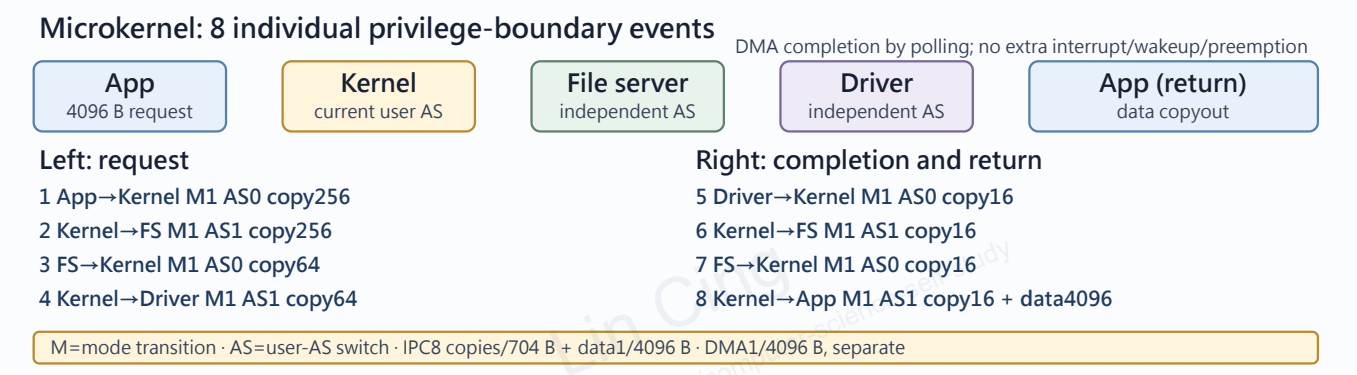
A fixed cold read in two structures (continued)

Event	Boundary	Cost
libc read	ABI call	0
sys_read	User→kernel, register args	Mode+1
VFS/filesystem/driver	Same-AS function calls	AS+0
Disk→page cache	4096 B DMA	CPU copies+0
Cache→user	4096 B CPU copy	CPU copies+1
Return4096	Kernel→user	Mode+1

Total2 mode transitions,0 AS switches,1 CPU copy/4096 B,1 DMA/4096 B. A cache hit removes driver/DMA but retains the cache-to-user copy.

Eight microkernel boundary events



App, FS, Driver are three user processes. Entering the kernel retains the current user AS; handing control to another user process changes AS.

Eight microkernel boundary events (continued)

Step	Boundary	IPC action	Control copy	Mode	AS	CPU copies
1	App→Kernel	SYS_IPC_SEND(read_req), register args	256 sender→kernel	1	0	1
2	Kernel→FS	Deliver request	256 kernel→receiver	1	1	1
3	FS→Kernel	SYS_IPC_SEND(block_req)	64 sender→kernel	1	0	1
4	Kernel→Driver	Deliver block request	64 kernel→receiver	1	1	1
5	Driver→Kernel	Completion send	16 sender→kernel	1	0	1
6	Kernel→FS	Deliver completion	16 kernel→receiver	1	1	1
7	FS→Kernel	Reply send	16 sender→kernel	1	0	1
8	Kernel→App	Deliver reply and copy cache	16 kernel→receiver plus4096data	1	1	2

Eight microkernel boundary events (continued)

Total8 modes,4 user-AS switches,9 CPU copies. Four messages256/64/16/16 B take8 copies and704 B. Add one4096 B data copy for4800 B. One 4096 B DMA remains separate. AS handoffs are App→FS, FS→Driver, Driver→FS, FS→App.

A function call is not automatically zero-copy, nor does all microkernel IPC require these eight transitions. Merging servers, private kernel AS, shared memory, or one-copy delivery requires a new ABI/deployment/count table.

DMA boundary and model sensitivity

Baseline counts cover only the listed request-chain handoffs. DMA completion uses polling, no other runnable work exists, and no timer preemption or completion interrupt occurs. Interrupt entry/exit, wakeup, and scheduling need separate rows if introduced.

Change	Mode	AS	CPU copies	Bytes copied
Monolithic, current-AS kernel	2	0	1	4096
Microkernel,256/64/16/16	8	4	9	4800
Microkernel request shrinks to128	8	4	9	4544
FS and driver in one process	4	2	5	4640
Monolithic, private kernel AS	2	2	1	4096
Microkernel, private kernel AS	8	8	9	4800

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DMA boundary and model sensitivity (continued)

The128 B request gives2×128+2×64+4×16=448 control bytes, plus4096=4544. Merged FS/driver retains256 B request and16 B reply,544 control bytes plus4096=4640. Private-kernel-AS changes AS at every mode transition in both architectures. These are model sensitivity results, not universal syscall costs.

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Four independent structural dimensions

Structure	Calls/dependencies	Privilege and AS	Extension	Failure contract
Monolithic	Mostly kernel function calls	Shared kernel privilege	May load modules	Bad pointer can corrupt kernel
Strict layers	Only lower interfaces	Layer need not be protection domain	Static or dynamic	Contracts support induction, not automatic isolation
Microkernel	Small kernel plus server IPC	Isolated user domains	Replace servers separately	Parsers, authority, recovery still need correctness
Modular kernel	Defined module interfaces	May share kernel privilege	Runtime load/unload	Maintainability is not fault isolation
Hybrid	Deployment varies by subsystem	Label each actual domain	Mixed module/server	Name determines neither cost nor security

Four independent structural dimensions (continued)

Layers describe dependencies, modules organization/linking, and protection domains access control. Separate files/modules create no automatic user/kernel boundary. Stacked boxes do not prove acyclic dependencies.

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Isolation and recovery are separate

A file server crashes after receiving append R but before replying. It may have crashed before commit or committed and lost the reply. Retrying repairs the first history but duplicates the second. Memory isolation alone cannot distinguish them.

Recoverable request IDs and dedup/result records can return the same result on retry. Records must persist atomically with the append or crashes can still break consistency. Limit device, memory, and handle authority. A server with unrestricted disk writes can corrupt disk data even when unable to overwrite kernel memory.

Module unloading separately must stop new entry and wait for references, DMA, callbacks. Removing a symbol-table entry does not stop old function pointers. This differs from restarting a user-space server.

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Historical hybrid systems

The textbook's Mac OS X has Aqua/Cocoa above a kernel environment containing Mach, BSD, I/O Kit/extensions. Mach's presence does not imply all BSD services are separate user processes. The historical iOS stack has Cocoa Touch, Media Services, Core Services, Core OS.

The historical Android stack is applications→framework→libraries/Dalvik→Linux kernel. Dalvik is the textbook runtime, not a timeless claim about all Android. Linux supplies process, memory, device services without implementing every Java API's high-level meaning. Compare the contract, domain, boundary of each component, not stack names or assumptions about later versions.

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Evidence at failure time

A user core preserves some process state; a kernel crash dump preserves kernel failure state. Logs contain selected events; traces expose recorded calls, scheduling, I/O ordering. None alone reconstructs every past event.

Evidence	Can answer	Cannot alone establish
PC, registers, stack	Failing instruction/call chain	First corruption happened there
Object/heap snapshot	Values at crash	Who changed them and when
Timestamped trace	Recorded order and timing	Unprobed events never occurred
CPU sampling	CPU-active stacks	Every long I/O wait was sampled

A broken filesystem may also fail when asked to store its kernel's dump. Reserve a reliable dump path/space and retrieval procedure. Collection failure does not establish absence of failure.

Decompose a 10 ms request

Assume comparable monotonic time and no lost events. Handler starts0, blocks for I/O at1, becomes ready at8, runs at9, returns 10. CPU[0,1)+[9,10)=2 ms; blocked[1,8)=7 ms; ready wait[8,9)=1 ms; total10 ms.

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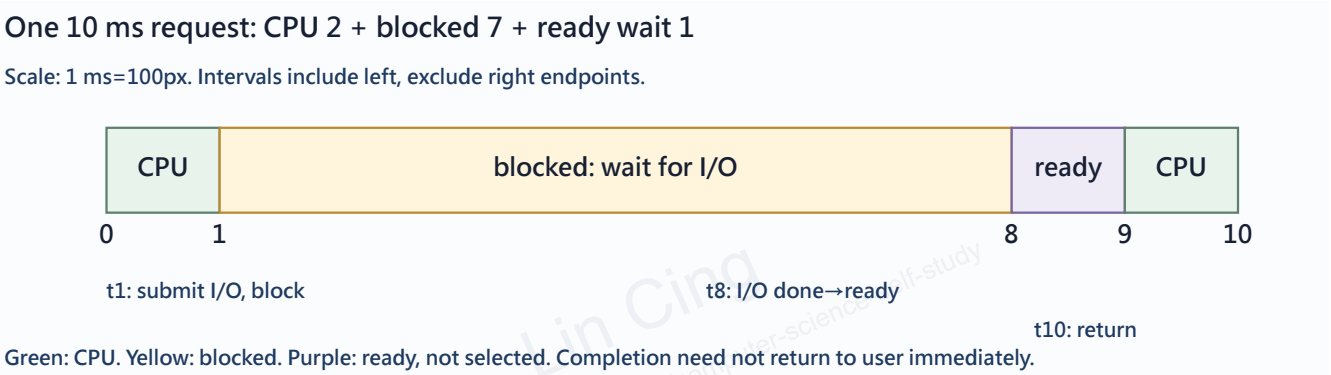
Decompose a 10 ms request (continued)



Seven blocked milliseconds are not disk CPU time. Device completion at8 is not return at8. Probes can change scheduling, so compare observable behavior with uninstrumented execution. For1,1,1,1,1,1,1,100 ms, mean107/8=13.375 ms. Nearest-rank p99 uses ceil(.99×8)=8 and yields100 ms. This small sample's order statistic is not a stable population-tail estimate.

Instrumentation can conceal a race

From x=0, T1: r0→T2: r0→T2: w1→T1: w1 produces1; serial increments produce2. printf can serialize the execution without repairing synchronization. Correctness requires indivisibility for all allowed interleavings, through a mutex covering the whole RMW or a suitable atomic RMW. Locking writes alone leaves both initial reads0 possible.

Dynamic probes must retain event IDs, time domain, drop counts, and necessary object identities. State assumptions about ordering per-CPU timestamps.

System generation and dependency closure

SYSGEN can select source at compile time, precompiled components at link time, or runtime functions through tables/discovery. These trade image size, portability, hardware-change cost, and early availability. Recompilation alone does not classify the design.

Configuration	Requirement	Failure if missing
CPU/ABI	Compatible kernel/user ISA/ABI	Unsupported instruction or exec format
RAM/reservations	Avoid firmware/device-reserved ranges	Corruption or invalid mapping
Boot storage	Controller driver reads root device	Root device unavailable
Filesystem	Root-format driver available early	Device exists but mount fails
Root/init	Correct root, executable init, loader	Mount works but userspace cannot start

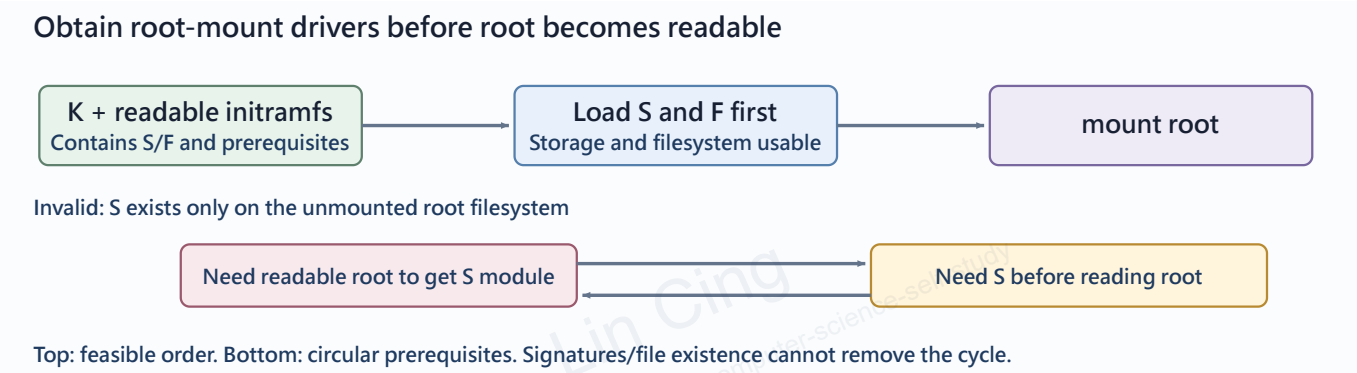
Discovery still needs a usable driver. A module's existence establishes neither version/ABI compatibility nor its availability before use.

Root drivers cannot exist only on inaccessible root

Root resides on storage S with filesystem F. Other dependencies are built in. B means built-in, I an accessible compatible initramfs module, R root-only.

S	F	Initramfs	Root mount reachable?
B	B	Neither needed	Yes
I	I	S, F, and all prerequisites	Yes, load first
R	B	S absent	No, need root to obtain root driver
B	R	F absent	No, device exists but format unavailable
I	I	Only F	No, closure incomplete

Root drivers cannot exist only on inaccessible root (continued)



Start from built-ins and readable initramfs components. Repeatedly add components whose prerequisites are all available. If mount-root never enters the set, a dependency is absent or cyclic. Closure only excludes this failure class; hardware, driver logic, and root contents still require correctness.

Reset to init

Assume firmware finds a loader, which loads kernel/initramfs and provides memory map/command line. These are roles; BIOS single-sector and UEFI implementations are not one identical flow.

Stage	Input	Next-stage premise
Reset/firmware	Reset entry, hardware	Usable CPU/RAM, executable loader
Loader	Configuration, images	Correct placement and parameters
Early kernel	Boot/hardware information	Tables, interrupts, scheduler, memory
Root setup	Storage/filesystem drivers	Mounted root, readable user files
Userspace init	Image, interpreter, libraries	Services/login under their policies

Linuxv3.2 start_kernel initializes early infrastructure; rest_init creates the task running kernel_init. It does not directly launch an arbitrary shell. Kernel_init completes initcalls/root preparation and tries the appropriate userspace init. Executable initramfs `/init` can run first; otherwise root namespace preparation is required. Failure points leave different diagnostics.

Locate the failed handoff

Use role gates F→L→K→R→I for firmware, loader, kernel/root access, readable init, and successful exec.

Fault	Last successful stage	First failing action	Incorrect diagnosis
Loader cannot find kernel	L	Load K	Init services failed
Missing S driver	K	Read root device	Only a misspelled root path
Missing F driver	K	Mount existing device	Disk absent
Missing init interpreter	R	Exec init	Kernel never started

This role table does not claim every Linux failure returns one exit code to firmware. Panic, loader error, and userspace nonzero exit are different failure paths.

Verification, measurement, and compatibility

Verified boot has a trusted predecessor authenticate/authorize the next stage and refuse execution on failure. Computing or recording a digest is measurement without automatic enforcement. Policy must name covered images/configuration/initramfs. A kernel filename alone is insufficient.

A validly signed old/revoked version may still be unacceptable, requiring rollback/revocation policy. Signatures do not remove authorized code's vulnerabilities. ABI compatibility and dependency closure support compatibility, not provenance. This abstract trust chain is separate from Linuxv3.2's control flow; it does not claim mainline3.2 implements every policy or that init immediately drops all capabilities. Service authority depends on actual work and security policy.

One read crosses several contracts

The4096 B cold-read API supplies fd, buffer, count; ABI selects registers; controlled entry establishes kernel context; validation and VFS/filesystem resolve objects; driver/DMA fills cache; the kernel copies4096 B and returns. Function calls, mode transitions, AS handoffs, CPU copies, and DMA are distinct events.

A microkernel can retain the API while adding internal IPC. GUI replaces the frontend without removing read protection. Modules change loading without automatically reducing privilege. Returned bytes mean this read's data is available, not that later writes are complete or durable. Pipe EOF requires empty buffers and no writers; missing server replies do not prove nonexecution.

Connect services through contracts/invariants, find the first broken condition using object IDs/timelines, and establish pre-handoff availability through boot dependencies. Buffer progress, fd references, address ranges, event tables, and module closure support these conclusions; architecture names alone do not.

References

- Operating System Concepts,9e, Chapter 2, printed pp. 55–93, especially Fig 2.6 API/syscall entry and Fig 2.13 layered dependencies.
- write(2),fsync(2),pipe(7).
- Linuxv3.2 init/main.c.
- The24 B argument block, RR workload, and root dependency table are assigned teaching models. Mac OS X, iOS, and Dalvik retain the textbook's historical scope.

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Processes

How does an OS represent, switch, and coordinate independent execution units that may cooperate? Prerequisites are CPU registers, address spaces, interrupts, and system calls. The goals are to trace process/PCB state, distinguish scheduling from dispatch, and specify IPC blocking, buffering, and termination.

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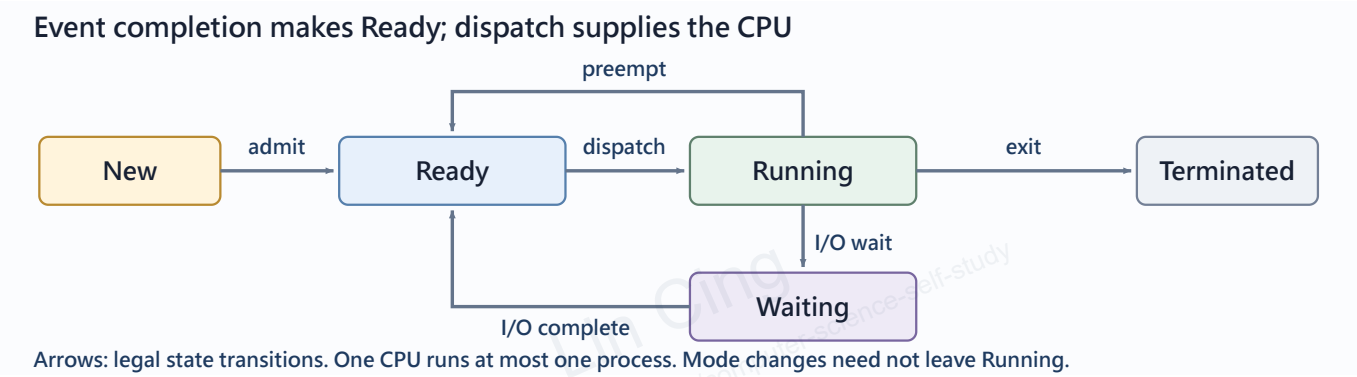
Program, process, and PCB

A program describes executable instructions/data. A process is one identified execution with resources and state. Two launches of a counter image, P100 and P101, can share read-only physical text while retaining private globals, heap, stack, registers, and descriptor tables.

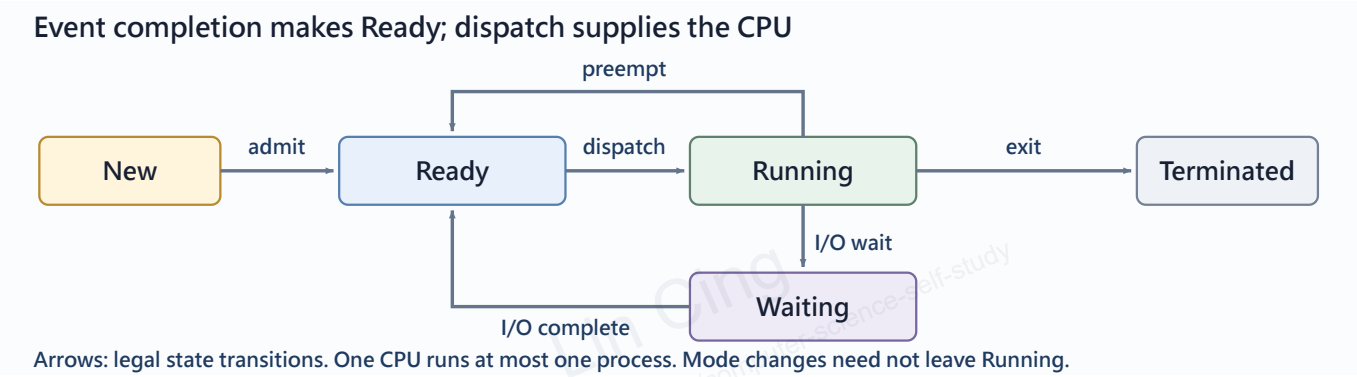
State	P100	P101	Sharing condition
PID, PC, r1	100,0x40,7	101,0x120,3	Execution context is independent
Private counter at VA0x5000	0→1	0→1	No shared mapping, no combined increment
fd3	OA→file A	OB→file B	Same number, different object
Code	Read-only image text	Same text	Frames can be shared without sharing execution

The PCB tracks identity, state, PC/registers, scheduling, and memory/I/O references. It can reference page tables and open descriptions without embedding the entire heap. Multithreaded execution separates additional thread context in Chapter 4.

Program, process, and PCB (continued)



Five states and their causes



Assume one CPU and one execution stream per process. At most one process is running. Ready means executable but awaiting CPU; waiting means an event is missing. This abstraction does not enumerate every OS substate or imply that terminated metadata has disappeared.

Five states and their causes (continued)

Admission is new→ready; dispatch ready→running; preemption running→ready; blocking running→waiting; event completion waiting→ready; exit running→terminated. I/O completion normally establishes readiness; policy determines immediate execution. Another CPU or interrupt can wake a waiter without first running the waiter itself.

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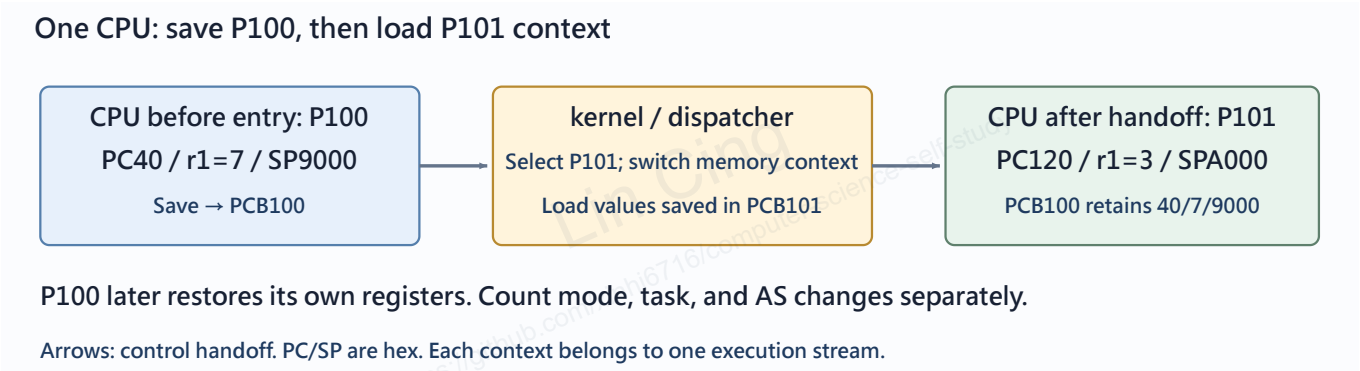
What a context switch preserves

P100 runs PC0x40, r1=7, SP0x9000. P101 has saved PC0x120, r1=3, SP0xA000. Entry first preserves ISA-required state. The kernel associates the resumable context with P100, then loads P101 and its memory context. Execution resumes at P101's PC, not P100's.



What a context switch preserves (continued)

Moment	CPU ownership	Saved/loaded state
Before entry	P100	40/7/9000
Save P100	Kernel	PCB100 retains resumable40/7/9000
Select/load P101	Kernel→P101	120/3/A000 and correct memory context
Later select P100	P100	Its own context, not P101's registers

A syscall can change mode without changing process. Switching threads in one process can change registers/stack without user-AS change. Saved PC/restart semantics depend on ISA and event; not every event repeats its instruction.

Queues, scheduler, and dispatcher

Ready queues contain runnable work; device/event queues contain waiters for predicates. Admission controls the resource-constrained job population. The short-term scheduler selects ready execution, and the dispatcher performs context/mode handoff. Long-term admission adjusts multiprogramming; medium-term suspension can remove work temporarily. An OS need not have separate components with all these names.

P1 runs, P2 is ready, and disk R1 is incomplete. P1 submits R1 and waits; P2 runs. Completion makes P1 ready, but a nonpreemptive policy can let P2 continue. Duplicate completion must not duplicate queue membership. A running task must not retain a dispatchable copy that another CPU could run simultaneously.

With10 ms useful work and1 ms scheduling/dispatch per cycle, the overhead fraction is1/(10+1)=1/11. The ratio1/10 instead measures added cost relative to useful work. This model does not measure cache/TLB disruption.

Sleep and completion must share one protocol

Initially done=false. Waiter and completion handler use lock L. Under L, the waiter checks done. If false, it joins the queue and sets WAITING before unlocking. It schedules sleep only if still WAITING. Under L, the handler sets done=true and removes a WAITING waiter and marks it READY.

Completion timing	Observation	Result
Before waiter locks	done=true	No queueing/sleep
While waiter holds L	Handler waits until registration	Handler finds waiter
After registration, before schedule	Handler changes state to READY	Scheduler must not sleep it unconditionally
After sleeping	Handler finds WAITING waiter	READY awaits dispatch

The incorrect order is check false, lose protection, completion sets true with no waiter, then enqueue/sleep. Without another completion, the wait is permanent. Saying register first is insufficient without the common lock, state transitions, and pre-schedule recheck.

CPU-bound and I/O-bound are demand patterns

A alternates CPU 1 ms/I/O9 ms; B needs CPU 10 ms. One CPU can run B during A's independent I/O. Admitting more work whenever CPU utilization is low can overload I/O queues or memory instead. Admission must consider complete demand and queueing.

A larger ready population need not improve throughput. Working sets exceeding frames can cause thrashing and more faults. Suspension/resumption is medium-term resource control, not each CPU selection. Chapter 6 defines waiting, response, and turnaround on common workloads.

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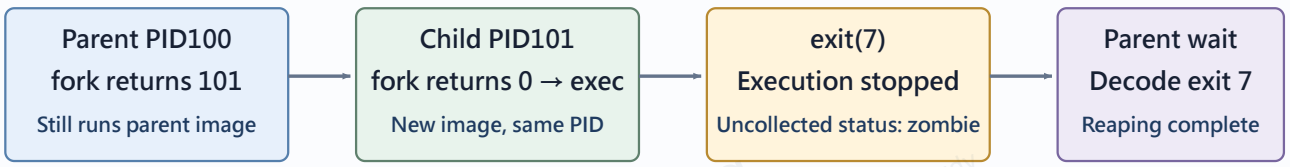
Fork, exec, exit, and wait

Fork creates a child with its own task/address-space metadata and required resource references. Private data pages can defer copying with COW. Exec replaces the caller's image; success does not return to its old call site and creates no additional process. Exit ends execution and releases most resources. Wait retrieves child status and completes reaping.

Parent100 forks101: parent returns 101, child0. Child execs worker but remains101. Worker exits7; collectible status remains until wait. The parent checks normal exit and decodes7. Zombie means terminated but unreaped; orphan means the parent relationship has ended, and the child may still run. Reparenting depends on OS, namespace, and subreaper policy. Parent termination can trigger cascading termination or reparenting under the actual system contract.

Fork, exec, exit, and wait (continued)

PID101: fork creates, exec replaces image, exit retains status, wait reaps



A living child after parent exit is orphaned. That differs from this terminated zombie.

Arrows: lifecycle order. Fork and exec succeed. These are not four processes.

Two unconditional forks produce four processes

Assume successful forks, no early exit, and every current process executes both unconditional lines. P0 first creates P1. On the second line, P0 creates P2 and P1 creates P3.

Completed point	Executing processes	New children	Alive
Initial	P0	None	1
Fork1	P0	P1	2
Fork2	P0, P1	P2, P3	4

For k such forks, N(k+1)=2N(k), N0=1; induction gives2^k. Parent-only continuation, child exec/exit, or failure breaks the premises. Only the original parent's second fork gives three total. The tree does not fix PID allocation or print order.

Inherited descriptors can share an open description

Parent fd3 points to O1 at offset 0. Fork copies the fd table entries but can preserve their shared O1 reference. Parent read 100 advances O1 to100; child read 50 starts100 and advances150. A separate open of the same inode creates O2 with an independent offset, possibly0. This differs from private-memory COW.

Fork failure is an error, not the child branch. Exec failure usually returns an error, but after an implementation's point of no return the old image cannot universally be guaranteed intact. Repeated unwaited child exits accumulate status records. General memory/handle release differs from retaining minimal reap metadata.

A multithreaded fork preserves only the calling thread in the child. Copied mutex state can have no surviving owner. Chapter 4 restricts operations before exec; the single-thread flow cannot simply be reused.

Shared memory and message passing

Shared memory establishes common backing/mappings and then exchanges data through loads/stores. It still needs authority and synchronization. Message-passing contracts determine copying, waiting, capacity, and delivery. They do not imply cross-machine execution, zero-copy, or exactly two copies.

Contract	Options to state	Risk if confused
Naming	Process/endpoint or mailbox	PID reuse, rights, cleanup
Synchronization	Each send/receive blocking or nonblocking	Send may await queueing, not processing
Capacity	Rendezvous, bounded, ideal unbounded	Backpressure or finite physical-resource limits
Data boundary	Messages or byte stream	One read need not be one record
Lifetime/failure	Close, unlink, remove, peer crash	EOF, error, and queue removal differ

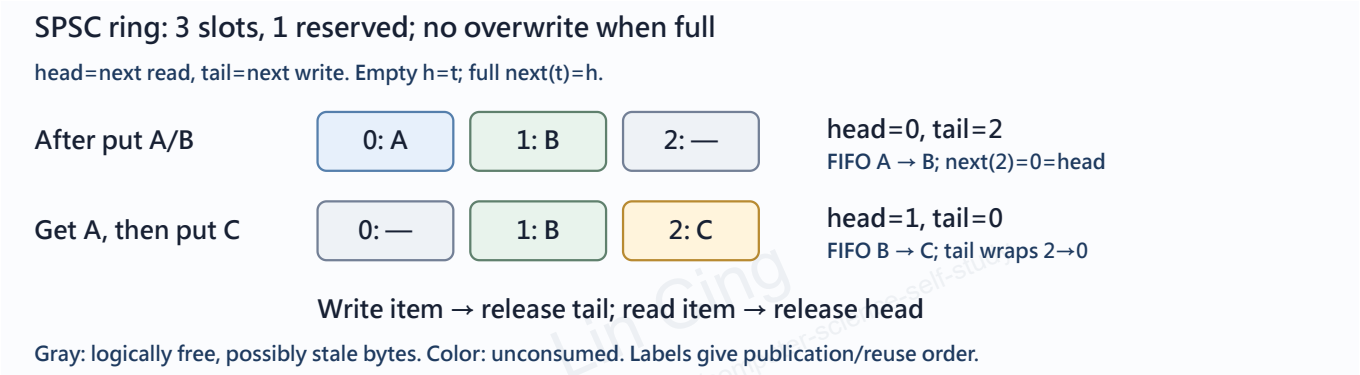
A capacity 2 channel holds complete J1/J2; J3 waits while full. Receiving J1 permits J3. Blocking APIs do not imply zero-capacity rendezvous if send completion means queue insertion.

Three physical ring slots hold two items

One producer and one consumer share slots[3] and atomic head/tail, initially0. Head is next to consume, tail next to fill. Reserve one slot. Empty means head=tail; full means next(tail)=head; next(x)=(x+1) mod3.

Event	Head	Tail	Pending FIFO	Result
Initial	0	0	[]	Empty
Put A at0	0	1	[A]	Publish tail1
Put B at1	0	2	[A, B]	Full, next2=0=head
Attempt C	0	2	[A, B]	Must not publish overwrite
Get A at0	1	2	[B]	Release head1
Put C at2	1	0	[B, C]	Wrapped, still full
Get B then C	0	0	[]	Empty

Three physical ring slots hold two items (continued)



Publication and reuse need two ordering arguments

Producer acquire-loads head, checks not full, writes slots[tail], then release-stores new tail. Consumer acquire-loads tail, checks not empty, reads slots[head], then release-stores new head. Producer alone writes tail; consumer alone writes head. Local indexes and cross-party loads follow the stated atomic rules. Process-shared atomics require platform support; mapping an arbitrary C++ atomic object across processes is not itself a portable guarantee.

First, slot write precedes release tail, and the consumer's corresponding acquire makes the completed item visible. Second, slot read precedes release head, and the producer's acquire permits reuse only after consumption. Successful put/get increases/decreases logical size by one. Empty/full checks preserve$0 \le \text{size} \le 2$ and FIFO.

Plain shared integers introduce language data races and invalid ordering assumptions. Publishing tail before data permits stale reads. Multiple producers compete for the same tail/slot and violate SPSC. Finite index traces do not replace the general release/acquire argument.

Frame records over a byte stream

Each record has a2 B big-endian length and payload, max8 B. Payloads ABC and DE form00 03 41 42 43 00 02 44 45,9 B. Reads can return[00],[03 41 42],[43 00 02 44 45]. These are not three messages.

Accumulate two header bytes to decode3, then collect ABC. Decode the next length 2 and collect DE. EOF inside a header or payload is truncation, not an empty completed message. Validate length≤8 and allocation arithmetic before receiving untrusted-sized data. Pipe/TCP ordered streams do not provide application records or atomic remote procedures. Datagram/message truncation and boundaries follow their own APIs.

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Shared names, descriptors, and mappings

Creator makes `/course-ring`, sets length 4096 B, and maps MAP_SHARED. Consumer opens/maps the name. P1's base0x100000 and P2's0x700000 can reference the same backing offset 16. Store offsets/indexes, not P1 absolute pointers intended for P2.

Closing a descriptor does not unmap existing mappings. shm_unlink removes the name, preventing new opens through it, while existing references/mappings can survive. MAP_SHARED does not publish completed initialization. A consumer accessing beyond valid object length before sizing is not merely reading an ordinary zero-filled buffer.

Put magic, version, length, and offsets in the region; complete initialization before a synchronized ready notification. Sleeping one second is no substitute for publication.

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Removal and drain differ

Object	Identifier	Removal	Existing users
POSIX message queue	Name/descriptors	mq_unlink removes name now	Open references survive until final close
System V queue	msqid/key	IPC_RMID removes queue	Blocked send/receive wake with EIDRM
Custom drain channel	Application handle	Close sends, drain, destroy	Only its specified state machine applies

Removal does not universally await consumers draining. Unlink does not automatically send every receiver EOF. Termination and errors require application protocols.

Sentinel termination

Use capacity 4 FIFO, one producer/consumer, positive jobs, and0 reserved as STOP. Send1,2,3, stop accepting new jobs, then send0. Queue states are[1]→[1,2]→[1,2,3]→[1,2,3,0]. Consumer completes the jobs before STOP, acknowledges termination, and manager cleans up after acknowledgment. This is not kernel-provided drain.

If0 is a legal job, use tagged messages. With two consumers, one STOP can leave the other waiting forever; use two sentinels or a closed/broadcast protocol. A crashed consumer can prevent acknowledgment forever, requiring recovery/reassignment policy. One normal trace does not establish exactly-once effects.

Mach port and reply rights

In the textbook's historical port/mailbox model, the server has receive right and clients need send rights. Headers specify destination/reply port; typed bodies carry data or controlled rights. Knowing a local port name grants neither global pointer identity nor receive authority.

A sends A1 then A2, and B sends B1. A1, B1, A2 or B1, A1, A2 are allowed, but A2 must not overtake A1 from the same sender under this model. Different senders lack automatic global order. Replying to A1 leaves other requests pending; send success does not equal reply success.

A full queue can wait, time out, or fail under its selected contract. Receiving requires rights. Out-of-line VM/COW can reduce large-data copies without making all inline headers or cross-host messages zero-copy. Rights and shared-memory lifetimes remain protocol obligations.

Historical Windows ALPC connection

A server publishes a connection port. An authorized client handle sends a connection request; acceptance creates private communication endpoints for both directions. Connection setup differs from repeatedly treating each payload as a new connection.

The textbook distinguishes copied small messages from shared-section large data using a historical256 B threshold. This is not a fixed modern Windows limit. A128 B request fits the small-message model; a4096 B payload can use a prepared section with offset/length control. Validate mapping bounds, rights, and writers; a client pointer is untrusted. Applications commonly reach this local mechanism through public RPC interfaces. ALPC, Mach, and POSIX queues share questions of naming, rights, waiting, data, and replies but not interchangeable lifetime contracts.

Sockets and RPC stubs

Sockets identify endpoints through address family, local/remote addresses, and ports. A stream server bind/listen/accepts; accept returns a connected endpoint while the listener accepts more. A port is not a PID. Connect success does not establish application-operation completion.

A client stub encodes procedure/request IDs and parameters. A server stub validates/decodes, invokes service, and encodes results. Native endianness, alignment, widths, and pointers are not a wire format. Use id32/opcode16/amount32, all big-endian. ID77, opcode1, amount25 produce00 00 00 4D | 00 01 | 00 00 00 19,10 B. Validate length/opcode before decoding amount. A transmitted client pointer does not grant memory access.

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One timeout, two histories

Balance starts100; request 77 debits25. In history A, the request never arrives and balance stays100. In B, the server commits75 but loses the reply. The client sees timeout in both. Retrying with a new ID makes B50; not retrying leaves A unperformed.

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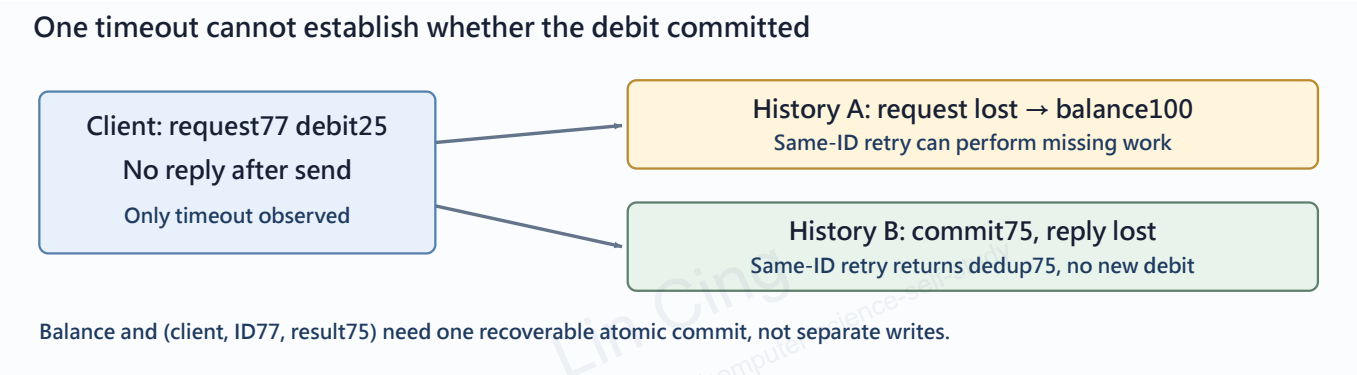
One timeout, two histories (continued)



Deduplicate by authenticated(client identity, request ID). Commit balance and recorded result atomically in one recoverable transaction. First77 changes100→75 and records75; duplicate77 returns 75 without another debit. Concurrent duplicates need a transaction/unique constraint, not independent absent checks followed by two debits.

At-most-once is separate from availability

Dedup records can bound an ID's effects to at most one during retention. Debit then separately record ID leaves a crash window. RAM-only records disappear on restart. Expiring a record while retries remain possible needs an epoch/retention agreement; identities and ID spaces cannot be arbitrarily reused.

At-most-once permits zero execution after permanent loss. At-least-once needs retries and eventual reachability but can repeat non-idempotent effects. Exactly-once requires explicit transaction, durable record, ID scope, failure, and communication assumptions. Reads or setting a value may be idempotent; debit 25 is not.

Anonymous pipes/named FIFOs support local stream inheritance/naming; sockets generalize endpoints across hosts. All still need framing, backpressure, and peer-failure handling. IPC costs also include waking, synchronization, references, and retries.

Execution and data lifetimes

Capacity2 holds J1/J2. P waits trying J3. C receives J1, frees a slot, and wakes P. P becomes ready and sends J3 after selection. FIFO/capacity remain valid; waking is not immediate execution. Receiving an item is not durable job completion; process exit is not parental reaping.

PID/PCB identifies execution, fd entries reference open descriptions, shared offsets identify data, and request IDs identify retry effects. Equal numbers or paths do not replace references/lifetime proofs.

Correctness requires matching saved/restored execution contexts; one synchronized wait predicate; write-before-publication and read-before-reuse; only complete valid framed records; and joint RPC effect/result commit. Numerical traces expose intermediate states. General invariants explain additional allowed events. Finite testing does not prove every interleaving/crash. New APIs, participant counts, memory orders, or persistence strategies require rechecking their premises.

References

- Operating System Concepts,9e, Chapter 3, pp. 105–147, Fig 3.4 context switch, Fig 3.13 bounded buffer. Mach/ALPC retain historical textbook scope.
- pipe(7),msgctl(2),mq_unlink(3),pthread_atfork(3).
- Process-shared atomic support, release/acquire, RPC commit/dedup, and framing are explicit premises. Examples simulate sequential state; ring methods are atomic model operations and ledger updates one recoverable commit.

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Threads

How can execution streams share process resources while preserving scheduling and data correctness? Prerequisites are address spaces, CPU scheduling, stacks, and registers. The goals are resource-boundary analysis, mapping-model comparison, and correct cancellation, signals, TLS, and fork.

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Private context and shared resources

A thread has its own PC, registers, stack, scheduling state, and identity. Threads usually share process code, globals, heap, address space, and open resources. One bad pointer can corrupt another thread's data. Separate stacks describe logical ownership, not hardware isolation within one address space. Another thread can access a passed stack pointer while its object remains alive.

Worker data	W1	W2	Coordination
Register/local index	Own index 0	Own index 0	Independent context
Scratch	B1 holds 110	B2 holds 120	Disjoint ownership
Shared queue	Takes job 11	Takes job 12	One protocol prevents duplicate removal
Global total	Adds 110	Adds 120	Atomic/mutex, or one reducer after joins

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Private context and shared resources (continued)

Responsiveness can come from a UI thread continuing while workers await I/O. Sharing can reduce some process-creation costs. Scalability needs cores and separable work. These are conditional benefits, not fixed speed factors. More threads do not remove locks, bandwidth limits, or serial dependencies.

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Globals, TLS, and lifetime

Unsynchronized increments of one non-atomic C/C++ counter are a data race and undefined behavior. Read 0/read 0/write 1/write 1 is an abstract lost-update witness, not the language's unique output.

With per-thread TLS counters, each independently goes 0→1. Main can collect and reduce to 2 after join. TLS persists across functions for the thread's lifetime, unlike one function's automatic local. A pool worker does not reset TLS per job automatically; initialize and clean it at job boundaries. A mutex cannot make a pointer to an ended thread's stack-local result valid.

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Concurrency and decomposition

Concurrency permits overlapping or interleaved work, even on one CPU. Parallelism requires simultaneous hardware execution resources. SMT contexts share core resources; logical CPUs are not necessarily independent full cores.

Data parallelism applies one operation to partitions. Task parallelism overlaps functions or stages. Decomposition must address partitioning, balance, dependencies, sharing, and testing. A 64 MiB array of 32-bit integers contains 16,777,216 elements. Eight partitions each contain 8 MiB or 2,097,152 elements. Each worker uses a sufficiently wide 64-bit accumulator. Seven additions combine eight partials. Every element must still be read.

Per-element global atomic addition instead performs 16,777,216 shared RMWs. Publishing eight partials changes operation counts, not a measured speedup. Bandwidth, caches, and NUMA placement still matter.

Amdahl uses baseline time

For the same fixed problem, T1=100 ms contains 20 ms serial and 80 ms ideally divisible among P cores, without overhead. With s=0.2, $T_P = 100[s + (1 - s)/P]$, and speedup is $1/[s + (1 - s)/P]$. At P=8, T8=30 ms, speedup 10/3, efficiency 5/12. As P grows, speedup$\leq$5.

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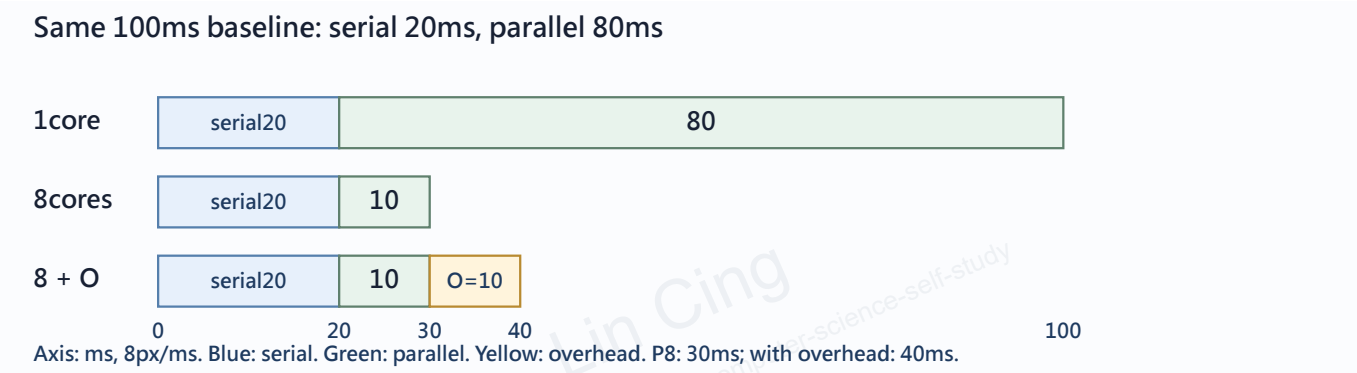
Amdahl uses baseline time (continued)

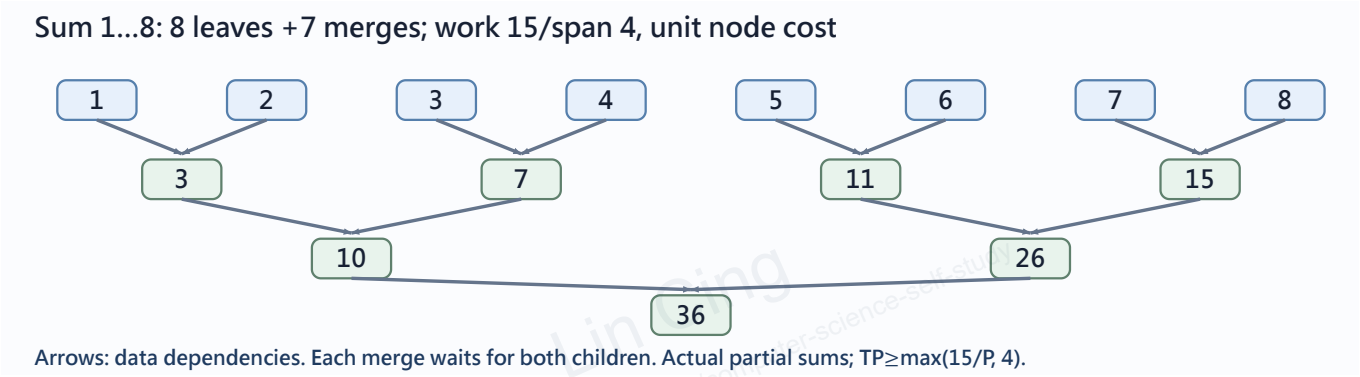


Adding 10 ms launch/synchronization overhead makes T8=40 ms and speedup 2.5. Eighty percent of operations need not mean 80% of baseline time. Eighty 1-cycle and twenty 10-cycle operations cost 280 cycles. Accelerating only the first group eightfold gives 10+200=210 cycles, speedup 4/3. Retain one denominator.

Work, span, and reduction

Eight leaves read values 1…8, and seven binary merges add them. Each leaf/merge costs one unit. Work=15. The longest path leaf→pair→quartet→root has span 4. Every P-processor schedule has $T_P \geq \max(15/P, 4)$, so speedup≤15/4. Work/span is average parallelism, not a theorem that processors beyond 3.75 cannot help.

Work, span, and reduction (continued)



Pair sums are 3,7,11,15; quartets 10,26; root 36. Exactly seven merges occur. Root must await both quartets. Unequal partitions impose a lower bound equal to the largest partition time, not the average.

False sharing differs from a data race

With 64 B lines, 4 B integers, and a 64 B-aligned base, W1 writes a[0] and W2 a[1]. Distinct objects may avoid a language data race while repeatedly transferring one coherence line's ownership. Separately aligned regions at least 64 B apart can reduce interference; measurement is required to claim improvement.

A mutex can add serialization. Replacing a shared counter with TLS changes abstract data and still needs reduction. Correctness, line-sharing cost, and result combination are separate obligations.

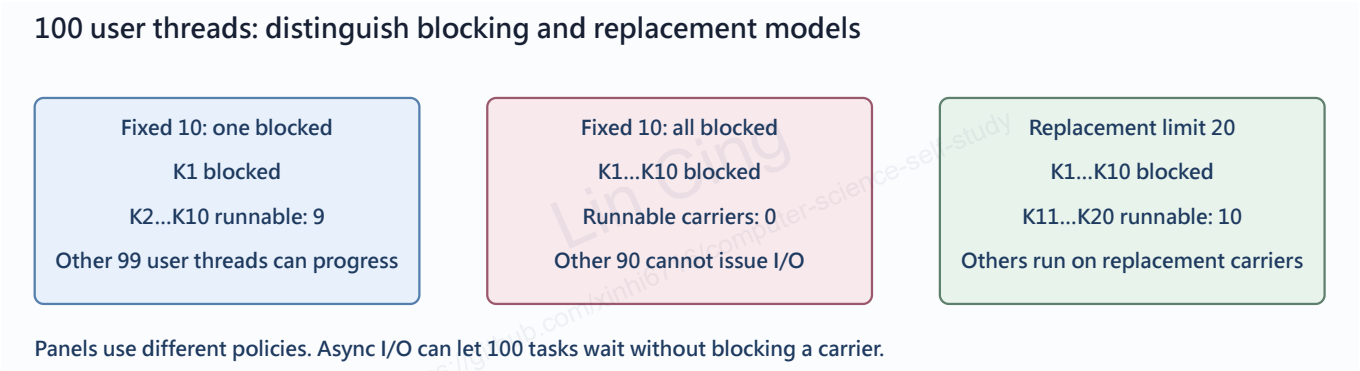
User-to-kernel mappings

Model	Mapping	Real blocking syscall	Parallelism
Many-to-one	Many user threads, one carrier	Without async substitution, blocks all execution there	One carrier
One-to-one	One kernel entity per thread	Other entities remain schedulable	Ready entities and cores
Many-to-many	Runtime multiplexes carriers	Depends on carriers and compensation	Runnable carriers and cores
Two-level	M:N with selected bound threads	Bound/unbound rules differ	Binding restricts runtime choices

A Pthreads API does not choose one universal mapping. Many user threads do not imply equal kernel-object counts. Nonblocking I/O and an event loop can leave many tasks logically waiting without blocking carriers.

One hundred user threads, ten carriers

Fix K1…K10, no compensation or async I/O, and 100 threads each wanting blocking I/O. After U1 blocks K1, nine carriers remain. After ten block, none runs. The other 90 threads stay ready inside the runtime and cannot issue their syscalls.



One hundred user threads, ten carriers (continued)

A separate policy provides replacement carriers up to 20. K11…K20 can run after K1…K10 block and can issue additional blocking calls. Async I/O can let all 100 tasks wait while no carrier blocks. A ten-waiter bound is conditional, not universal M:N behavior.

Historical scheduler activations/upcalls coordinate kernel resource notifications with runtime selection. They need rules for runtime locks, notification context, and allocation. They create no CPU capacity. Binding does not let one thread run on two carriers simultaneously.

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Creation, completion, and join

Main owns Work{input14,output}; worker writes output=3×input=42. Initialize input before create, keep Work alive until successful join, and do not read output before completion synchronization. After create, the worker may already have started or finished.

Stage	Main	Worker	Result rule
Before create	Initialize 14	Absent	Valid object and arguments
After success	Nonconflicting work	May start, finish, write 42	No fixed parent/worker order
Worker complete	May await join	No longer uses Work	Publication follows library contract
Join success	Read 42, then end lifetime	Finished	No overlapping write

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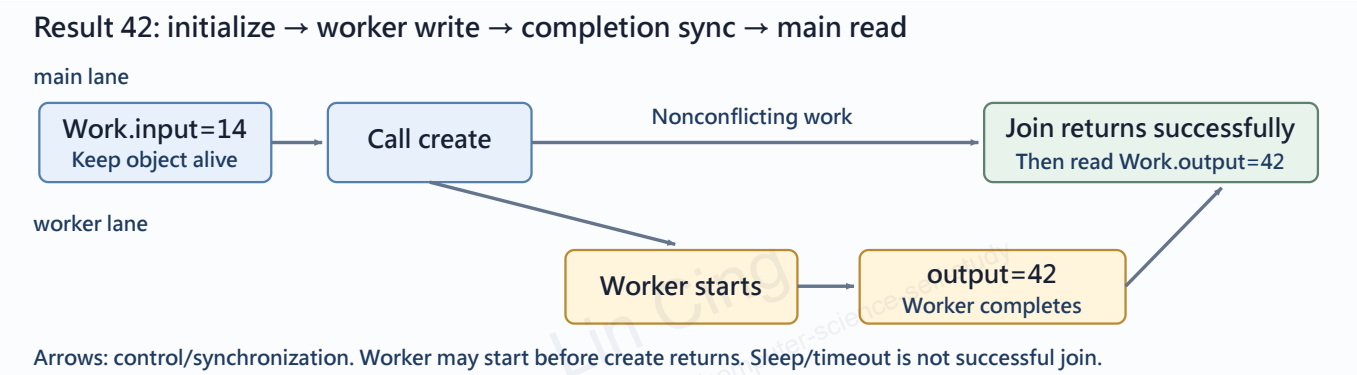
Creation, completion, and join (continued)



Join is not a fixed sleep. Sleeping 10 ms proves no completion. Detached threads need another lifetime/notification protocol. Detach does not authorize immediate parameter free.

Pthreads, Windows, and Java

Pthreads uses pthread_create with a routine and void* argument. Successful pthread_join waits for a joinable target and obtains its result. Handle its returned error number, not arbitrarily errno. Define join ownership; multiple joiners are not a generally safe broadcast. Never return a pointer to the worker's dead stack local.

Windows uses CreateThread and a handle. Read 42 after successful signaled WaitForSingleObject, not timeout or WAIT_FAILED. CloseHandle remains necessary and does not terminate the thread. Synchronization supplies ordering; volatile is insufficient.

Java 8 Thread.start creates execution; directly calling run is an ordinary method call on the current thread. Pre-start actions happen-before the worker. Worker actions happen-before successful join return. InterruptedException does not establish completion. API/language contracts and host mapping are separate layers.

Pools and fork-join

Eight 1-unit jobs on four FIFO workers with no overhead run J1–J4 in [0,1), J5–J8 in [1,2), makespan 2. Worker reuse avoids per-job creation, but bounded queues, backpressure, cancellation, and errors remain. Worker count and queue capacity limit different resources.

Fork-join splits independent subtasks and combines after completion. The sum tree gives 3/7/11/15→10/26→36. Too-fine granularity can make task, queue, and synchronization costs exceed useful work.

In a two-worker pool, A and B each submit A1/B1, then block waiting. Children are ready but no worker is free. This starvation deadlock can occur with an acyclic task DAG. Help-join/work stealing may execute dependencies while waiting, but arbitrary pools and Futures do not promise it.

OpenMP and GCD

OpenMP parallel-for partitions iterations. Independent c[i]=a[i]+b[i] can use distinct i; shared sum+=a[i] needs reduction/synchronization. Four workers on 1…8 produce private 3/7/11/15, then 10+26=36. Reduction is not arbitrary concurrent plain-sum writes. Floating-point reassociation can change final bits.

Historical GCD queues submit blocks to managed workers. A serial queue completes one at a time. Concurrent queues allow overlap; FIFO enqueue does not imply FIFO completion. A block on serial Q that synchronously submits another to Q and waits creates self-wait. Different queues do not automatically exclude shared-object access.

Runtime scheduling leaves dependencies, shared/private data, lifetime, and unsafe blocking to the application. A directive cannot erase loop-carried dependencies.

Multithread fork and signals

T2 holds M when T1 forks in a POSIX-style process. Child retains only T1, but copied M can appear owned by absent T2. A pre-exec library call needing M can stall. Before exec, use the required async-signal-safe operations. Atfork requires coordinated library/lock design and cannot universally repair unknown third-party locks.

Disposition is generally process-shared; mask is per thread. Synchronous faults target their causing thread. Process-directed signals go to an eligible thread under mask/implementation rules, not once per thread. Dedicated sigwait needs coordinated masks elsewhere. Signals and condition variables are different objects.

Successful exec replaces the image; other threads do not continue the old image. Ordinary thread creation shares it. These choices determine lock/resource inheritance.

Deferred cancellation

Successful pthread_cancel queues a request. The target may disable cancellation or not reach a point. Deferred cancellation acts at specified points. Asynchronous cancellation can interrupt broken invariants or lock ownership.

Worker owns M and buffer B and waits on a stable predicate in pthread_cond_wait. With cancellation disabled, establish resources and push unlock-M then free-B cleanup handlers. Enable deferred cancellation. Cancellation during wait first reacquires M, then runs LIFO cleanup, freeing B and unlocking M. Main successfully joins before reclaiming joinable state. Do not enable arbitrary cancellation before registration.

Cleanup invariants

State	M	B	Free count	Unlock count
Resources/handlers ready	Owned	Live	0	0
Cond wait	Atomically released	Live	0	0
Cancellation handled	Reacquired	Live	0	0
First cleanup	Owned	Freed	1	0
Second cleanup	Unlocked	Freed	1	1

Normal execution also pops or cleans exactly once. Normal free followed by cancellation free is wrong. Releasing locks cannot repair arbitrary partial updates. Disable cancellation or avoid points while invariants are broken, or provide rollback. Include normal, error, and timeout ownership paths.

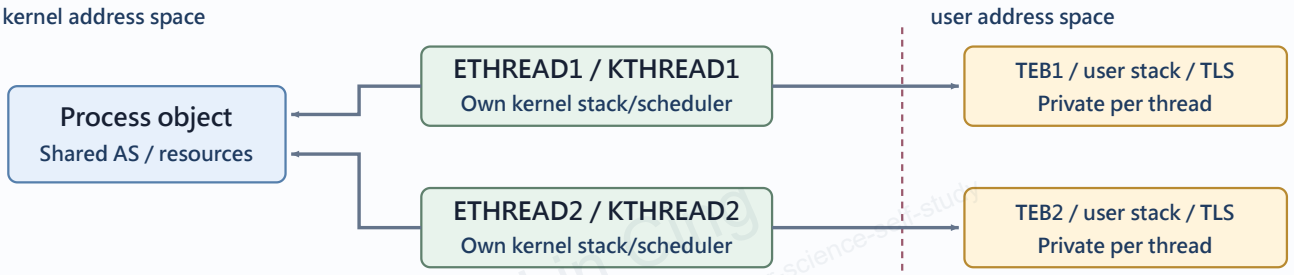
Windows 7 and Linux 3.2 objects

Fix the textbook snapshot. Windows 7 processes own address spaces/resources; threads have ETHREAD, KTHREAD, and user TEB. CreateProcessW creates an initial thread; one additional CreateThread gives two one-to-one threads. Session 0 system threads are outside this ordinary interactive-process example.

Linux 3.2 task_struct represents every schedulable task. Fork's copy_process→dup_task_struct creates a new task/kernel stack. Without CLONE_VM, copy_mm→dup_mm creates independent mm/page tables. Clone flags choose sharing of fs_struct, files_struct, sighand_struct, and mm.

Windows 7 and Linux 3.2 objects (continued)

Windows 7: 1 process, 2 threads, each with kernel objects and user TEB



Arrows: process/TEB references. CreateProcessW supplies the initial thread; one CreateThread makes two.

Windows 7 and Linux 3.2 objects (continued)

Aspect	Windows 7	Linux 3.2
Process creation	CreateProcessW	fork
Thread-like creation	CreateThread	clone with VM,FS,FILES,SIGHAND,THREAD
Execution objects	Process and thread	One task_struct per stream
Private state	Registers, stacks, TEB, TLS	Kernel stack, scheduling, registers/stack
Shared state	AS and process resources	mm, files, fs, handlers per flags
Scheduling unit	Kernel thread	Task

One process/two streams gives Windows 1 process, 2 ETHREAD, 2 KTHREAD, 2 TEB, 1 AS, 2 scheduling entities. Ordinary Linux fork gives 2 tasks, 2 mm, 2 page tables. Specified thread-like clone gives 2 tasks, 1 mm, 1 file table, 1 handler table, 2 kernel stacks.

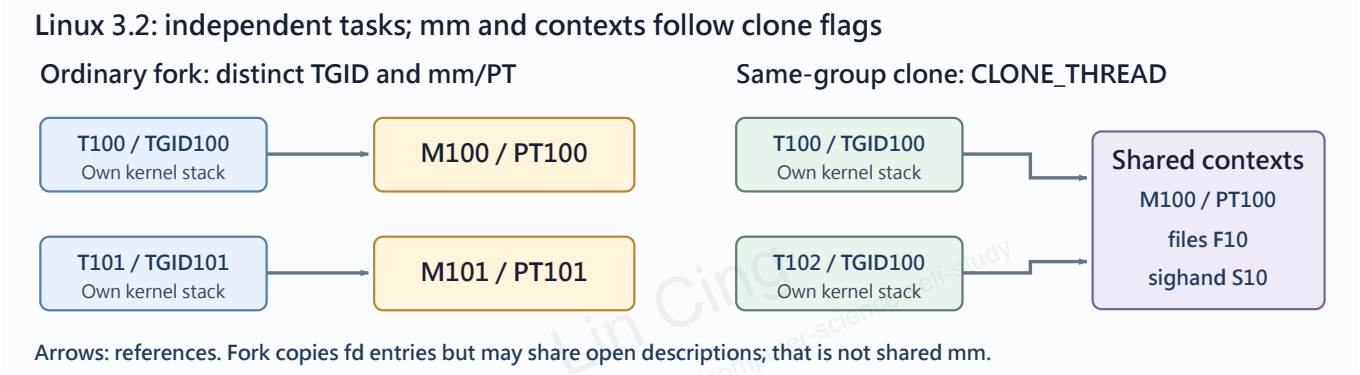
References and version counterexamples

ETHREAD references its process; KTHREAD tracks scheduling/synchronization; TEB is user space. The second thread owns independent stacks, registers, and TEB while referring to the same process/AS. Identity remains independent.

Fork creates child PID, state, stack, mm, and tables before private-page copying. COW sharing concerns physical frames, not a shared task object. Exec retains identity and changes image. Failure before destructive transition usually returns an error; some v3.2 failures after flush_old_exec terminate execution, so the old image is not unconditionally preserved.

Thread-group clone includes CLONE_THREAD, different TIDs, and one TGID. THREAD requires SIGHAND, which requires VM. VM/FS/FILES/SIGHAND with SIGCHLD may share resources without one thread group and is not a complete pthread implementation. Shared heap access still needs synchronization; ordinary fork private pages separate on first write.

References and version counterexamples (continued)



These counts describe fixed historical models, not today's kernel ABI guarantees.

Results, completion, and dependencies

Sum 1…8=36 can use one thread, four partition workers, or fork-join. Scheduling, objects, and synchronization costs differ. Read partials after joins, combine with the stated operation, and preserve lifetime.

Amdahl uses baseline-time fractions; work/span bounds a DAG; carriers bound kernel-schedulable streams. These are different bottlenecks, not proof that threads=cores is optimal or processors above average parallelism never help.

Changing 1:1 to M:N requires blocking/compensation analysis. Changing scratch to global requires conflict analysis. Detach replaces join's lifetime/completion proof. Multithread fork adds inherited-lock risk. Cancellation adds resource and abstract-state recovery.

For measurements, fix inputs/results, cores/carriers, memory placement, and time denominator. Retain failures and wait locations; mean throughput can hide lock contention, pool exhaustion, and cleanup tails.

References

- Operating System Concepts, 9e, Chapter 4, pp. 163–191. Fig. 4.6/4.7 mappings; Fig. 4.14/4.15 Windows objects/Linux flags. Historical scope is Windows 7, Linux 3.2, and textbook GCD/OpenMP.
- Linux v3.2 kernel/fork.c.
- pthread_join, pthread_cancel, cleanup, pthread_cond_wait.
- Java 8 happens-before, Windows synchronization ordering.

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Process synchronization

Shared-state races motivate critical-section safety, progress, and bounded waiting. Hardware atomics, mutexes, semaphores, monitors, and classical coordination problems show how these guarantees depend on ordering and resource protocols.

Prerequisites. Process and thread states, Shared memory, Interleaving and atomic operations.

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Learning objectives

- Reason about mutual exclusion, progress, and bounded waiting
- Use semaphores for ordering and backpressure
- Identify deadlock, starvation, priority inversion, and lost wake-ups

Central question. How can arbitrary interleavings preserve shared state while allowing progress and bounded waiting?

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5.1. Background

- A race condition makes correctness depend on uncontrolled event ordering. Shared mutable state requires a protocol for conflicting accesses. A language-level data race has a more specific definition and may make program behavior undefined.
- Atomic operations provide indivisible state changes with specified ordering. Libraries build locks and condition waits from such mechanisms.

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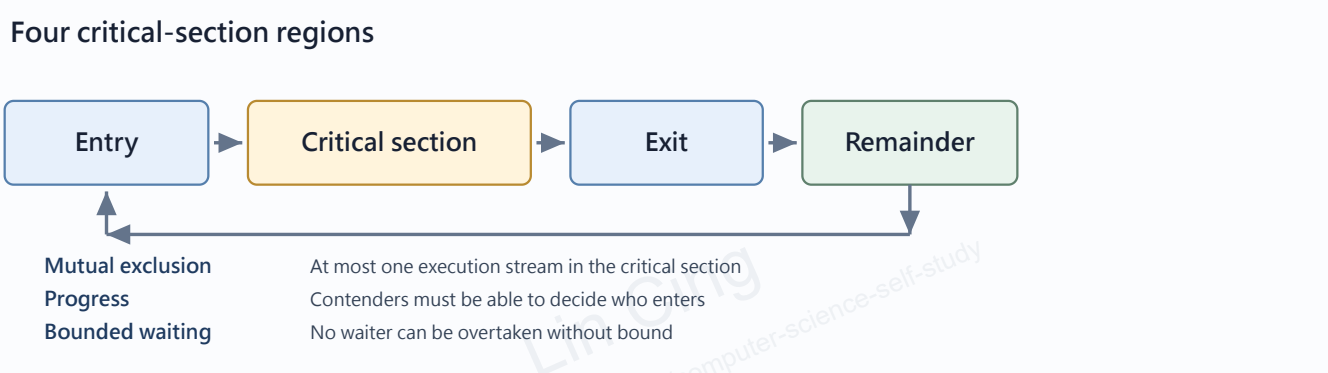
5.1. Background (continued)



Original vector diagram for this course.

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5.1. Model, ordering, and boundary cases

Model assumptions. Multiple execution contexts access shared mutable state. Atomicity and visibility are defined by the architecture and language model.

Invariant and ordering. Intermediate invalid states remain inaccessible to observers protected by the protocol. Publicly observable states satisfy the shared invariant.

Counterexample or boundary. Individually atomic changes to a list's next pointer and size do not automatically make the combined list update atomic.

Worked check. In an abstract load-add-store trace, two writers read balance 100, each adds 50, and each stores 150. The required result was 200. The rerunnable example models this interleaving without relying on an actual undefined-language race.

Reasoning prompt. Why cannot one atomic instruction always protect an invariant spanning several fields?

5.1 Race invariants and the complete counter trace

Producer executes R1=counter; R1=R1+1; counter=R1. Consumer executes R2=counter; R2=R2−1; counter=R2. The net serial effect is zero, but interleavings can leave 4,5,6 from initial 5. Treating counter++ as one step hides the conflict.

Event	Producer	Consumer	Counter	Observation
T0	R1=5		5	Old value loaded
T1	R1=6		5	Only register changed
T2		R2=5	5	Same old value
T3		R2=4	5	Only register changed
T4	counter=6		6	Producer stores
T5		counter=4	4	Consumer overwrites

5.1 Race invariants and the complete counter trace (continued)

Swapping T4/T5 leaves 6; the intended result is 5. Mutual exclusion covers only agreed regions. Visibility, reordering, non-atomic widths, and failure retries need the same synchronization contract.

Identify all shared mutable locations, state cross-field invariants such as $0 \leq size \leq capacity$ or readers=0 or writer=0, identify operations temporarily breaking them, and define which observers must be excluded plus acquire/release visibility. Atomic stores to a list's next and size do not make their combined update indivisible.

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5.2. The critical-section problem

- Mutual exclusion allows at most one participant in a critical section. Progress prevents indefinite postponement of entry selection when the section is free and contenders exist. Bounded waiting bounds how many entries by others may overtake a participant after its request.
- These are distinct safety, liveness, and fairness requirements. A mutually exclusive protocol can deadlock, and a progressing system can starve one participant.

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5.2. Model, ordering, and boundary cases

Model assumptions. Participants interleave at arbitrary relative speeds within the model. A participant that enters must eventually execute its exit for the usual progress argument.

Invariant and ordering. Entry preserves mutual exclusion, and the protocol cannot let noncontending remainder-section activity prevent a contender from entering.

Counterexample or boundary. Repeatedly selecting favored contenders can preserve system-wide progress while one waiting participant is overtaken without bound.

Worked check. Counter starts at 0. P1 and P2 each load, add 1, and store. P1.load=0, P2.load=0, P1.store=1, P2.store=1 leaves 1 after two increments. Protecting each full read-modify-write with the same mutex produces 2.

Reasoning prompt. Which counterexample refutes each requirement, and which requirements cannot imply the others?

5.2 Safety, progress, and bounded waiting

Each of n processes P0…P(n−1) has entry, critical, exit, and remainder sections.

Requirement	Definition	Failure
Mutual exclusion	While Pi is in its critical section, no other participant is in its critical section	Concurrent writers break one invariant
Progress	If the section is empty and contenders exist, only non-remainder contenders participate in selection, and selection cannot be postponed indefinitely	No contender is selected
Bounded waiting	Every requester has a finite bound on other entries allowed before its admission	Global progress while one requester starves

An unfair exclusive lock can repeatedly admit high-priority arrivals without bounding a lower-priority wait. Progress with exclusion still permits indefinite overtaking. Conversely, admitting everyone can be fast yet violate exclusion. Safety must cover every reachable interleaving; progress and fairness need separate arguments.

5.2 Safety, progress, and bounded waiting (continued)

For enter_i/leave_i, prove exclusion at arbitrary relative speeds; finite-step admission progress when free with contenders; a finite overtaking bound per requester; and the atomicity, ordering, and failure recovery of entry/exit themselves. A finite sample of schedules can find counterexamples, not prove arbitrary n and speeds.

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5.3. Peterson's solution

- For participant i and the other participant j, set flag[i]=true, set turn=j, wait while flag[j] and turn==j, enter the critical section, then set flag[i]=false. Flags announce interest and turn resolves contention.
- With atomic sequentially consistent reads and writes, the two-participant protocol satisfies the classical requirements. Ordinary unsynchronized variables in a modern language do not automatically meet those assumptions.

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5.3. Model, ordering, and boundary cases

Model assumptions. There are exactly two participants. Shared flag and turn accesses are atomic and sequentially consistent, and participants eventually continue executing.

Invariant and ordering. If both flags remain true, the single value of turn excludes one participant. On reentry the winner sets turn to its opponent; a waiting opponent cannot be overtaken repeatedly.

Counterexample or boundary. Compiler or hardware reordering and visibility differences invalidate the sequential-consistency argument unless suitable language atomics and ordering are used.

Worked check. Suppose both could be in the section. Both flags must be true. P0's successful test requires turn not equal to 1, while P1's requires turn not equal to 0 during unresolved contention. Considering the last turn write leaves its writer waiting, contradicting simultaneous entry. With only one flag true, that participant proceeds; with both true, one turn value permits one. The waiting participant is bypassed at most once after the other's relevant exit and reentry under these assumptions.

5.3. Model, ordering, and boundary cases (continued)

Reasoning prompt. Which steps of the proof fail when writes and reads no longer have one sequentially consistent order?

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5.3 Peterson's complete protocol

There are exactly two processes, j=1−i, atomic flags initially false, and shared turn.

```
P_i:
  flag[i] = true
  turn = j
  while (flag[j] && turn == j) { wait }
  critical_section_i
  flag[i] = false
  remainder_i
```

Flag means competing. Turn resolves simultaneous contention, not permanent lock ownership. The slogan that the last turn writer always loses omits required contention premises.

5.3 Peterson's complete protocol (continued)

Assume atomic sequentially consistent reads/writes. Suppose the two current critical-section attempts overlap. Let W0 be P0's turn=1 and W1 P1's turn=0. If W0<W1, P0's flag0=true preceded W0 and W1. Before P0 leaves, flag0 remains true; neither participant writes turn again inside this attempt. P1 therefore faces flag0=true and turn=0 when it would need to pass, making its while condition true, a contradiction. If W1<W0, symmetry blocks P0. Thus overlapping critical sections are impossible. This compares the current overlapping attempts' final turn writes, not arbitrary past iterations.

For progress, one contender sees the other's false flag and enters. With both competing, turn selects one side whose while is not blocked by turn. A participant already inside eventually clears its flag on exit. For bounded waiting, blocked Pi has flag[j]=true and turn=j. Pj does not change turn in its critical section. After leaving it clears flag; any new attempt first writes turn=i, giving Pi priority. Pi can be overtaken by Pj at most once, assuming the threads continue taking steps and critical sections terminate.

5.3 Peterson's complete protocol (continued)

The proof needs shared order/visibility. Use corresponding atomics with memory_order_seq_cst, or a proved equivalent fencing protocol. Renaming ordinary variables flag/turn supplies no such proof.

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5.4. Synchronization hardware

- Test-and-set and compare-and-swap atomically test and change a value. A spinlock repeatedly tries or observes a lock; it is appropriate only when the owner can make progress and the expected wait justifies CPU use.
- Spinning consumes execution resources. High contention or a long critical section can make blocking preferable. Kernel spin protocols also constrain preemption and interrupt contexts.

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5.4. Model, ordering, and boundary cases

Model assumptions. The atomic primitive follows its specified ordering and the lock owner can eventually run.

Invariant and ordering. Only a successful acquisition owns the lock. Release and subsequent acquire provide the required publication of protected updates.

Counterexample or boundary. On one CPU, an interrupt handler that spins on a lock held by the interrupted code prevents that owner from resuming. A preempted owner can likewise suffer priority-related starvation if the spinner never yields. Calling this a nonpreemptive scheduling example would be inconsistent.

Worked check. Two cores attempt CAS from zero to one. Only one succeeds; the other uses bounded backoff or a defined wait strategy. Release stores the unlocked state with suitable ordering.

Reasoning prompt. When is spinning cheaper than sleep and wake, and when does it consume a CPU needed by the owner?

5.4 Atomic hardware and spinning

Test-and-set atomically returns old L and sets true. CAS(&L,0,1) sets 1 only when L was 0 and returns the old value whether it succeeds or not.

```
acquire(L):
  repeat old = test_and_set(L)
  until old == false
release(L):
  L = false
```

One acquire sees false and owns the lock; others see true and retry. Owner-only release with proper publication preserves exclusion. Basic test-and-set does not bound waiting. Queueing, such as a circular waiting array with explicit successor handoff, adds fairness.

Spinning avoids sleep/wakeup and context costs for short sections. Long sections, one core, or a spinner competing with its owner can waste CPU. An interrupt handler spinning on a lock held by the interrupted task can prevent that owner resuming, producing livelock/deadlock. Interrupt contexts need controlled non-sleeping primitives and suitable interrupt/spin rules.

5.4 Atomic hardware and spinning (continued)

Instruction indivisibility does not alone provide language-level acquire/release. Later owners must see prior protected writes; writes before release must remain before publication; compilers must not move critical accesses outside the lock. Use atomic acquire/release semantics or corresponding fences. CAS alone does not solve ABA, reclamation, or lock-free progress.

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5.5. Mutex locks

- A mutex defines acquisition, ownership, and release. A blocking mutex sleeps when it cannot acquire, reducing CPU waste during long waits.
- A global lock order can eliminate circular waiting among those locks. Error and cancellation paths must also release resources and restore invariants.

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5.5. Model, ordering, and boundary cases

Model assumptions. Each successful acquisition has an authorized release. Waiters can be awakened under the API's policy.

Invariant and ordering. Protected updates finish consistently before release. All conflicting accesses participate in the same locking or equivalent synchronization rule.

Counterexample or boundary. One thread holds A and waits for B while another holds B and waits for A. Each currently holds one lock, yet their requested acquisitions form a cycle.

Worked check. Require every thread to acquire A before B. Then a cycle would require both increasing and decreasing order, which is impossible. Reverse-order release is conventional, though the absence of acquisition cycles follows from acquisition order. A timeout stress run can find failures but cannot prove all schedules safe.

Reasoning prompt. How do exceptions and cancellation retain the release guarantee?

5.5 Mutex ownership and ordering proof

Every participating critical section uses acquire(m), protected work, release(m). Acquisition/release transitions are atomic. A mutex usually requires its owner to release it; a semaphore need not enforce ownership. Error, cancellation, and exception paths depend on that distinction. Blocking mutex contenders wait; spinners continue consuming CPU and require short sections with a runnable owner.

Assign a strict global lock order, ord(A)=1, ord(B)=2. Acquire a new lock only above every held lock. Any resource-wait cycle would require ord(A)<…<ord(A), a contradiction. In particular holding B while acquiring A violates ord(B)<ord(A). Callbacks, condition waits, upgrades, exceptions, and cancellation must obey the same rule. Sort dynamic locks by a stable address or unique ID, not transaction argument order such as from/to accounts.

5.6. Semaphores

- A counting semaphore represents permits. In one blocking implementation, wait decrements an internal signed value and blocks if it becomes negative; signal increments it and wakes a waiter when required. Other implementations keep a nonnegative permit count and a separate wait queue.
- A binary semaphore can represent an event or support an exclusion protocol, but it does not inherently have mutex ownership. Extra signals can corrupt a resource-accounting protocol.

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5.6. Model, ordering, and boundary cases

Model assumptions. Wait and signal, including their queue transitions, are atomic under the semaphore abstraction.

Invariant and ordering. For a fixed pool of C resources, available resources plus outstanding allocations equal C. A signed implementation value may be negative to count waiters; that value is not a negative number of physical resources. Event semaphores need not be bounded by their initial value.

Counterexample or boundary. A producer that holds the buffer mutex while waiting for an empty slot can block the consumer that must create that slot.

Worked check. A three-slot buffer starts with empty=3 and full=0. After three completed productions and no consumption, full=3 and no empty permit exists. A fourth producer blocks before taking the buffer mutex.

Reasoning prompt. Which error paths become possible when a semaphore does not enforce ownership of permits?

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5.6 Permit abstraction and signed implementation

A resource semaphore counts permits; a binary abstraction has 0 or 1 free permits. A signed blocking implementation also encodes waiters. Counter and queue actions must be atomic with blocking/wakeup.

```
busy_wait(S): while (S <= 0) {} ; S--
busy_signal(S): S++

wait(S):
  S--
  if S < 0:
    enqueue current on S.queue
    block current
signal(S):
  S++
  if S <= 0:
    dequeue one waiter
    wake it
```

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5.6 Permit abstraction and signed implementation (continued)

The busy-wait notation requires indivisible successful test/decrement, not two racing ordinary operations. Negative signed state represents waiters, not negative resources. At quiescent boundaries, free=max(S,0), queued=max(−S,0). A handed-off permit is allocated even if its recipient has not received CPU; newcomers cannot take it. Resource semaphores preserve free+allocated=capacity. Event semaphores need not have fixed capacity.

For capacity 1: T0 wait 1→0 obtains the permit; T1 wait 0→−1 queues/sleeps. T0 signal−1→0 wakes T1 because the incremented value≤0 and hands it the permit. T1 resumes after its original wait without decrementing again. T1 signal0→1 leaves a free permit because nobody waits. Waking only on0→1 strands T1.

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5.6 Permit abstraction and signed implementation (continued)

For capacity 3, empty=3, full=0, mutex=1:

Completed operation	Empty	Full	Buffer
Initial	3	0	[]
Put A	2	1	[A]
Put B	1	2	[A, B]
Put C	0	3	[A, B, C]
Get A	1	2	[B, C]
Put D	0	3	[B, C, D]

Producer uses wait(empty), wait(mutex), update, signal(mutex), signal(full). Consumer first waits full. Holding mutex while awaiting capacity can prevent the consumer releasing capacity.

5.6 Permit abstraction and signed implementation (continued)

If P0 takes S then Q and P1 Q then S, each initially1, P0 can hold S and P1 Q before waiting forever for the other. Starvation instead leaves some work progressing while a waiter is indefinitely skipped; LIFO arrivals can do that. FIFO removes one starvation pattern but not incorrect pairing.

For priority inversion, low-priority L owns H's needed resource while medium M preempts L. Inheritance temporarily raises L to H's priority until the section completes, then restores it. The historical Mars Pathfinder VxWorks incident illustrates this mechanism.

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5.7. Classical synchronization problems

- A bounded buffer uses empty and full for capacity and a mutex for its representation. Producers wait for empty before the mutex, publish an item under the mutex, release the mutex, and signal full. Consumers symmetrically wait for full before the mutex and signal empty after removal.
- Readers-writers permits simultaneous readers and exclusive writers, but fairness policy matters. Reader preference can starve a writer. Dining philosophers demonstrates circular waiting during incremental resource acquisition.

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5.7. Model, ordering, and boundary cases

Model assumptions. Acquisitions are not automatically all-or-nothing. Scheduling and admission assumptions must be stated separately from deadlock prevention.

Invariant and ordering. Buffer occupancy matches published item and free-slot accounting, including in-flight reservations. A philosopher eats only while holding both required forks and releases them afterward.

Counterexample or boundary. If all five philosophers hold their left fork and wait for their right, none can proceed.

Worked check. Admit at most four philosophers to the five-fork table. The chain of occupied left forks has a boundary next to a free fork, so some admitted philosopher can acquire its second fork and finish, assuming it is scheduled. This breaks the all-five cycle but does not itself guarantee fair admission or starvation freedom.

Reasoning prompt. Why does deadlock freedom fail to imply starvation freedom?

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5.7 Classical coordination protocols

Empty prevents producing into a full buffer, full prevents consuming an empty one, and mutex protects compound slot/head/tail/size updates. Capacity waits occur outside mutex. Otherwise producer waits for consumer capacity while consumer waits for producer's mutex. The occupancy safety condition is $0 \leq \text{items} \leq n$. General capacity accounting is $E+F+P+C=n$. E and F are the nonnegative, unreserved free permits of empty and full. P counts producers already granted an empty permit that have not yet signaled full; C counts consumers already granted a full permit that have not yet signaled empty. A permit handed directly to an awakened waiter counts as its P or C reservation even before it runs.

Every atomic protocol transition preserves the sum. Successful producer wait moves E to P; producer signal(full) moves P to F, or directly to a consumer's C on handoff. Consumer wait moves F to C; signal(empty) moves C to E or directly to another producer's P. Writing or removing the item does not change this capacity classification. With capacity 3 and an empty buffer, $(E,F,P,C)=(3,0,0,0)$. After only one successful wait(empty), it is $(2,0,1,0)$, so $E+F=2$, even before insertion. $E+F=n$ holds at quiescent boundaries with no reservations, $P=C=0$.

5.7 Classical coordination protocols (continued)

E/F are not signed semaphore raw counters. A negative raw value also encodes queued waiters and cannot be substituted into permit conservation. Raw zero can coexist with an outstanding handoff grant. Modify the structure under mutex, publish full only after insertion, and publish empty only after removal. These rules preserve occupancy safety; F need not equal the number of stored items during an operation.

Reader-preference allows new readers while writers wait, but not while a writer owns access; writers may starve. Writer-preference stops new readers after a writer starts waiting; readers may starve. Both allow concurrent readers when permitted. In the first protocol, read_count is protected by mutex; first reader acquires rw_mutex, last releases it. A lock protects its actual state, not everything its name suggests.

5.7 Classical coordination protocols (continued)

Five philosophers each take chopstick[i], then chopstick[(i+1)%5], eat, release right then left. All can hold left and wait right, producing a five-edge cycle. Alternatives limit simultaneous contenders to four, acquire both atomically when available, or use opposite acquisition orders for odd/even philosophers. They exclude this deadlock pattern without automatically proving starvation freedom; waiter selection still matters.

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5.8 Wait for a predicate, not a notification ticket

Use Mesa semantics, one mutex `m`, condition variable `not_empty`, and an initially empty shared queue `q`. P inserts A once. C1 and C2 each call `get()` once. Every queue mutation and `q.empty()` check holds the same mutex.

```
get():                          put_A():
  lock(m)                         lock(m)
  while q.empty():                q.push_back(A)
    wait(not_empty, m)            signal(not_empty)
  x = q.pop_front()               unlock(m)
  unlock(m)
  return x
```

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Wait combines lock release and wait registration atomically relative to another thread acquiring m and notifying; successful return reacquires m. Signal does not transfer the mutex or reserve A for the waiter. With no waiter, no permit is accumulated.

5.8 Notification does not determine the next lock owner

Time flows down; P inserts one A; no spurious wakeups in this trace.

	Consumer C1	Producer P	Consumer C2	mutex / queue
T0	wait: unlock + enqueue			m=—, q=[]
T1		lock; enqueue A		m=P, q=[A]
T2	notified; no lock yet	signal C1; keep m		m=P, q=[A]
T3		unlock →	lock; take A; unlock	m=—, q=[]
T4	relock; q is empty			m=C1, q=[]
T5	while → wait again			m=—, q=[]

T3 orders P's unlock, C2's lock and removal of A, then C2's unlock. The mutex is not assumed FIFO. C1's T4 wait return owns m, but the predicate is false again. No spurious wakeup occurs in this counterexample: `if` would allow an empty pop; `while` sends C1 back to waiting at T5.

5.8 Track three distinct forms of state

CV holds condition waiters. E contains notified waiters awaiting m, **not a kernel ready queue**. Notification changes eligibility, not ownership or predicate truth. States follow each event.

Event	m owner	q	CV	E
T0 C1 waits	None	`[]`	`{C1}`	`{}`
T1 P inserts A	P	`[A]`	`{C1}`	`{}`
T2 P signals C1	P	`[A]`	`{}`	`{C1}`
T3 C2 removes A and unlocks	None	`[]`	`{}`	`{C1}`
T4 C1 reacquires	C1	`[]`	`{}`	`{}`
T5 C1 waits again	None	`[]`	`{C1}`	`{}`

5.8 Lost wakeup comes from a gap before waiting

Broken: separate unlock and wait

1. C1: sees empty, unlocks; not yet queued
2. P: inserts A, signals; nobody waiting
3. C1: queues and sleeps too late; q=[A]

Data exists, but C1 sleeps; signals do not accumulate.

Correct: atomic wait contract

1. C1: sees empty; wait unlocks + registers
2. P: locks, inserts A, signals, unlocks
3. C1: relocks, rechecks, then takes A

If P arrives first, C1 sees A and skips wait.

The left side is a broken abstract wait implementation: unlock first, register and block later. It is **not legal `pthread_cond_wait` usage**. P inserts and signals in the gap while CV is empty; C1 then sleeps despite queued data. Atomic wait excludes that gap. If P completes first instead, C1 sees a nonempty queue under m and skips waiting.

5.8 Prove safety and liveness separately

Safety. Every pop checks `!q.empty()` while holding m. No other thread can acquire m and change q between this check and removal. Wait return guarantees reacquisition, not predicate truth.

Liveness. The while loop prevents an invalid removal, but does not promise data for every consumer. There is only one A; once C2 consumes it, C1 may legitimately continue waiting. Scheduler fairness, mutex fairness, and future arrivals require separate assumptions.

If broadcast wakes two consumers for one A, the first owner takes A; the other returns to the while loop. Waking more threads creates neither data nor a bounded-waiting guarantee.

5.8 Handoff contracts: Hoare, Mesa, and signal-and-exit

The trace uses Mesa semantics: the signaler continues, and the waiter reacquires later. Hoare's handoff-time condition guarantee must not be imported into that model.

Handoff	Lock/execution after signal	Waiter reasoning
Hoare signal-and-wait	Immediate handoff; signaler waits	May use the condition established at handoff under that language's contract
Mesa signal-and-continue	Signaler retains the lock; waiter reacquires later	Recheck the predicate
Concurrent Pascal signal-and-exit	Signaler leaves, then hands off	Language-defined

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5.9. Synchronization examples

- The book's Solaris, Windows, Linux, Java, and Pthreads cases are versioned illustrations. Similar names do not establish identical ownership, fairness, or cancellation behavior.
- Java synchronized uses an object's monitor. Its wait and notification operations require ownership of the corresponding monitor. Other APIs have their own contracts.

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5.9. Model, ordering, and boundary cases

Model assumptions. Each API's semantics and permitted calling contexts are explicit.

Invariant and ordering. Check ownership, wake-up behavior, interrupt context, priority rules, and acquisition order as one protocol.

Counterexample or boundary. Taking an ordinary sleeping mutex in a non-sleepable interrupt context can deadlock or trigger a kernel failure.

Worked check. Windows 7 dispatcher mutex M starts signaled. T1 acquires it, making it nonsignaled. T2 waits on its queue. T1 releases, making M signaled; the kernel selects one waiter, T2 moves waiting→ready, then acquires M, which becomes nonsignaled again.

Reasoning prompt. Which guarantee would require a version-specific specification before using it in a current implementation?

5.9 Historical synchronization environments

Historical case	Mechanism	Boundary
Windows 7 kernel	SMP spinlocks for short global-resource sections	Holder not preempted
Windows 7 user services	Dispatcher mutex, semaphore, event, timer	Signaled/nonsignaled states; mutex grants one owner
Linux 2.6 onward textbook snapshot	Preemptible kernel, atomic_t, mutex, spinlock, semaphore	Short SMP spin regions; preempt_count>0 prevents safe preemption
Solaris textbook snapshot	Adaptive mutex, RW lock, turnstile	Spin while holder runs, otherwise sleep; turnstile supports inheritance
Pthreads/POSIX	Mutex, CV, RW lock; POSIX SEM semaphores	API determines waits, wakeups, errors

5.9 Historical synchronization environments (continued)

The textbook Windows critical-section object first spins in user mode and uses kernel assistance under prolonged contention. The cited uniprocessor Linux case replaces ordinary SMP-style spinning with preemption control. Solaris selects spin/sleep according to holder execution. These are versioned cases, not all-current-system claims.

Before substitution, identify user/kernel/interrupt context, scheduling entity, ownership and process-sharing support, fairness/inheritance/cancellation/wakeup order, and versioned memory ordering. Similar names do not establish equivalence.

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5.10. Alternative approaches

- Transactional memory commits a region atomically or aborts it. Retrying requires that effects are replayable and that external side effects obey the transaction contract.
- Message passing and immutable data can reduce shared mutable state. Lock-free structures still share mutable state but provide a different progress guarantee. All designs retain costs in conflict handling, scheduling, or reclamation.

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5.10. Model, ordering, and boundary cases

Model assumptions. Conflict detection, retries, progress, and memory reclamation have explicit rules.

Invariant and ordering. Successful commit exposes the specified atomic result. Abort leaves no visible partial transaction effects under the model.

Counterexample or boundary. A lock-free queue that immediately frees a node still referenced by another thread can cause use-after-free despite atomic pointer updates.

Worked check. Start A=100, B=200. T1 computes A−=25, B+=25. A pre-commit read-set conflict aborts, leaving 100/200. A successful retry atomically publishes 75/225. The abort and commit cannot each leave a partial debit or credit; an external payment effect also cannot be blindly repeated.

Reasoning prompt. Why does a low conflict rate alone fail to establish efficient transactions?

5.10 Alternatives and their remaining boundaries

A memory transaction makes a sequence of shared accesses all-or-nothing. Commit publishes it; abort rolls it back, excluding partial observation. For atomic{x=x−25; y=y+25}, conflicting read/write sets can force one transaction to retry. Aborting can avoid lock-order deadlock but still permits starvation and must handle I/O, irreversible effects, and retries. STM instruments accesses through compiler/runtime; HTM uses coherent-cache mechanisms. Capacity and failure paths differ.

OpenMP critical regions of the same name exclude one another; distinct names identify distinct regions. The programmer still identifies shared conflicts. Immutable functional values avoid in-place writes and permit sharing old values, but files, networks, mutable databases, and mailboxes remain shared resources. Message copying/ownership transfer reduces shared mutation without eliminating capacity, ordering, backpressure, or retry coordination.

5.11. Summary

- Synchronization combines state invariants with the required progress and fairness guarantees. Higher-level operations must connect predicates, publication, waiting, and wake-up correctly.
- Final values alone can miss illegal intermediate states and timing failures. Model interleavings and check the observable invariant.

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5.11. Model, ordering, and boundary cases

Model assumptions. Analysis includes every access and failure path. Lock-free algorithms also specify reclamation and their precise progress property.

Invariant and ordering. Operations preserve the abstract state, and the required progress holds under stated scheduling assumptions.

Counterexample or boundary. Releasing a lock only on normal return leaves later operations blocked after an exception.

Worked check. Two threads each execute load, increment, store, giving C(6,3)=20 order-preserving interleavings. Loading 0 in both before either stores leaves 1; other interleavings give 1 or 2. With the same mutex protecting the whole RMW, only P1-then-P2 or P2-then-P1 remains, both giving 2. This finite model does not replace the general lock proof.

Reasoning prompt. How can cancellation and exception transitions be included in a synchronization check?

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5.11 Choosing a synchronization primitive

Need	Required model	Check
One critical section	Length, sleepability, ownership	Paired acquire/release
Fixed resources	Capacity, initial permits, authorized signalers	No excess signal
Predicate wait	Variables, notification, recheck	One protecting mutex
Multiple locks	Global order, dynamic key, cancel paths	Every acquisition follows order
Readers/writers	Preference/fairness rules	Reader/writer starvation

Trace modes, owned resources, waiters, and predicates after selection. Typical successful tests do not exclude rare schedules. Supply general proofs and rerunnable counterexamples separately.

Chapter conclusions

- Critical-section requirements separate safety, progress, and fairness
- Semaphore and monitor errors often arise from predicates and wait order
- Classical proofs require their stated memory-model assumptions

References

A. Silberschatz, P. B. Galvin, and G. Gagne, *Operating System Concepts*, 9th edition, Wiley, 2013. Historical system examples use the versions identified in this chapter.

POSIX.1-2024 condition-wait ownership, atomic release, and reacquisition contract.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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CPU scheduling

CPU and I/O bursts provide a model for comparing fairness, throughput, response, priorities, deadlines, and multiprocessor costs. A useful scheduling claim specifies both its workload and the guarantees it promises.

Prerequisites. Process states and queues, Context switching, Probability and expectation.

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Learning objectives

- Derive waiting, turnaround, and response times
- Compare FCFS, SJF, SRTF, RR, priority scheduling, and feedback queues
- Analyze preemption, starvation, priority inversion, and real-time schedulability

Central question. Under which workload and execution assumptions can a policy allocate finite CPU capacity correctly and predictably?

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6.1. Basic concepts

- A process alternates between CPU execution and I/O waits. A CPU burst is computation between blocking events; scheduling may interrupt that burst without completing its CPU demand.
- The short-term scheduler selects a runnable kernel execution context. The dispatcher saves and restores context, performs required mode transitions, and resumes execution. Dispatch latency is overhead rather than application progress.

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6.1.1. The scheduler's time scales

- Long-term admission controls which jobs enter and affects multiprogramming. Short-term scheduling selects the next execution context. Classical medium-term scheduling suspends or swaps work to reduce memory pressure.
- Admission changes the future workload. Batch throughput and interactive response therefore cannot be optimized by treating every decision as one CPU-selection policy.

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6.1.2. Preemptive and nonpreemptive policies

- A nonpreemptive policy switches when the running task blocks, terminates, or yields. A preemptive policy can also switch on a quantum expiration or a newly ready higher-priority task.
- Preemption can improve responsiveness but adds state and run-queue work and permits more interleavings of shared kernel data.

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6.1. Model, ordering, and boundary cases

Model assumptions. Jobs have defined arrivals and CPU/I/O demands. Context can be saved and restored, and queue transitions are synchronized.

Invariant and ordering. Each modeled CPU runs at most one task. A task cannot be enqueued twice, run before arrival, or use an unreleased resource.

Counterexample or boundary. Ignoring switching cost can make an extremely short quantum look fair while useful CPU time collapses.

Worked check. With P1 burst 5 before I/O, P2 burst 2, one CPU and switch cost 1, FCFS runs P1[0,5), switch[5,6), P2[6,8). Useful-work fraction is 7/8=87.5% and switching overhead is 1/8=12.5%. The CPU is never idle, so busy-time CPU utilization is still 8/8=100%. These are different measures; omitting switching also incorrectly changes elapsed time from 8 to 7.

Reasoning prompt. Which event-order bug could both duplicate a queue entry and permit execution in the wrong state?

6.2. Scheduling criteria

- Utilization and throughput generally favor more completed useful work. Turnaround, waiting, and response favor lower delays. These objectives can conflict.
- Waiting time accumulates only in the ready state, excluding CPU service and blocked I/O. Response must be defined as first dispatch or first externally visible output; the two need not coincide.

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6.2.1. Formulas and observation intervals

- Let arrival be a_i, first dispatch s_i, and completion f_i. Turnaround_i=f_i-a_i. Under the first-dispatch convention, response_i=s_i-a_i. Waiting_i is the sum of ready intervals, not generally turnaround minus CPU time when blocked I/O exists.
- A window that omits unfinished long jobs systematically favors short completions. Report the treatment of censored jobs rather than silently dropping them.

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6.2. Model, ordering, and boundary cases

Model assumptions. All jobs use the same time units, arrival definition, and completion criterion. The observation includes jobs still outstanding at its endpoint.

Invariant and ordering. Turnaround equals completion minus arrival. With compatible definitions, response does not exceed turnaround. Ready waiting excludes blocked time.

Counterexample or boundary. Maximizing utilization by indefinitely deferring short interactive work can violate response requirements.

Worked check. Arrival a=2, first CPU s=5, first visible output o=9, completion f=14, with [2,5) in the ready queue and no other ready waiting. Waiting=3, scheduling response=3, visible response=7, turnaround=12. They describe overlapping intervals and cannot be added.

Reasoning prompt. How can equal mean waiting coexist with very different maximum response?

Ready waiting and two response events

Let arrival be a_i, total ready intervals W_i, first CPU s_i, first visible output o_i, and completion f_i. Then

$$waiting_i = W_i, \quad turnaround_i = f_i - a_i, \quad schedule\ response_i = s_i - a_i, \quad visible\ response_i = o_i - a_i.$$

Blocked I/O is neither service nor ready waiting. Task A arrives at 0, is ready[0,2), runs[2,4), blocks[4,8), is ready[8,11), then runs[11,12). Turnaround=12, ready waiting=2+3=5, service=2+1=3, blocked=4. The decomposition 5+3+4=12 closes. Counting blocked time as waiting gives 9 and violates the definition.

Equal mean waiting need not imply equal maximum response. A and B can each accumulate 4 ready units, yet first run at 4 and 8 under different arrivals/blocking. First visible output can occur still later. Metric names, observed events, units, and percentiles must accompany comparisons.

6.3. Scheduling algorithms

- Define the comparison key, preemption rule, tie-break, and queue update. A lower mean wait does not establish a per-job bound.
- Nonpreemptive SJF minimizes mean wait for simultaneously available, known single CPU bursts on one CPU without switching cost. SRTF preempts according to remaining demand. Unknown bursts require estimates, so the ideal result no longer follows automatically.
- An exchange proof explains the SJF bound. For adjacent available jobs x>y, running x before y contributes x to their combined waiting; swapping contributes y. Later jobs see the same total x+y. Removing every inversion therefore cannot increase total waiting. Different arrivals or deliberate idling require a different argument.

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6.3.1. FCFS and the convoy effect

- FCFS serves arrival order without preemption. A long first job delays many short jobs and can create a convoy through CPU and I/O resources.
- With simultaneous arrivals and bursts 120,10,10 in that order, waits are 0,120,130. Their mean is 250/3, approximately 83.33 time units.

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6.3.2. SJF and SRTF

- SJF selects the shortest available next burst when the CPU becomes available. SRTF can preempt when a job with shorter remaining demand arrives.
- Continually arriving short jobs can starve a long job. An arbitrary tie-break is not a fairness guarantee.

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6.3.3. Round Robin

- RR rotates a ready queue using quantum q. A quantum large enough to finish every burst approaches FCFS; very small q increases timer and possible switching overhead.
- For arrivals zero, bursts 24/3/3, and q=4, the timeline is P1[0,4), P2[4,7), P3[7,10), P1[10,14), P1[14,18), P1[18,22), P1[22,26), P1[26,30). The total is 30. Once only P1 remains, a quantum boundary need not switch to a different thread.

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6.3.4. Priorities and multilevel queues

- Priority policies can starve lower-priority work. Aging increases effective priority with waiting, but its rate, saturation, ties, and arrival assumptions determine whether waiting is bounded.
- Multilevel queues separate classes such as foreground and background. Feedback queues allow class migration according to behavior; quanta and promotion rules are part of the policy.

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6.3. Model, ordering, and boundary cases

Model assumptions. Arrivals, actual or estimated bursts, preemption, and ties are specified.

Invariant and ordering. Timeline intervals do not overlap. Each execution segment consumes no more than remaining demand, and arrivals and completions update the ready set.

Counterexample or boundary. A stream of underestimated short jobs can postpone a long job indefinitely.

Worked check. Simultaneous bursts 8/4/2 give FCFS waits 0/8/12, mean 20/3, and SJF execution order P3/P2/P1 with waits 0/2/6 in execution order, mean 8/3. RR q=2 gives completions P1/P2/P3=14/10/6; total useful execution remains 14.

Reasoning prompt. Which optimality statement must change if the burst ordering uses inaccurate predictions?

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SJF and RR on the same 5/2/3 input

P1/P2/P3 all arrive at 0, bursts are 5/2/3, and switch cost is zero. SJF runs P2[0,2), P3[2,5), P1[5,10). In process order, completions are 10/2/5 and waits 5/0/2.

RR q=2 starts queue P1, P2, P3. It runs P1[0,2), P2[2,4), P3[4,6), P1[6,8), P3[8,9), P1[9,10). P2 completes at 4 and is never requeued. Completions are 10/4/9, waits 5/2/6, and first responses 0/2/4. Both consume 10 useful units; mean waits are 7/3 and 13/3.

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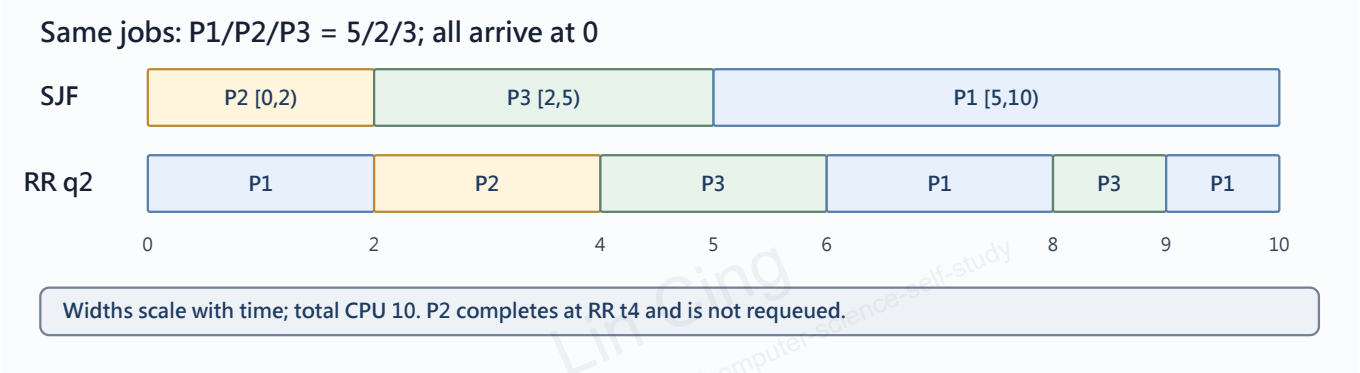
SJF and RR on the same 5/2/3 input (continued)



The separate 8/4/2 case instead has RR q=2 segments P1[0,2), P2[2,4), P3[4,6), P1[6,8), P2[8,10), P1[10,12), P1[12,14), and completions 14/10/6. Mixing those inputs changes the problem.

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6.4. Thread scheduling

- A library can schedule user threads within process-contention scope. Kernel-visible entities compete in system-contention scope.
- One-to-one mapping normally exposes each thread to kernel scheduling. In many-to-one, a genuinely blocking kernel call can stall the process's other user threads unless the runtime avoids that blocking.

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6.4. Model, ordering, and boundary cases

Model assumptions. The mapping and the runtime's handling of blocking calls are known.

Invariant and ordering. One kernel entity runs at most one context at a time. A runtime cannot promise independent progress without an available execution carrier.

Counterexample or boundary. The sole carrier blocking on I/O stops all other runnable tasks in the many-to-one example.

Worked check. T1 awaits network, T2 needs 4 CPU units. With one-to-one mapping T2 can run [0,4) while T1 blocks. With one carrier and a genuinely blocking T1 syscall, T2 has no guaranteed start, so completion at 4 is unjustified.

Reasoning prompt. Which mapping and blocking assumptions support a thread API's concurrency promise?

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6.5. Multiprocessor scheduling

- Affinity can preserve cache and TLB locality. Excessive binding can leave CPUs idle while one queue is overloaded, so balancing trades locality against parallel use.
- Shared queues, migration, and work stealing must prevent a runnable thread from being dispatched on two CPUs simultaneously.

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6.5. Model, ordering, and boundary cases

Model assumptions. Queue ownership and migration synchronization are defined, with cache and communication costs.

Invariant and ordering. A thread has only one actual running owner. Balancing respects required affinity constraints.

Counterexample or boundary. Binding every task to one core preserves some locality while wasting other cores.

Worked check. Fix eight cores, all jobs ready at t=0, indivisible jobs, and no job running on two cores at once. A/B each need 8 ms, C/D each need 4 ms, and E/F/G/H/I/J each need 2 ms: 36 ms of useful work. CPU0 initially queues [A,I,J]; CPU1 through CPU7 initially hold B,C,D,E,F,G,H respectively.

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6.5. Model, ordering, and boundary cases (continued)

Without migration CPU0 runs A[0,8), I[8,10), J[10,12), so makespan is 12 ms. At t=2 CPU4/5 finish E/F. Under queue synchronization they each remove one still-queued job I/J and transfer its unique ownership. Charge each migration 0.5 ms for queue/transfer and 0.5 ms for cache warmup. Both destination cores pay overhead [2,3), then run I/J [3,5). CPU0 finishes A at 8; makespan becomes 8 ms, speedup=12/8=1.5. Useful work remains 36 ms; added CPU work is 2 ms. These are specified model costs, not hardware measurements.

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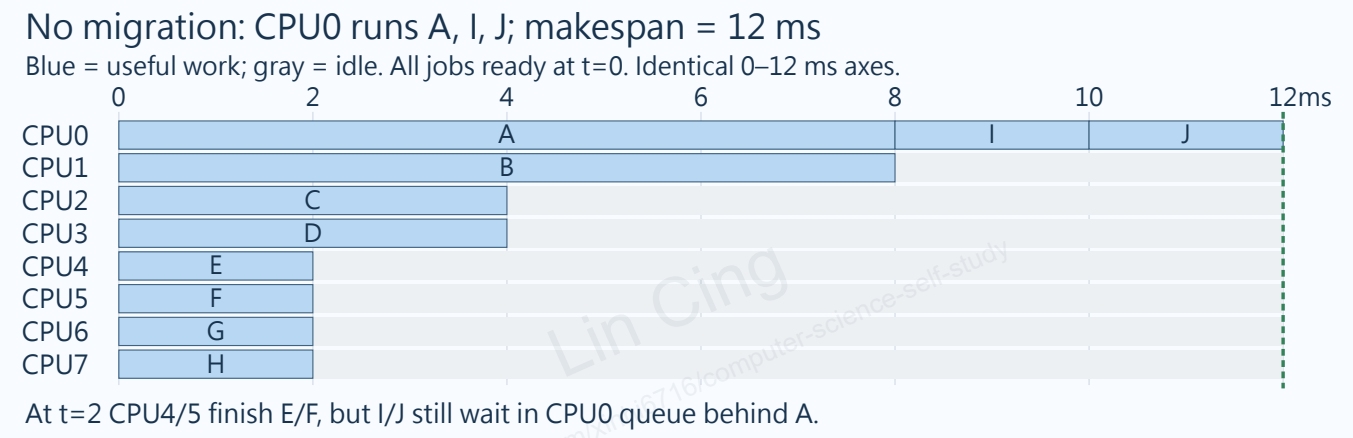
6.5. Model, ordering, and boundary cases (continued)



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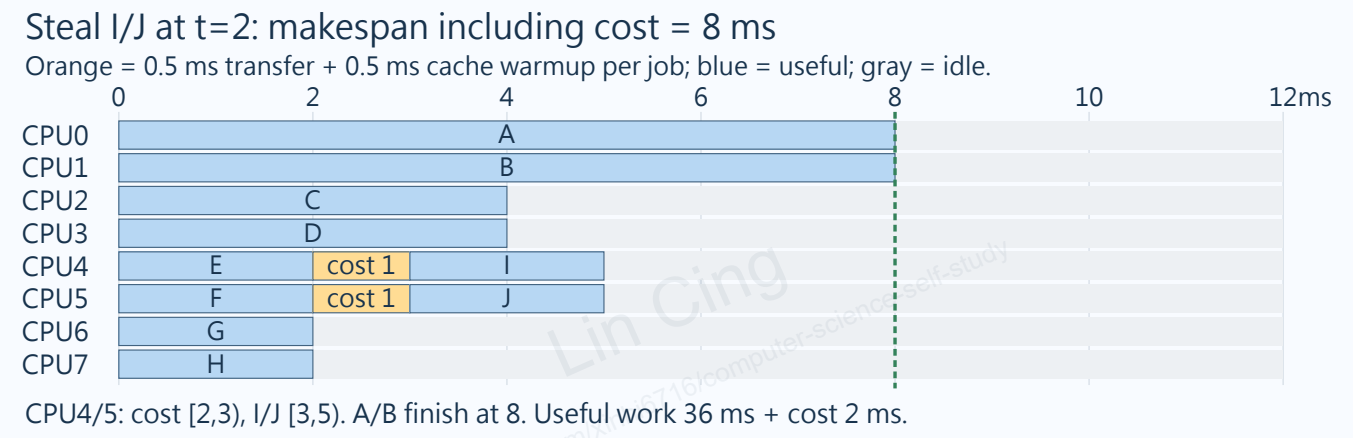
6.5. Model, ordering, and boundary cases (continued)



Indivisible A gives the lower bound $T \geq 8$, reached here. With per-migration cost c instead, makespan=max(8,2+c+2)=max(8,4+c). Strict improvement over 12 ms requires $c<8$. Dependencies, mandatory affinity, or different cache costs change the model; idle cores alone do not prove migration beneficial.

6.6. Real-time CPU scheduling

- Real-time requirements concern deadlines. Soft real-time permits defined deadline misses; hard real-time requires a guarantee under its specified workload and failure model.
- Describe a periodic task by execution upper bound e_i, period p_i, and relative deadline d_i. U=sum(e_i/p_i) is load. For independent, preemptible tasks on one ideal CPU, d_i=p_i, and no blocking or overhead, RM has sufficient bound U<=n(2^(1/n)-1), while EDF is schedulable exactly when U<=1 in this model.
- With e1=2, p1=d1=5 and e2=3, p2=d2=10, U=0.7 is below 2(sqrt(2)-1), approximately 0.828. RM gives T1[0,2), T2[2,5), T1[5,7), idle[7,10). Both tasks meet deadlines in the hyperperiod, repeated under the stated assumptions.

6.6.1. RM, EDF, and resource priorities

- RM assigns shorter periods higher fixed priority. Its classical sufficient bound relies on independent tasks with deadlines equal to periods.
- EDF chooses the earliest absolute deadline dynamically. Its ideal one-CPU utilization result does not specify graceful behavior under overload.
- Priority inheritance temporarily raises a blocking owner's priority. Ceiling protocols provide different restrictions and bounds. Neither replaces analysis of execution and blocking costs.

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6.6. Model, ordering, and boundary cases

Model assumptions. Release patterns, deadlines, execution upper bounds, processor count, switching, and blocking are defined.

Invariant and ordering. Each job completes by its absolute deadline. A blocking bound must be justified; merely enabling priority inheritance does not prove schedulability.

Counterexample or boundary. U<=1 alone does not prove arbitrary hard-real-time workloads meet deadlines when overhead or blocking is omitted.

Worked check. The two-task 0.7 example is schedulable in its ideal model. Add bounded overhead and resource blocking before transferring that conclusion to an implementation.

Reasoning prompt. Which violated assumption prevents utilization alone from establishing a deadline guarantee?

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Historical scheduler mechanisms

Solaris converts class priorities to global priorities. TS/IA uses feedback; priority 0 has a 200 ms quantum, priority 59 a 20 ms quantum. CPU exhaustion and sleep return use different new priorities. Windows 7 uses 32 ready queues with versioned boosts. These are historical mechanisms rather than present-day specifications.

Linux 3.2 CFS stores ready entities in a red-black tree keyed by vruntime; current is outside the tree. The increment is actual execution × 1024/weight. Nice 0 has weight 1024; nice 5 has 335. Select the smallest vruntime, subject to the version's bounded buddy preferences.

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Historical scheduler mechanisms (continued)

Same workload, three ready structures

Solaris TS/IA

global priority

- p20 → q120 ms
- sleep return → p50
- highest global priority
- equal priority: round robin
- IA favors window apps

Windows 7

32 ready queues

- p1–15 variable / p16–31 RT
- time quantum expires → lower
- I/O / foreground → boost
- foreground quantum ≈ 3×
- scan highest nonempty queue

Linux 3.2 CFS

red-black tree

- key = vruntime
- nice 0/5 weight 1024/335
- 2 runnable → 3 ms slice
- tick: delta_exec > slice
- HRTICK queues exact slice

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A fixed Linux 3.2 event model

One CPU, P1 burst 8 ms, P2 burst 4 ms, both runnable at 0. P3 burst 2 ms sleeps until 9. All have nice=0, weight=1024, and normalized initial vruntime=0. P1 enters the tree first on equal keys. Fix GENTLE_FAIR_SLEEPERS=true, wakeup granularity=1 ms, CONFIG_SCHED_HRTICK=y. Switch cost, interrupt cost, and timer jitter are zero. Normalization does not claim a real boot-time START_DEBIT offset; one common shift preserves relative comparisons.

For n runnable entities,

$$period = \begin{cases} 6\ \text{ms}, & n \le 8, \\ 0.75n\ \text{ms}, & n > 8, \end{cases} \qquad ideal_i = period \frac{weight_i}{\sum_j weight_j}.$$

Ideal runtime is a wall-time slice, not a global quantum. Two equal tasks get 3 ms each; three get 2 ms each. Current leaves the ready tree and returns with updated vruntime on preemption; blocked/completed tasks are not ready. HRTICK, a satisfied wakeup condition, or completion must justify a switch. No event at t=8 justifies splitting P1's [6,9) slice.

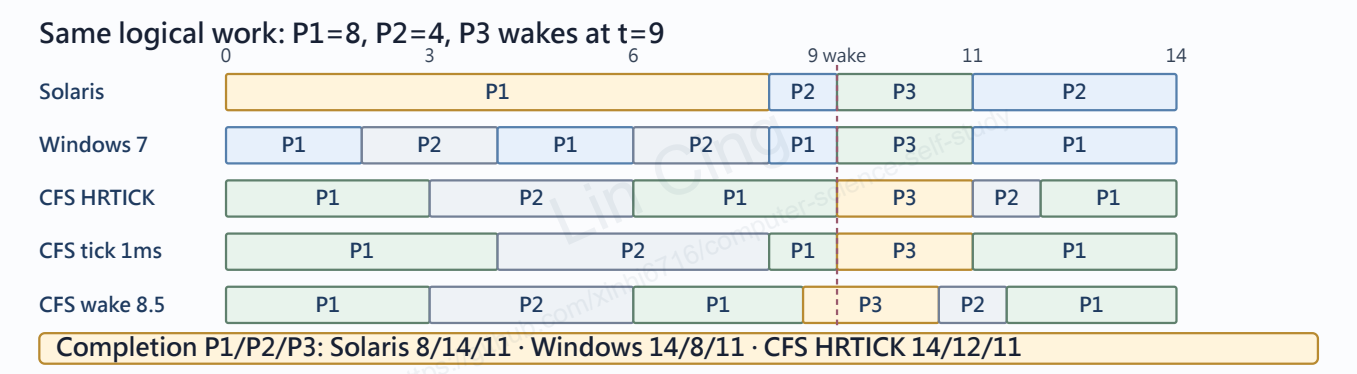
CFS event trace

States below are after the event and next-current selection. Ready excludes current; min_vruntime never moves backward.

Time/event	Current	Other ready	min_vruntime	New ideal	Reason
0 initial	P1	P2@0	0	3	Equal keys, P1 inserted first
3 HRTICK	P2	P1@3	0	3	P1 used slice
6 HRTICK	P1	P2@3	3	3	Equal keys, P1 already leftmost
9 HRTICK/wake	P3	P2@3, P1@6	3	2	P3 placed at 0, lead 6>1
11 P3 complete	P2	P1@6	3	3	P2 key 3 precedes 6
12 P2 complete	P1	None	6	6	Only P1, remaining 2
14 P1 complete	None	None	8	None	No work remains

CFS event trace (continued)

P3 placement at 9 is $\max(v_{sleep}, min_vruntime - 3) = \max(0, 3 - 3) = 0$. P1's key is 6. Wakeup compares current−wakeup>1 ms, so 6−0>1 and P3 immediately runs at 9, not earlier. Gantt is P1[0,3), P2[3,6), P1[6,9), P3[9,11), P2[11,12), P1[12,14). Completions are 14/12/11, not 14/9/11.



Tick, HRTICK, and wakeup sensitivity

Settings	Gantt	Completion P1/P2/P3
HRTICK, P3 wake 9	P1 0–3, P2 3–6, P1 6–9, P3 9–11, P2 11–12, P1 12–14	14/12/11
No HRTICK, 1 ms tick, wake 9	P1 0–4, P2 4–8, P1 8–9, P3 9–11, P1 11–14	14/8/11
HRTICK, wake 8.5	P1 0–3, P2 3–6, P1 6–8.5, P3 8.5–10.5, P2 10.5–11.5, P1 11.5–14	14/11.5/10.5

check_preempt_tick uses strict delta_exec>ideal_runtime. At t=3, 3>3 is false; the ordinary tick first preempts at t=4. Queued HRTICK instead directly requests rescheduling at the programmed slice boundary, without reapplying that strict comparison. Without HRTICK, P2 finishes at 8. At t=9 min_vruntime=5, so P3 returns at max(0,5−3)=2; P1 is 5, and 5−2>1 gives the wake preemption.

Tick, HRTICK, and wakeup sensitivity (continued)

The periodic tick also checks, after minimum granularity, whether current leads the tree's leftmost by more than ideal runtime. This handles fairness discrepancies but does not replace wakeup comparison. Last/next/skip buddy are locality/scheduling hints subject to virtual-time fairness. In this example n<8 before wake, so there is no ordinary next buddy from the sleeping P3; a successful wakeup comparison can nominate it. A buddy too far ahead cannot override the fairness bound.

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Nice weights and historical comparisons

With only A nice 0 and B nice 5 runnable, total weight=1359 and period=6 ms.

Task	Weight	Ideal runtime	vruntime per real ms	CPU share
A	1024	$6\times1024/1359\approx4.521$ ms	1 ms	75.35%
B	335	$6\times335/1359\approx1.479$ ms	$1024/335\approx3.057$ ms	24.65%

Weights are not linear nice percentages. Above eight runnable tasks the period grows to preserve minimum granularity instead of shrinking slices without bound.

Fixed case	Rules	Logical workload completion P1/P2/P3
Solaris TS/IA	p20/q120 ms, sleep return p50	8/14/11
Windows 7	32 queues, normalized 2/6 ticks, I/O boost	14/8/11
Linux 3.2 CFS	HRTICK, nice 0, 6 ms period, P3 wake 9	14/12/11

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Nice weights and historical comparisons (continued)

These are specified event models, not product benchmarks. The same workload alone does not determine one Gantt. Timer mode, wake time, insertion ties, and placement are inputs; changing them yields the alternate completions above.

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6.8. Algorithm evaluation

- Deterministic models calculate a trace exactly. Queueing models use arrival and service distributions. Simulation supports more complex policies. An implementation captures actual hardware and OS costs.
- Disclose workload distribution, processors, switching, and observation window. Means can hide starvation and long tails.

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6.8. Model, ordering, and boundary cases

Model assumptions. Methods use comparable inputs and costs. Random simulations retain seeds and an appropriate uncertainty analysis.

Invariant and ordering. If models disagree on the same simplified workload, investigate omitted costs and distribution assumptions. Agreement is not itself an algorithmic invariant.

Counterexample or boundary. One deterministic SJF success does not establish superiority under every burst-estimation error distribution.

Worked check. Use an exactly replayable deterministic comparison: one CPU; P1/P2/P3 arrive at 0 with bursts 5/2/3 ms; no I/O or switching cost. Response means time to first CPU. Define p99 by nearest rank: with three observations, rank ceil(0.99×3)=3 is this batch maximum. Waiting=completion−burst.

6.8. Model, ordering, and boundary cases (continued)

Policy	Completion P1/P2/P3	Waiting P1/P2/P3	Mean waiting	First response P1/P2/P3	Batch p99 response
FCFS, initial P1/P2/P3	5/7/10	0/5/7	4	0/5/7	7
SJF, P2/P3/P1	10/2/5	5/0/2	7/3	5/0/2	5
RR q=2, initial P1/P2/P3	10/4/9	5/2/6	13/3	0/2/4	4

All use window [0,10], so throughput=3/10 jobs/ms. SJF has smaller mean waiting while RR has smaller batch first-response tail. Neither metric substitutes for the other. These are exact values for three specified jobs, not random simulation or population-p99 estimates; no 95% confidence interval is claimed. Random experiments require arrival/service distributions, seeds, independent replication units, and an estimator. A confidence interval for a mean is not a range containing 95% of individual waits.

6.8. Model, ordering, and boundary cases (continued)

Reasoning prompt. How would an evaluation expose the tail rather than only mean throughput?

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6.9. Summary

- Scheduling correctness includes queue ownership and no double execution. Service quality adds explicitly required waiting bounds and response distributions.
- High CPU utilization does not imply short user delay. Additional runnable work can increase shared-resource contention and reduce useful throughput.

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6.9. Model, ordering, and boundary cases

Model assumptions. Correctness, quality, and cost receive separate evidence.

Invariant and ordering. If starvation freedom is required, establish it under admission and arrival assumptions. Fixed priorities alone may not provide it.

Counterexample or boundary. A 99% busy batch CPU that indefinitely postpones interactive work fails an interactive-service requirement.

Worked check. Retain RR q=2, all arrivals 0, and initial order P1/P2/P3; multiply every burst by k. Pop the queue head, run min(2,remaining), and append it only if unfinished. Exact replay gives:

6.9. Model, ordering, and boundary cases (continued)

k; bursts	Completion P1/P2/P3	Waiting P1/P2/P3	Throughput jobs/ms	Batch p99 first-CPU response	Maximum waiting
1; 5/2/3	10/4/9	5/2/6	3/10	4	6
2; 10/4/6	20/10/16	10/6/10	3/20	4	10
3; 15/6/9	30/16/25	15/10/16	1/10	4	16

For k=2 the complete trace is P1[0,2), P2[2,4), P3[4,6), P1[6,8), P2[8,10), P3[10,12), P1[12,14), P3[14,16), P1[16,18), P1[18,20). First responses remain 0/2/4 and busy utilization remains 100%, yet maximum ready waiting grows 6→10→16. First-CPU response, throughput, and waiting measure different effects. A finite batch does not prove starvation freedom under unbounded arrivals.

Reasoning prompt. Which per-task fairness measure reveals starvation hidden by aggregate utilization?

Chapter conclusions

- Scheduling names require arrival and burst assumptions
- Response and turnaround are different measures
- Multiprocessor and real-time analysis must specify locality, deadlines, and failure assumptions

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Exercises and sources

- the separate exercises PDF.
- the separate solutions PDF.
- Coverage and source limitations.

Reference: A. Silberschatz, P. B. Galvin, and G. Gagne, *Operating System Concepts*, 9th edition, Wiley, 2013. These are original explanations and original exercises. Historical product cases refer to the historical versions in the edition. No publisher solution text is used.

- Linux v3.2 sched_fair.c defines period/slice, sleep placement, tick, HRTICK, and wakeup comparisons.
- Linux v3.2 sched.c weight table.
- Microsoft priority boosts and Solaris time-sharing class.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Deadlocks

Deadlock differs from starvation and ordinary waiting. Necessary conditions, resource graphs, safe-state analysis, detection, and recovery require different information and impose different costs.

Prerequisites. Process states, Mutexes and semaphores, Resource-allocation graphs, Basic vector comparisons.

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Learning objectives

- Distinguish necessary from sufficient deadlock conditions
- Prove how a global lock order excludes circular waiting
- Apply the Banker's safety test using complete resource vectors

Central question. When can resource waiting become permanent, and which prevention or recovery strategy is justified by available information?

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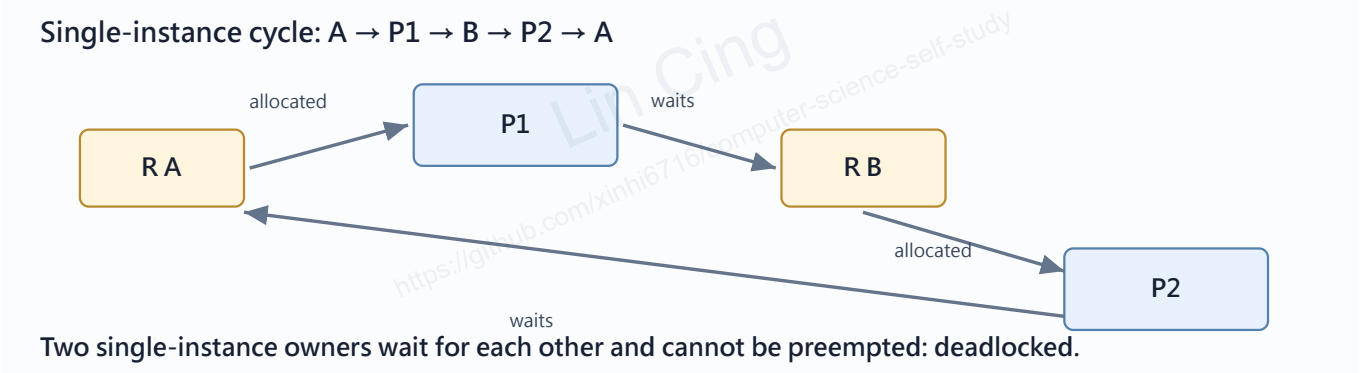
7.1. System model

- A resource type has a defined number of instances. Processes request, use, and release instances; an unsatisfied request may block.
- In a resource-allocation graph, process-to-resource edges represent requests and resource-instance-to-process edges represent allocations. The graph records specified ownership and waiting, not every possible future action.



7.1. Model, ordering, and boundary cases

Model assumptions. Instance counts are fixed in this model. Ownership and requests are observed consistently.

Invariant and ordering. Each exclusive instance has at most one owner. Every edge corresponds to a real request or allocation.

Counterexample or boundary. An unrecorded available instance can allow progress despite an apparent cycle.

Worked check. R1 has two instances held by P1 and P2. P1 waits for R2; P2 requests another R1. P3 holds R2 but can finish and release it. P1 then finishes and releases R1, allowing P2 to proceed. The partial waiting picture was insufficient to prove deadlock.

Reasoning prompt. Which additional instance or releaser can break an apparent cycle?

Resource state and concrete deadlocks

A resource type groups interchangeable instances. Two printers with incompatible location/function requirements belong to different types. Tables record type totals, free/allocated instances, owners, and each outstanding requester/type. A deadlocked set has every member waiting for an event only another waiting member can cause.

With three CD-RW drives, P1/P2/P3 each holds one and requests one more. Available=0 and none can finish, so all three deadlock. With one printer and one DVD, P1 holds DVD and requests printer; P2 holds printer and requests DVD. Neither can release, so their distinct resource types still deadlock.

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7.2. Deadlock characterization

- The classical resource deadlock requires mutual exclusion, hold and wait, no preemption, and circular wait to hold together. These conditions guide prevention; a cycle alone is sufficient only under the relevant single-instance model.
- With multiple instances, another runnable owner can release an instance and break the apparent dependency. Analyze whether any legitimate completion sequence remains.

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7.2. Model, ordering, and boundary cases

Model assumptions. Resources are exclusive, held while other requests wait, and not forcibly reclaimable. The system's permitted release behavior is specified.

Invariant and ordering. A deadlocked set cannot progress through its outstanding requests under those rules. Mere delay or lack of current CPU execution does not establish this.

Counterexample or boundary. Let R1 have one instance held by P1, and R2 two held by P2 and P3. P1 waits for R2 and P2 for R1. If P3 finishes, it frees R2 and breaks the cycle. This argument cannot be applied to two single-instance mutexes by inventing another owner.

Worked check. For single mutexes A and B, A is allocated to P1, P1 waits for B, B is allocated to P2, and P2 waits for A. If neither owner can proceed or release and neither lock is preemptible, the four-node cycle is deadlocked.

Reasoning prompt. What future-release information does a snapshot omit?

Exact graph criteria and a multiple-instance counterexample

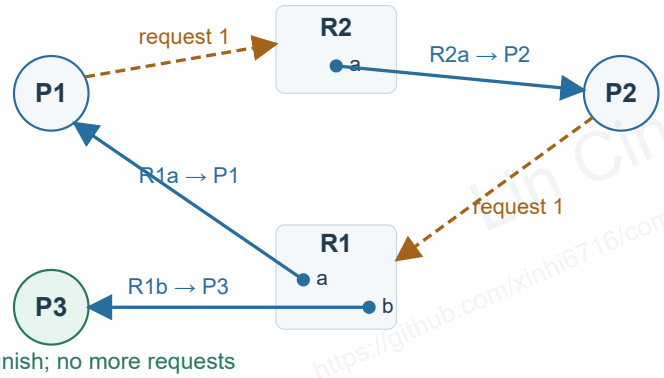
Mutual exclusion makes at least one resource exclusive. Hold-and-wait keeps an allocation while another request waits. No preemption forbids forced reclamation. Circular wait gives a dependency chain returning to its start. Circular wait already implies hold-and-wait; the four conditions are not logically independent.

A process node is a circle, resource type a box, instances dots. Request Pi→Rj becomes assignment Rj→Pi when granted; release removes the assignment. No graph cycle implies no deadlock. When every resource on a cycle has one instance, the cycle is necessary and sufficient for deadlock. With multiple instances it is only a necessary warning.

7.2. Why can a cycle still make progress?

R1 has instances R1a/R1b; R2 has R2a. P3 needs nothing more. Each process finishes and returns all holdings after its current request is granted. Available=(0,0).

Initial state: Total=(2,1), Available=(0,0)



Can finish; no more requests

Read arrow direction

Solid: instance → owner

Dashed: waiter → resource type

R1 contains two distinct instances.

Cycle: P1 → R2 → P2 → R1 → P1

P3 holds the other R1 instance.

Let P3 finish to unlock progress.

P3 can release R1b even while P1 and P2 wait on the cycle.

7.2. Finish P3, then unblock P2 and P1

Completion order P3 → P2 → P1; vector order (R1,R2)

Event	R1a owner	R1b owner	R2a owner	Available
Initial: P1 and P2 wait	P1	P3	P2	(0, 0)
P3 finishes; returns R1b	P1	free	P2	(1, 0)
Grant R1b to P2	P1	P2	P2	(0, 0)
P2 finishes; returns R1b, R2a	P1	free	free	(1, 1)
Grant R2a to P1	P1	free	P1	(1, 0)
P1 finishes; returns R1a, R2a	free	free	free	(2, 1)

A grant decreases Available; an actual completion returns holdings and increases it. The completion order P3→P2→P1 alone omits the grants to P2 and P1. The figure includes both intermediate deductions.

7.2. Detection's Work is a simulation, not live inventory

Vectors use order (R1,R2). Current outstanding requests and existing allocations give:

Process	Allocation	Request	Work before	Work after simulated completion
P3	(1,0)	(0,0)	(0,0)	(1,0)
P2	(0,1)	(1,0)	(1,0)	(1,1)
P1	(1,0)	(0,1)	(1,1)	(2,1)

Check Request≤Work before each step, then set Work←Work+Allocation. This combines granting Request with later returning Allocation+Request, for a net increase of Allocation only. Work may include returns by processes simulated as finished. **The simulation does not execute them; initial Available remains (0,0).**

This test uses **current outstanding Request**, not the Banker's maximum remaining Need. Section 7.6 gives the complete detection algorithm.

7.2. Change only P3's next action to obtain deadlock

Keep every allocation and the requests of P1 and P2. Now require P3 to obtain one R2 before it can finish, adding P3→R2.

Initial state	P3 Request	Eligible at Work=(0,0)	Result
Original	(0,0)	P3	Completion order P3→P2→P1
Modified	(0,1)	None	P1, P2, P3 are deadlocked

P1 and P3 now both wait for R2a, held by P2. Both instances of R1 needed by P2 are held by those waiting processes. Every instance is trapped within the waiting set, with no owner able to return one first. This assumes no preemption, cancellation, extra instances, or early release before completion. Merely delaying P3's CPU service does not change resource feasibility in the original case; scheduling starvation and resource deadlock are different questions.

7.2. A single-instance lock cycle

Thread A acquires `first` then `second`; B uses the reverse order.

Time	Action	Holdings and waits
t0	A acquires first	A holds first
t1	B acquires second	A holds first; B holds second
t2	A requests second	A holds first and waits for B
t3	B requests first	B holds second and waits for A

Both locks have one instance. Neither owner can proceed or release, and no external holder can help. This interleaving deadlocks. If A finishes before t1, that schedule avoids the cycle, but one successful stress schedule cannot exclude the bad schedule.

7.3. Methods for handling deadlocks

- Prevention structurally removes a necessary condition. Avoidance examines whether a proposed allocation preserves safety using declared future demands. Detection permits dangerous allocations, then identifies deadlock and recovers.
- A general-purpose system may leave some deadlocks to applications because global prevention or detection is costly. That choice does not remove application lock-order obligations.

7.3. Model, ordering, and boundary cases

Model assumptions. Maximum claims, runtime costs, and the consequences of aborting work are known to the degree the selected policy requires.

Invariant and ordering. Every accepted allocation follows that policy's guarantee. Avoidance cannot rely on demands absent from its model.

Counterexample or boundary. A safety test using an understated maximum claim can approve allocations that are unsafe for the real demands.

Worked check. With one R1 and one R2, prevention forbids reverse-order acquisition. Avoidance checks declared claim edges before granting the dangerous request. Detection permits the state and later scans the wait-for cycle to choose a victim. These mechanisms act at different times with different information.

Reasoning prompt. Which dominates the design choice, missing future information or the cost of recovering a failure?

7.4. Deadlock prevention

- Exclusive devices may inherently require mutual exclusion. Remove hold-and-wait by acquiring everything together or releasing before another request. Preemption works only when state can be safely saved and restored. A global acquisition order removes circular waiting.
- All-at-once allocation can waste capacity. Preemption can require rollback. Ordering is often practical for software locks but must include dynamic objects and indirect calls.

7.4.1. Lock-order proof

- If a thread may request a new lock only above every lock it already holds, each dependency follows an increasing rank. A cycle would require rank(A)<...<rank(A), a contradiction.
- Callbacks, lock upgrades, condition-wait reacquisition, and exceptional paths must obey the same rule. Checking only the normal path is incomplete.

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7.4. Model, ordering, and boundary cases

Model assumptions. All acquisition paths, including callbacks, upgrades, and errors, obey the model. Reclaimed resources have a consistent restoration protocol.

Invariant and ordering. Strictly increasing acquisition ranks cannot form a directed cycle. State must be consistent before a resource becomes available to another participant.

Counterexample or boundary. A normal A-then-B path and a repair B-then-A path can deadlock despite the usual path's ordering.

Worked check. Order A before B. T1 can hold A and wait for B; a T2 holding B cannot acquire A. The checker rejects that second request before waiting, reporting expected A<B and actual B→A. Rejection does not release B: the caller must explicitly restore a consistent state, release B, and retry in A-then-B order. Ignoring the error while retaining B forever can still leave T1 waiting indefinitely.

Reasoning prompt. How can dynamic resources receive a stable total order without depending on mutable names?

Policy information and cost

Policy	Information	Guarantee	Cost
Prevention	Restricted request protocol	Removes a necessary condition	Underutilization, refused work, rollback
Avoidance	Trustworthy declared maxima	Every grant preserves a safe sequence	Claims and runtime safety checks
Detection/recovery	Current ownership/request snapshot	Finds and repairs allowed deadlocks	Delay, scanning, lost victim work
Ignore	No general deadlock protocol	No general guarantee	Application timeout or restart

For one R1 and one R2, prevention orders F(R1)=1<F(R2)=2; a holder of R2 cannot request R1. Avoidance uses prior claim edges and refuses a grant that would create a cycle. Detection grants first and scans the wait-for cycle later. Their decision times differ.

Policy information and cost (continued)

Sharing read-only files removes exclusion only for those resources. Acquiring everything initially removes hold-and-wait but can reserve an idle printer until the final stage. Releasing everything before a new request may destroy uncommitted state. Preemption is useful only for restorable resources; it can reclaim a blocked process's allocations and restart when all needed resources become available. Strict ordering removes circular wait.

With injective F:R→N, a cycle would require F(R0)<F(R1)<…<F(Rn)<F(R0), hence F(R0)<F(R0), a contradiction. Sort dynamic account locks by stable ID before acquiring them; from/to parameter order can reverse across transactions. Callback, upgrade, wait reacquisition, and error paths participate in the same proof.

7.5. Deadlock avoidance

- The Banker model uses Available, Allocation, Max, and Need=Max-Allocation. Claims are upper bounds, each no greater than total resources, and requests stay within those claims. A single-instance claim graph is a related but different representation.
- A state is safe if some order can satisfy each remaining Need from current Work, then recover the completed process's Allocation. Safe does not mean nobody waits. Unsafe means that such a worst-case guarantee is absent, not that deadlock has already occurred.

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7.5. Model, ordering, and boundary cases

Model assumptions. Resource totals are fixed, snapshots are consistent, and participants eventually finish and return allocations once their needs are met.

Invariant and ordering. Initialize Work=Available. Repeatedly choose unfinished i with Need_i<=Work componentwise, set Work=Work+Allocation_i, and mark i finished. Safety requires all processes to finish in the reduction.

Counterexample or boundary. A process in an unsafe state might use less than its maximum or release resources early. That possible future does not establish the Banker's worst-case safety guarantee.

Worked check. Starting Work=(2,1,0), choose a process whose complete Need fits. Add only its existing Allocation on simulated completion; do not add imaginary resources or double-count the temporary grant, which is returned at completion.

Reasoning prompt. Why does deadlock imply unsafety in this model while unsafety does not imply current deadlock?

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Claim edges and the complete Banker algorithm

For single instances, a claim Pi→Rj means Pi may request Rj later. A request changes claim to request, then trial assignment. Grant only if the resulting graph remains acyclic. Claims must be known before start, or added only while that process holds no assignments. Missing claims invalidate the guarantee. An adjacency-matrix cycle method costs $O(n^2)$.

For n processes and m resource types, Need=Max−Allocation and comparisons are componentwise. Legal inputs have nonnegative integers and 0≤Allocation≤Max≤Total.

```
Work = Available
Finish = [false] * n
while there is unfinished i with Need[i] <= Work:
    Work += Allocation[i]
    Finish[i] = true
safe iff all Finish
```


Claim edges and the complete Banker algorithm (continued)

Once Need_i≤Work, lending Need_i and receiving Allocation_i+Need_i has net effect Work−Need_i+(Allocation_i+Need_i)=Work+Allocation_i. It neither manufactures Need nor double-counts the temporary loan. Assume participants finish and return everything once their declared remaining needs are met.

The greedy choice needs no backtracking. If a safe order exists and a different currently feasible Pi is chosen first, move Pi to the front and keep all other processes in their original order. Returning Pi's nonnegative allocation only increases resources faced by every remaining step. If the algorithm stops unfinished despite a supposed safe order, take the first unfinished Pj in that order. Every predecessor has already been removed by the algorithm; current Work contains their allocations and possibly more. Thus Need_j≤Work, contradicting the absence of a candidate. Completion provides a safe-order certificate; stopping unfinished proves unsafety within this model.

Resource-request transaction

Validate an m-component nonnegative integer Request_i. If any component exceeds Need_i, reject as exceeding the claim. Otherwise if any component exceeds Available, wait. Negating componentwise $\leq$ means some component is greater, not all components.

Trial Available−=Request, Allocation_i+=Request, Need_i−=Request. Run safety. Commit only if safe; otherwise restore all three arrays and wait. The update is transactional. Typical safety complexity is $O(mn^2)$, at most n scans of n rows with m-component tests.

Counterexample: Total=(2,1), Allocation=(1,0), Need=(1,1), Available=(1,1) is legal. Request=(2,0) exceeds Need and Available in the first component. It must error before trial. Requiring both components to be greater would wrongly allow Available=(−1,1).

Banker T0 and every Work state

Total=(10,5,7), Available=(3,3,2).

Process	Allocation	Max	Need
P0	(0,1,0)	(7,5,3)	(7,4,3)
P1	(2,0,0)	(3,2,2)	(1,2,2)
P2	(3,0,2)	(9,0,2)	(6,0,0)
P3	(2,1,1)	(2,2,2)	(0,1,1)
P4	(0,0,2)	(4,3,3)	(4,3,1)

Each next Need fits. Work proceeds (3,3,2)→P1 (5,3,2)→P3 (7,4,3)→P4 (7,4,5)→P0 (7,5,5)→P2 (10,5,7). Thus ⟨P1, P3, P4, P0, P2⟩ is safe.

P1 requests (1,0,2), within Need(1,2,2) and Available. Trial gives Available=(2,3,0), Allocation1=(3,0,2), Need1=(0,2,0). The same order yields Work (2,3,0)→(5,3,2)→(7,4,3)→(7,4,5)→(7,5,5)→(10,5,7), so grant.

Banker T0 and every Work state (continued)

After that grant, P0 requests (0,2,0). Although available, its trial leaves Work=(2,1,0). No remaining Need fits, including P0's updated Need=(7,2,3), so rollback and wait. Unsafe means the declared worst-case completion guarantee is missing. Voluntary early release or smaller actual demands can still avoid a real deadlock.

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7.6. Deadlock detection

- A consistent wait-for graph detects cycles for exclusive single-instance resources. With multiple instances, reduce a snapshot using current outstanding Request rather than maximum future Need.
- Detection frequency trades checking cost against time spent blocked. The result describes the snapshot rather than every future execution.

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7.6. Model, ordering, and boundary cases

Model assumptions. Available, current requests, and ownership form one consistent snapshot.

Invariant and ordering. Repeatedly satisfy a current Request_i<=Work and return Allocation_i to Work. Interpret the remaining holders according to the detection algorithm's initialization rules, including processes holding no resources.

Counterexample or boundary. Observing ownership before a release but availability after an unrelated update can misclassify the state.

Worked check. With Work=(1,0,0), one request (1,0,0) can finish. If its allocation includes a second-type resource, returning that resource can permit the next request (0,1,0).

Reasoning prompt. What snapshot or locking protocol makes the reduction meaningful?

Detection reduction and recovery states

For single instances, replace Pi→Rq→Pj by Pi→Pj. A cycle is deadlock exactly under the exclusive single-instance assumptions. Any multiple-instance resource requires the allocation/request analysis.

```
Work = Available
Finish[i] = (Allocation[i] == zero_vector)
while there is i with not Finish[i] and Request[i] <= Work:
    Work += Allocation[i]
    Finish[i] = true
remaining false holders form the detected blocked set
```

Detection uses current Request, not maximum Need. It optimistically assumes a satisfiable process completes without additional requests. New requests require a later snapshot. A consistent snapshot and $O(mn^2)$ reduction are required.

Detection reduction and recovery states (continued)

Available=(0,0,0). The complete base snapshot is:

Process	Allocation	Request
P0	(0,1,0)	(0,0,0)
P1	(2,0,0)	(2,0,2)
P2	(3,0,3)	(0,0,0)
P3	(2,1,1)	(1,0,0)
P4	(0,0,2)	(0,0,2)

Work transitions P0→(0,1,0), P2→(3,1,3), P3→(5,2,4), P1→(7,2,4), P4→(7,2,6). All Finish become true, so no deadlock in this snapshot.

Detection reduction and recovery states (continued)

Change only Request2 to (0,0,1). Simulating P0 gives Work=(0,1,0); no other Request fits, so Finish=(true, false, false, false, false) and the detected set is {P1, P2, P3, P4}. Simulation does not actually complete P0 or change Available. If P0 has not finished, aborting P1 and safely reclaiming Allocation1=(2,0,0) changes actual Available from (0,0,0) to (2,0,0).

Restart reduction with Work=Available, excluding the terminated P1:

Simulated completion	Current Request	Work after returning Allocation
Initial; P1 terminated	—	(2,0,0)
P0	(0,0,0)	(2,1,0)
P3	(1,0,0)	(4,2,1)
P2	(0,0,1)	(7,2,4)
P4	(0,0,2)	(7,2,6)

Detection reduction and recovery states (continued)

Check Request≤Work before each row and add only the existing Allocation. This certifies that survivors can finish; actual Available rises only when they really return resources. Starting at actual Available=(2,1,0) is valid only if P0 has already completed outside the simulation.

Frequent scans reduce detection delay and can identify the latest requester, but cost more. Periodic scans can see several cycles together. A snapshot does not guarantee every future state.

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7.7. Recovery from deadlock

- Recovery can abort all or selected participants or reclaim suitable resources. Victim choice considers work lost, allocations, priorities, and repeated victimization.
- Taking a lock away does not restore partially modified data. Recovery may need transactions, checkpoints, or a restart of the affected subsystem.

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7.7. Model, ordering, and boundary cases

Model assumptions. The recovery unit and all resource ownership can be reconstructed. If rollback is required, a usable boundary exists.

Invariant and ordering. Recovery exposes a consistent shared version. Victim selection must satisfy any required starvation restriction.

Counterexample or boundary. Releasing a lock over a half-updated tree lets the next owner observe broken links.

Worked check. Track checkpoint positions, rollback costs, and prior victim counts. Prefer low-cost victims subject to a rule that prevents unlimited repeated selection.

Reasoning prompt. Which evidence distinguishes a committed success from work discarded during recovery?

Recovery unit and victim fairness

Abort all deadlocked processes to break every cycle at the cost of all unfinished work, or abort one, reclaim its resources, rerun detection, and stop once survivors can progress. Victim policy considers priority, interactivity, executed/remaining time, held/needed resources, reclamation difficulty, and how many aborts are needed.

Files and devices can retain partial external effects. Recovery needs undo, journals, checkpoints, or device reset, not merely unlocked mutexes. Preemption must specify victim, consistent rollback point, and protection against repeated victimization.

T2 has checkpoint cost 2 and rollback count 0; T4 cost 3/count 0. Choose T2, roll back and release 2 resources. Rerun detection; if T4 now completes, stop aborting. Increment T2's rollback count. A cost such as execution_i+k×rollback_count_i penalizes repeated selection, but a formal finite bound also needs workload assumptions. An explicit policy can exclude a process after at most r consecutive victimizations, for finite r.

Recovery unit and victim fairness (continued)

A single-instance P1/P2 wait cycle is deadlock. Separately, a ready P3 selected zero times while P4 completes five admissions is evidence of possible starvation; only an unbounded execution or policy argument proves no finite bound. No fabricated wait cycle is needed. Ordering can remove deadlock while a CPU policy still starves the lock holder.

Observation	Conclusion	Additional evidence
Four conditions	Deadlock possible	No remaining completion transition
Single-instance graph no cycle	No deadlock in snapshot	Validated graph
Single-instance cycle	Deadlock	All cycle resources truly single-instance
Multi-instance cycle	Deadlock possible	Availability/reduction
Unsafe	No worst-case avoidance guarantee	Actual current requests
One long wait while others finish	Possible starvation	Admission policy and waiting bound

7.8. Summary

- The four conditions guide prevention. Allocation graphs, avoidance, and detection answer different questions under different information assumptions.
- Deadlock prevents a dependent set from progressing. Starvation can leave one participant waiting while others continue.

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7.8. Model, ordering, and boundary cases

Model assumptions. Deadlock, unsafety, cycles, and starvation are distinct classifications.

Invariant and ordering. Allocation and recovery conserve resources and specify the scheduling assumptions needed for progress.

Counterexample or boundary. An ordered locking protocol can exclude cycles while a lock owner is still indefinitely denied CPU time by another policy.

Worked check. Apply a necessary-condition check, a resource graph, and a per-participant progress analysis separately.

Reasoning prompt. Give a progressing execution in which one participant nevertheless starves.

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Chapter conclusions

- Necessary conditions alone do not establish every deadlock claim
- Avoidance depends on valid maximum claims
- Detection, recovery, and data consistency must be designed together

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Exercises and sources

- the separate exercises PDF.
- the separate solutions PDF.

Reference: A. Silberschatz, P. B. Galvin, and G. Gagne, *Operating System Concepts*, 9th edition, Wiley, 2013. Chapter 7, printed pages 315–339; resource-allocation graph criteria in Section 7.2, detection in Section 7.6.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Main memory

Relocation, segmentation, paging, and page-table structures connect address translation to protection, fragmentation, TLB behavior, and effective access time.

Prerequisites. Binary addresses, Registers and the memory interconnect, Process isolation.

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Learning objectives

- Derive base-and-limit, segment, and page translations
- Distinguish internal and external fragmentation
- Calculate TLB costs and explain page-table levels

Central question. How can a logical address space map safely onto finite, physically scattered memory?

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8.1. Background

- The MMU translates CPU-generated logical addresses to physical addresses under protected configuration. Ordinary processes do not gain unrestricted physical access by choosing an integer address.
- In a base-and-limit model, a legal offset d satisfies 0<=d<limit and translates to base+d. Hardware checks accesses; untrusted software's prior check cannot enforce the boundary.

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8.1. Model, ordering, and boundary cases

Model assumptions. The CPU uses logical addresses and hardware enforces the selected relocation or paging model.

Invariant and ordering. A segment offset is less than its limit. A k-bit page offset has numeric range 0 through 2^k-1; comparing an address value with the bit count k would be a unit error.

Counterexample or boundary. Trusting a user program to validate its own pointer lets hostile code bypass the check.

Worked check. With base=0x8000 and limit=0x2000, offset 0x1FFF is legal and resolves to 0x9FFF. Offset 0x2000 is outside the half-open range.

Reasoning prompt. How does placing protection in trusted hardware rather than untrusted software affect the trust model?

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8.2. Swapping

- Classical swapping moves an entire process image to backing storage to free RAM. Restoration needs the saved image, address-space metadata, and an appropriate placement or relocation mechanism.
- Demand paging selectively loads pages rather than requiring the whole image. Its granularity and cost differ from whole-process swapping.

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8.2. Model, ordering, and boundary cases

Model assumptions. Backing storage has sufficient capacity and the manager retains the required process state.

Invariant and ordering. Only one image is authoritative during a move. A frame used by outstanding DMA cannot be reassigned while the device may still access it.

Counterexample or boundary. A DMA buffer swapped out and reused for another process can receive a later device write into unrelated data.

Worked check. An 80 KiB image fits a 100 KiB free backing area. Before resuming the restored process, install the correct mappings and invalidate translations as required by the architecture.

Reasoning prompt. Which control points and costs differ between a whole-image swap and a demand-page transfer?

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8.3. Contiguous allocation

- Each allocation occupies a contiguous physical region in this model. First-fit selects the first sufficient hole in traversal order; best-fit selects the smallest sufficient hole; worst-fit selects the largest.
- Allocations and frees create external fragmentation. Compaction moves allocated regions and requires relocation support and coordination with address-dependent activity.

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8.3. Model, ordering, and boundary cases

Model assumptions. The free list contains only usable holes. Updates are synchronized, and growth or maximum-size rules are explicit.

Invariant and ordering. Allocated ranges do not overlap and allocated plus free ranges account for the managed region. Movement preserves references and protected access.

Counterexample or boundary. Four separate 5 KiB holes total 20 KiB yet cannot satisfy one contiguous 10 KiB request.

Worked check. In free-list order 100,500,200 KiB, a 180 KiB request uses the 500 KiB hole under first-fit, leaving 320. Best-fit uses 200, leaving 20. Worst-fit also uses 500. Selecting 200 for first-fit would ignore the given traversal order.

Reasoning prompt. Which allocation pattern leaves many unusably small holes under best-fit?

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8.4. Segmentation

- A logical address contains segment number and offset. A segment-table entry supplies base, limit, and applicable rights. A legal access translates to base+offset after bounds and permission checks.
- Segments represent logical units with variable lengths and can suffer external fragmentation. Fixed-size paging avoids that placement problem at page granularity but can waste space inside a final page. The mechanisms can be combined.

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8.4. Model, ordering, and boundary cases

Model assumptions. Segment bounds and lifetimes are tracked, and hardware faults on unauthorized or out-of-range accesses.

Invariant and ordering. Offset is below limit and the resulting access respects the authorized segment mapping. Explicitly shared segments are a separate permitted case.

Counterexample or boundary. Checking base without checking the requested offset and size permits an access into an adjacent segment.

Worked check. Segment 2 has base=0x40000 and limit=0x3000. Offset 0x2FFF maps to 0x42FFF; 0x3000 faults for a single-byte access.

Reasoning prompt. Why can a logical segmentation model still use paging to simplify physical placement?

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8.5. Paging

- Split a virtual address into page p and offset d. If an authorized resident entry maps p to frame f, physical address=f*page_size+d. The offset is unchanged.
- Smaller pages reduce tail waste but increase the number of entries for a fixed range. Larger pages increase TLB coverage per entry and may reduce table work, while increasing allocation and transfer granularity.

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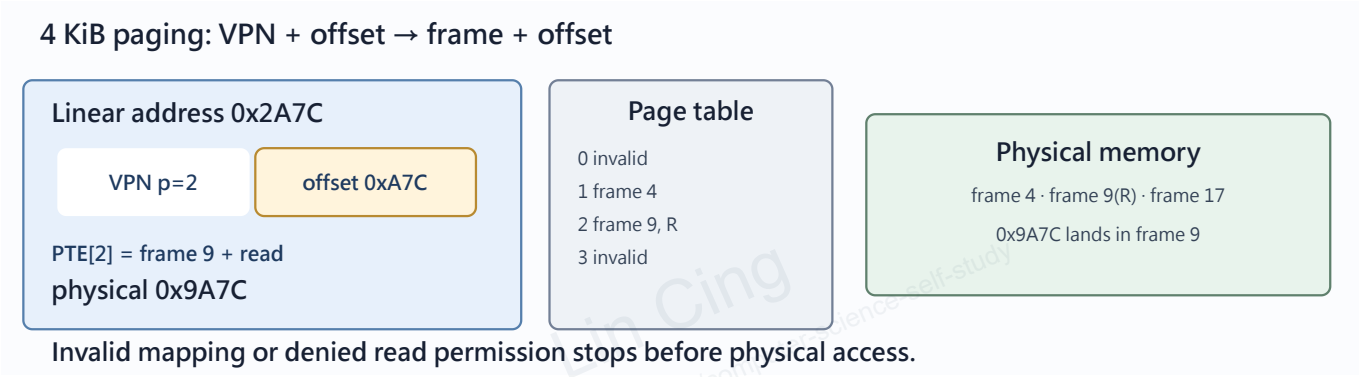
8.5. Paging (continued)



Original vector diagram for this course.

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8.5.1. Internal and external fragmentation

- Internal fragmentation is unused space inside an allocated unit that exceeds the request. Fixed pages and filesystem blocks can create it.
- External fragmentation leaves enough total free space in separate holes but no contiguous region large enough for one request.

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8.5.2. TLBs and effective access time

- A TLB caches translations. In a serialized lookup model, a hit costs t+m. A miss requiring k page-table memory reads costs t+k*m+m, including the eventual data access.
- With hit fraction h, EMAT=h(t+m)+(1-h)(t+k*m+m). A parallel TLB/cache lookup changes the timing model; page faults and additional cache levels require separate terms.

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8.5.3. Protection and sharing

- Entries may encode read, write, execute, privilege, and presence rules. Different address spaces can share a physical frame through explicitly authorized mappings.
- Read-only code sharing saves frames. An accidentally writable or executable mapping can break isolation. Presence alone does not grant every operation.

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8.5. Model, ordering, and boundary cases

Model assumptions. Page size is 2^k and descriptors contain architecture-specific mapping, presence, and permission state. Translation caches follow an invalidation protocol.

Invariant and ordering. Successful access has a valid translation and sufficient rights. Revocation or frame reuse cannot leave an old translation usable in an affected address space.

Counterexample or boundary. Unmapping a page without the required TLB invalidation may let a CPU reach a frame already reused by another process.

Worked check. With 4 KiB pages, frame 9 and offset 0xA7C give 0x9000+0xA7C=0x9A7C.

Reasoning prompt. Which mapping changes also require TLB or cache maintenance?

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8.6. Page-table structures

- Large virtual ranges make dense linear tables expensive. Hierarchical tables omit unused lower levels; hashed tables and inverted tables organize lookup differently.
- More levels save sparse-table storage but can add work on a TLB miss. A 64-bit instruction set need not implement all 64 bits of virtual address space.

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8.6. Model, ordering, and boundary cases

Model assumptions. Each hierarchical entry points to the next valid level or an allowed leaf. Address fields follow the chosen mode.

Invariant and ordering. A page walk must resolve every required level before reaching a frame. A missing intermediate mapping cannot be treated as a usable translation.

Counterexample or boundary. Building a fully dense table for every possible 64-bit byte address ignores both sparsity and implemented address-width limits.

Worked check. Linear 0x12345678 with 4 KiB pages splits PDI 0x048, PTI 0x345, offset 0x678. If outer[0x048] points to T and T[0x345] to frame 0xABC, PA=0xABC678. An absent inner entry faults; VA cannot be used as PA.

Reasoning prompt. Which miss costs accompany the storage saved by a hierarchical table?

IA-32 linear-address walk

Fix protected mode, ordinary 4 KiB pages, and linear address 0x12345678 after segmentation. The selector first locates an LDT/GDT descriptor; segment base/limit and rights produce the linear address. Paging translates that result, so the given number is not the pre-segmentation logical address.

A 4 KiB directory/table has 1024 four-byte entries, giving 10/10/12 bits. For linear L,

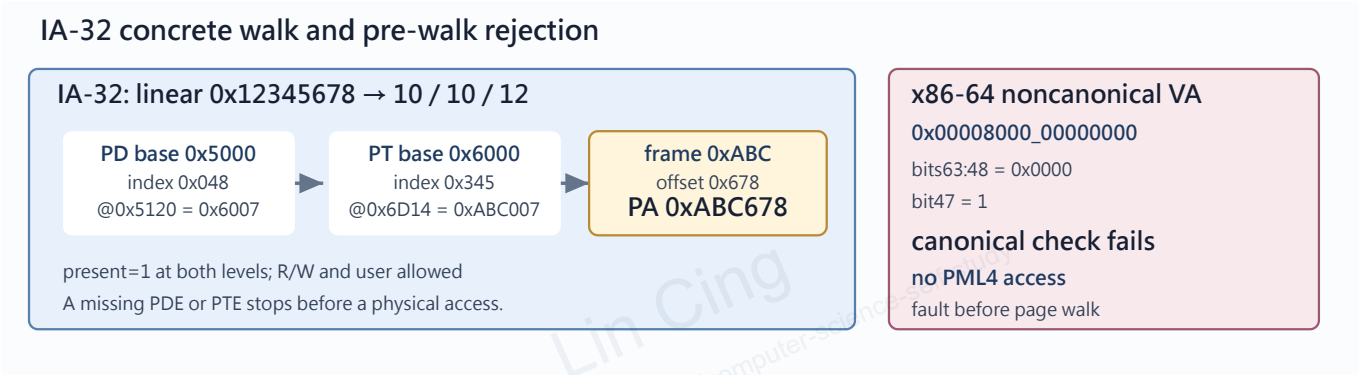
$$PDI = (L \gg 22)\ \&\ 0x3FF, \quad PTI = (L \gg 12)\ \&\ 0x3FF, \quad d = L\ \&\ 0xFFF.$$

CR3 base 0x5000 and next PT base 0x6000 yield:

Level	Index	Entry address	Entry value	Meaning
Directory	0x048	0x5000+4×0x048=0x5120	0x00006007	Present, next base 0x6000, RW/user
Table	0x345	0x6000+4×0x345=0x6D14	0x00ABC007	Present, frame 0xABC, RW/user

Offset 0x678 gives PA0xABC000+0x678=0xABC678. This path uses directory, table, and target page, each 4 KiB. Table-allocation costs are separate from an individual walk.

IA-32 linear-address walk (continued)



An absent PDE stops before reading the PT; an absent PTE cannot supply a usable frame. A4 MiB PDE with Page Size set terminates at the directory and uses 22 offset bits instead. That is a different encoding from this fixed trace.

Canonical x86-64 addresses and four levels

Fix 48-bit canonical virtual addresses, four levels,4 KiB pages,8-byte entries. Index fields are[47:39], [38:30],[29:21],[20:12], with offset[11:0]. Each table has 512 entries. First require 0≤v<2^64, then require every bit 63:48 to repeat bit 47.

$$canonical(v) \iff (v \gg 48) = \begin{cases} 65535, & ((v \gg 47)\ \&\ 1) = 1, \\ 0, & ((v \gg 47)\ \&\ 1) = 0. \end{cases}$$

Comparing only bit 63 to bit 47 is insufficient. v=0x0001_0000_0000_0000 has both 0 but upper 16 bits=1, so is noncanonical. Flipping any one sign-extension bit of a legal address invalidates it. Ranges 0…0x00007FFFFFFFFFFF and 0xFFFF800000000000…0xFFFFFFFFFFFFFFFF are legal; the interval between is illegal.

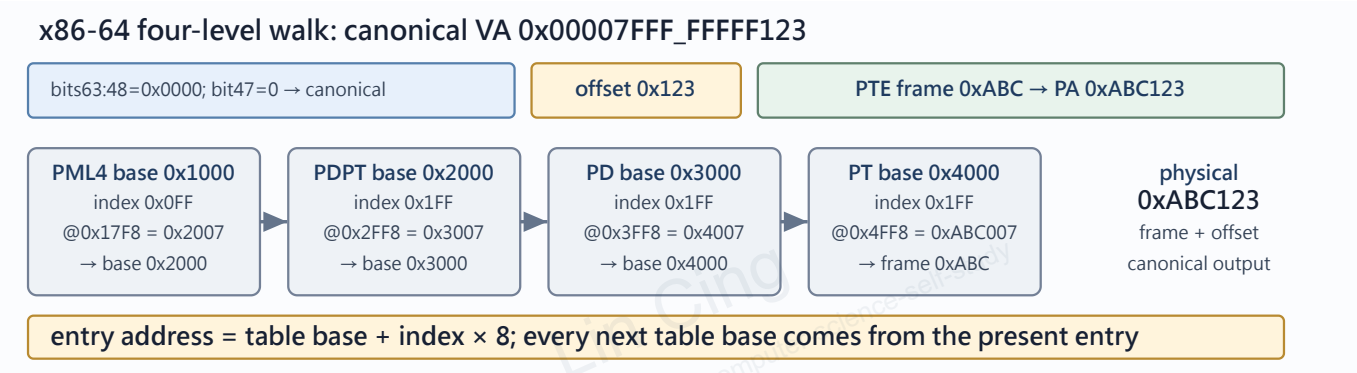
For VA 0x00007FFF_FFFFF123, bit 47=0 and upper 16 bits=0. CR3=0x1000; subsequent bases are 0x2000,0x3000,0x4000.

Canonical x86-64 addresses and four levels (continued)

Level	Index	Offset	Entry address	Entry value	Next base/frame
PML4	0x0FF	0x7F8	0x17F8	0x2007	PDPT 0x2000
PDPT	0x1FF	0xFF8	0x2FF8	0x3007	PD0x3000
PD	0x1FF	0xFF8	0x3FF8	0x4007	PT0x4000
PT	0x1FF	0xFF8	0x4FF8	0xABC007	Frame 0xABC

Every address is base+index×8; every next base comes from the present entry. Offset 0x123 is retained, so PA=0xABC123.

Canonical x86-64 addresses and four levels (continued)



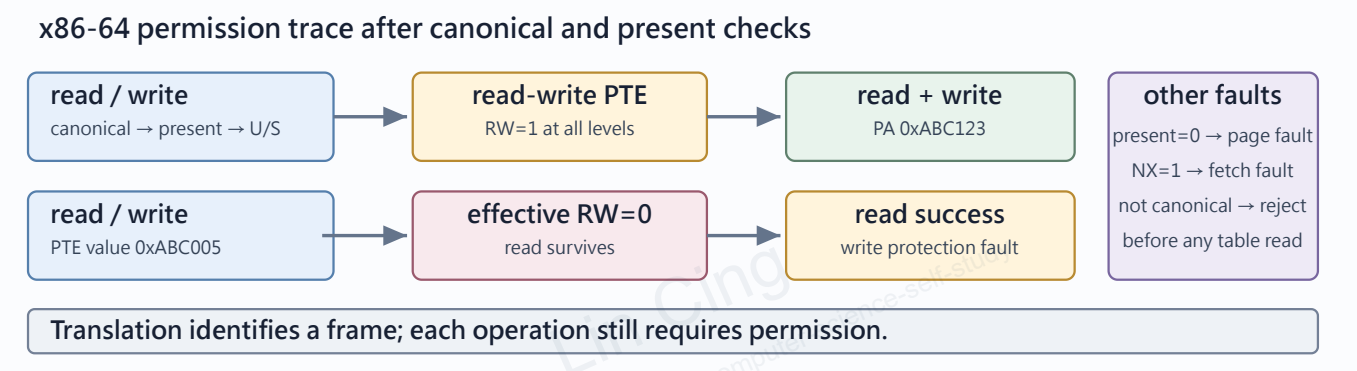
2 MiB and 1 GiB encodings end higher in the walk and require other formats/alignment and larger offsets. The 9/9/9/9/12 split applies only to this 4 KiB configuration.

Mapping and operation permission

Canonical check precedes all table memory accesses. Present establishes usable entries, U/S restricts user access at every level, effective RW restricts writes, and NX restricts instruction fetch.

Case	Canonical/mapping	Permissions	Read	Write	Fetch
Normal trace	Canonical, four present levels	All user/RW, NX0	0xABC123	0xABC123	Allowed
PTE 0xABC005	Same present path	User, RW0, NX0	0xABC123	Protection fault	Allowed
PML4[0x0FF] absent	Canonical, no mapping	No effective mapping	Page fault	Page fault	Instruction page fault
PTE NX1	Canonical, present	User/RW, NX1	Allowed	Allowed	Fetch fault
VA 0x00008000_00000000	Noncanonical	No walk	Reject	Reject	Reject

Mapping and operation permission (continued)



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Mapping and operation permission (continued)

The last address has bit 47=1 but upper 16 bits=0. The canonical address 0x00007FFF_FFFFF000 still page-faults if any required level is absent. The illegal interval ends at 0xFFFF7FFFFFFFFFFF, immediately below the upper canonical range. A root, frame, and valid canonical input do not alone authorize every operation. PAE, five levels, large pages, or another physical width require the matching masks and entries. Updated entries need required TLB invalidation; a stale cached permission can outlive a changed PTE.

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ARMv7-A short descriptors

Fix AArch32, short descriptors, TTBCR. N=0, SCTLR. AFE=0, SCTLR. TRE=0, PL1 table construction, and 4 KiB small pages. This is not AArch64 long-descriptor translation. N=0 selects TTBR0 for all input VAs. The complete L1 has 4096 four-byte entries,16 KiB, aligned 16 KiB. Sparse use does not permit allocating only the populated entries; other entries need valid architectural encodings, including invalid ones.

$$L1addr = TTBR0 + ((VA \gg 20)\ \&\ 0xFFF) \times 4.$$

A coarse L1 descriptor points to a256-entry L2,1 KiB, aligned 1 KiB; bits 31:10 supply its base.

$$L2addr = L2base + ((VA \gg 12)\ \&\ 0xFF) \times 4, \qquad PA = frame + (VA\ \&\ 0xFFF).$$

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ARMv7-A short descriptors (continued)

Storage	Base	Inclusive range	Size/alignment
P1 root	0x8000	0x8000–0xBFFF	16 KiB, base mod 0x4000=0
P2 root	0xC000	0xC000–0xFFFF	16 KiB, base mod 0x4000=0
P1 L2	0x10000	0x10000–0x103FF	1 KiB, base mod 0x400=0
P2 L2	0x10400	0x10400–0x107FF	1 KiB, base mod 0x400=0

All four ranges are distinct. Only the final data frame is shared.

Process/VA	L1 index/address	L2 base	L2 index/address	Frame	PA
P1 0x00101000	0x001 /0x8004	0x10000	0x001 /0x10004	0x40000	0x40000
P2 0x00202000	0x002 /0xC008	0x10400	0x002 /0x10408	0x40000	0x40000
P1 0x00101A5C	0x001 /0x8004	0x10000	0x001 /0x10004	0x40000	0x40A5C

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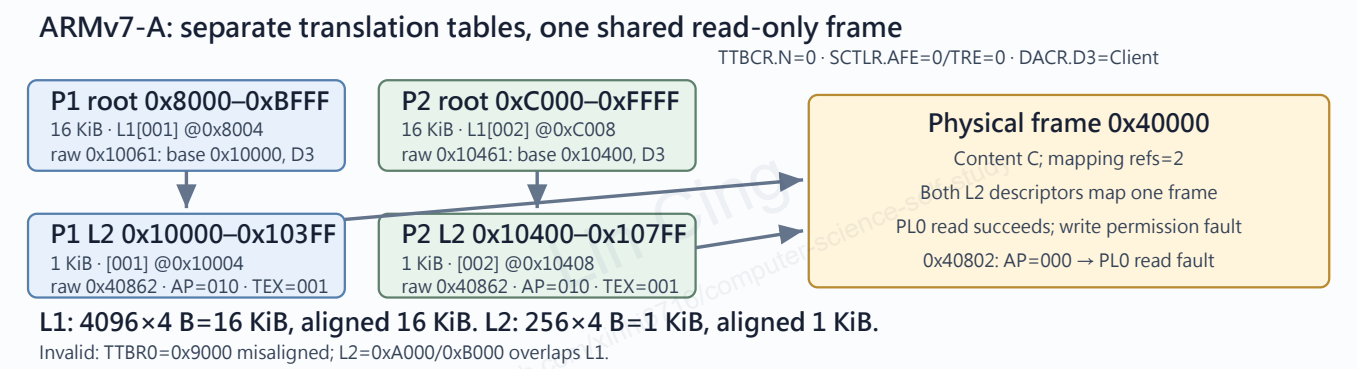
ARMv7-A short descriptors (continued)

Offset[11:0]=0xA5C does not enter either table index. Keeping only 8bits would incorrectly give 0x4005C.



Raw ARM words, domains, AFE and AP

Both L1 descriptors use domain 3; DACR. D3=01 means Client. Domain is in L1 and inherited by L2. With AFE=0, small-page AP[2] is bit 9 and AP[1:0] bits 5:4, not section bits 11:10.

Descriptor	Raw	Decode
P1 L1	0x10061	Type 01 coarse, base 0x10000, domain 3
P2 L1	0x10461	Type 01 coarse, base 0x10400, domain 3
Both L2	0x40862	Frame 0x40000, type 10, XN0, nG1, AP010, TEX001, C=B=0

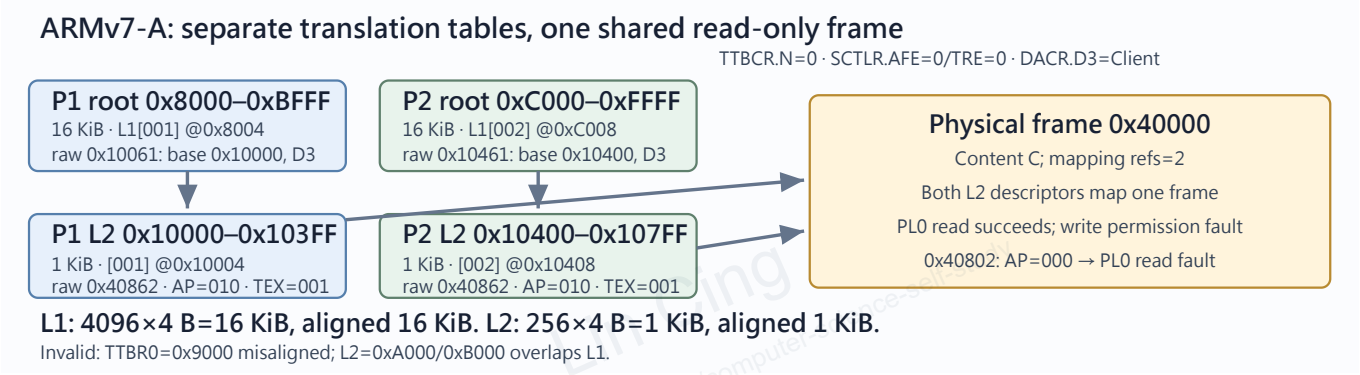
Decode field=(raw>>shift)&mask. AP2=(raw>>9)&1, AP1=(raw>>5)&1, AP0=(raw>>4)&1, TEX=(raw>>6)&7, C=(raw>>3)&1, B=(raw>>2)&1, XN=raw&1. AP010 gives PL1 read/write and PL0 read-only. With TRE0, TEX001/C0/B0 means Normal non-cacheable. Fetch additionally assumes WXN=UWXN=0, readable mapping, Client domain, and no other restriction. XN0 alone cannot prove executable Strongly-ordered memory.

Raw ARM words, domains, AFE and AP (continued)

0x40802 instead decodes AP000 and denies PL0 read/write even though the frame is valid. 0x40022 has AP010 but TEX/C/B=0/0/0, Strongly-ordered; it only illustrates data AP, not guaranteed fetch.

Mode	PL0 read	PL0 write	Reason
AFE0, Client, AP010	Success	Permission fault	Read-only PL0
AFE0, No access domain	Domain fault	Domain fault	Domain rejects first
AFE0, Manager domain 11	Not rejected by AP	Not rejected by AP	Manager bypasses AP; other faults remain possible
AFE1, Client, same 0x40862	Access-flag fault	Access-flag fault	AP0 is now access flag 0; software-managed flag model
AFE1, Client,0x40872	Success	Success	Flag 1, AP[2:1]=01 grants PL0 RW

Raw ARM words, domains, AFE and AP (continued)



Changing DACR or AFE changes the interpretation, so the AFE0 read-only result cannot be reused unchanged.

ARM unmap, reference counts and stale translations

Initially the frame contains C, mapping refcount 2, both entries 0x40862, Client parents, and no cached translation.

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ARM unmap, reference counts and stale translations (continued)

State/operation	Translation/permission	Result	Content/references
S0 P1 read	0x00101000→0x40000, AP permits	Success	C, refs 2
S0 P1 write	Valid mapping, AP denies	Permission fault	C unchanged
S0 P2 read	0x00202000→0x40000, AP permits	Success	C, refs 2
S0 P2 write	Valid mapping, AP denies	Permission fault	C unchanged
S1 remove P1	L2 invalid, required TLB maintenance complete	P1 translation fault	P2 remains, refs 1
S2 P2 read	Independent tables still valid	Success	refs 1
S3 remove P2	L2 invalid, maintenance complete	P2 translation fault	refs 0; reclaim only after all stale translations are unusable

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ARM unmap, reference counts and stale translations (continued)

An absent mapping is not a valid-but-read-only mapping. Descriptor changes need architecture-specific VA/ASID TLB maintenance and barriers. nG1 associates translations with their ASIDs; ASID does not determine domain or AP. Clearing a PTE alone cannot authorize frame reuse while an old TLB can still access it.

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Table storage must be valid before sharing is analyzed

Invalid layout	Failure
P2 TTBR0=0x9000, N0	0x9000 mod 0x4000=0x1000, root misaligned
P1 root 0x8000, P2 root 0x9000	Ranges 0x8000–BFFF and 9000–CFFF overlap
P1 L2=0xA000	Inside P1 root
P2 L2=0xB000	Inside P1 root
Only 4 KiB allocated for L1	4096×4 entries require 16 KiB

Legal sharing has independent roots/L2 storage and one shared frame. Verify page identity, storage, domain/AP/XN, references, and TLB state together. A writable P1 mapping would modify the frame P2 reads; that is writable sharing, not read-only code. Preserving P2's old data while P1 changes requires COW or a separate frame, not merely changing permissions in one entry.

8.9. Summary

- Segmentation organizes logical objects; paging simplifies physical placement. Both require defined protection and sharing semantics.
- TLB lookup, page walks, and fault handling form one access path. Mapping changes must invalidate obsolete interpretations before unsafe reuse.

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8.9. Model, ordering, and boundary cases

Model assumptions. Hardware checks accesses and fault entry preserves the architecture's recovery state.

Invariant and ordering. Mapping, presence, rights, and cached translations agree at the points required for successful access and safe reclamation.

Counterexample or boundary. Changing a frame number while using the wrong offset width produces a different physical address even if the frame exists.

Worked check. An invalid range is rejected, a forbidden write is denied or handled under an explicit mechanism such as COW, and a TLB miss may simply trigger a successful page walk.

Reasoning prompt. Which faults can the OS service, and which accesses must it reject?

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Chapter conclusions

- A mapping and its permissions are distinct parts of access validation
- Classify fragmentation before comparing allocation schemes
- Effective access time is a model with explicit cache and fault assumptions

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Exercises and sources

- the separate exercises PDF.
- the separate solutions PDF.
- Coverage and source limitations.

Reference: A. Silberschatz, P. B. Galvin, and G. Gagne, *Operating System Concepts*, 9th edition, Wiley, 2013. These are original explanations and original exercises. Historical product cases refer to the historical versions in the edition. No publisher solution text is used.

- Intel SDM.
- ARMv7-A/R manual DDI0406C. d, B3.5 and B3/B4 descriptor, domain, and AFE definitions.
- Linux v3.2 ARM helpers. Fixed physical bases here are educational layouts rather than machine constants.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Virtual memory

When working sets can exceed RAM, fault service, replacement, and allocation must preserve isolation while controlling latency. Prerequisites are paging, protection faults, and storage latency. The chapter traces demand/COW faults, compares FIFO/OPT/LRU/CLOCK, and relates fault frequency to residency and thrashing.

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Legality, mapping and residency

Virtual memory separates logical ranges, software mappings, and physical residency. A legal virtual page need not be resident. Present/resident bits have architecture-specific meanings; valid is not one universal cross-architecture name. A handler must separately establish legal range/backing, translation state, and read/write/execute/user permission.

Fix 4 KiB pages and processes A/B with identical virtual layouts.

Virtual range	Legality/rights	Content source	A mapping	B mapping
[0x0000,0x1000)	Legal RX, not W	Same program code	Shared F0	Shared F0
[0x1000,0x2000)	Legal RW private	Zero-fill-on-demand	Nonresident	Nonresident
[0x2000,0x3000)	Legal RW private	Separate data	F1	F2
[0x3000,0x4000)	Guard, no mapping	None	None	None

Legality, mapping and residency (continued)

Each has 3 legal pages and 2 resident mappings. Four resident mappings consume only 3 unique frames because code F0 is shared. Adding per-process resident counts double-counts shared frames.

A first read 0x1004 can allocate zeroed storage and restart. Writing 0x0004 violates code permission. Reading 0x3004 is guard access, not an invitation to create a mapping merely because a free frame exists. Nonpresent PTEs need VMA/backing metadata to distinguish these outcomes.

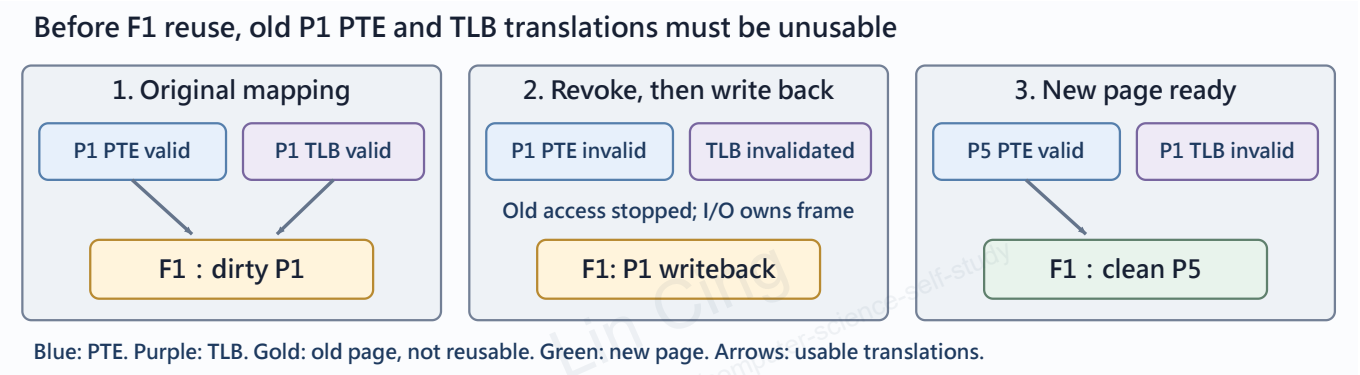
Reservation, committed backing, and residency are different quantities. A VMA can reserve addresses before policy promises RAM/swap. A virtual range larger than RAM does not guarantee all pages can be written simultaneously or that allocation can never fail.

Demand-fault service and frame reuse

Fix 4 KiB pages and full frames F0→P0, F1→dirty P1, F2→P2. P5 is readable in a legal VMA/backing object but nonresident. Replacement chooses F1.

Step	State/action	Can P5 be read?
0	CPU accesses nonpresent P5 and traps	No
1	Validate VMA, backing, and read rights	No
2	Revoke P1 PTE; complete old TLB invalidation and prevent further writes to F1	No
3	Write back dirty P1; I/O still owns F1, so cannot overwrite it	No
4	Load 4096 bytes of P5 into F1; mapping not yet published	No
5	Publish P5→F1, perform required translation maintenance, restart instruction	Yes

Demand-fault service and frame reuse (continued)



This is1 fault,1 disk write,1 disk read, and one old-translation maintenance phase. On one CPU that can be local invalidation; other CPUs caching the old translation require completed shootdown. Installation may require additional architecture-specific invalidation/barriers, so this is not a claim of exactly one TLB instruction everywhere. Nonoverlapped 8 ms write plus 8 ms read costs 16 ms without other overhead. A clean victim removes the write, leaving 8 ms.

Demand-fault service and frame reuse (continued)

$$EAT = (1-p)m + pF, \qquad p \le \frac{EAT_{max} - m}{F - m}.$$

For m=200 ns and clean F=8,000,000 ns, EAT≤400 requires p≤200/7,999,800≈2.5001×10⁻⁵. Dirty write+read F=16,000,000 reduces it to≈1.2500×10⁻⁵. The same fault ratio can therefore have different service cost. Another fixed baseline m=100 ns, F=8 ms, EAT≤200 gives p≤100/7,999,900≈1.25×10⁻⁵; keep its denominator separate.

Fault classification and stale-TLB counterexample

The handler checks legal VMA/backing, software mapping versus nonresident hardware translation, requested rights or authorized COW exception, free/victim eligibility including dirty/writeback/pin state, then complete contents/permissions before retry.

CPU0 clearing P1 PTE does not revoke CPU1's cached writable P1→F1. Loading P5 into F1 immediately lets CPU1's next P1 store corrupt P5. Reuse requires old translation revocation, necessary shootdown acknowledgments, and completed old I/O/ownership. Conversely, loading P5 without installing its mapping makes the restarted instruction fault again.

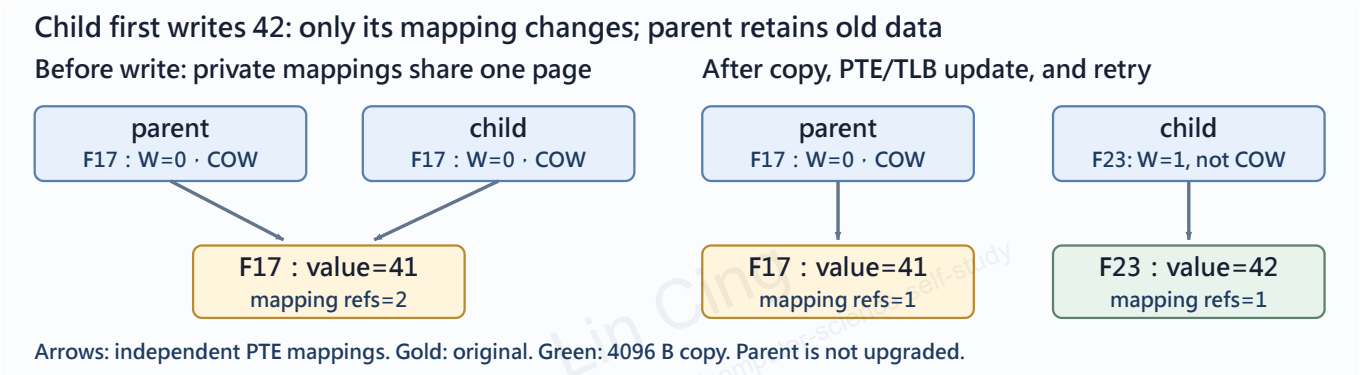
Read-ahead may bring P5/P6 in one transfer, changing fault counts and frame pressure. Every prefetched page still needs legality, backing, a frame and a consistent mapping. A batch read cannot authorize an adjacent guard page.

COW is a write-permission fault

Fork preserves separate tasks and page-table metadata while parent/child PTEs initially share F17. Hardware marks both read-only; software records COW. The translation exists, so the child's failed store is not a missing-backing-page fault.

State/event	Parent PTE	Child PTE	F17	F23
S0 fork complete	F17, W0, COW	F17, W0, COW	41, refs 2	Absent
S1 child tries write	Unchanged	Unchanged	41, refs 2	Absent
S2 allocate/copy 4096 B, not published	F17 RO	Still F17	41, refs 2	41, handler-owned
S3 publish, maintain TLB, retry 42	F17, W0, COW	F23, W1, non-COW	41, refs 1	42, refs 1, dirty

COW is a write-permission fault (continued)



Validate COW→allocate→copy→publish child PTE/TLB→retry. Making both mappings writable or clearing only the parent's COW lets the child change the parent's original data. Here refs counts user mappings, not the handler's temporary hold.

COW is a write-permission fault (continued)

The child handler need not visit or immediately upgrade the parent's PTE. On the parent's later write fault, an exclusively owned private anonymous page with no other sharing/pin restrictions may be made writable without copying. One mapping alone does not authorize writing a shared file-cache page in place. Fork followed by exec can unmap untouched COW pages without copying. Zero-fill must clear a reused frame before exposing it to another process; demand paging, COW, and zero-fill have different fault causes and data sources.

Replacement policies and full trace

FIFO uses load order; LRU last use; OPT farthest next use, requiring an offline future; CLOCK gives referenced pages another chance. Exact LRU ordering on every access has hardware/concurrency cost. Reference-bit histories or clock scans approximate it and have their own sampling/hand rules.

Fix 3 frames, initially empty, references 1,2,3,4,1,2,5,1,2,3,4,5. H/F means hit/fault. FIFO is load order, LRU oldest→newest, CLOCK physical slots; these are policy metadata rather than one shared frame numbering.

t/page	FIFO	LRU	OPT	CLOCK
1/1	[1]F	[1]F	[1]F	[1]F
2/2	[1,2]F	[1,2]F	[1,2]F	[1,2]F
3/3	[1,2,3]F	[1,2,3]F	[1,2,3]F	[1,2,3]F
4/4	[2,3,4]F	[2,3,4]F	[1,2,4]F	[4,2,3]F
5/1	[3,4,1]F	[3,4,1]F	[1,2,4]H	[4,1,3]F

Replacement policies and full trace (continued)

t/page	FIFO	LRU	OPT	CLOCK
6/2	[4,1,2]F	[4,1,2]F	[1,2,4]H	[4,1,2]F
7/5	[1,2,5]F	[1,2,5]F	[1,2,5]F	[5,1,2]F
8/1	[1,2,5]H	[2,5,1]H	[1,2,5]H	[5,1,2]H
9/2	[1,2,5]H	[5,1,2]H	[1,2,5]H	[5,1,2]H
10/3	[2,5,3]F	[1,2,3]F	[2,5,3]F	[5,3,2]F
11/4	[5,3,4]F	[2,3,4]F	[5,3,4]F	[5,3,4]F
12/5	[5,3,4]H	[3,4,5]F	[5,3,4]H	[5,3,4]H

Totals FIFO 9, LRU 10, OPT 7, CLOCK 9. Victims, with dash for free-frame insertions/hits:

Replacement policies and full trace (continued)

Policy	Victims
FIFO	–,–,–,1,2,3,4,–,–,1,2,–
LRU	–,–,–,1,2,3,4,–,–,5,1,2
OPT	–,–,–,3,–,–,4,–,–,1,2,–
CLOCK	–,–,–,1,2,3,4,–,–,1,2,–

FIFO hits do not reorder; LRU hits move to most recent. OPT evicts the latest next use or never-used-again page. Aging can shift the reference bit into a history byte each sampling period, with smaller values suggesting older usage. Changing sample periods, hand positions, or bit clearing changes the approximation.

Every CLOCK hand and reference bit

Start hand 0. Fill free slots 0/1/2 with R1 without moving hand. Hits set the page's R1 without moving hand. When full, clear each encountered R1 and advance; replace first R0, set new R1 and advance once.

t/page	Slots 0/1/2	R0/R1/R2	Next hand	H/F, victim
1/1	[1]	[1]	0	F,–
2/2	[1,2]	[1,1]	0	F,–
3/3	[1,2,3]	[1,1,1]	0	F,–
4/4	[4,2,3]	[1,0,0]	1	F,1
5/1	[4,1,3]	[1,1,0]	2	F,2
6/2	[4,1,2]	[1,1,1]	0	F,3
7/5	[5,1,2]	[1,0,0]	1	F,4

Every CLOCK hand and reference bit (continued)

t/page	Slots 0/1/2	R0/R1/R2	Next hand	H/F, victim
8/1	[5,1,2]	[1,1,0]	1	H,–
9/2	[5,1,2]	[1,1,1]	1	H,–
10/3	[5,3,2]	[0,1,0]	2	F,1
11/4	[5,3,4]	[0,1,1]	0	F,2
12/5	[5,3,4]	[1,1,1]	0	H,–

At4 it clears a full circle then evicts 1 in slot 0; at 7 another circle evicts 4. The initialization and hit behavior are necessary to reconstruct these states.

FIFO anomaly with both capacities

t/page	3-frame queue	4-frame queue	Cumulative faults 3/4
1/1	[1]F	[1]F	1/1
2/2	[1,2]F	[1,2]F	2/2
3/3	[1,2,3]F	[1,2,3]F	3/3
4/4	[2,3,4]F	[1,2,3,4]F	4/4
5/1	[3,4,1]F	[1,2,3,4]H	5/4
6/2	[4,1,2]F	[1,2,3,4]H	6/4
7/5	[1,2,5]F	[2,3,4,5]F	7/5
8/1	[1,2,5]H	[3,4,5,1]F	7/6
9/2	[1,2,5]H	[4,5,1,2]F	7/7
10/3	[2,5,3]F	[5,1,2,3]F	8/8

FIFO anomaly with both capacities (continued)

t/page	3-frame queue	4-frame queue	Cumulative faults 3/4
11/4	[5,3,4]F	[1,2,3,4]F	9/9
12/5	[5,3,4]H	[2,3,4,5]F	9/10

After 7 the smaller set contains 1 but the larger does not; access 8 faults only with 4 frames. FIFO's capacity-dependent fault history determines load order, so increasing capacity can raise faults. This Belady anomaly does not apply to every policy.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

General LRU inclusion proof

Form one recency stack of every seen page from newest to oldest, independent of capacity. On every access move that page to the front. Inductively, exact k-frame LRU holds the first k entries, or all entries when fewer exist. Initially empty; a hit reorders without changing membership of those k; a miss inserts at front and pushes out old entry k, exactly the LRU victim.

Thus $R_k(t) \subseteq R_{k+1}(t)$. If k+1 faults at time t, its requested page was absent from R(k+1) before access and therefore absent from subset Rk. Fault indicators satisfy the inequality pointwise, and summing gives $F_{k+1} \leq F_k$. The finite 12-access example cannot replace this induction. FIFO load histories and CLOCK hand/reference scans depend on capacity and do not supply that same stack invariant.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

OPT exchange proof and its cost boundary

Fix an offline finite string, identical initial residents, unit miss cost, and no dirty cost difference. At a full-frame miss OPT evicts q with latest next use, treating never again as infinity. Take an optimal schedule's first disagreement, evicting p instead. Modify it to retain p and evict q. Before next(p), q is not requested earlier, so other replacements can track the original with only p/q differing. If the original evicts q sooner the states can align. Otherwise at next(p), the modified schedule already has p while the original faults. Use that saved transfer to bring q in and remove the original's chosen victim, aligning states without more transfers. If neither page is reused, the difference costs nothing.

Any early transfer in that alignment can be delayed until the page's first request without increasing cost, giving a legal demand schedule. Hence an optimal schedule agrees with this OPT step; eliminate first disagreements successively. This is a unit page-in proof, not a proof for heterogeneous page sizes, dirty page-out, or unequal I/O weights. More frames can simulate any smaller-capacity optimum by leaving space unused, so OPT's minimal cost cannot rise. This simulation does not require every OPT tie-break to produce nested residents.

OPT exchange proof and its cost boundary (continued)

Fault service is trap+dirty swapout+swapin. Half dirty victims gives an expectation between read-only and write+read paths. Minimum fault count need not minimize a dirty-weighted cost.

Lin Cing
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LFU/MFU frequency lifetimes

Start 3 empty frames, count 1 on load, increment on hit, discard count on eviction, and break ties by earliest load. References 1,1,1,2,3,4 give counts{1:3,2:1,3:1} before the final access. LFU evicts 2→{1:3,3:1,4:1}; MFU evicts 1→{2:1,3:1,4:1}. Old high counts can preserve 1 after locality shifts to 2/3/4. Aging counts can reduce stale history but adds a decay parameter. MFU's belief that low counts are newly loaded future candidates is a workload assumption, not universal optimality.

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Quotas, rounding and victim pools

Equal allocation first gives floor(F/m) each and needs a remainder rule. Proportional allocation floors $a_i = Fs_i / \sum s_j$, then distributes remaining frames by specified remainder/priority/order.

For F64 and sizes A100/B200/C300, raw shares 10.667/21.333/32, floors 10/21/32 total 63. Largest remainders 0.667/0.333/0 give the last frame to A, final 11/21/32 total 64.

In a separate six-frame model, A={1,2,3}, B={4,5}, C={6}; last-use A1=10, A2=5, A3=8, B4=2, B5=7, C6=9. A faults 7 with no free frame.

Policy	Eligible victim	Victim	New residents
Local	Only A	A2	A={1,7,3}, B={4,5}, C={6}
Global	All processes	B4	A={1,2,3,7}, B={5}, C={6}

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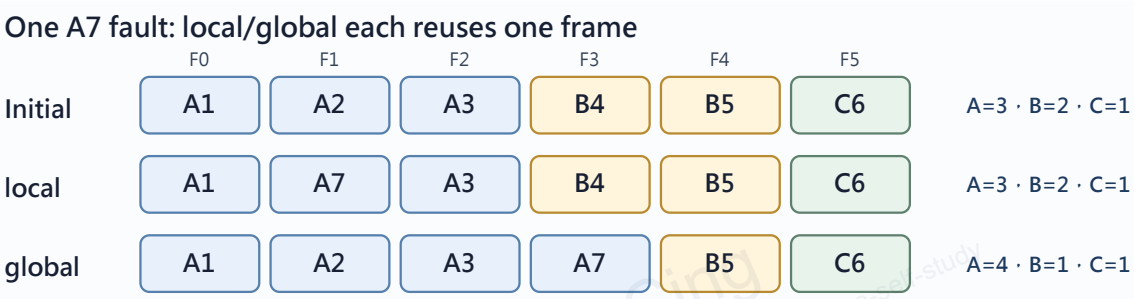
Quotas, rounding and victim pools (continued)



Global raises A's residency 3→4, lowers B2→1, and retains 6 total. It neither deletes A3 nor invents B8. B's next access 4 now faults, spreading pressure. Local limits interference but cannot borrow an idle other's frame. Allocation defines quotas/bounds; replacement defines eligible victims. Global need not mean unrestricted admission.

Minimum frames versus a working set

Instruction code/source/destination pages and restart rules create a minimum progress requirement. Evicting an earlier required operand to load the last can fault forever when an instruction cannot partially complete. That is different from WSS, a performance-demand estimate over a chosen window. Fault-handler and table kernel memory are separately guaranteed system costs.

If lower bounds l_i apply, require $\Sigma l_i \le F$. Grant them first, then split $F - \Sigma l_i$ with explicit rounding. Taking $\max(\text{proportional_share}, l_i)$ independently can exceed F. For F6 and three processes each requiring 3 unshared frames, total minimum 9 is impossible regardless of fairness. Suspend at least one, change execution, or add frames. Pins and kernel reserves also reduce the user pool.

Working sets and evidence of thrashing

Thrashing means paging dominates useful execution. A working set uses pages referenced within a defined recent window. For the last 4 references:

Process	References	Working set	WSS
A	1,2,3,1,2,4,5,1	{1,2,4,5}	4
B	6,7,6,8,7	{6,7,8}	3
C	9,9,9	{9}	1

Distinct backing pages give D8>m6. If all remain active, demand exceeds residency and predicts thrashing risk. Shared pages require a union of unique backing identities rather than summing WSS. A separate 20/24/28 example totals 72>64, an8-frame demand gap, not proof of measured thrashing.

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Working sets and evidence of thrashing (continued)

Short windows miss reused pages; long ones retain stale/rare pages. A blocked lock, DMA, or one slow I/O can lower CPU use without paging thrash. Diagnose with high fault rates, low useful CPU utilization and growing paging queues, rather than D>m alone.

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PFF hysteresis controller

For fixed 5-access windows and upper 0.4/lower 0.1, FFFFF gives 1.0 and requests another frame; FHHHH gives 0.2 and holds; HHHHH gives 0 and may release a spare under pressure. Two thresholds avoid allocation/reclaim oscillation near one threshold.

```
for each process every sample_window:
    if accesses_in_window == 0: continue
    rate = faults_in_window / accesses_in_window
    if rate > upper and free_frames > 0:
        grant_one_frame(process)
    else if rate > upper and free_frames == 0:
        choose one process to suspend
    else if rate < lower and memory_pressure and quota > minimum_required:
        reclaim_one_spare_frame(process)
```

High PFF with no free frame calls for reduced multiprogramming, swapping/reallocation, or rejected admission. More CPU priority cannot manufacture resident pages. Short windows are noisy, long windows react slowly; thresholds, window and suspension rule jointly define the controller.

Mapped files, visibility and durability

MAP_SHARED processes map the same file-cache page; shared updates still require application synchronization. MAP_PRIVATE writes COW into private pages without writing the original file. Creating a mapping is not residency; a store is not persistence.

These experiments independently start at 7, with no external file modification.

Shared event	P1	P2	Cache F40	File
S0	F40	F40	7, clean	7
S1 P1 writes 8 and synchronizes	F40	F40	8, dirty	7
S2 P2 reads after synchronization	F40	F40	Reads 8, not durable	7
S3 completed required flush	F40	F40	8, clean	8

Mapped files, visibility and durability (continued)

Private event	P1/value	P2/value	Cache	File
V0	F40/7, COW	F40/7, COW	F40/7	7
V1 only P2 writes 9	F40/7, uncopied	New F53/9	F40/7	7
V2 P1 writes 8	New F52/8	F53/9	F40/7	7

V1 copies 4096 B; cumulative V2 copies 8192 B. P2's write does not copy the still-unwritten P1 mapping. MAP_PRIVATE does not guarantee whether later external file changes appear, so this is not an automatic file snapshot.

msync/platform counterparts trigger writeback/order under their actual API/filesystem/cache/flush contracts. Dirty shared file pages need safe writeback before reclaim. Dirty private COW pages use swap/backing rather than overwriting the source file. Failed writeback must retain error/dirty state instead of declaring clean and reusing data.

mmap and read/write costs

Fix cold cache, one complete 4 KiB page, no read-ahead, successful operations. Disk→cache I/O is different from cache↔user CPU copy.

Path	Calls	CPU copies	Translation	Write durability
read	1 syscall	Cache→user 4096 B	Ordinary	No write promise
write	1 syscall	User→cache 4096 B	Ordinary	Requires specified sync/commit
First mapped read	Earlier mmap call; current fault/trap	No cache→user copy; disk still reads 4096 B	Install/maintenance	No write
Mapped shared store	No syscall until flush	Zero bytes in this model	Access/dirty tracking	Specified msync/flush needed

read updates file offset; mapped loads use MMU after faults create PTEs. A page fault is an exception, not an explicit user syscall. Both write and mapped store can finish with only dirty cache data.

Kernel allocation and buddy splitting

Kernel interrupt/I/O paths may need nonsleeping, contiguous, or specially owned memory, so ordinary user demand-paging assumptions do not suffice. Buddy allocators split power-of-two blocks; slabs cache fixed-size objects.

A256 KiB region with 4 KiB minimum rounds 21 KiB up to 32. Ranges below use KiB.

Step	Split/result
0	Free[0,256)
1	[0,128),[128,256)
2	Split left into[0,64),[64,128)
3	Split left into[0,32),[32,64)
4	Allocate[0,32),21 used and 11 internal waste

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Kernel allocation and buddy splitting (continued)

Free blocks[32,64),[64,128),[128,256) total 224 KiB. A64 KiB request can use an existing 64 block; coalescing[0,64) specifically requires releasing its allocated 32 first. Blocks have unique order/buddy ownership and never overlap. For appropriately aligned a and block size 2^k bytes, buddy=a xor 2^k. Freeing[0,32) merges with[32,64), then[64,128), then[128,256). Each step checks only the matching-order buddy; maximum tree height log2(256/4)=6.

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Slab object counts and the denominator

Ignoring metadata, a4096 B slab page holds 16 objects of256 B. This is a model, not an exact count for every slab implementation. Allocate 2 and free 1: live objects 1, free objects 15, live bytes 256, retained slab bytes 4096. Process RSS cannot be inferred from kernel object count because it uses process mappings as its denominator.

Retaining free objects saves initialization and page-allocation work, so footprint need not shrink with live count. NUMA caches add node locality and remote-reclaim/allocation cost.

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https://github.com/xinhi6716/computer-science-self-study

Page size, prepaging and pins

One MiB needs 256 PTEs with 4 KiB pages or16 with 64 KiB pages. Larger pages give 16 times coverage per TLB entry but larger tails, transfer sizes, and read amplification. More page faults or more PTEs can offset small-page tail savings.

An8-page workload uses locality{P1, P2, P3}. Prepaging those 3 does 3 reads and the later first 3 accesses fault 0 times. Pure demand does the same 3 reads but 3 faults. The difference is trap/latency placement, not eliminated data I/O. Guessing unused P4 adds a needless read.

DMA stage	F40 pin/state	Reclaimable?
t0 user buffer	Pin 0	Yes if otherwise eligible
t1 device queued	Pin 1, DMA pending	No
t2 device completes	DMA 0, pin not released	No
t3 completion releases pin	Pin 0, eligible LRU	Can be considered

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Page size, prepaging and pins (continued)

Reusing at t1 lets DMA overwrite a different page. Device completion alone does not end the completion protocol; pins need normal/error release paths. A pin, lock bit and mlock are different API mechanisms, but none allows duplicate allocation of a still-used frame. Long pins shrink the evictable pool.

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Access order changes page locality

C/C++ row-major int A[4][1024],4-byte int,4 KiB-aligned base,4 KiB pages makes one row per page. Count only array references,2 initially empty exact-LRU frames; code/stack/cache-line costs are separate.

Loop order	Page references	Faults
Rows outside, columns inside	Each 0,1,2,3 repeated 1024 times	4, first access per row
Columns outside, rows inside	0,1,2,3 repeated 1024 rounds	4096, each reuse already evicted

First 8 column states oldest→newest are[0]F,[0,1]F,[1,2]F,[2,3]F,[3,0]F,[0,1]F,[1,2]F,[2,3]F, then repeat. Identical 16 KiB and operation count produce different fault counts under this residency model. Real OS frame grants/prefetch/policies can change that difference; it is not a measured universal loop speed ratio.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Linux 3.2 reclaim branches

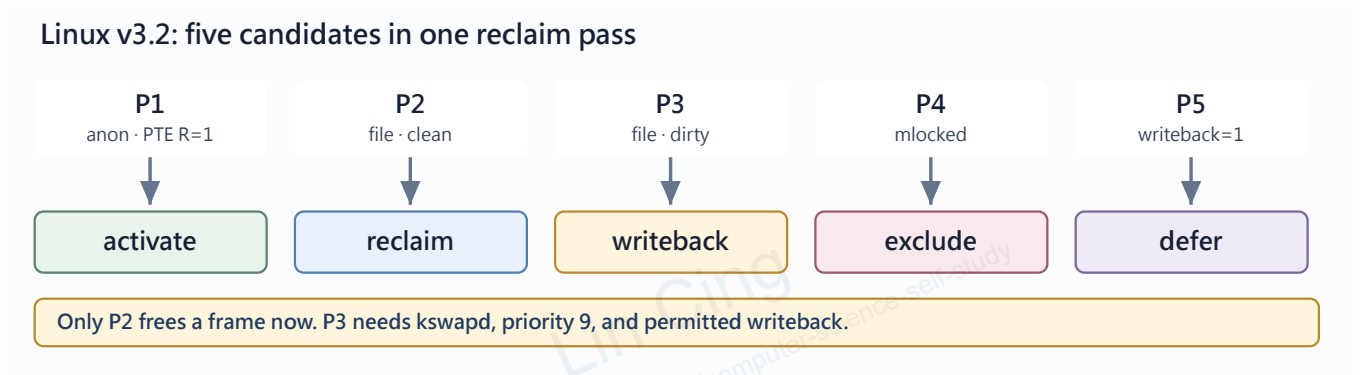
Fix asynchronous order-0 Linux v3.2. Free pages below the zone low watermark can wake kswapd; scanning does not guarantee every candidate frees memory. It scans inactive anon/file lists and balances active/inactive. page_check_references distinguishes PTE references and PageReferenced; shrink_page_list then tests unmap, dirty, writeback, mapping/refcounts. This is not modern multi-gen LRU.

Fix current_is_kswapd=true, priority 9, DEF_PRIORITY12, may_writepage=true, filesystem/I/O/unmap allowed, successful page locks, no concurrent rereference/mutation, removable P2/P3 mapping/refcounts, and successful asynchronous P3 pageout.

Linux 3.2 reclaim branches (continued)

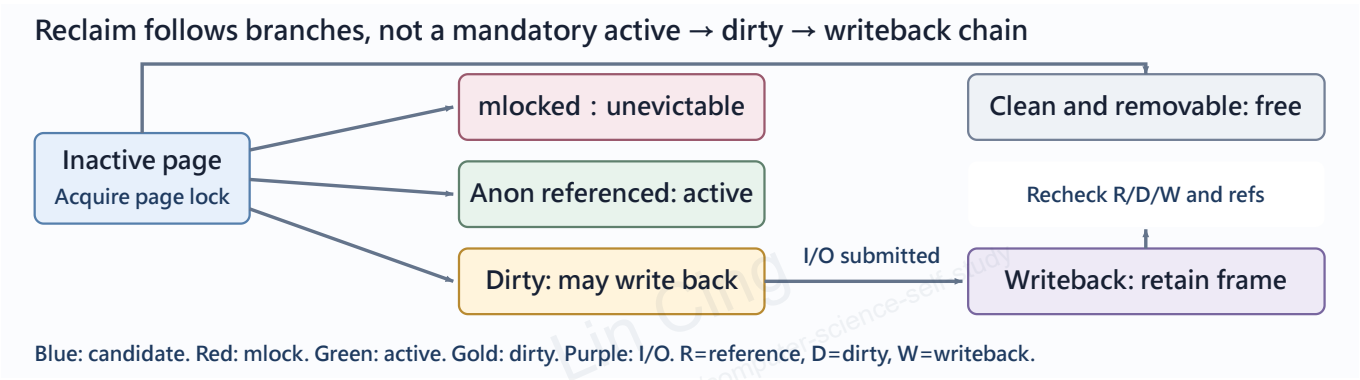
Page	Type	PTE referenced	Dirty	Writeback	mlocked	Action	Free delta
P1	Anon	1	0	0	0	Activate	0
P2	File	0	0	0	0	Reclaim after final removal	+1
P3	File	0	1	0	0	Start async writeback	0
P4	Anon	0	0	0	1	Cull to unevictable	0
P5	File	0	1	1	0	Defer	0

Linux 3.2 reclaim branches (continued)



Only P2 frees a frame now. P3 is a selected candidate but remains owned by I/O. An extra DMA pin/reference is not mlock; it may remain on ordinary LRU and fail final mapping/reference removal rather than become unevictable.

Recheck after writeback and changed conditions



Writeback completion does not guarantee no subsequent dirty write. Before reuse recheck dirty 0, writeback 0, removable mappings/references. Mlocked pages require authorized unlock; time alone cannot clear ownership. Free count increases only after actual reclaim.

Recheck after writeback and changed conditions (continued)

Changed condition	v3.2 branch/result
P1 instead file, first PTE reference, PageReferenced 0	Set PageReferenced, PAGEREF_KEEP; remain inactive rather than unconditional activate
P3 priority 10	priority≥DEF_PRIORITY−2; retain/mark reclaim, other writeback path
P3 direct reclaim, not kswapd	Dirty file-cache restriction blocks this pageout
P3 may_writepage=false	No pageout, retain dirty, free+0
P2 extra pin/reference	Final removal fails, retain; not synonymous with unevictable
P5 completed I/O then redirtied	Dirty branch again; no clean reuse

A single if-referenced-activate/if-dirty-writeback sketch omits page type, reference source, scan context, and final ownership. Windows working-set limits, Solaris lotsfree/desfree/minfree, and Linux watermarks are distinct version-specific controls.

Historical Windows trimming

Use the OSC 9e simplified Windows working-set model, cross-read with Chapter 19's Windows 7 snapshot. It is not a guarantee for every current patch. Min/max control residency; low free memory can trigger trimming above minima.

Set threshold 3, free 1; A(min 2, resident 4, max 4), B(2,3,4), C(2,2,4). All selected pages are clean/unlocked; visit A→B→C and stop only once free>3.

Action	A	B	C	Free
Initial	4	3	2	1
Trim 2 from A	2	3	2	3
Trim 1 from B	2	2	2	4
Do not trim C	2	2	2	4

Historical Windows trimming (continued)

A later fault can grow its set below maximum if free pages exist; at maximum this model uses local replacement. Clustered paging can add neighbors, so exactly one resident page per fault belongs only to the earlier single-page model.

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Historical Solaris two-hand window

In the OSC 9e snapshot free<lotsfree starts the scanner, whose scanrate increases from slowscan under pressure. desfree/minfree prompt stronger reclaim/swap measures. These are not Linux's same low watermark.

Front clears R; back revisits after h/s seconds for hand distance h pages and steady rate s pages/s. h4, s100 gives 0.04 s, illustrative rather than a default.

Page	After front clears R	Back observes	Action
Q1	Read again, R1	R1	Retain locality
Q2	Unused, clean	R0, clean	Unmap, then reclaim/cache list
Q3	Unused, dirty	R0, dirty	Write back; cannot reuse during I/O

Raising s to 400 with h4 shortens the window to 0.01 s, making absence of a recent reference more likely. An unoverwritten clean cache-list page may be recovered; invalid free-list contents are not a retained-data guarantee.

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Complete access decision and cost model

Stage	Required evidence	Invalid inference
Range/backing	Authorized mapping and content source	Every nonpresent fault authorizes page-in
Rights/COW	Requested operation allowed or valid COW	Every forbidden write gets a new frame
Residency/PTE	Frame, standby, or storage requirement	Every fault means one disk I/O
Victim/writeback	Unmap eligibility and safe dirty handling	Pinned/dirty frames can be reused immediately
Restart	Contents, mapping, rights coherent	This page can never refault

Capacity counts unique frames; COW/unmap/replacement must track final mapping, TLB, and I/O users. With the fixed 12 references,3 frames,100 ns hits, and 8 ms total fault service including that access:

Complete access decision and cost model (continued)

Policy	Faults	Hits	Total cost
FIFO	9	3	72 ms+300 ns
LRU	10	2	80 ms+200 ns
OPT	7	5	56 ms+500 ns
CLOCK	9	3	72 ms+300 ns

OPT uses future information for a lower bound. Different fault costs, prefetch, or clustering invalidate this table's common multiplier. Working sets, quotas, PFF, and scanning jointly affect residency; evaluate useful throughput and waits as well as fault counts.

References

- Operating System Concepts,9e, Chapter 9, pp.397–448 for mechanisms and the chapter's broader printed range 397–461. Original traces, thresholds, and examples are explicit model inputs, not benchmarks.
- Linux v3.2 vmscan. c and memory. c.
- mmap(2), shared/private and external-file-change semantics.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Mass-storage structure

Storage media, attachment, scheduling and redundancy jointly determine latency, throughput and recovery. Prerequisites are blocks/sectors, interrupts, and average latency. The chapter derives disk service costs, compares scheduling paths, and analyzes RAID capacities, write costs and persistence.

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Media mechanisms and metric denominators

HDDs use magnetic surfaces, rotating platters and moving heads. SSDs use nonvolatile semiconductor media with controllers/FTLs. Tape supports streaming backup/archive with long positioning costs. Hosts address logical blocks rather than controlling every physical location. SSD costs cannot be modeled as head seek plus rotation.

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Media mechanisms and metric denominators (continued)

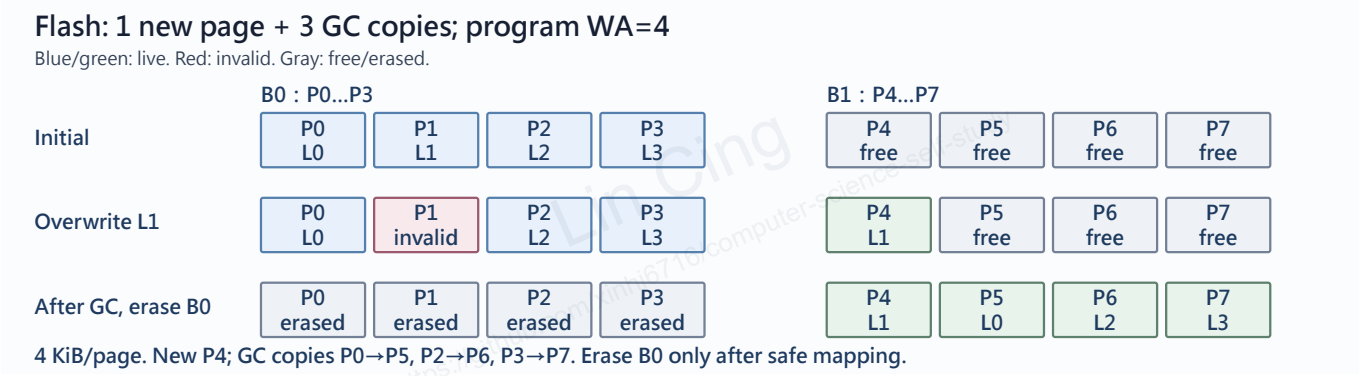
Metric	Denominator/boundary	Distinct quantity
Bandwidth	Completed bytes/elapsed seconds	Interface/media peak need not equal end-to-end
IOPS	Completed requests/second	Different request sizes have different byte rates
Service time	Actual device service interval	Excludes time not yet selected
Response latency	Arrival→specified completion	Can include queue, scheduling, copy
Durability	Failures tolerated after acknowledged success	Completion may still be volatile

Media mechanisms and metric denominators (continued)

1000 completed 4 KiB requests/s gives 3.90625 MiB/s; 1000 of64 KiB gives 62.5 MiB/s. The same IOPS does not mean the same bandwidth, and this arithmetic is not a product measurement. Comparisons fix request size, random/sequential pattern, R/W mix, queue depth, cache, and synchronization contract.

Flash out-of-place writes and GC

Fix 4 KiB NAND pages and four-page 16 KiB erase blocks. B0 P0/P1/P2/P3 hold L0/L1/L2/L3; B1 P4… P7 are erased. Overwriting L1 programs new P4, sets L1→P4, and invalidates P1. An unerased NAND page cannot be arbitrarily overwritten in place.



Flash out-of-place writes and GC (continued)

GC copies live L0:P0→P5, L2:P2→P6, L3:P3→P7. Only after live contents and necessary mapping are safe can it erase B0. One host page causes 1 new+3 copied physical page programs, write amplification 4 in this crash-free model. If L2 is discarded first, only 2 copies remain,3 programs and WA=3. Mapping writes and other controller overhead are excluded.

Discard/TRIM says logical data is no longer needed; it does not promise immediate erase or secure deletion. Wear leveling distributes erase load. GC, over-provisioning, and controller caches change latency. Power-loss correctness additionally needs recoverable mapping/data commit order; a normal-path diagram does not prove it.

Geometry, LBA and access costs

Tracks at the same radius form a cylinder. Seek changes radius, rotation brings a sector under the head, and transfer moves bytes. Zones and remapping alter geometry. LBA is a one-dimensional logical index, not a permanent physical cylinder; SSDs have no head radius for SSTF to optimize.

With 512-byte logical sectors, LBA 4096 starts at 2 MiB. With 4 KiB blocks, the same index starts at 16 MiB. Unsigned 32-bit LBA addresses $2^{32} \times 512 = 2^{41}$ bytes=2 TiB; this is an address-space calculation, not a universal partition/API limit.

A512-byte upper write on a4 KiB physical/atomic unit may need read-modify-write or device translation. Filesystem block size does not establish atomic sector size; alignment and crash granularity follow device/driver contracts. Bad-sector remapping can preserve the LBA while adding latency.

A7200 RPM,64 KiB read

Fix seek 5 ms, uniform rotational phase,7200 RPM, transfer 128 MiB/s, request 64 KiB, no cache/controller overhead. A rotation is60,000/7200=25/3 ms; mean rotation 25/6 ms; transfer 64 KiB/(128 MiB/s)=125/256 ms.

$$service = 5 + 25/6 + 125/256 = 7415/768 \approx 9.65495 \text{ ms}.$$

With queue wait 15 ms, response≈24.65495 ms. One-at-a-time service without overlap gives 1000/9.65495≈103.574 IOPS, not a multiqueue/cache/adjacent-track prediction. Sequential 1 GiB has transfer-only lower bound 1024/128=8 s, with positioning/queue added. Random small-I/O positioning dominates differently from streaming bandwidth.

Host, NAS and SAN responsibilities

Path	Client abstraction	Filesystem metadata owner	Additional failure boundary
Host-attached block	Local driver/controller blocks	Local host filesystem	Controller/device
NAS	Remote file/directory operations	File server	Network, server, cache/retries
SAN block	Logical blocks over storage fabric	Host using the LUN	Fabric, target, multipath

For a4 KiB NAS read, the client sends handle/offset/length, the server resolves filesystem metadata and issues block I/O. For SAN, the local filesystem first maps offset to LBA, then block transport reaches the target. The same array can sit below both while pathname/cache/retry responsibility differs.

Host, NAS and SAN responsibilities (continued)

A SAN LUN shared by two hosts does not provide filesystem distributed locking/coherence. Concurrently mounting an ordinary exclusive-host filesystem for writes can corrupt metadata. A remote mount success proves neither all future I/O availability nor durable completion.

Paths A/B to one target can survive A's link loss while B/target remain; target loss defeats both. Two paths are not two independent data copies. Retries preserve request identity/completion semantics; an unknown prior write commit cannot authorize blindly repeating nonidempotent upper operations. Latency depends on protocol, size, network, cache and queues, not the attachment name alone.

Scheduling on one fixed batch

FCFS follows arrivals; SSTF chooses nearest pending cylinder; SCAN serves one direction to a physical endpoint then reverses; C-SCAN wraps and serves only one direction. LOOK reverses at the last request; C-LOOK wraps between extreme requests.

Fix cylinders 0…199, head 53, all arrivals 0, requests 98,183,37,122,14,124,65,67. Cost is absolute head movement until the last request completes, including turn/wrap, excluding rotation, transfer and final empty travel. SSTF ties choose smaller cylinder.

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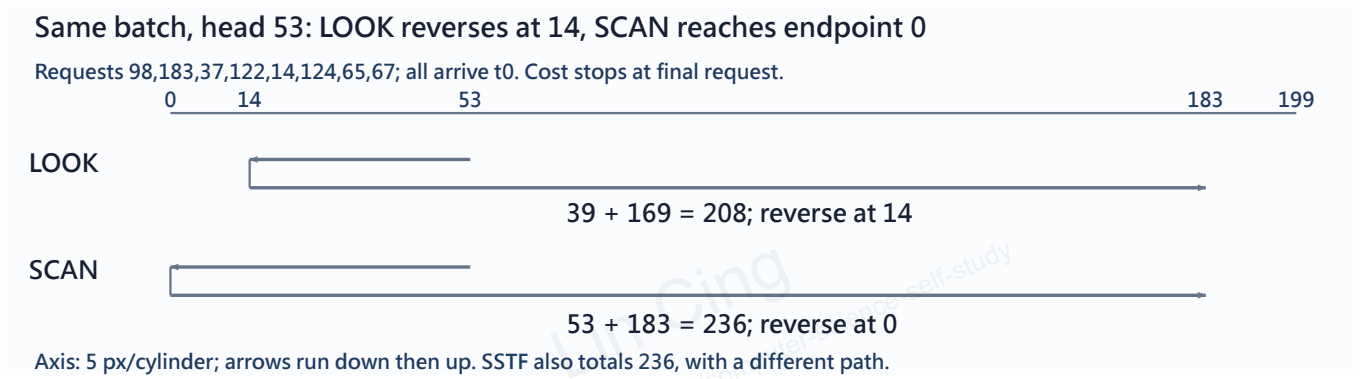
Scheduling on one fixed batch (continued)

Policy	Path including turn/wrap	Movement
FCFS	53→98→183→37→122→14→124→65→67	640
SSTF	53→65→67→37→14→98→122→124→183	236
LOOK down	53→37→14→65→67→98→122→124→183	208
SCAN down	53→37→14→0→65→67→98→122→124→183	236
C-SCAN up	53→65→67→98→122→124→183→199→0→14→37	382
C-LOOK up	53→65→67→98→122→124→183→14→37	322

Circular returns count physical movement but serve no requests during wrap. A problem excluding that cost must explicitly use another accounting model.

Scheduling on one fixed batch (continued)



LOOK=|53−14|+|183−14|=39+169=208. SCAN=|53−0|+|183−0|=53+183=236. SSTF's236 is coincidental, not the same path. FCFS differences 45+85+146+85+108+110+59+2=640; SSTF 12+2+30+23+84+24+2+59=236. LOOK 208 disproves SSTF's global-batch optimality.

Arrival assumptions and waiting guarantees

A finite batch has no infinite starvation. SSTF's starvation risk needs continuing nearby arrivals that indefinitely defer a distant request. SCAN/LOOK reduces that bias, but finite wait needs bounded range, per-sweep service and admission/track rules; an unbounded stream at one position cannot silently be served in finite time.

Controller reordering, merging and parallel queues alter actual device behavior. SSD erase/parallelism differs from seek geometry. Movement/request is an average positioning proxy, not actual response time. Deadlines, fairness and throughput require their own cost model.

Disk layout and management

Physical formatting organizes media sectors/error controls, often by device manufacturing/management. Partitioning divides logical address ranges. Logical formatting creates filesystem metadata/free-space structures. Mount attaches a recognized filesystem to a namespace. Formatting is neither secure erase nor automatic mount.

For a teaching 1024 MiB disk with 512-byte sectors, reserve[0,1) MiB for abstract boot/partition metadata. A=[1,101) MiB, B=[101,301) MiB.

Partition	First LBA	Sector count	Exclusive end
A100 MiB	2048	204800	206848
B200 MiB	206848	409600	616448

Ranges do not overlap. This is not a complete GPT specification. B mounting does not establish A's formatting/data integrity. A common mountpoint does not prove device/UUID identity. Bootloader kernel discovery and kernel root-mount need early driver dependencies from Chapter 2.

Disk layout and management (continued)

Controller remapping keeps an LBA while moving a defective physical sector to a spare. It cannot reconstruct already unreadable bytes by itself. Checksums/redundancy can detect/rebuild only with trustworthy data and error propagation. Boot-block damage, filesystem-metadata damage, and data-block damage require different recovery objects/versions. A partition existing is not proof that the disk is repaired. Scrubbing competes with foreground I/O and depends on spare/detection capacity.

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Swap backing and slot lifetime

Anonymous pages may need swap; clean file pages can be reread from their file. Swap partition/file are allocation and access-path choices. Prebuilt extent maps can reduce normal filesystem lookup, so an old swap-file overhead is not universal. There is no unconditional twice-RAM sizing formula.

Fix 4 KiB pages, slot S7/value 41, and two private-COW logical pages backed by it. Count only surviving swap-PTE references, not Linux swap_map encodings or all swapcache holds.

Event	P1	P2	S7 swap-PTE refs
Initial	Swap S7	Swap S7	2
P1 reads/private-writes 42	Resident F23/42	Swap S7, expects 41	1
P2 reads back	F23/42	Resident F24/41	0
No I/O/cache ownership remains	Resident	Resident	Only now reclaim S7

Swap backing and slot lifetime (continued)

P1's new value cannot overwrite S7 while P2 needs 41; allocate new backing if required. This does not fix Linux fault counts or swapcache-sharing choices. Existing swap PTEs, I/O or cache holds prevent early slot reuse after one process exits. Counts need explicit scope/increment/decrement rules. Swap extends current execution's memory, not backup; hibernation has a different recovery contract. More swap can help commit but does not fix physical working-set thrashing.

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RAID capacities and device-loss assumptions

Fix N equal-capacity D devices, ignore metadata/spares. Striping places data; mirror/parity adds specified redundancy. RAID is not historical backup against deletion, ransomware, silent overwrite or common-controller/site failure.

Level	Organization	Usable capacity	Healthy-array device-loss guarantee
0	Stripe, no redundancy	ND	No tolerance for loss of needed stripe data
1 two-copy pairs	Mirrors	D per pair	One loss per pair; failure grouping matters
2	Bit striping/ECC	Depends on code/check disks	Depends on ECC assumptions
3	Fine-grained stripe, dedicated parity	(N−1)D	One disk; many disks participate per request

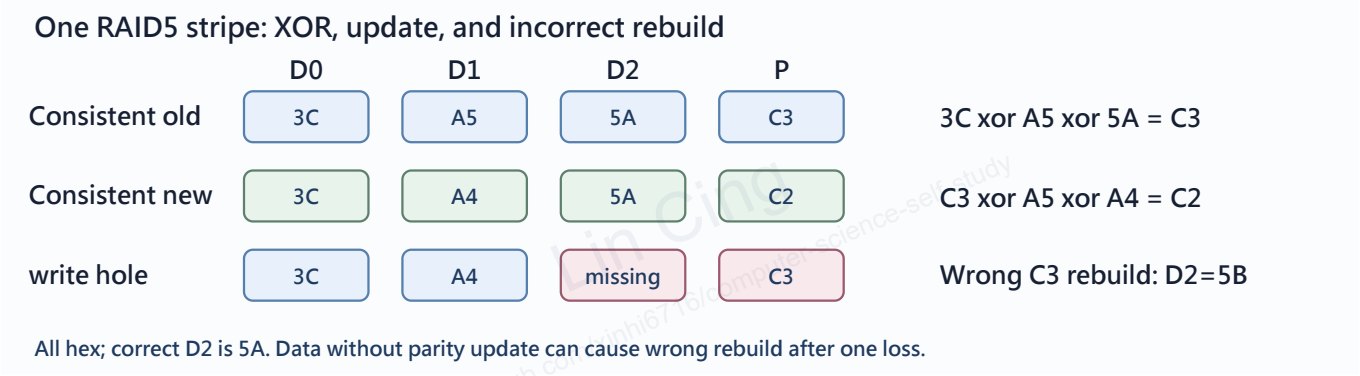
RAID capacities and device-loss assumptions (continued)

Level	Organization	Usable capacity	Healthy-array device-loss guarantee
4	Block stripe, dedicated parity	(N−1)D	One disk; parity write bottleneck
5	Block stripe, distributed parity	(N−1)D	Any one disk, parity location rotates
6	Two independent parity relations	(N−2)D	Any two disks for suitable code/layout

Two two-disk mirrors on4 disks give 2D total; one four-copy mirror givesD. Topology precedes capacity claims. Rebuild reads surviving devices, raising load and exposure to further loss; its throughput is not foreground bandwidth.

RAID5 XOR and the write hole

One stripe has hex D0=3C, D1=A5, D2=5A, P=3C xor A5 xor 5A=C3. Missing D1 reconstructs C3 xor 3C xor 5A=A5. Two unknown data values cannot be uniquely solved from one XOR relation.



Change D1 to A4. New parity C3 xor A5 xor A4=C2. Cold uncached read-modify-write reads old D1 and P, writes new D1/P',2 reads+2 writes excluding log/cache. A full three-data stripe can calculate parity directly and do 4 writes without old-stripe reads.

RAID5 XOR and the write hole (continued)

If A4 is durable but parity remains C3, all data disks can still be directly read but the relation is inconsistent. Then losing D2 and rebuilding with old parity yields C3 xor 3C xor A4=5B, not 5A. Single-device tolerance assumes coherent redundancy; it does not include arbitrary interrupted parity update plus a later loss.

Properly protected write-back cache, journaling/parity protocols or COW can protect that transition under implemented commit/recovery rules. Mirrors, checksums or a RAM flag alone do not solve it. Redundancy and persistence are separate contracts.

Why RAID6 needs an independent second equation

Over a suitable binary field, P=ΣDi, Q=ΣαiDi. Missing locations a/b have distinct coefficients. Subtract known values to get A+B=Pmissing and αaA+αbB=Qmissing. Substitute A=Pmissing+B:

$$(\alpha_a + \alpha_b)B = Q_{missing} + \alpha_a P_{missing}.$$

Distinct coefficients make their sum nonzero/invertible, so B then A is unique. Duplicating the same XOR gives dependent equations and cannot solve two unknowns. Codes specify field polynomial, layout, and error/erasure model.

Use GF(2^8), modulus 0x11D, coefficients 1/2/4, data 3C/A5/5A. Q=3C xor(2・A5) xor(4・5A)=3C xor 57 xor 75=1E. Lose D1/D2: Pmissing=FF, Qmissing=22. Then 6B=22 xor(2・FF)=22 xor E3=C1. Inverse 6=7A; B=7A・C1=5A, A=FF xor 5A=A5. Multiplication is carryless with field reduction, not integer multiplication.

0+1 and 1+0 failure sets

0+1 stripes(A, B) and(C, D), then mirrors the stripes. 1+0 mirrors(A, B) and(C, D), then stripes the pairs. After A fails,0+1 depends on complete(C, D); 1+0 uses B and C/D. A second C loss breaks 0+1 but leaves 1+0 withB/D. A second B loss instead leaves 0+1's complete CD, but destroys 1+0's AB mirror. Neither layout tolerates every arbitrary two-device loss.

With independent per-device loss probability p over one interval and no repair, a two-device mirror survives $1-p^2$. Four-disk RAID5 survives$(1-p)^4+4p(1-p)^3$. Common power/firmware/environment and repair/rebuild violate those independence/no-repair assumptions. These are model probabilities, not measured lifetimes.

Checksums, storage pools and reservations

The book's ZFS example stores child addresses/checksums in parent metadata and validates down the tree. A checksum inside the same corrupt block is not equivalent to a trusted parent's end-to-end check. Detection requires the stated corruption model; repair additionally requires a correct redundant source. A checksum is not malicious-authentication protection.

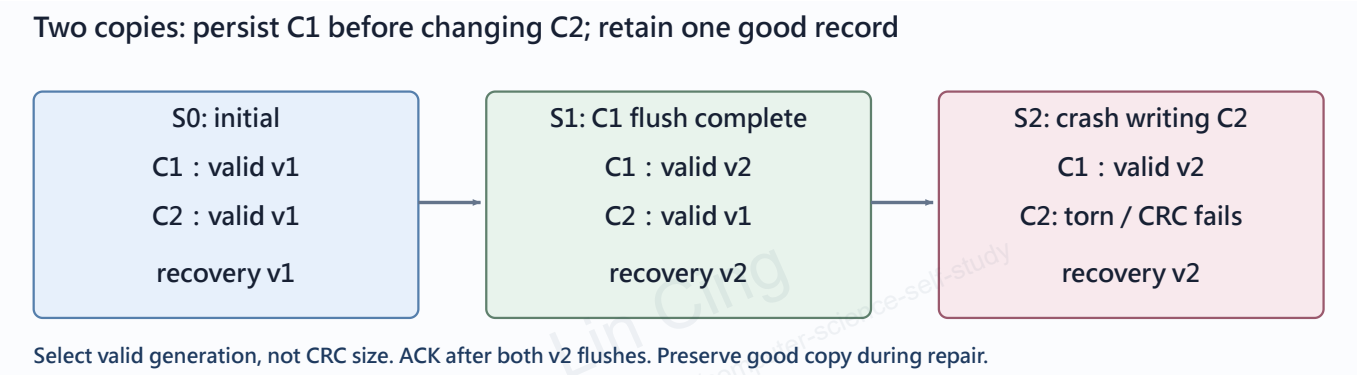
A storage pool supplies several filesystems; quotas limit growth, reservations protect shares. In an abstract 100 GiB pool, A uses 30, B20, free 50, A quota 60, B reservation 40 still needs 20 protected. Unreserved free=30, so A can grow min(quota remaining 30, unreserved 30)=30. Snapshots, dedup and compression are excluded; this is not all ZFS accounting or unbounded capacity.

Two-copy stable storage

Ideal stable storage never loses information; implementations approximate it within a failure model. Fix serialized 512-byte records, two device copies, nonwrapping generation, CRC detecting the injected tears, and trustworthy durable flush. Write only one copy at a time. A crash/torn write affects the currently written copy; the untouched and previously flushed copies stay valid. Common loss of those copies is outside this invariant.

Record format is4-byte big-endian generation+504-byte payload+4-byte CRC32 over the first 508 bytes. V1 payload is ASCII V1 then 502 zeros, CRC BF4CA2E8; V2 similarly gives 4A112E80. Recovery validates CRC then selects highest generation, not the numerically largest CRC. CRC is neither collision-free arbitrary-corruption proof nor authentication.

Two-copy stable storage (continued)



Every crash prefix and the invariant

Write/flush C1, then write/flush C2, then ACK.

Crash prefix	C1	C2	Recoverable version
Initial	Valid v1	Valid v1	v1
During C1, no confirmed flush	v1/v2/torn	Valid v1	At least v1; complete valid v2 may win
C1 v2 flushed, before C2	Valid v2	Valid v1	v2
During C2	Valid v2	v1/v2/torn	v2, not v1
C2 v2 flushed, then ACK	Valid v2	Valid v2	Acknowledged v2

Induction starts with two valid copies. During C1 the untouched C2 is valid old; before C2 is modified, C1 is durable new. Hence at least one valid copy survives every prefix. Recovery preserves the selected good copy, repairs/flushes the other first, and still retains a good copy if repair crashes. Concurrently overwriting both, untrustworthy flush, or common loss breaks the premises.

Every crash prefix and the invariant (continued)

An unacknowledged operation may recover old or complete new; no reply does not mean no execution. One-record recovery does not make a multi-record filesystem transaction atomic.

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Flush, FUA, ordering and atomicity

Volatile device cache can complete before nonvolatile media. Flush asks specified prior writes to reach persistence; FUA makes that request's completion satisfy the durable contract. Native support, emulation, and propagation through layers follow driver/device rules. Modern Linux REQ_PREFLUSH/REQ_FUA are concrete examples, not every historical OSC 9 device's flag names.

CPU memory barriers are not device-cache flushes. Ordered writes can still tear. Two-copy record recovery is distinct from RAID5 parity consistency. The guarantee must connect host buffer, device cache, persistent media, and ACK under the same failure model.

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Positioning, redundancy and durable completion

HDD 53→14 versus 53→0 changes positioning. FTL maps L1 to a new page and moves live data. RAID reconstructs under a specified loss set. Stable-write ordering decides which version survives interrupted updates. Success in one layer cannot replace another layer's contract.

Keep queue wait, seek/rotation, transfer and completion scheduling separate. State request bytes with IOPS; host/physical bytes with WA; topology with RAID capacity; failure correlation/repair with reliability; flush/atomicity with ACK. Changing media, topology or cache strategy requires revisiting that layer's premises. Detection, repair sources and historical backups each serve a different need.

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References

- Operating System Concepts,9e, Chapter 10, pp.467–496; Figures 10.5/10.6 scheduling,10.12 RAID0+1/1+0, and § 10.8 stable writes.
- Linux RAID arrays, dirty/degraded parity recovery risks.
- Linux volatile write-back cache control, modern flush/FUA propagation.
- RPM/transfer, FTL four-page blocks, CRC records, and GF256 values are explicit teaching models, not product measurements or guarantees under arbitrary failures.

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References and attribution

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Chapter route

How does a file-system interface express persistent-data access through composable objects, names, and rights?

Prerequisites are block storage, processes and file descriptors, and basic access control.

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Learning objectives

- Distinguish file metadata, data, and handle state.
- Compare sequential, direct, and indexed access.
- Analyze boundaries between mounting, sharing, and protection.

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File concepts

- Names, inodes, and open file descriptions have different identities and lifetimes.
- Descriptors inherited through dup or fork share an OFD. Independent opens do not share their offsets.

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Files, names, and descriptors have different identities

Use ordinary local Linux/POSIX regular files, excluding special devices and additional LSM revocation. Object identity is a pair such as (filesystem F1, inode 42), not the string `/data/a`. Two hard-link names can refer to one object. The same pathname can refer to different inodes at different times.

Layer	State	Sharing
Directory entry	Name → inode; the name lives here	A hard link adds another entry
Inode/file object	Size, owner, mode, data mapping, link count	All opens of that object
Open file description, OFD	Current offset, access mode, status flags	Descriptors shared through dup/fork
Process descriptor table	Integer fd → OFD; FD_CLOEXEC	Each process has its own table in this fork example

Files, names, and descriptors have different identities (continued)

Open checks permission and creates an authorized OFD. Subsequent read/write still checks the descriptor's access mode. Changing DAC permissions with chmod does not automatically invalidate an already open local descriptor. Other protection mechanisms have their own rules.

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Shared offsets with data ABCDEFGHIJ

P first obtains O1 through `fd3=open(F)`, then calls `fd4=dup(fd3)`. Q independently obtains O2 through `fd8=open(F)`. P then forks C, whose fd3/fd4 also refer to O1. Operations below execute in the specified sequence; this avoids claiming historical Linux concurrent-offset behavior as a guarantee.

Step	Returned data	O1.offset	O2.offset
Initial	None	0	0
P read(fd3,3)	ABC	3	0
P read(fd4,2)	DE	5	0
Q read(fd8,2)	AB	5	2
C read(fd3,2)	FG	7	2
P pread(fd3,2, offset=1)	BC	7	2

A successful read of k bytes advances its OFD offset by k. Pread uses an explicit position without changing O1. Sharing an inode does not combine independently opened offsets.

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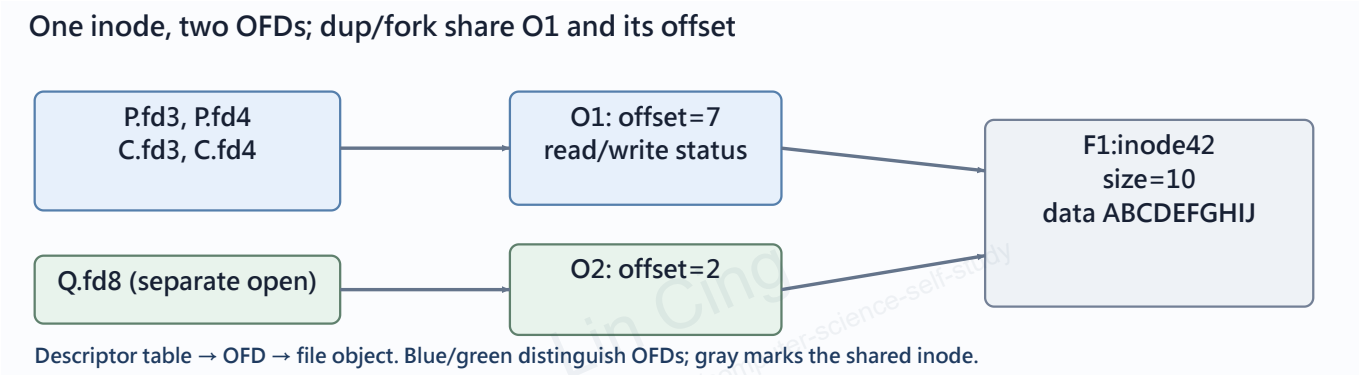
Shared offsets with data ABCDEFGHIJ (continued)



What survives unlink, truncate, and rename?

Regular object F1:42 initially has names `/a` and `/b`, link count 2, and open description O1. Unlinking a reduces the count to 1; unlinking b reduces it to 0, but O1 can still read the object. Reclamation requires the last close and no other references. Keep name references separate from open references; real systems may also have mmap references.

`truncate(F,0)` retains the object and names but changes size to zero. Existing OFD offsets need not reset. Reading an empty file at offset 7 returns zero bytes without moving the offset back to zero.

A same-filesystem `rename(tmp, target)` atomically replaces the name. New lookups of target reach the new inode; descriptors already open on the old target keep the old inode. Atomic name visibility during execution is separate from surviving power loss.

Filename extensions are application conventions. They prove neither content format nor permission to execute. Linux execution additionally checks permissions and the executable/interpreter format. Renaming arbitrary bytes to `.exe` does not create a valid executable.

Access methods

- Sequential, direct, and indexed access supply different positioning information.
- Partial operations preserve a completed-prefix invariant based on actual returned lengths.

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Logical access methods do not specify physical positions

Method	Caller specifies	Required state
Sequential	The next portion	Current position and rewind/seek rules
Direct	Relative record/block number	Fixed length or computable record boundaries
Indexed	Key	A consistent key-to-record/block index

For fixed record length L and zero-based record j, the starting byte is jL. Relative block numbers start at the beginning of the file. They are not disk LBAs; Chapter 12 handles physical mapping.

With L=32 B and blocks of 1024 B, each block holds 32 records. Record 77 starts at byte 2464, file block 2, displacement 416. A thousand records occupy 32000 B, requiring $\lceil 32000/1024 \rceil = 32$ blocks. The final block holds eight records, or 256 B, leaving 768 B of internal fragmentation. Exclude sparse files, compression, and inline data in this example.

Sparse indexes and I/O counts

The thousand records are sorted by unique integer keys 0 through 999, still at 32 records per block. The index has 32 entries of (first key, block pointer), each 16 B, totaling 512 B. For key 77, choose the greatest first key no larger than 77. Entry 64 identifies block 2; inspect keys 64 through 95 there.

If block i covers $[k_i, k_{i+1})$, sorted keys and a correct index restrict lookup to that candidate block. A gap still requires comparing the actual key. A predecessor-index hit does not prove that a record exists.

With the index in RAM, binary search takes at most $\lfloor \log_2 32 \rfloor + 1 = 6$ comparisons plus one data-block I/O. If the uncached entire index occupies one block, the bound is one index read plus one data read. This is neither a universal fixed two-I/O cost nor constant-time CPU search.

Short transfers preserve a completed prefix

Write 6000 bytes from initial file offset 100, with successful results 2500,1500,2000:

Call	Buffer start	Requested	Returned	Total done / next offset
1	0	6000	2500	2500 / 2600
2	2500	3500	1500	4000 / 4100
3	4000	2000	2000	6000 / 6100

Maintain $0 \le done \le N$: buffer[0:done] has been delivered in order, and only [done:N] remains. A successful k>0 advances done by k and reduces the remainder. If every iteration makes positive progress, at most N one-byte successes complete the request. Errors do not guarantee eventual success. Retry EINTR only under its no-data-transferred condition; wait for readiness on EAGAIN; never spin indefinitely on zero progress.

Short transfers preserve a completed prefix (continued)

A read near EOF can return a short result; a later positive-count request returns zero. A zero-count request returning zero does not prove EOF. After a read error, the offset may not be inferable. The table assumes successful outcomes and does not invent offset advances for errors.

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Append atomicity and positioned I/O

Separate `lseek(SEEK_END)` and write calls are not one atomic step. P and Q can both observe size 100 and write at offset 100, allowing the later write to overwrite the earlier one. O_APPEND combines EOF selection with that write within the interface's implementation scope. In particular, an NFS emulation need not provide the same promise as a local filesystem.

Even if each append write is atomic, a record split into A 1 and A 2 can interleave as A 1, B1, A 2. Whole-record guarantees require framing and locking or a single writer. Regular files are not pipes, so PIPE_BUF is not their atomicity theorem.

Pread/pwrite carry an explicit offset and avoid another thread changing a shared cursor between lseek and read. They do not serialize overlapping data writes. The pwrite scenario here excludes O_APPEND; Linux has a special append behavior for their combination.

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Directories and disk structure

- Trees, DAGs, and general graphs have different sharing, termination, and reclamation conditions.
- Process `..` after symlink resolution using the resolved object, not prior textual cancellation.

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Directory structures and their costs

A directory maps names to objects. Names are unique within one directory but can repeat across directories. Lookup, create, unlink, rename, and enumeration are interfaces, not a requirement to use a linear list.

Structure	Capability	Cost or limitation
Single level	One shared naming domain	Collisions and no grouping
Two level	A directory per user	Awkward sharing across users
Tree	Paths and subtree grouping	No direct shared child under a unique-parent rule
DAG	Multiple paths/names to one object	Reference counting and duplicate visits
General graph	Cycles	Termination, reclamation, cycle detection

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Directory structures and their costs (continued)

A hard link adds an entry to the same inode within one filesystem, with one copy of the data. A symlink is a separate object storing a pathname; it can cross filesystems, dangle, or cycle. Ordinary users cannot create arbitrary directory hard links. A conceptual DAG does not authorize unrestricted Linux directory hard links.

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Why reference counts cannot reclaim cycles

Consider only the abstract name/edge graph, excluding held opens. Let A→B, B→A, and root→A. Counts are A=2, B=1. Removing root→A leaves A=1, B=1. Both objects are unreachable, but neither count becomes zero.

For a finite DAG, an unreachable subgraph remains acyclic and has an indegree-zero vertex. It cannot have an incoming edge from a reachable vertex, or it would be reachable. Its reference count is therefore zero. Remove it and apply induction to reclaim every unreachable vertex. A cycle lacks that starting vertex.

A visited set prevents repeated traversal. DFS on a finite graph takes O(V+E), but traversal alone is not reclamation. Mark/sweep marks from roots and reclaims unmarked objects, requiring coordination with concurrent graph changes. A symlink traversal limit terminates one resolution; it does not prove all data reachable.

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Path resolution cannot start with textual cancellation

Let `/a/link` store absolute target `/x/y`; all required directories exist and are searchable. Resolve `/a/link/../b`:

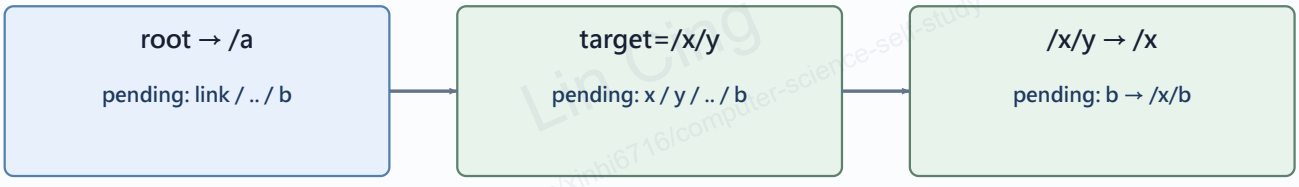
Step	Current object	Pending components	Action
0	Root	a, link,.., b	Start at process root
1	/a	link,.., b	Read link as a symlink
2	Root	x, y,.., b	Absolute target resets root; preserve suffix
3	/x/y	.., b	Resolve x and y
4	/x	b	Apply .. to resolved object
5	/x/b	Empty	Complete

Path resolution cannot start with textual cancellation (continued)

Canceling `link/..` first incorrectly gives `/a/b`. Resolve a relative symlink target from the directory containing that link, not the process cwd. Each next component requires search permission on the current directory. Whether to follow a final symlink depends on the operation.

/a/link/../b: expand the symlink before resolving ..

Target=/x/y. Blue = original path; green = expansion. Preserve the suffix.



A symlink is a separate object. An absolute target resets root; append the original suffix.

Wrong: cancel link/.. to get /a/b. Correct: /x/y/.. reaches /x, then /x/b.

TOCTOU can change the object behind a name

P checks that `/tmp/out` is a safe ordinary file. An attacker replaces it with a symlink to a sensitive file before P opens the same string. Two resolutions can reach different objects because the name graph changed; symlinks themselves are not universally invalid.

A trusted directory descriptor with openat fixes the relative starting point. O_NOFOLLOW constrains only the final symlink and does not authenticate intermediate components. Later Linux openat2 resolution restrictions can constrain the whole walk; do not assign that later interface to Linux 3.2.

Complete the operation through the same protected reference or appropriately restricted resolution. After open, fstat examines the opened object. Still define acceptable owner/type and allowed open side effects. Checking safety after O_TRUNC is too late to prevent truncation.

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File-system mounting

- Mount changes namespace resolution without moving old data.
- Trace mount crossings separately from open-reference lifetime.

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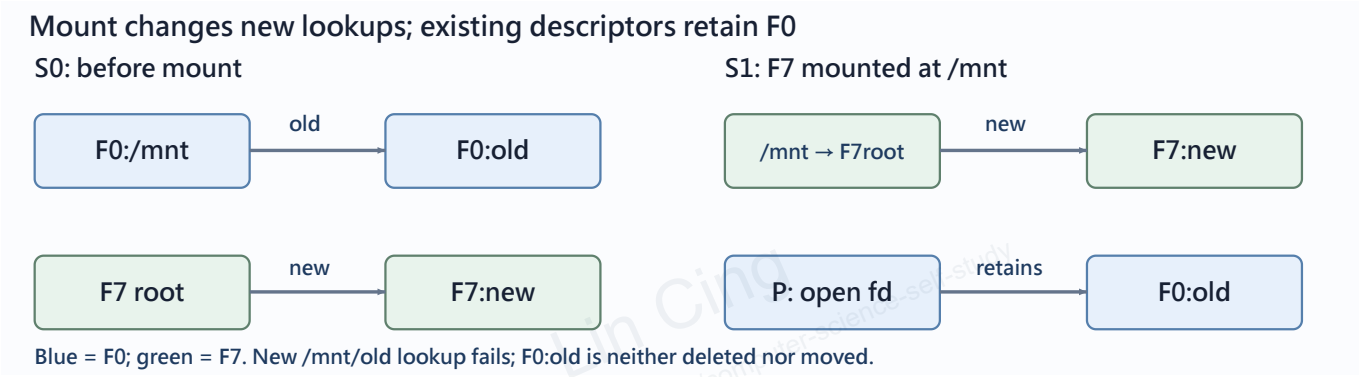
Mount changes a resolution edge

F0's `/mnt` initially contains old; F7's root contains new. After mounting F7 at /mnt, a new lookup of `/mnt/new` reaches F7:new, while `/mnt/old` returns ENOENT. F0's old remains allocated and becomes visible after unmount.

Observation	Before mount	After mount
New lookup /mnt/old	F0:old	ENOENT in F7
New lookup /mnt/new	ENOENT	F7:new
Previously opened F0:old descriptor	F0:old	Still F0:old

The namespace selects a mount map. The same string can reach different objects in different namespaces without copying their data. Blue denotes F0, green F7, and gray the old content hidden from new lookups.

Mount changes a resolution edge (continued)



Mount crossings, .., and busy references

In this Linux-style model, crossing a mount point enters the mounted root. Applying `..` there returns to the mount point's parent, rather than arbitrarily remaining at the underlying filesystem root. At the process root itself, `..` stays at root.

Let `/a/link` target `../b` and mount F7 at `/b`. Resolving `/a/link` follows root→a→link target→a/..=root→b→F7 root, then stops. Without an input filename, it cannot invent `/file` as a suffix.

Normal unmount can report busy for active cwd, open-file, or submount references under the system's rules. Linux lazy detach first removes the namespace attachment, then reclaims after references disappear. It neither immediately invalidates every descriptor nor guarantees pending I/O success.

A mount point is not a symlink. The namespace mount map controls mounting; a symlink is a directory object storing a target string. Both can occur in one path walk without becoming the same entry type.

File sharing

- Specify identity, visibility, ordering, and durability separately.
- Advisory-lock exclusion assumes that every conflicting participant follows the protocol.

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Identity, ordering, visibility, and durability

The same user name on two hosts need not map to the same UID. Inconsistent AUTH_SYS-style numerical mappings can authorize the wrong principal. A network address or declared UID does not replace trusted authentication. NIS, LDAP, and Active Directory are naming/identity infrastructure examples. LDAP is a directory protocol; its use alone proves neither encryption nor correct authentication.

A file protocol must separately define who may operate, operation order, when another client rereads new data, and what survives a crash. NFSv3 basic file operations do not need OPEN state to identify each READ/WRITE, but still have caches and stable-write semantics. NFSv4 adds stateful mechanisms. Stateless does not mean no state of any kind or automatic security.

Equal paths can reach different mounted objects. Clients of one object can temporarily observe different cached/session versions. Successful write, successful lock acquisition, another client's visibility, and media durability are four different events.

One history under immediate and session visibility

Start at value 0. A and B open, A writes 1, B reads before A closes, A closes, B reads within its existing session, then newly opened C reads. Execute in order and ignore errors/data races. The immediate column is a single coherent image abstraction, not a claim about every network cache.

Event	Immediate shared image	Session snapshot
A/B open	Both see 0	Each holds snapshot 0
A write 1	Shared image becomes 1	A private image 1; B still 0
B read	1	0
A close/publish	Still 1	Published image becomes 1
B read without reopen	1	Still 0
C open/read	1	1

One history under immediate and session visibility (continued)

Session snapshots do not change an already open instance when another closes. Multiple-writer conflict and last-close rules need separate definitions. Immutable sharing prohibits changing contents or reusing the unique name, simplifying caching but expressing updates as new objects/names.

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Advisory locks provide conditional exclusion

A holds an exclusive lock on [0,100). Cooperating B waits for a shared or exclusive lock on the same range; a nonoverlapping lock on [100,200) can proceed. Half-open ranges overlap exactly when $\max(start) < \min(end)$, so these two ranges do not overlap.

Time	A	Cooperating B	C without a lock
0	Acquires [0,100) X	None	None
1	Writes version 8	Waits for overlapping lock	Can directly write the same data
2	fsync succeeds	Still waits	May violate A/B expectations
3	Unlocks	Acquires, then validates/reads under cache protocol	None

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Advisory locks provide conditional exclusion (continued)

Assume every conflicting operation first obtains a compatible lock. If two writers executed inside the same range simultaneously, both would hold incompatible X locks, contradicting the lock service. C's uncooperative access breaks that premise. Locks alone establish neither cache invalidation nor durability; the shared-filesystem contract must provide those rules.

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Atomic publication is separate from durable publication

Assume a local Linux filesystem supporting same-filesystem atomic replacement and directory fsync, and a device honoring flush. Tmp and target are in one directory. Preserve the old target while preparing the new version:

1. Create a unique tmp, write all data with short-write handling, and set required metadata.
2. Successfully fsync(tmpfd) to persist contents and necessary metadata.
3. Successfully rename(tmp, target). New lookups reach the new inode; existing old descriptors retain the old version.
4. Successfully fsync the parent-directory descriptor before promising that the new name survives a crash. Handle every error.

Step 3 avoids a runtime interval in which target is absent. Step 4 concerns the durable name. Before it finishes, this chapter does not claim that every filesystem must recover to a specified old or new name. Two fsync calls are not a cross-filesystem transaction; rename can fail with EXDEV.

Protection

- Directory permissions govern name operations; file permissions govern data operations.
- ACL decisions depend on class selection, the complete requested set, and the mask.

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File and directory rwx govern different operations

First exclude root/capability privilege, ACLs, sticky rules, and other controls. Select one mode class: owner if matched, otherwise a matching effective group, otherwise other. Do not combine all three classes. With r=4, w=2, x=1, every requested bit must be present in the selected class.

Right	Regular file	Directory
r	Read data	Enumerate names
w	Change data	Add, remove, rename entries, normally also requiring x
x	Request execution with valid format and other conditions	Search/traverse a known name

With D mode0711 and F mode0644, a nonowner can traverse D and read known F but cannot list D. With D mode0733, other has wx and can add/remove known names without listing. Unlink primarily checks parent-directory wx, not the file's w bit. Sticky directories add identity restrictions.

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Linux POSIX-style ACL mask and selection order

Use Linux access ACLs, not Windows deny ACLs, excluding privilege/LSM. Owner alice and file group dev have entries user::rw-, user:bob:-w-, group::r--, group:qa:-w-, mask::rw-, other::r--.

1. If UID matches owner, check user:: directly without the mask.
2. Otherwise, a matching named-user entry must contain the entire requested set, as must the mask. Failure denies without falling back to groups/other.
3. Otherwise consider matching owning/named groups. At least one matching entry must contain the whole requested set, and the mask must contain it. A matched but insufficient group class denies without falling back to other.
4. Only if no preceding identity matches, use other without the mask.

Bob has only w even if he belongs to dev; group r does not supplement his named entry. Changing mask to r leaves him no rights, without falling back to other r. Alice keeps rw. With a mask entry, displayed mode group-class bits represent the mask, not necessarily group::.

Multiple groups do not authorize every combined request

Carol belongs to dev and qa. Here dev=r, qa=w, mask=rw. A read request passes through dev; a write passes through qa. A combined rw request, such as O_RDWR, fails because no single matching group entry contains both rights.

Subject/request	Selected path	Decision
Alice/rw	Owner rw	Allow without mask
Bob/r	Named user w	Deny without group fallback
Bob/w	Named user w and mask rw	Allow
Carol/r	dev r and mask rw	Allow
Carol/w	qa w and mask rw	Allow
Carol/rw	Each matching group lacks a right	Deny
No matching identity/r	Other r	Allow

Multiple groups do not authorize every combined request (continued)

Changing mask to r also denies Carol's write. Apply the algorithm to the complete current request rather than OR-ing every displayed entry. Named-user precedence is not a general deny-entry composition rule.

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Default ACLs, umask, and existing handles

Without a default ACL, creating with mode0666 and umask0027 gives $0666\ \&\ \sim 0027 = 0640$. Umask removes rights and does not add unrequested execute permission.

A directory default ACL is inherited as the new object's access ACL and restricted by the supplied create mode for owner/group-class/other. Do not additionally apply umask unconditionally. A default ACL initializes new objects without retroactively changing old files or replacing the ordinary access-check algorithm.

Opening a writable descriptor before chmod0000 generally retains write authority for an ordinary local Linux regular file, while a new open is denied by DAC. Immediate revocation requires a separately defined mechanism or closing/revoking references. Changing displayed permissions does not recall every authorized reference. See acl(5).

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Four traces for the same file

Trace	Record	Typical mistake
Names	Root/cwd, components, symlink suffix, mount map	Textually cancel link/..
Objects	Filesystem/inode, links, open references	Free after unlinking the last name
Access	OFD offset/access mode, actual transfer results	Assume dup offsets are independent
Authority	Directory search, selected ACL class, mask, authorization time	Treat a displayed w as unconditional write access

Atomic publication adds a persistence trace: complete tmp data→file fsync→rename→directory fsync. The name graph and durable storage have different publication points; one success label cannot represent all of them.

Choose operations from their counterexamples

Requirement	Insufficient approach	Required condition
Read a specified offset without changing shared cursor	lseek then read	Pread with short-read/error handling
Append an entire record without interleaving	Several O_APPEND writes	Framing and whole-record locking or one writer
Remove a known name	Check file w	Parent wx, sticky and other policies
Confine resolution to a trusted subtree	Check only final symlink	Trusted root/reference and whole-path constraints
Consistent cross-client update	Acquire advisory lock alone	Cooperating writers plus cache/visibility protocol
Find the new version after a crash	Successful rename alone	The stated flush contract and both file/directory persistence

Choose operations from their counterexamples (continued)

Chapter 12's directories, inodes, and caches must preserve these identity, naming, and handle contracts. Physical blocks and cache states cannot substitute different user-visible semantics.

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References

- Silberschatz, Galvin, and Gagne, *Operating System Concepts*, 9th edition, Chapter 11, printed pages 503–541. Numerical cases, exercises, and diagrams here are original.
- Ordinary local Linux regular-file examples use open, read, pread, unlink, rename, and fsync. Special/network filesystems and additional controls have their own contracts.
- path_resolution(7) defines component, symlink, and mount traversal. Openat2 is a later interface, not a Linux 3.2 feature.
- acl(5) covers access/default ACLs, masks, requested sets, and mode correspondence.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Chapter route

How can a filesystem maintain accessible, recoverable persistent objects using finite blocks, metadata, and recovery protocols?

Prerequisites are the file interface, logical/physical blocks, and basic journaling.

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Learning objectives

- Trace inodes, directories, open tables, and page caches.
- Compare contiguous, linked, and indexed allocation.
- Analyze metadata/data consistency across failures.

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File-system structure

- Layers translate logical files into fixed-size blocks. A superblock describes global metadata, free-space structures track allocation, and directories map names to control blocks.
- Caches absorb I/O, while crash consistency requires an explicit metadata/data commit order.

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Logical files and recoverable disk structures

Use a teaching volume with 4 KiB blocks:

Block	Persistent structure	Contents
0	Superblock	Block size/count, inode/free-space locations
1	Inode bitmap	Allocated inodes
2	Data-block bitmap	Free data blocks
3	Inode table	Type, size, mode, direct/indirect pointers
4	Root directory	home→inode 3
5	/home directory	u→inode 5
6	/home/u directory	notes.txt→inode 7
8–10	File data	Three 4 KiB blocks of notes.txt

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Logical files and recoverable disk structures (continued)

The basic filesystem translates logical blocks into device blocks; file organization manages free space and logical-to-physical mapping; the logical filesystem manages directories, FCBs/inodes, and APIs; I/O control and drivers perform transfers. Caches may reside above these layers, but the system contract must define the durable commit point.

An append must preserve

$$\text{name} \to \text{valid inode} \to \text{consistent block set}, \qquad \text{allocated} \cap \text{free} = \varnothing.$$

A normal write return does not make the directory entry, inode, bitmap, and data simultaneously durable. Unmount, successful flush, and transaction completion are different events.

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File-system implementation

- An inode/FCB stores file identity, size, and block mapping. Descriptor tables refer to OFDs, which store offsets and status flags.
- Independent opens share an inode but create descriptions O1/O2. Dup/fork can share one description and its offset.

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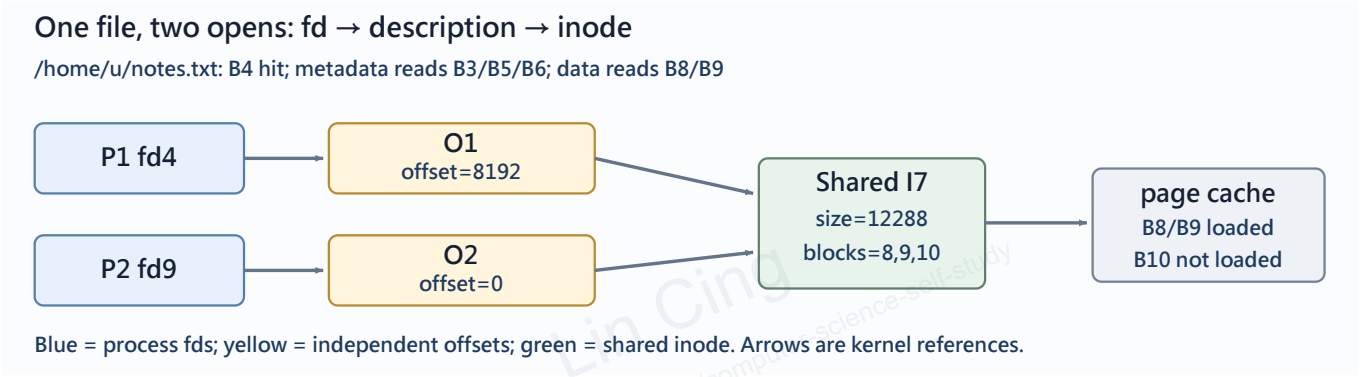
Complete path walk, open, and read states

Use `/home/u/notes.txt`, initially caching only root-directory block 4:

Step	Lookup	Read or hit	Result
1	home in /	Root block 4 hit	Inode 3
2	Inode 3	Cold inode-table block 3	Directory block 5
3	u in /home	Cold directory block 5	Inode 5
4	notes.txt in /home/u	Cold directory block 6; inode table cached	Inode 7
5	Open inode 7, permission succeeds	No further disk read	O1→I7, P1 fd4→O1, offset 0
6	P1 reads 8192 B	Data blocks 8,9	Data loaded, offset 8192
7	P2 independently opens same file	Inode/cache hit	New O2→I7, P2 fd9→O2, offset 0

Complete path walk, open, and read states (continued)



The cold walk costs three metadata reads, inode table plus directories 5/6, and two data reads, totaling five. With everything cached it costs zero disk reads. Directory cache, inode cache, system-wide open-file table, and page cache are different layers. A hit in any layer proves no durability property.

Complete path walk, open, and read states (continued)

The process fd table maps integers to descriptions and contains descriptor flags such as close-on-exec. An OFD is the kernel object created by open, containing offset, access/status flags, and an inode reference. An inode/FCB contains identity, size, permissions, and mapping and is associated with shared page-cache state. Independent opens do not merge offsets merely because the inode is shared.

The textbook's teaching representation places some fields in per-process tables and has a system-wide FCB table. That abstraction must still identify the object owning a shared offset. This example uses the Linux/POSIX open/dup three-layer representation. Two descriptions now refer to I7; this is not one description with reference count two.

Independent-open and dup offset invariants

Continue with P1 fd4→O1, P2 fd9→O2, I7 size 12288. Reads succeed completely, without signals or other writers.

Event	Descriptor relation	O1 offset	O2 offset	Returned range
After opens	P1:4→O1; P2:9→O2	0	0	None
P1 read(fd4,8192)	Unchanged	8192	0	[0,8192)
P1 fd6=dup(fd4)	fd6 also points to O1	8192	0	No read
P1 read(fd6,4096)	fd4/fd6 share O1	12288	0	[8192,12288)
P2 lseek(fd9,4096, SEEK_SET)	Only O2 changes	12288	4096	Contents unchanged
P1 read(fd4,1)	O1 is at EOF	12288	4096	Zero bytes

Independent-open and dup offset invariants (continued)

For a successful r-byte read, $o' = o + r$, with $0 \le r \le \min(n, \max(0, size - o))$. Dup and fork inheritance of the same description share o; independent opens do not. Closing fd6 decreases O1's references while fd4 remains usable. Closing the last description does not remove a directory name. Unlink and final-open-reference release have different lifetimes.

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Mount and VFS dispatch

Mount filesystem B's root inode 2 at A's /mnt. Resolving /mnt/log/x first finds A's mount point, crosses to (B, inode 2), then uses B's lookup for log and x. A and B may both have inode number 2, so identity requires filesystem/mount context as well as inode number.

VFS resolves dentry/inode and mount relationships, then dispatches a regular-file read through that filesystem's operations. A local filesystem maps blocks and submits block I/O; NFS forms RPCs. A common interface does not require every lower layer to use the same on-disk index block. Handle permissions, symlinks, .., and mounts during each resolution step. The five-read example explicitly excludes those extra traversals and permission-data I/Os.

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Directory implementation

- A linear list supports simple insertion but O(n) name lookup. Hashing narrows lookup on average, with chaining or probing for collisions.
- A directory is itself a file-like block structure requiring metadata consistency, protection, and crash recovery.

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Directory lookup costs

For equally likely successful positions in a linear list, average name comparisons are $(n+1)/2$. A hash table maps a name to a bucket and compares entries there. Fix

$$h(name) = \left(\sum_i code(char_i) \right) \bmod 8.$$

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Directory lookup costs (continued)

Bucket	Entries	Chain length
0	Empty	0
1	Empty	0
2	delta, epsilon, eta	3
3	gamma	1
4	beta, zeta	2
5	Empty	0
6	alpha	1
7	Empty	0

Directory lookup costs (continued)

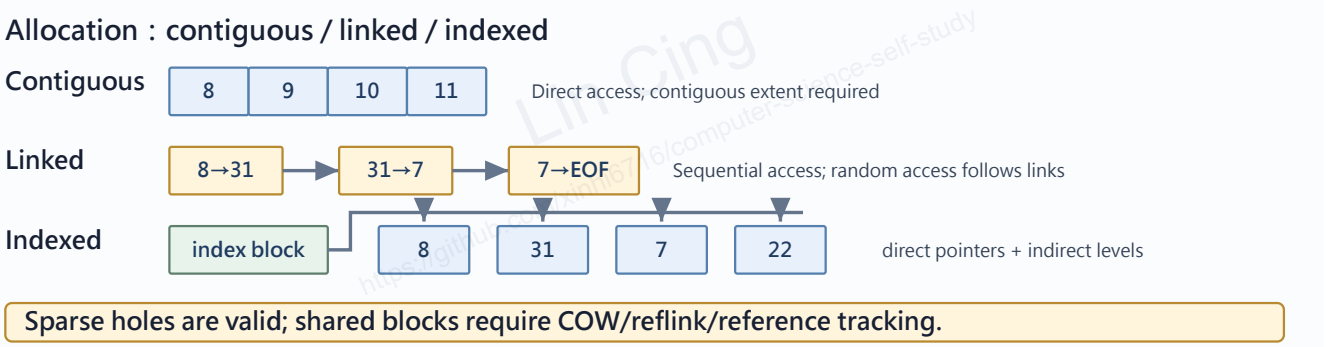
Load is $\alpha = 7/8 = 0.875$. Uniform queries over these seven stored keys give $(1 + 1 + 1 + 1 + 2 + 2 + 3)/7 = 11/7 \approx 1.571$ comparisons. This averages query targets for a fixed function, not random functions.

Under an additional uniform pair-collision model over m buckets with fixed insertion order, insertion i has expected successful cost $1 + (i - 1)/m$. Uniformly averaging stored keys gives $1 + (n - 1)/(2m)$, or 1.375 for n=7, m=8. The common $1 + \alpha/2$ omits the finite-n correction. Observed 1.571 and expected 1.375 use different randomness and denominators. All keys colliding still yields O(n) worst case.

Eight chaining buckets can store more than eight names; a ninth name does not inherently overflow the table. Resize policy can depend on load and latency. Open addressing instead has at most one entry per slot. A visible intermediate rehash can omit an old entry or duplicate a name. Locks, versions, or a specified incremental migration protocol must preserve a coherent lookup result.

Allocation methods

- Contiguous allocation supports direct access but has external fragmentation and growth constraints. Linked allocation stores successor locations and favors sequential access over random access.
- Indexed allocation centralizes block pointers. Direct, single-, double-, and triple-indirect pointers extend addressability.



Levels of indexed allocation

- Direct pointers identify data blocks. Indirect blocks contain data pointers; double/triple indirection adds levels. Addressability depends on block size and pointer width.
- A 4 KiB block holds 1024 four-byte pointers. Twelve direct pointers are followed by 1024 single-indirect blocks and another 1024^2 through double indirection.

Direct, linked, and indexed access costs

Fix 4 KiB blocks and four-byte pointers, giving 1024 entries per pointer block. Retrieve zero-based logical block 2:

Method	Mapping	Disk reads	Main cost
Contiguous	Start11, length3	One data read	Fragmentation and relocation on growth
Linked	11→12→13	Three block reads	Follow links for random access
Indexed, index cached	Index entry→data13	One data read	Index space
Indexed, index cold	Index block, then data13	Two reads	Large-index memory/TLB costs

Contiguous mapping is start+i. Linked access starts at the head. Indexed access selects pointer i, with cost determined by its direct/single/double/triple level.

Direct, linked, and indexed access costs (continued)

Region	Logical blocks	Data capacity
Direct	0–11	48 KiB
Single indirect	12–1035	4 MiB
Double indirect	1036–1,049,611	4 GiB
Triple indirect	Following region	4 TiB

Offset 5 MiB exceeds the direct-plus-single region and requires double indirection. A sparse hole need not allocate a data block and reads as zero even though logical size covers it. A zero inode pointer is therefore not automatically corruption.

Derive indirect indices from a byte offset

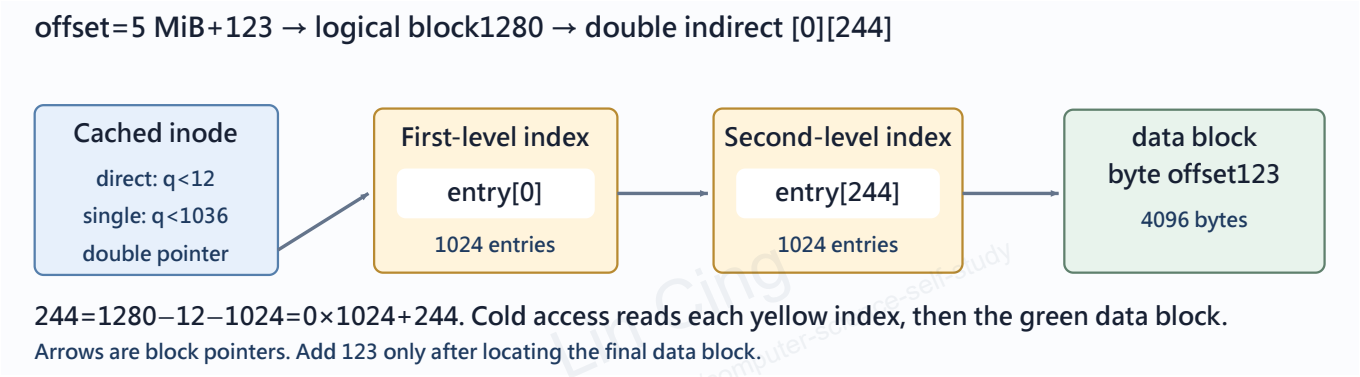
Let $q = \lfloor offset/4096 \rfloor$, $d = offset \bmod 4096$. Twelve logical blocks are direct, the next 1024 single-indirect, then 1024^2 double-indirect and 1024^3 triple-indirect. Use half-open ranges to avoid an extra endpoint.

For offset=5 MiB+123=5,243,003, q=1280, d=123. Subtract 12 direct and 1024 single blocks to obtain double-region index 244:

$$244 = 0 \cdot 1024 + 244.$$

Follow inode.double→first-level entry0→second-level entry244→data frame, then byte123. With the inode resident and both index levels/data cold, this costs two index reads plus one data read. A cached first index saves one metadata I/O.

Derive indirect indices from a byte offset (continued)



Total data addressability is $4096(12 + 1024 + 1024^2 + 1024^3)$ bytes, excluding index blocks from user data. Actual file limits also depend on size fields, block-number ranges, and filesystem rules. A hole pointer can return zeros without allocating a data block; returned zeros do not prove a zero-filled disk block exists.

Derive indirect indices from a byte offset (continued)

Allocation invariants must admit sparse holes and reflink/COW sharing. A hole has no data block. Reflink lets different inodes temporarily reference a COW-protected block, tracked by reference/version metadata and split on modification. Duplicate allocation means treating one block as independently writable without a sharing protocol. Removing one inode reference cannot free a block still used by a snapshot/reflink.

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FAT versus pointers embedded in data blocks

For chain 11→12→13→EOF, embedding a four-byte next pointer in every 4096 B data block leaves 4092 B payload. Reaching the third chunk reads 11 and 12 for links, then 13, totaling three reads. Do not simultaneously count 4096 B payload and ignore the embedded pointer.

FAT stores FAT[11]=12, FAT[12]=13, FAT[13]=EOF in a separate table. If resident, locating the third block still needs two pointer lookups but only one data I/O. A cold FAT adds the table blocks actually read. FAT improves chain-metadata locality without changing CPU traversal from O(q) to O(1). Indexed allocation directly selects entry q.

An extent stores (logical_start, physical_start, length). Extent (0,8,6) maps logical 4 to physical12 by displacement. Occupied adjacent space can require another extent rather than moving the entire file, but extent-tree lookup and metadata costs remain separate.

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Free-space management

- The textbook bit-vector convention is 1=free,0=allocated. For 16 blocks, 0000000100011111 has free blocks 7 and 11–15. First-fit allocation of three consecutive blocks selects 11–13, yielding 0000000100000011 and free blocks 7,14,15.
- Bitmap/list state must agree with inode pointers. A crash between changes can create duplicate allocation or lost clusters. A 1 TiB volume with 4 KiB blocks requires at least a 32 MiB bitmap.

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Bitmap convention and a concrete allocation

Use 1=free,0=allocated, with block 0 first:

```
block:  0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
bits:   0 0 0 0 0 0 0 1 0 0  0  1  1  1  1  1
```

The string 0000000100011111 represents free set {7,11,12,13,14,15}. Inode7 already owns data8–10. Appending 12 KiB needs three blocks. A single contiguous extent starts at 11, because block 8 prevents a three-block run starting at 7.

```
after: 0000000100000011
free: {7,14,15}
```

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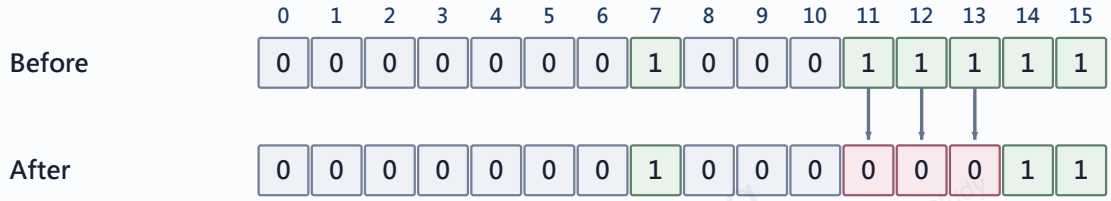
Bitmap convention and a concrete allocation (continued)

First-fit allocates consecutive blocks 11, 12, 13: free → used



inode7: blocks [8,9,10] → [8,9,10,11,12,13]; size 12 KiB → 24 KiB

Green 1 = free; gray 0 = already allocated; red 0 = newly claimed. Free set = {7,14,15}.

Preserve that allocated blocks are not free and valid allocated inode pointers identify allocated blocks. Selecting volatile bits and updating disk bitmap/inode separately can crash into duplicate allocation or an orphan. A memory lock prevents concurrent races, not power-loss inconsistency.

$$\#bits = \#blocks, \qquad \#bytes = \left\lceil \frac{\#blocks}{8} \right\rceil.$$

Bitmap convention and a concrete allocation (continued)

A 1 TiB volume has 2^{28} four-KiB blocks, requiring 2^{25} bitmap bytes, or 32 MiB. Storage I/O still operates in blocks.

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Four representations of the same free set

For {7,11,12,13,14,15}, a bitmap uses 16 bits. A free list can be head7→11→12→13→14→15→null. Allocating one block pops head7 and sets head=11; it need not scan the entire list. Finding a contiguous three-block run needs additional search/organization.

Grouping places several free addresses in a management block. With three addresses per group, the first two identify allocatable pages and the third links the next group. One read obtains two candidates. The format must define when the next-group management page itself becomes free.

Counting stores (start, length), here (7,1),(11,5). Allocating three from (11,5) returns [11,14), leaving (14,2). Freeing [11,14) merges it with (14,2) into (11,5). Free intervals are disjoint, adjacent intervals coalesced, and their union exactly the free set. A balanced offset tree supports O(log r) insertion/coalescing for r runs. Finding a sufficiently large run additionally needs a size index or subtree maximum; an offset tree alone does not make every query logarithmic.

Four representations of the same free set (continued)

The textbook ZFS space-map model logs ordered extent allocations/frees. Starting at (11,5), replay alloc(11,3) to obtain (14,2), then free(12,1) to obtain (12,1),(14,2). Rebuild from a checkpoint and remaining log instead of rewriting an entire bitmap in place. Preserve event order and transaction identity; replaying one allocation twice is not two new allocations.

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Efficiency and performance

- Caches, read-ahead, block size, and scheduling affect throughput; consistency and crash safety can increase write costs.
- Measure application bytes separately from device bytes. A cache hit is not durability. Larger blocks reduce mapping overhead but increase internal fragmentation.

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Cache, read-ahead, and writeback

The cold example costs three metadata plus two data reads. Fully cached directory/inode/data state gives zero disk reads. Report hit rates by layer instead of replacing the model with a generic reduction percentage.

1. A cache hit avoids disk transfer but still costs lookup, copying, locking, and memory work.
2. Read-ahead loads consecutive blocks 8/9 to reduce seeking/latency; incorrect predictions perform extra reads.
3. Asynchronous writeback can return from write while data remains in dirty page cache. Flush/commit records determine the later durability point.

Fix 10,000 files, each exactly 33 KiB, without compression, deduplication, or holes. Four-KiB units allocate nine units, 36 KiB per file, wasting 3 KiB. Sixty-four-KiB units allocate one unit, wasting 31 KiB. The difference is 28 KiB per file, or 280,000 KiB=273.4375 MiB total. An average size alone cannot justify taking a ceiling after averaging because generally $E[\lceil S/B \rceil] \neq \lceil E[S]/B \rceil$.

Cache, read-ahead, and writeback (continued)

In the simplified model of one four-byte mapping entry per unit, entries decrease from nine to one, saving only 32 bytes/file and 320,000 bytes total. These pointer bytes are separate from the 28 KiB data-space difference. Actual inode/extent/index metadata must follow its own structure.

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Pointer sizes, unified cache, and synchronous writes

A four-byte pointer addresses 2^{32} block positions, theoretically $2^{32} \cdot 4\ \mathrm{KiB} = 16\ \mathrm{TiB}$. Actual file size also depends on indirection, block counts, and filesystem fields. Eight-byte pointers expand addressing but halve pointers per index block, so they do not automatically save space.

A unified page cache shares pages between mapped files and read/write. Consistency depends on PTEs, dirty tracking, writeback, and locking. Synchronous writing waits for its specified durability point. An asynchronous write may return in cache; its success alone does not prove survival of a later power loss.

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Recovery

- A journal persists descriptors and transaction metadata/data before committed recovery. COW writes new structures and publishes a new root last.
- Fsck can repair orphan/inconsistent metadata without recovering user data never made durable. Consistency and durability are different properties.

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Journal recovery at each failure point

Transaction T7 appends 12 KiB:

```
bitmap 11,12,13: free→used
I7.blocks += 11,12,13
I7.size = 24 KiB
data: blocks 11,12,13
```

Use a redo-only, no-steal ordered metadata journal. Descriptors, metadata after-images, and commit identify the transaction. A complete commit is atomically recognizable; torn/incomplete records do not count. Durable ordering is descriptor→data completion→metadata records→commit→checkpoint→transaction clearing. Before commit, uncommitted metadata cannot reach home locations. Device volatile caching must honor the required flush/barrier order.

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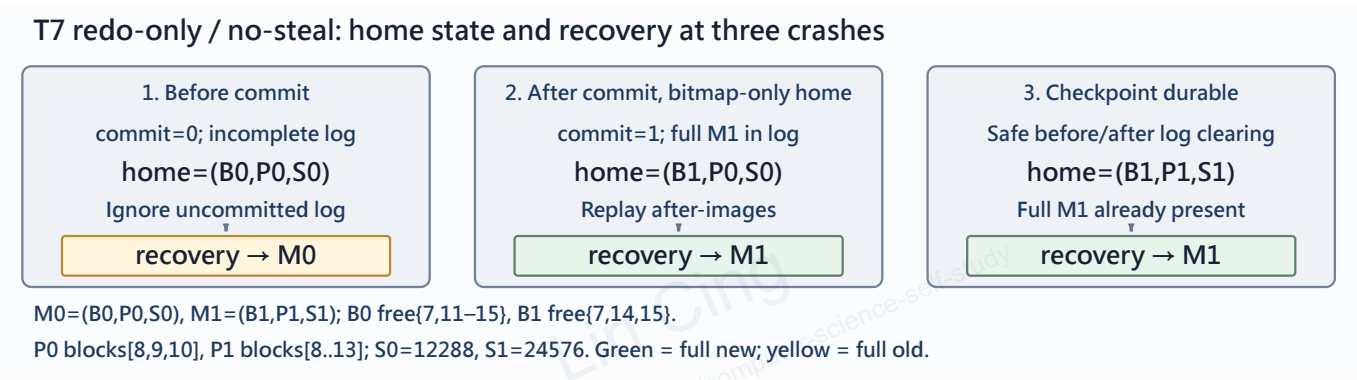
Journal recovery at each failure point (continued)

Crash point	Recovery evidence	Action	Visible state
Before descriptor	No T7	Discard	Old, size 12 KiB
After descriptor, before commit	Uncommitted	Ignore	Old
After commit, incomplete home	Committed	Idempotent redo	New, size 24 KiB
Home complete, log not cleared	Committed, possibly applied	Redo or recognize completion	Still new

Written data alone is not the commit criterion. This ordered model completes data before commit so metadata never publishes unwritten data. Metadata journaling alone does not give all user data the same transaction atomicity/durability.

Fsck scans directory/inode/bitmap relationships for orphan or duplicate blocks. It cannot reconstruct a user write that never persisted.

Journal recovery at each failure point (continued)



Every legal crash prefix recovers to old or new

Let metadata $M = (bitmap, pointers, size)$. Old M_0 has pointers [8,9,10] and size 12288; new M_1 has [8,9,10,11,12,13] and size 24576.

Before commit, no-steal keeps home at M_0. New data may exist in blocks 11–13 but no published inode points there. Ignoring uncommitted records preserves M_0, and those free blocks can be allocated again.

After commit but before checkpoint completion, full after-images are already durable. They overwrite any partially updated home into M_1. If bitmap is new while inode pointers are old, recovery must redo all three fields before mounting. Redo assigns size←24576 and the entire new pointer array; repeating it still yields M_1, rather than appending again to 36 KiB.

Log clearing is allowed only after durable checkpoint, so a cleared transaction has full M_1 at home. These branches cover every legal crash prefix and prove atomic metadata visibility. Allowing precommit home changes would require undo or another protocol; discarding the uncommitted log would then be insufficient.

Journal, snapshot, and backup solve different failures

A journal repairs interrupted updates, not legitimate deletion followed by overwrite. A snapshot retains old reachable blocks but can fail with the same destroyed device. A backup stores a separate copy; recovery requires that copy to exist.

Let Sunday full backup be B0, Monday incremental B1 contain changes since Sunday, and Tuesday B2 changes since Monday. Recovering Tuesday after Wednesday failure requires B0→B1→B2, including deletion/rename records. B0+B2 omits files changed only Monday. Only a differential B2 containing all changes since B0 permits B0+B2. Untrusted mtimes or a changing live source require a consistent snapshot/point and reliable change records.

Network File System

- NFS accesses a server through RPC, with client metadata/data caches. Handles, validation, and protocol versions govern access and visibility. Traditional Solaris NFS is not a specification for NFSv4 leases/delegations.
- Timeout means unknown outcome. Retries can repeat effects; distinguish idempotent lookup/read from non-idempotent mutations.

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NFSv3 handles, validation, and retries

Fix NFSv3, RFC1813, without importing NFSv4 OPEN/session state. Mount supplies a remote root handle. LOOKUP(directory_handle, name) obtains opaque child handles; READ carries handle, offset, count. NFSv3 has no remote OPEN/CLOSE procedure. Client fd/offset management is not a server POSIX OFD. A handle identifies an object, not an arbitrary resolvable filename.

Given /export handle H0, LOOKUP(H0, u)→H1, LOOKUP(H1, notes.txt)→H7, then READ(H7,4096,4096). Retrying READ adds no repeated mutation, but another client may have changed its bytes. Idempotent effects do not imply eternally identical results. If REMOVE's response is lost and another client recreates the same name, executing the old REMOVE as a new request can delete the wrong object.

Step	Client	Server	Evidence
1	Sends create X, id12	Creates X; reply lost	Request ID/effect/result
2	Times out, retries id12	Finds prior record	Returns prior result without another create

NFSv3 handles, validation, and retries (continued)

This requires a duplicate-request cache or the corresponding version's transaction/session mechanism, with explicit retention and crash behavior. Expiring finite caches do not establish permanent exactly-once execution. Client caches must validate relevant version/mtime/handle state; a cache hit is not a server commit.

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UNSTABLE writes, COMMIT, and server restart

NFSv3 WRITE returns its actual stability level and a write verifier. UNSTABLE can leave data only in volatile memory. A client requiring durability sends COMMIT and compares its verifier with prior WRITE. A restart losing volatile data changes the verifier, requiring retransmission of unconfirmed data.

Event	Server state	Client obligation
WRITE offset 0, A returns UNSTABLE, v7	A may only be in RAM	Retain A and uncommitted range
Restart changes verifier to v8	A may be lost	Prior WRITE success proves no durability
COMMIT returns v8 instead of v7	A's persistence unproven	Retransmit A, then request stability
Retransmitted write completes stably with matching verifier	Protocol-stable storage reached	Release retransmission obligation

UNSTABLE writes, COMMIT, and server restart (continued)

Attribute/data caches can still be stale. Close-to-open and timeout validation are client policies, not global linearizability implied by the NFSv3 name. A remote mount does not automatically import the server's other mounts; clients resolve their own mount tables.

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WAFL example

- WAFL uses COW snapshots and new blocks to preserve consistent versions. Unchanged blocks remain shared and space usage grows with modifications.
- COW needs allocation tracking, old-version reclamation, and write-amplification management.

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COW root publication and snapshot sharing

Version7 has R7→M7→D7, reachable from both active and snapshot roots. Counting two retained roots is a reachability model, not a claim that WAFL stores integer refcount2 in every block. The textbook's snapshot bitmap records snapshot usage.

Stage	Active root	New blocks	Old-tree reachability
v7	R7	None	Active/snapshot reach R7/M7/D7
Prepare v8	R7	D8, M8, R8	Transaction reserves unpublished R8; old tree retained
Durable	R7	D8/M8/R8 durable	Crash before switch still selects v7
Atomic switch	R8	Published	Active reaches new tree; snapshot reaches old

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COW root publication and snapshot sharing (continued)

Write new data and all new metadata before recoverable atomic root publication. This does not assume arbitrary-size hardware root writes are intrinsically atomic. A snapshot retaining R7 prevents reclaiming D7. Only removal of all retaining roots permits reclamation. Transactions also reserve unpublished new blocks so other allocators cannot reuse them merely because no public root reaches them yet.

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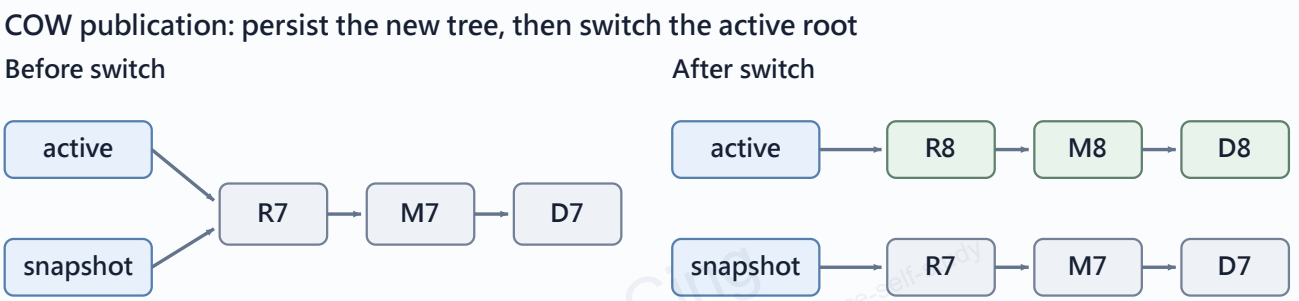
COW root publication and snapshot sharing (continued)



One data-block update can require new data, metadata, and root blocks. Reclamation must account for active and snapshot references instead of freeing a block merely because it is old.

Four mappings from names to blocks

Mapping	Invariant	Failure
Path/directory→inode	Namespace reaches intended object	Textual symlink-suffix cancellation
fd→OFD→inode	Correct offset/reference ownership	Treat dup offsets as independent
Logical→physical block or hole	Mapping agrees with allocation	Treat hole as disk block 0
Pointers/size↔free space	No incompatible reallocation	Mark free while old pointer remains usable

Memory locks coordinate concurrent updates. Crash consistency of several metadata fields additionally requires journaling/COW. Valid, dirty, writeback, and device-completed cache states are not one written state.

Journal and COW publication points

In T7's redo-only/no-steal model, home remains old, size 12288 until durable commit. After full after-images and commit persist, even a bitmap-only home update can be redone into size 24576 and matching pointers/bitmap.

Crash prefix	Recovery
Commit not durable	Home remains old; no uncommitted home writes
Commit durable, home partial	Replay every after-image into consistent new
Home/checkpoint durable	Reclaim log only under the proved truncation rule

After-image assignments are idempotent. Precommit home writes require undo or another mechanism. COW instead persists new data/metadata before recoverable root publication; retained snapshots forbid premature reclamation.

Both protocols define incomplete work, publication, and reclamation. Metadata consistency alone supplies neither full user-data transactions, backups, nor NFS cross-client coherence.

References

- *Operating System Concepts*,9th edition, Chapter12, printed pages543–585. Sections12.1–12.10 cover structure543–546, implementation546–552, directories552–553, allocation553–561, free space561–564, performance564–568, recovery568–571, NFS571–577, WAFL577–580, and summary580.
- The path, bitmap, hash, journal, and COW cases use the fixed models stated here. Their numerical results are not filesystem benchmarks.
- Linux open(2) distinguishes descriptors, descriptions, and independent offsets.
- RFC1813 specifies NFSv3 LOOKUP/READ/WRITE/COMMIT, verifiers, and duplicate requests. NFSv4 OPEN/session state is not imported.

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Chapter route

How can an OS control heterogeneous devices through standard interfaces while controlling latency, throughput, isolation, and errors?

Prerequisites are interrupts, DMA, memory mapping, blocks, and files.

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Learning objectives

- Trace an application request through device completion.
- Distinguish blocking, nonblocking, asynchronous, and buffered I/O.
- Locate bottlenecks beyond the CPU.

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Overview

Track the thread, request, and buffer ownership separately. Return, completion notification, and persistence each have their own conditions.

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Three I/O timelines

A read(fd, buf,4096) can hit page cache or start a device. The thread can be running, blocked, or ready; the request queued, in flight, or complete; and the buffer owned by CPU or device. These are separate state machines.

Assume one blocking thread, cache miss, and successful DMA. At t=0 the syscall starts; at 2 µs validation/submission finishes; at 3 the thread sleeps; at 102 DMA completes; at 104 the handler records completion and wakes the thread; at 110 scheduling runs it; at 112 it returns to user mode. Device service takes about100 µs, call latency112 µs, and wakeup-to-execution wait6 µs.

Before successful data is observed, it must be CPU-readable. Ready does not mean returned. Scheduling, notification, and visibility have separate synchronization conditions.

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One interface, several contracts

A descriptor denotes an opened object and access authority, not a hardware address. Regular files support seek; pipes/sockets normally do not. Terminals can apply line discipline; ioctl performs type-specific control. A common entry point does not enable every operation.

Distinguish acceptance into a buffer, driver submission, controller completion, media persistence, and remote application processing. A write returning 4096 does not universally establish all five. Durability needs a contract supported by filesystem and device; transmission success does not prove a receiving application committed. Numerical cases here are explicit teaching models, not machine measurements.

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I/O hardware

PIO, IRQ, and DMA address command/data access, notification, and transfer. Cancellation cannot free a DMA buffer while hardware may access it.

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Ports, MMIO, controllers, and buses

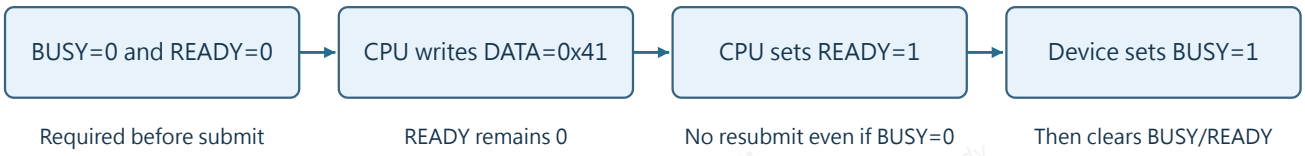
A device is the peripheral; its controller decodes commands, retains status, and transfers data. CPUs access ports through dedicated instructions or MMIO registers through protected mappings. MMIO is not ordinary RAM; caching, combining, and reordering follow architecture/driver I/O-accessor and barrier rules.

The teaching controller has STATUS, COMMAND, and DATA. STATUS bit0=BUSY, bit1=ERROR; COMMAND bit0=READY. Require BUSY=0 and READY=0, write DATA=0x41, select output, then publish READY. Publishing READY first can expose old data. Multiple CPU submitters additionally need a lock/queue; BUSY alone is not mutual exclusion.

A bus arbitrates shared transfers. Daisy-chain position/topology affects access and interrupt routing without removing permission checks.

Polling handshake and invariant

PIO output 0x41: prepare data before publishing



can_submit = !BUSY && !READY. Do not overwrite DATA before completion.

Polling handshake and invariant (continued)

Step	Actor	BUSY	READY	DATA	Next action
0	Initial	0	0	Old	Driver may submit
1	CPU writes data	0	0	0x41	Publish command
2	CPU sets READY	0	1	0x41	Controller accepts
3	Controller accepts	1	1	0x41	Output data
4	Completion clears state	0	0	0x41	CPU checks ERROR

can_submit=(BUSY==0 && READY==0). READY=1, BUSY=0 means published but not yet accepted. Even a single producer immediately attempting another byte must not overwrite this window. Testing only BUSY can replace0x41 with 0x42 before consumption.

Polling handshake and invariant (continued)

The correct trace is (0,0, old)→publish(0,1,0x41)→reject another submit, retaining0x41→accept(1,1,0x41)→complete(0,0,0x41)→publish(0,1,0x42). Under one producer, both-bit checking, and required ordering, can_submit stays false from publication through completion, preserving DATA. A device that never completes defeats termination; timeout does not prove it stopped using buffers.

Interrupts notify without moving every byte

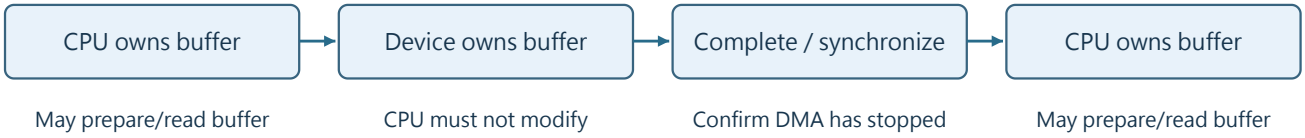
The controller raises an IRQ. At a permitted boundary the CPU saves required state and dispatches by vector. The handler checks source/status, acknowledges, updates the request, queues deferred work or wakes waiters, and returns. Shared IRQs require source identification rather than blindly completing the queue head.

Maskable interrupts, NMI, and synchronous exceptions differ; a low vector range is not uniformly NMI. Short handlers and sleepable later processing follow OS rules. Interrupts change CPU state/mode; switching threads additionally requires a scheduling context switch.

Lost wakeup occurs when a waiter reads done=false, IRQ sets true and wakes nobody, and the waiter subsequently sleeps. Under one lock, check the predicate, enqueue, and atomically release the lock while sleeping. Recheck in a while loop after waking; the IRQ publishes done under corresponding synchronization. The predicate stores completion evidence, while notification alone need not accumulate.

DMA addresses, coherence, and lifetime

4096 B receive: address mapping and ownership



Before submit: map/cache maintenance → descriptor/ownership → doorbell

After terminal: synchronize → publish → unmap/free only after hardware stops accessing

Prepare descriptor {device_addr, dma_addr, length=4096, direction=FROM_DEVICE}, mapping, and required cache maintenance before publication. Hardware transfers and notifies completion. IOMMU translation means a DMA address need not equal user VA or physical address. IOMMU confinement does not replace descriptor authorization.

DMA addresses, coherence, and lifetime (continued)

After handing the receive buffer to the device, CPU must not access it until completion and synchronization. Retain descriptor, mapping, and pages until hardware can no longer access them. Unmapping user VA does not cancel DMA. Noncoherent systems especially require the specified DMA API; a CPU barrier does not automatically invalidate caches.

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Transfer and notification costs

For4 KiB, PIO moving4 B per80 ns needs1024 operations, or81.92 µs CPU time. DMA submission2 µs plus IRQ/cleanup1 µs costs 3 µs CPU time, while device transfer still takes 100 µs.

A 100 ns polling check every5 µs performs about20 checks during service, costing2 µs CPU, with 0–5 µs detection delay, averaging2.5 µs under uniform phase. At2000 completions/s and1 µs/IRQ, demand is0.002 s/s=0.2% CPU. Continuous5 µs polling needs200000 checks/s, or2% CPU.

DMA still consumes memory/bus bandwidth. Overlapped CPU work does not remove system cost. Small transfers can favor PIO's lower setup cost. Compare fixed bytes, latency, and CPU denominators.

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Application I/O interface

Device classes expose different operations. Short transfers, EOF, EAGAIN, and asynchronous completion require distinct branches.

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Device classes, mapping, and timers

Block devices address blocks and can support random access. Character devices provide sequential data with device-specific record/seek rules. File mmap maps contents through faults/page cache; controller MMIO accesses registers.

Raw/direct I/O bypasses selected services while retaining permission, alignment, lifetime, and isolation requirements. Sockets add endpoints/connections/transports. Streams do not preserve write boundaries; datagrams have message/truncation rules. Readiness transfers no bytes and does not prevent a competing thread consuming them first.

Timers provide clocks, interrupts, and deadlines. A deadline queue at now7 ms contains10,13,13. Program10, remove that expiration, program13, then remove both equal deadlines. A trigger deadline does not promise immediate callback execution.

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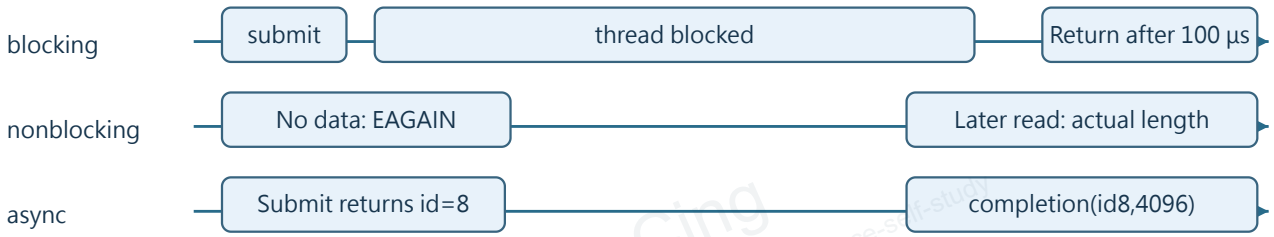
Blocking, nonblocking, and asynchronous control flow

One read, three API contracts



Device service time is unchanged; call control flow and notification timing differ.

Blocking can sleep and returns an ended operation, possibly short. Nonblocking returns EAGAIN without available data; read after poll/select still handles a competing consumer's EAGAIN. Async submission and completion differ; completion carries ID, status, actual length. Failed submission may create no request.

Blocking, nonblocking, and asynchronous control flow (continued)

A 512 B request with 200 B available can return200. Then done=200, remaining=312, dst=buf+200. Another112 gives done=312, remaining=200; subsequent zero means EOF. Deliver the short312 B or report protocol truncation. Never add -1 to done. Handle EINTR/EAGAIN under the API and progress already made.

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Partial vector I/O

For writev lengths[100,200,300], successful250 B consumes100 then150. Retry [base1+150,50], [base2,300], totaling350 B. Reusing the original vector duplicates250 B.

Vectors reduce calls and copies into contiguous staging storage. They do not universally guarantee device zero-copy or arbitrary-length atomicity. Atomicity, order, and durability are independent. Delayed async buffers live through completion; a kernel copy changes cost/lifetime. Failed cancellation or an already-completed response does not authorize freeing memory hardware may still use.

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Kernel I/O subsystem

Scheduling, buffering, caching, spooling, errors, protection, and dispatch jointly constrain I/O.

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Queues, buffers, caches, and spools

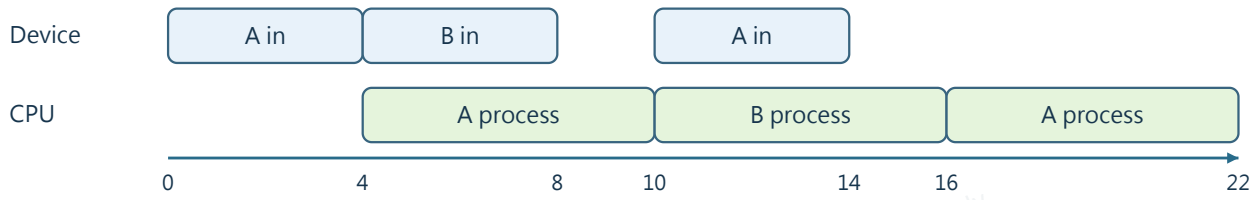
Scheduling selects service order; buffering absorbs rate/unit mismatches and preserves copy semantics; caching retains reusable data; spooling queues complete jobs for relatively exclusive devices. One region can be buffer and cache, with distinct correctness conditions.

Reads arrive for 40,10,41. Reordering10→40→41 can reduce HDD movement but cannot cross dependencies/barriers. This mechanical-seek model is not SSD latency.
Fairness/deadlines/throughput can conflict.

A dirty write-back cache can hold the only newest copy. Cache is not always an identical duplicate elsewhere. Accepted buffered write, device completion, and durability differ.

Double buffering with occupied slots

Alternate two slots; hand only a free slot to the device



ms. Input 4 ms/block; CPU 6 ms/block. B waits at 8–10; A waits at 14–16.

Fill 0–4; overlap 4–16; drain 16–22. Three blocks finish at 22 ms, not 18 ms.

Input takes 4 ms/block and CPU6 ms. One buffer takes 30 ms for three blocks. With two, A inputs0–4 and processes4–10. B inputs4–8, waits8–10, processes10–16. A inputs10–14, waits14–16, processes16–22. Total22 ms; steady period=max(4,6)=6 ms.

At4–8 the units overlap, but at 8 hardware cannot overwrite A while CPU owns it. For N blocks with full overlap and two sufficient slots,

Double buffering with occupied slots (continued)

$$T = t_{in} + t_{cpu} + (N-1)\max(t_{in}, t_{cpu}).$$

Finite batches retain fill and drain time.

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Copy semantics, printing, and errors

An application writes ABC then changes its buffer to XYZ. If write copies/finishes consumption before returning, output remains ABC. Zero-copy async may forbid modification until completion. State the applicable contract.

The spooler persists three-page job J17 and prints. Page2 prints but acknowledgment is lost. Resending can duplicate pages; deleting can omit pages. A persistent host job ID prevents duplicate host job creation, not necessarily duplicate physical effects. Without device progress/deduplication, report outcome unknown.

Transient failures can permit retry, while non-idempotent operations cannot be blindly resent. Length/failure conventions determine valid errno use; stale errno does not make a successful call fail. Controller sense/error logs provide detail.

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Permissions and kernel objects

P fd4→process table→open-file object{offset 8192, readable=true, ops=regular_file_ops} →inode{logical 2→physical70}→block-driver queue. Some descriptors share an open offset; independent opens need not.

Validate fd/mode/range/user pointer, then VFS dispatches by type. Page-cache hit copies directly; miss enters block/driver. Socket ops receive network bytes rather than using block 70. Name resolution and read dispatch differ; read need not walk the pathname anew.

Privilege/mapping permissions protect ports, MMIO, and control registers. Validate/appropriately pin user buffers. Device authority does not authorize arbitrary-memory DMA. Reservations can deadlock when A holds scanner waiting printer while B holds printer waiting scanner. Ordered acquisition, joint allocation, or recovery must break the cycle.

From request to hardware

Translate fd→inode→logical block→LBA→controller separately. One terminal publication does not prove exactly-once external effects.

Complete address translation

Mount /data selects device(8,2); /data/log.bin has inode 37; open creates fd4. Read4096 B at offset 8192 gives logical 2, displacement0 for 4096 B blocks. Inode maps logical 2 to device block 70. With512 B sectors, partition LBA=560, count8. Major 8/minor 2 selects driver/partition; startLBA 2048 gives whole-disk LBA 2608. The controller additionally uses a DMA mapping. Fd, inode, user VA are not LBAs. This fixed teaching model does not describe every UNIX implementation identically.

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Completion, timeout, and abort races

NEW→QUEUED→IN_FLIGHT→DONE or ABORTED publishes once. Timeout requests abort without proving ABORTED. Release mappings/pages only after normal completion or confirmed quiescence/abort acknowledgment.

Request8 submits0, times out100, requests abort101, completes102, and receives already-completed abort reply103. A lock/CAS protects terminal state. Completion102 wins; abort103 only releases bookkeeping without waking/unmapping twice. A caller may already have received timeout while resources remain until hardware stops.

Only a branch atomically changing nonterminal to terminal obtains publication/unmapping authority. All others see terminal and refrain. Two winners would require a forbidden second terminal transition, a contradiction. This proves at-most-once completion, neither eventual response nor exactly-once external effects.

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Buffer and control-block lifetimes

Stopped DMA permits buffer synchronization/unmapping, but timers/IRQs/abort callbacks may still read request state. The winner cannot immediately free its control block. Every queued/active callback already holds a stable reference and releases on exit. Canceling a timer cannot drop its reference without proving callback exit.

At101 owner, IRQ, abort hold count3. IRQ wins102, cleans DMA, exits3→2; owner delivers result and exits2→1; abort103 reads DONE, cleans bookkeeping, exits1→0, then free. An in-flight timer adds a reference. Acquisition itself requires lifetime protection; incrementing through a freed raw pointer is unsafe.

While any callback can read control state, a positive reference preserves it. Together with terminal checking this prevents double completion and use-after-free. Slot/ID reuse additionally requires generation or stable event identity against late events.

STREAMS

Each component has opposite read/write queues. High/low watermarks propagate backpressure through finite buffers.

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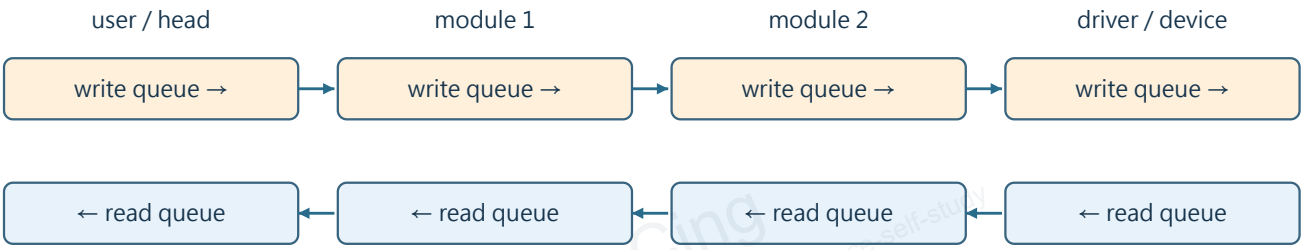
Bidirectional message queues

Historical STREAMS: two queue directions per component



Write moves right; read moves left. Full ordinary data queues propagate backpressure upstream.

Use historical System V/Solaris STREAMS. Head meets users, driver meets devices, ioctl pushes/pops modules. Every component has read/write queues; read moves toward head, write toward driver. Data/control messages do not require every module to copy payload.

Bidirectional message queues (continued)

Put procedures can process/forward immediately or enqueue. Service procedures drain runnable queues. Read/write expose byte streams; getmsg/putmsg can preserve control/data portions. Blocking at the user head does not mean synchronous waiting through every module.

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Numerical backpressure

Q capacity 1000 B, high 800, low 300, messages 200. Starting600, one accepted message gives800 and FULL, stopping upstream. Service yields600,400,200; below300 back-enables upstream. Hysteresis avoids switching repeatedly at one threshold.

Two upstream messages still consume memory. Full upstream queues eventually block the head writer. An unpausable hardware source may drop when all fill; IRQ cannot sleep indefinitely. Actual STREAMS priority/accounting is richer. This model covers ordinary fixed200 B data messages only.

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Performance

Account for CPU, memory bytes, and device service per request. Little's relation needs the same boundary and finite steady-state means.

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Resource throughput limits

Requests4 KiB; device100000/s; CPU5 µs/request gives200000/s on one core. Memory1 GiB/s with 12 KiB traffic/request, 4 KiB DMA plus copy read 4/write 4, gives87381.33/s. Thus overall throughput≤min(100000,200000,87381.33)=87381.33/s, a bound rather than achieved performance.

Truly removing that copy gives4 KiB traffic and262144/s memory bound; device100000/s becomes limiting. Count any remaining traffic. More cores cannot overcome the original memory bound.

Batching eight arrivals10 µs apart adds70 µs for first,0 for last, average35 µs. At1 µs/IRQ, CPU cost/request falls from 1 to0.125 µs at a latency cost.

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Latency and Little's relation

For finite steady-state means over one boundary, $L = \lambda W$. At80000/s and250 µs queue-plus-service response, L=20. With100 µs service, in-service count8, queue12, waiting150 µs. Queue count must not be paired with total response.

Minimum100 µs latency and100000/s throughput require average10 in flight, but queue depth10 does not guarantee that rate. Beyond saturation, more queueing can raise W without raising λ. At110000 arrivals/s and at most100000 completions/s without dropping/throttling, queue grows and finite steady-state L is unavailable.

This explicit conservation-model extension assumes neither exponential service nor the M/M/1 formula $W = 1/(\mu - \lambda)$. The latter needs additional distribution assumptions.

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One read across layers

Fd4, offset 8192→inode 37 logical 2→partitionLBA 560→whole-diskLBA2608→DMA4096 B. Thread blocked, request in flight, device owns buffer. Completion establishes CPU visibility, publishes state, readies thread; scheduling later returns it.

Cache hits skip lower layers. Nonblocking can return EAGAIN without bytes; async can return submission first. Short reads, errors, permissions, and lifetime remain. Check pre/postconditions along naming/authorization→queue→device→publication→use→reclamation rather than infer durable effects from one callback.

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Costs and observable guarantees

PIO/DMA choose transfer, poll/IRQ detection, blocking/async call contract. These can combine. Buffer/cache/spool solve different problems; copies can improve performance while separating return from durability.

State bytes/request, CPU time/request, bus bytes/request, service time, and observation boundary. Compare equal denominators; retain finite fill/drain and steady-state stability. Exercises retain intermediate states and general arguments rather than equating a few traces with proof.

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References

Silberschatz, Galvin, and Gagne, *Operating System Concepts*,9th edition, Chapter13, printed pages587–618. Cases, figures, and exercises are original. STREAMS uses the historical System V/Solaris model. Little's relation and DMA lifetime analysis are extensions under explicit assumptions.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Chapter route

How can enforceable structures limit a subject's authority over objects and preserve provable consistency after revocation?

Prerequisites are mode bits, access control, graphs, and matrices.

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Learning objectives

- Distinguish subjects, objects, domains, and rights.
- Compare ACLs and capabilities.
- Analyze revocation of existing handles.

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Goals of protection

Policy specifies permitted behavior; mechanisms enforce it without bypass. Complete mediation can issue protected handles, but must also define their revocation.

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Who may perform which operation on an object?

A subject/domain issues requests; an object is a resource; a right denotes an operation. Protection represents and enforces this relation. Security also addresses authentication, attacks, faults, auditing, and trust. An obscure pathname is not access control.

A policy permits assistants to read submissions and only the instructor to change grades. ACLs, capabilities, and protected syscalls implement it. Changing people/courses should change policy data rather than kernel code. Complete mediation requires effective authorization for every use. Checking an ACL at open and a protected handle's mode on each read/write can satisfy that design, with consequences for later revocation.

A caller's claim of authority is insufficient. A reference monitor must resist bypass, protect its own state, and remain small enough to inspect. Hardware mode/address-space isolation supports these properties without proving application policy correct.

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Principles of protection

Least privilege limits object scope, operations, and duration. A small privileged service must still validate the caller's arguments and target objects.

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Least privilege includes scope and time

A backup job needs read access to /course/submissions and write access to its backup target, not grade changes or module loading. Running the whole parser with full authority lets a malicious input exploit every permission. A low-privilege parser plus a small authorized service limits exposure, provided that service validates parameters and scope.

Granted rights R(t) must include currently necessary N(t), while minimizing the excess $R(t) \setminus N(t)$. Counting permissions alone is inadequate. Reading every student's personal data can be riskier than writing one temporary file. Scope, duration, and delegation affect the damage bound.

Fail-safe defaults deny absent explicit authorization. Audit subject, object, operation, decision, time, and correlation ID; logging does not itself block an attack. Separation of duty can require two roles to publish grades, but fails if one person can freely activate both roles.

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Domains of protection

Domains, processes, users, and CPU modes do not correspond one-to-one. UNIX effective-UID transitions and Multics gates impose different entry conditions.

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A domain is a set of rights

Let D={⟨F1,{read}⟩,⟨F2,{read, write}⟩}. Two processes of one user can have different domains; a protected call within one process can switch domains. User/kernel modes alone do not separate user-mode processes, which also need address spaces and authorization.

A static domain unions phase requirements, such as reading source during compilation and writing binaries during linking, exposing both rights throughout. A dynamic domain narrows each phase, but its switches must be authorized and preserve/restore caller context correctly while validating arguments.

A traditional UNIX setuid executable can change effective UID to its owner on successful exec under specified conditions. It does not necessarily change real UID or act like an ordinary function call. With real UID1001 and owner0, an allowed setuid exec can produce euid0 while retaining real UID1001. The program must validate input/environment and drop privileges correctly. OS-specific suppression conditions still apply.

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Multics rings and controlled gates

In the historical abstraction, ring 0 is more privileged. An access bracket (b1, b2) and gate limit b3≥b2 allow direct execution from ring i in [b1, b2]. For i>b2, entry into a more privileged ring requires i≤b3 and a designated gate. For i<b1, transfer toward lower privilege must not let parameter references expose memory unavailable to that callee.

With bracket[1,3], limit5, and gate entryA, ring 2 may use an ordinary entry and remain ring 2. Ring4 may enter ring 3 through entryA, but entryB is denied. Ring6 is denied even at entryA. Return restores caller ring/stack. Full Multics also has read/write brackets beyond this model.

Nested privilege cannot express incomparable requirements. If one domain needs only F and another only G, forcing either to contain the other grants an unnecessary object. A general access matrix supports this separation.

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Access matrix

Rows are domains and columns objects. Changes themselves require authorization. Copy/owner affect an object column; control permits deletion from a domain row.

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The initial matrix

Access matrix and domain-switch reachability

domain	F	G	P	D2	D3
D1	r*	ø	ø	switch	ø
D2	w	r	ø	ø	switch
D3	ø	ø	print	ø	ø

Switch path: D1 → D2 → D3

Current rows are not unioned

r* = read plus same-object delegation; ø = no rights. Switch paths do not authorize arbitrary entry points.

A[D, O] is a set of operations; a blank is ∅. The process uses its current row. D1 may read F, not write it; D2 may write F and switch to D3; D3 may print P. Because D1 can switch to D2 then D3, its initial row does not prove it can never print.

The initial matrix (continued)

Switch is a right over a domain object. Switching selects another row rather than unioning both. A controlled service also needs entry/parameter restrictions against jumping into arbitrary privileged code.

Copy, transfer, limited copy, owner, and control

Right `r*` permits r and its propagation within the same object column. F:`r*` cannot authorize G:r. Ordinary copy retains the source; transfer removes it; limited copy grants r without `r*`.

Initially A[D1, F]={`r*`, owner}, A[D2, F]={w}, A[D3, F]=∅. Limited copy gives D3={r}, unable to delegate again. As F's owner, D1 can add/remove specified rights throughout F's column, including removing D2's w. Ownership of F is not ownership of every object.

Control applies to domain objects. With control in A[D1, D2], D1 may remove rights from D2's row, not freely add them. Removing D2's switch to D3 breaks D1→D2→D3, requiring a new reachability analysis.

Rights propagation does not prevent information flow

D3 without copy(F) can still read F and write each byte to G if it holds those data rights. Copy rights govern delegation, not information copying. Revoking F:read also cannot ordinarily recall bytes already read into user memory.

Consider a restricted system where the only grant rule attenuates existing source rights in the same column, with no ownership or other creation of rights. Every right in every successor state originates in an initial right for that same object. The initial state satisfies the claim. Inductively, each new r has a source r, whose provenance reaches the initial state. This proves the claim for arbitrary finite transition sequences. It neither prevents data leakage nor applies when an owner can create rights freely.

Implementing the access matrix

ACLs index columns; capability lists index rows. Both must preserve rights semantics. Deleting an ACL need not invalidate cached open handles.

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Sparse representations

Representation	Index	Convenient query	Protection/update requirement
Global triples (D, O, R)	Global or dual indexes	Any pair	Protect table and avoid a full scan per use
ACL(O)=[(D, R)]	Object column	Who may use O?	Central deletion; explicit matching/default rules
Capability list(D)	Domain row	What authority does D hold?	Unforgeable references; controlled transfer/attenuation
Locks/keys	Object locks and domain keys	Is a key accepted?	Protect lists; change keys to revoke

A capability is a protected object reference with rights. A numeric fd indexes the current process's table; guessing another process's number does not acquire its capability. Kernel tables, tagged hardware, or another protected representation can ensure unforgeability. Capabilities need not universally be absent from user address spaces.

Convert columns to rows without changing rights

F has ACL[D1:r, D2:rw]; G has[D2:r, D3:w]. Corresponding lists are D1=[F:r], D2=[F:rw, G:r], D3=[G:w]. Both represent one matrix at one instant, but delegation/revocation costs differ. Hash-indexed pair queries can have expected O(1) cost; a linear list of length k costs O(k). The representation's name does not imply constant time.

A hybrid checks ACL at open and issues a kernel handle{F, read}; write later fails its mode check. If the ACL subsequently removes read, continued use depends on revocation policy. UNIX generally does not promise immediate invalidation of every open fd. Per-use generation checks can change this contract, with mappings/in-flight operations handled separately.

This chapter's simplified ACL uses allow entries with union and default denial. Deny ACEs, ordering, and group rules in real systems need their own algorithm. Do not substitute this model for Windows DACL evaluation.

Access control

Keep the historical Solaris10 case separate from the simplified RBAC model. Active roles are a subset of assigned roles; define whether exclusion applies per session or per user.

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Users, roles, and task permissions

Solaris10 separates roles, rights profiles, authorizations, and process privileges. A role groups administrative tasks; a profile can specify commands/execution attributes; an application can check authorizations; the kernel checks privileges. Role authority is not merely a UID, and an authorization string is not itself a kernel privilege.

In the original teaching model, U assumes BackupRole and executes BackupCmd with only source-read and target-write rights, unable to change GradeDB. V assumes GradeRole but cannot approve its own publication. Role activation still needs authentication/auditing. User-to-role assignment alone does not allow every request.

Formally $UA \subseteq Users \times Roles$, $PA \subseteq Roles \times Permissions$, and $activeRoles(session) \subseteq assignedRoles(U)$. A request passes iff some active role contains the required permission, subject to activation constraints such as exclusion. This simplified RBAC equation is not all Solaris privilege-set computation.

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Revocation

Define effect boundaries for cached handles and reused slots. Slot reuse must distinguish object versions and synchronize resolution.

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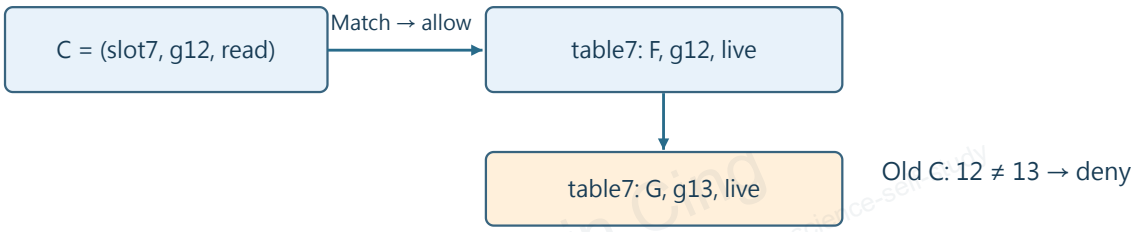
Four independent revocation dimensions

Immediate/delayed specifies when; selective/general specifies holders; partial/total specifies operations; temporary/permanent specifies future regranting. Immediate commonly means later checks deny new operations. It does not cancel already-started I/O or erase data already read.

An ACL centralizes removal, but hybrid handles/caches can delay effect. Distributed capabilities can use reacquisition/leases, back-pointers, indirection, or keys/versions. Reacquisition bounds delay; back-pointers track all copies; indirection centralizes checks at a revocable slot. A master-key change revokes every capability using it; selective revocation needs separate proxies or key groups.

Indirection and generation numbers

Reusing a slot: index 7 alone is insufficient



Change generation on reuse. Synchronize checking and retaining the object during resolution.

Capability C={slot7, generation12, rights r}; table[7]={F, g12, live}. Resolve checks liveness, generation equality, and rights, then obtains a stable object reference under the same synchronization. Revocation clears live; reuse for G increments generation to13. Old C=(7,12) fails against(7,13).

Indirection and generation numbers (continued)

Assume generations are never reused while old tokens may exist, and revocation/reallocation cannot interleave between checking and retaining the object. Then every accepted C identifies the object version for which it was issued. Finite wrapping counters with indefinitely retained old tokens eventually permit ABA. Prevent reuse or prove old-token quiescence before wrapping. Checking only slot7 demonstrably grants G to F's old capability.

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Capability systems

Hydra/CAP support abstract operations through protected typed procedures. Amplification is limited by type/procedure, not requested arbitrarily by the caller.

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Hydra and restricted amplification

Historical Hydra supports auxiliary rights for user-defined types. A client capability for Account A contains deposit, not raw representation read/write. Calling trusted deposit(A,10) checks type and operation rights; the protected procedure temporarily obtains representation access, changing balance40→50. The client still holds only deposit afterward.

Only approved procedures, types, and rules amplify authority. Arbitrary routines that turn read-only arguments into writable ones would violate callers' protection. Mutually distrustful subsystems must constrain amplifiable rights, retained references, and parameter scope; encapsulation alone is insufficient.

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Cambridge CAP data and software capabilities

Machine/microcode enforces data-capability segment read/write/execute rights. Protected procedures interpret software capabilities' custom meanings; the lower mechanism prevents forgery/modification. Seal/unseal-style authority limits representation inspection to the matching type/environment.

Both CAP and Hydra give clients abstract operations while implementations require lower-level access through a protected typed boundary. Their rights interpretation/support libraries differ. Running arbitrary user code does not itself prove safety. A Bank procedure that corrupts balance still breaks Bank semantics; isolation limits damage to its environment without proving business invariants. Account values here are an original model, not executable Hydra/CAP APIs.

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Language-based protection

Languages rely on trusted runtime, loader, and hardware. Historical Java stack inspection checks a privileged frame's own permission and, after ordinary frames, inherited context.

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Compile-time, load-time, and runtime guarantees

Types/modules/interfaces declare access boundaries. Compilers reject statically invalid operations; runtimes check dynamic indices, types, and capabilities. If an attacker can alter machine code, forge references, or bypass checks through native entries, source-level private fields alone cannot isolate it.

Type safety constrains memory/type behavior, not business authorization. A valid Account reference can still be misused to allow withdrawals or leak information. Kernel isolation separates processes; language isolation can separate classes/objects within a process, adding runtime, loader, and native interfaces to the trusted computing base.

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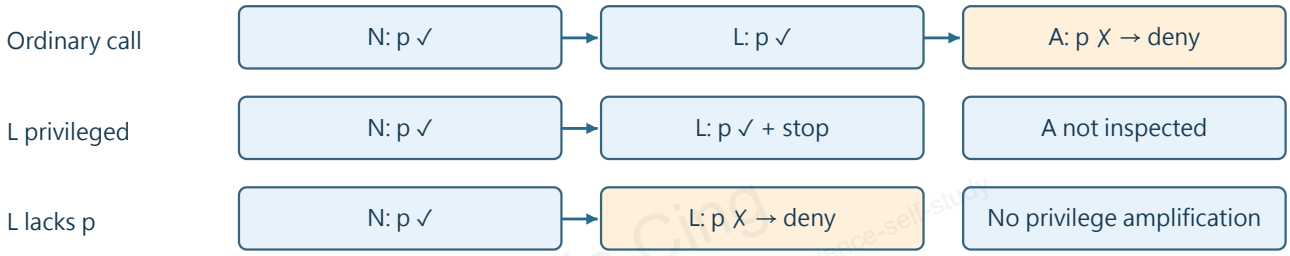
Historical Java SE 8 stack inspection

Java SE8: doPrivileged stops inspection; it grants no new rights



Inspect newest N → oldest A. Check each reached frame before considering a privileged stop.

Use traditional Java SE 8 AccessController, without explicit context or limited-permission doPrivileged. The newest-to-oldest stack is Networking N→Loader L→Applet A. For p=connect(proxy.example:80), N/L hold p and A does not. Ordinary inspection passes N, L and rejects A. If L uses doPrivileged in its authorized domain, N passes, L first passes its own p check, then stops inspection before A.

Historical Java SE 8 stack inspection (continued)

If L lacks p, privileged marking still fails there; it grants no new right. Passing every explicit frame also requires checking inherited access-control context, rather than an arbitrary decision after scanning. This historical SecurityManager sandbox is not presented as a modern Java recommendation.

Confused deputies and validated targets

Loader can connect to a fixed proxy for a low-privilege applet within a controlled privileged block. Passing arbitrary applet addr directly into that block lets the caller use Loader to scan an internal network. Restrict validated destination/port, operation, and returned data; do not execute an entire untrusted callback with elevated authority.

Inspection maintains that every checked frame holds p. A privileged stopping frame must also hold p. Thus this authorization never exceeds that boundary caller's existing p. Correct object/argument selection still depends on the caller; the stack theorem does not prove the service policy.

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Equivalence from matrix to implementation

The matrix states current permission; ACLs/capabilities index it; dynamic operations change it. Preserve object identity, right types, and domain context with unforgeable, non-bypassable checks. Cached decisions require explicit revocation semantics.

Owner changes a column; control deletes from a domain row; switch changes current domain; copy/transfer/limited copy change same-object grants. Their composition changes reachable domains, so one static row is not a whole-program authority bound.

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Three distinct safety claims

Authorization safety asks whether each operation has a right. State safety asks whether revocation/reuse/races identify the wrong object. Information/business safety asks whether legal reads/writes or privileged services still leak data or violate policy. Each needs different evidence.

State allowed transitions before proving a general claim. Attenuation induction excludes arbitrary owner grants; generation proofs exclude finite wrap with everlasting old tokens; Java privilege does not excuse parameter validation. Exercises use full columns, versions, and stacks to expose these conditions.

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References

Silberschatz, Galvin, and Gagne, *Operating System Concepts*,9th edition, Chapter 14, printed pages625–650. Matrices, examples, models, and exercises are original.

Java SE 8 AccessController specifies frame checks, privileged stopping, and inherited context. Solaris10 RBAC guide distinguishes roles, profiles, authorizations, and privileges. Historical cases retain their version boundaries.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Chapter route

Which cryptographic and system mechanisms provide verifiable guarantees when identities, programs, and networks can be attacked?

Prerequisites are protection, basic networking, probability, and hashing.

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Learning objectives

- Distinguish confidentiality, integrity, availability, and authentication.
- Analyze symmetric/asymmetric encryption and hashes.
- Evaluate defenses under explicit attack models.

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The security problem

Specify the threat model and CIA goals. Identity, authorization, and freshness differ. Valid messages still require receiver state against replay.

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Assets, adversaries, and goals

Confidentiality restricts unauthorized reading; integrity restricts unauthorized modification; availability requires authorized service under stated conditions. A threat can violate a goal, a vulnerability is an exploitable weakness, and an attack is an actual attempt. Theft followed by encryption/deletion can violate several goals.

The teaching grade service protects grades, credentials, and availability. The adversary controls the network and submits arbitrary forms, but cannot modify the kernel or obtain the server's private key. If key theft is later allowed, earlier authentication conclusions no longer follow. Every guarantee depends on its adversary, trust roots, keys, and endpoint integrity.

Physical, human, OS, and network factors affect the boundary. TLS does not protect input captured by an endpoint keylogger. ACLs restrict unauthorized use without proving every valid-account action follows business rules.

Impersonation, replay, and intermediaries

Masquerading impersonates an identity; replay resends old valid messages; a man-in-the-middle controls what both endpoints observe. Source IP is not cryptographic identity, and receiving a public key does not bind it to a service.

A valid MAC on transfer(A, B,100, id17) can be replayed even without understanding its content. Verification proves a key holder produced that content, not fresh authorization. Authenticate session, sequence, nonce, and operation ID, and maintain receiver usage state. Returning an earlier result on retries requires durable deduplication atomically linked to the transaction.

Distinguish inability to forge a new valid operation from inability to execute one authorization repeatedly. Encryption alone establishes neither combined claim.

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Program threats

Separate execution authority from infection, triggering, and concealment. Memory-safety claims need explicit byte bounds.

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Authority and propagation

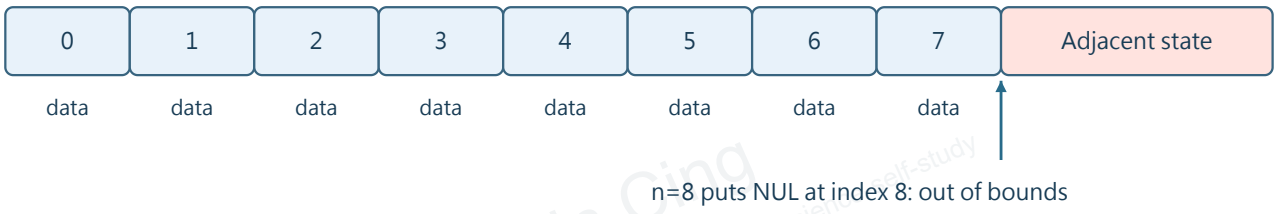
A Trojan abuses execution authority behind apparent useful behavior. A backdoor bypasses normal entry; a logic bomb triggers under a condition. Viruses infect host files/programs, while worms propagate through services/networks. Categories overlap and do not restrict viruses to one OS vendor.

An untrusted directory in PATH can substitute a namesake tool. Malicious code acquires the victim program's actual rights, not automatic root authority from its label. A compromised compiler can produce malicious output from inspected source. Executables, loaders, compilers, and update channels belong in the trust boundary.

File, boot, macro, and source viruses describe infection locations. Polymorphic, encrypted, stealth, and armored describe evasion; multipartite affects several locations. Byte signatures and behavior analysis complement one another. No signature match does not prove absence.

Buffer bounds precede control-flow claims

Abstract byte layout: the string terminator occupies a byte



Capacity 8 holds at most 7 data bytes + NUL. Binary buffers use explicit lengths.

An eight-byte buffer has indices 0–7. A C string with n data bytes requires $n+1 \leq 8$ for complete copying. At n=8 the data fits, but its NUL at index 8 overflows. Actual layout depends on ABI/compiler; this abstract figure illustrates adjacent corruption without platform addresses.

Buffer bounds precede control-flow claims (continued)

One safe contract rejects overlength input. Another explicitly truncates to k=min(n,7), copies k bytes, and sets buf[k]=0. Strncpy(...,7) alone need not append NUL. Binary data requires explicit lengths rather than strlen, with range and integer-overflow checks.

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Different defenses prove different properties

Bounds/type/lifetime checks address invalid accesses. Canaries detect some overwrites before return; NX/W⊕X restrict execution from data pages; ASLR complicates address prediction; control-flow restrictions constrain targets. NX neither prevents out-of-bounds writes nor alone excludes reuse of existing executable code.

Corrupting is_admin, length, or an object pointer can violate authorization without changing return addresses. Least privilege limits a damaged component's reach but cannot replace fixing the overflow. A managed safe subset eliminates some memory errors while native/unsafe code, runtime flaws, and business policy remain boundaries. Type safety does not prove every security goal.

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System and network threats

Model exposure, propagation, and denial of service through resources/state. Pending allocation rate and retention time jointly determine occupancy.

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Worm propagation and service exposure

The historical 1988 Morris worm exploited service weaknesses and remote trust, then sought reachable hosts. Repeated infection consumed resources, demonstrating that a security event need not steal data.

In the original discrete model, each infected host infects at most two fresh hosts per round, without duplication or removal. $I_0=1$ and $I_{t+1}=3I_t$ give 1,3,9,27. This ideal early upper-bound model is not Morris measurement. A closed 100-host network instead satisfies $I_{t+1} \leq \min(100,3I_t)$; it cannot contain 243 infected hosts.

Scanning observes reachable services. An open port does not prove a vulnerability, and banners need not be truthful. Authorization, patching, and disabling unnecessary services reduce exposure; removing one entry does not fix input bugs in remaining services.

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Resource exhaustion and defensive side effects

A service has 1000 pending slots retained 10 s per incomplete handshake. Without early cleanup, 100 arrivals/s fills 1000 on average under $L=\lambda W$. Attackers need not log in if they can force scarce-state allocation. Cookies, quotas, and delayed allocation change that model without creating infinite upstream bandwidth.

Legitimate 80/s plus attack 40/s exceeds 100/s service by at least 20/s. Rate limits must name their key, such as user, IP, tenant, or global quota. IP limits can penalize users sharing NAT. Five wrong passwords causing a ten-minute lock blocks guessing but lets attackers lock others out. DDoS uses coordinated sources; a flash crowd is legitimate excess demand. High load alone does not identify malice. Responses trade availability against false blocking.

Cryptography

Each primitive supplies a specific guarantee; identity binding and freshness require further arguments. RSA correctness, nonce reuse, and historical TLS require explicit derivations.

Primitives and key roles

Tool	Confidentiality	Integrity/source	Creation and verification
Symmetric encryption	With a suitable scheme	Not from encryption alone	Shared-secret holders
Public-key encryption	With a suitable scheme	Does not establish sender	Anyone encrypts; private key decrypts
Hash	No	Mismatch against trusted digest	Anyone computes
MAC	No	Shared-key origin/integrity	Key holders generate/verify
Signature	No	Private-key origin/integrity	Private signs; public verifies

Correctness means $D_k(E_k(m))=m$. Security is a computational claim against specified distinguishing/forgery attacks, not logical impossibility of guessing m. Algorithms can be public; secret keys remain secret. Public keys need authenticated integrity of their identity binding.

Reused keystream algebra

For $c=m \oplus s$, $m=c \oplus s$. Let m_1=0x41, m_2=0x42, s=0xA6; c_1=0xE7, c_2=0xE4. Reuse gives $c_1 \oplus c_2 = 0x03 = m_1 \oplus m_2$. Knowing m_1 reveals s and then m_2. This identity does not break the underlying cipher.

Block ciphers process fixed blocks; longer messages need correct modes and nonce/IV rules. AES's 128-bit block differs from 128/192/256-bit keys. DES, 3DES, RC4, WEP are historical context, not new-deployment recommendations. Use a complete analyzed confidentiality/authentication scheme. An unprotected public hash appended to ciphertext can be recomputed after tampering.

A complete small RSA computation

This number-theory example is not secure encryption. Set p=5, q=11, n=55,ϕ=40, e=3, d=27, ed=81=1+2ϕ. Public key=(55,3), private=(55,27). For m=12, 12^2 mod55=34 and 12^3 mod55=34 • 12 mod55=23.

Decrypt with square table 23^1→23,23^2→34,23^4→1,23^8→1,23^{16}→1. Since 27=16+8+2+1, 23^{27} mod55=1 • 1 • 34 • 23 mod55=12. Raw RSA is deterministic and algebraically structured. Deployable encryption/signatures additionally require specified encoding/padding, key sizes, and purpose separation. These small primes provide no confidentiality.

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RSA correctness for every message

For every m∈{0,…, n−1}, ed=1+k(p−1)(q−1). Modulo p, if p divides m, both m and m^ed are 0. Otherwise Fermat gives m^(p−1)≡1, so

$$m^{ed} = m(m^{p-1})^{k(q-1)} \equiv m \pmod{p}.$$

The same branches hold modulo q. Distinct primes and CRT yield m^ed≡m modpq, including messages not coprime with n.

Euler applied only when gcd(m, n)=1 would omit valid multiples of p/q. This proves mathematical decryption correctness, not RSA hardness, chosen-ciphertext security, or side-channel resistance.

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Hashes, MACs, and signatures

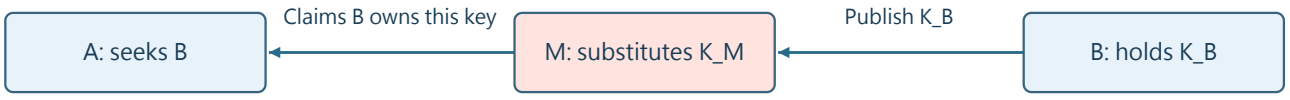
An n-bit hash has 2^n outputs, so collisions exist by the pigeonhole principle for a larger message space. Collision resistance concerns finding any distinct pair with equal digest; second-preimage resistance fixes one message first. Digest equality is computational evidence, not a theorem that messages are equal.

Anyone can replace (m, H(m)) with (m′, H(m′)); public hashes do not authenticate. MAC verifiers also know the secret, so MAC alone cannot prove to a third party that only the other participant generated a tag. Signatures use private signing and public verification, with trusted key binding, complete encoding, and key protection. They are not arbitrary encryption run backward.

HMAC is a standard keyed construction, not arbitrary H(key ∥ m). Include context, roles, session, and sequence against cross-purpose reuse.

Key distribution and substitution

Public keys need trusted identity binding, not secrecy



Without binding: A encrypts under K_M → M decrypts. Correct cipher, wrong identity.

Validate chain, name, purpose, validity, and handshake proof of private-key possession.

A seeking B receives an unauthenticated claim "B, key K". M substitutes K_M; A encrypts to M, which can decrypt although the cipher works correctly. Certificates bind service identity and key. Validate signature chains to trusted roots, target name, purpose, time, and relevant revocation policy. Certificate presence alone is insufficient.

Key distribution and substitution (continued)

Pairwise symmetric keys for N users require N(N−1)/2 keys. Six users need 15; adding a seventh needs six more. A KDC reduces long-term secrets per user but centralizes trust. Public-key techniques reduce secret distribution without eliminating public-key authentication or private-key protection.

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Endpoints determine protection scope

IPsec protects IP traffic between its network endpoints; TLS protects its session endpoints; encrypted mail can span store-and-forward hops. TLS termination at a reverse proxy exposes plaintext there and does not automatically protect proxy-to-backend traffic.

Link/network protection does not directly identify an application user. Disk encryption protects offline media, not a logged-in process authorized to read plaintext. For each layer state key holders, plaintext observers, and trusted components. Layer counts do not sum into unconditional safety.

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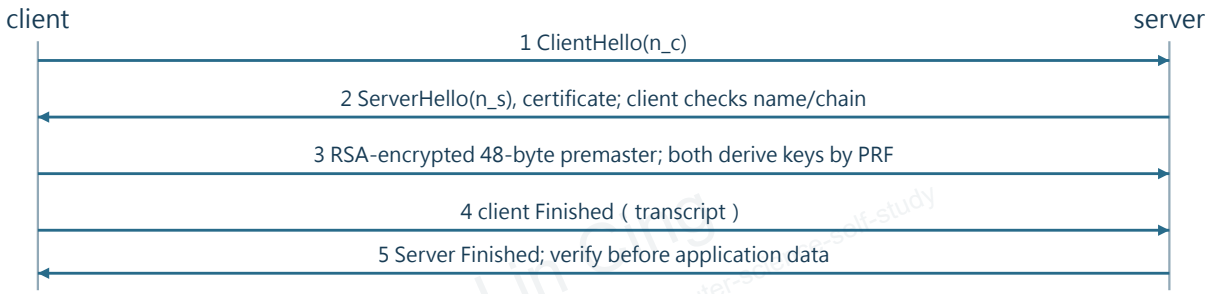
Historical TLS 1.2 RSA handshake

Historical TLS1.2 RSA full handshake (format/alert/CCS details omitted)



Use RFC5246 full handshake with RSA key exchange, not TLS1.3. ClientHello carries n_c; ServerHello carries n_s and certificate. Client validates chain, target name, purpose, and time, constructs a 48-byte premaster secret, two version bytes plus 46 random bytes, and sends its RSA-encoded encryption. Both sides use the PRF to derive master and directional traffic keys.

Historical TLS 1.2 RSA handshake (continued)

Both must verify Finished authentication of the prior handshake transcript. A certificate alone proves a CA bound that key, not current private-key possession or untampered handshake execution. With static RSA exchange, later private-key compromise and a recorded handshake can reveal old premaster secrets even after traffic keys were deleted. It lacks forward secrecy.

SSL3 and TLS are not synonyms. RFC7568 prohibited SSL3 in 2015. This historical example explicitly limits its version and omitted details.

Freshness and atomic nonce consumption

Server issues nonce 42 with pending[42]=(U, expiry). Client returns MAC_k("login" ∥ U ∥ n ∥ serverID). Receiver validates tag and pending/expiry, and atomically changes n to used. First succeeds; an identical valid replay fails because n is used.

MAC integrity cannot reject a copied tag without freshness state. Consume only after successful verification and make checking/consumption atomic; otherwise two threads can both observe pending. Random nonces need entropy/nonreuse; counters need restart persistence or a new session. Payments additionally link consumption and business effects transactionally or through recoverable deduplication, preventing crash-induced loss/repetition.

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User authentication

Passwords, OTPs, and biometrics use different error/cost denominators. Salt separates record computations without eliminating offline guesses.

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Credentials and sessions

Knowledge, possession, and inherence are distinct factor types. Two passwords remain one type; MFA uses at least two types, not necessarily all three. A stolen authenticated session token can bypass the original factors, so session protection, invalidation, and reauthentication matter.

An abstract verifier record is {user, salt, cost, KDF(password, salt, cost)}. Login recomputes and compares. Public salt prevents sharing one precomputed dictionary across records, not guessing one record or mathematically eliminating collisions. Stolen verifiers still permit offline guesses and need access protection. Historical eight-character UNIX limits belong to particular old schemes, not every UNIX or a modern policy.

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Guess spaces and computation

Uniform random four-digit decimal PINs, including 0000, have 10000 possibilities. Distinct unbiased guesses have mean rank 5000.5. At 1 ms/guess, mean time is 5.0005 s. This is not every four-character password or a human-biased dictionary.

For 1000 distinct salts and 100000 candidates, testing every record costs about $1000 \cdot 100000$ KDFs rather than 100000 reusable hashes. At 5 ms each, one serial work unit takes 500000 s$\approx$5.79 days. Parallel resources, memory constraints, and prior distributions change actual cost. Online limits do not constrain stolen-verifier offline work; lockouts can become DoS.

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OTP and biometric errors

OTP/challenge-response adds freshness. One-time use is a receiver acceptance rule, not merely changing displayed digits. TOTP windows need explicit replay/used-code handling and remain vulnerable to real-time phishing relay.

Biometric thresholds trade false acceptance and rejection. FAR uses impostor attempts; FRR uses genuine attempts. With 100000 impostor attempts, FAR 0.1% means 100 false accepts; with 10000 genuine attempts, FRR 2% means 200 false rejects. Threshold, population, device, and environment change estimates.

Biometrics are not freely replaceable secrets. Templates, liveness/attack detection, lost devices, and fallback policy affect security. Factors sharing a compromised endpoint/recovery path can fail together; multiplying failure probabilities requires independence.

Implementing defenses

Connect controls to attack paths. Interpret IDS alarms with Bayes. Baseline, checker, and log integrity are trust assumptions.

Policy, assessment, and layers

Policy specifies assets, operations, responsibilities, and response. Assessment checks configuration, code, and exposure; authorized penetration tests examine exploit paths. A passed scan supports its time/scope/tools/known checks, not universal absence of vulnerabilities.

Layers sharing a stolen key or bad identity header have common failure. Patching, least privilege, memory/input safety, segmentation, authentication, monitoring, and recovery affect different probabilities/damage. Name the path each blocks.

An original model has loss 100000 currency units and annual probability 0.02, expected loss 2000/year. Reducing probability to 0.005 leaves 500, saving expected 1500/year. Tail disasters, correlated events, and regulatory obligations are absent and can still determine policy.

IDS/IPS and base rates

Signatures detect encoded patterns; anomaly detectors find baseline deviations. Unknown attacks need not all evade signatures; anomalous events need not be malicious. IDS usually alerts; IPS can block, directly risking availability on errors.

Of 1000000 events, 100 are attacks. TPR=0.9 gives TP=90, FN=10. Normal=999900 with FPR=0.001 gives expected FP=999.9, TN=998900.1. Alarm precision=90/(90+999.9)≈8.26%, not 90%. TPR divides by attacks; precision divides by alarms.

$$\text{precision} = \frac{\pi TPR}{\pi TPR + (1-\pi)FPR}.$$

At π=0.0001, TPR=0.9, precision≥0.5 requires FPR≤0.000090009…. Baseline contamination, drift, and correlated-event aggregation affect estimation. Development data cannot be relabeled independent acceptance data.

Integrity monitoring and recovery

Compare content/metadata with a trusted baseline. Compromising both checker and baseline removes that trust root. Hash comparisons require suitable collision resistance and protected original digests; equality is not absolute identity proof.

Tripwire-style monitoring distinguishes authorized updates from unexpected changes. Changing logs still need protected remote/append-only records and correlation. A honeypot must not become an unrestricted internal attack relay. Antivirus signatures, behavior, and sandboxes have limited coverage; a clean scan does not prove supply-chain innocence.

Response includes isolation, evidence preservation, root-cause repair, trusted-backup recovery, and credential rotation. Audit subject, object, action, decision, time, and correlation ID while protecting log integrity/availability.

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Firewalls

Specify zones, direction, port, state, and first-match rules. Vulnerabilities inside allowed services remain exploitable.

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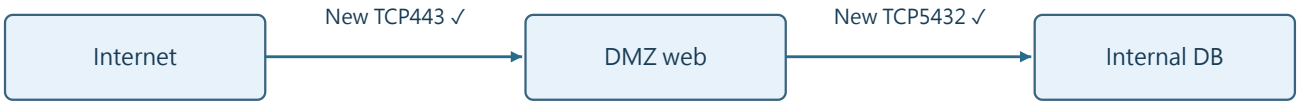
Allowed connections before filter rules

Zones are not full trust: allow only specified directions/services



Deny new Internet → DB and web → DB22. Validate return packets against existing state.

Compromised web still reaches DB5432. DB needs authentication and minimum query authority.

Allow new Internet→DMZ web TCP443, web→internal DB TCP5432, and internal→Internet TCP443. Deny other new connections. Permit returns only if they match existing state. DMZ does not gain all internal access. A compromised web server can abuse its required DB5432 path, requiring database authentication/query authorization.

Allowed connections before filter rules (continued)

Stateless filters inspect headers; stateful filters track connections; proxies terminate/understand applications; host firewalls govern local ingress/egress. Without TLS termination, packet filters cannot inspect encrypted application meaning. Allowing 443 does not make its payload safe.

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Rule order and limits

Rules2–4 implement the three new-connection paths: Internet→web, web→DB, and internal→Internet. Rule1 admits subsequent packets in either direction of an existing allowed connection. New TCP service rules match destination ports.

Order	Match	Action
1	Packet matching an existing allowed connection state, either direction	Allow
2	New TCP, Internet→DMZ web, destination443	Allow
3	New TCP, DMZ web→internal DB, destination5432	Allow
4	New TCP, internal→Internet, destination443	Allow
5	All remaining packets	Deny

Rule order and limits (continued)

Packet	First matching rule	Result
New Internet→web TCP443	2	Allow
New web→DB TCP5432	3	Allow
New internal→Internet TCP443	4	Allow
New Internet→DB TCP5432	5	Deny
New web→DB TCP22	5	Deny
DB→web return actually matching existing state	1	Allow
web→DB subsequent data in an existing allowed connection	1	Allow

Match source/destination addresses, zones, destination port, direction, and state. An ACK bit or a claimed established status without a matching state-table entry does not satisfy rule1.

Rule order and limits (continued)

Putting deny-all first makes later permits unreachable. Actual products have specific chain/priority semantics. Firewalls do not repair allowed-protocol bugs, authenticate hosts merely by IP, or restore saturated upstream bandwidth through local dropping. Tunnels, egress, and dependencies such as DNS need explicit policies.

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Computer-security classifications

Historical TCSEC evaluates features/assurance under specified policies/configurations. TCB enforcement does not prove every business operation correct.

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Historical TCSEC assurance

The DoD Orange Book is not a universal modern product score. D fails higher classes. C1/C2 emphasize discretionary access/accountability; C2 adds individual control, auditing, and object reuse. B1–B3 add mandatory labels and stronger structure/analysis. A 1 adds formal design/verification and development assurance to B3 functionality.

B2 covert-channel requirements identify/analyze paths bypassing intended information flow, rather than support exfiltration. The TCB includes hardware, software, and firmware enforcing policy. If an outside component can break that policy, the boundary is wrong.

Evaluation applies to its version, configuration, policy, and environment. A 1 does not eliminate business errors/all attacks or automatically certify later versions. Formal proofs depend on their models, assumptions, and specifications.

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Windows 7 example

Separate tokens, ordered DACLs, MIC, and UAC. Remaining-mask checks and operation-specific integrity rules have different branches.

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Tokens and security descriptors

The historical primary token holds user/group SIDs and privileges. An impersonating server thread can use client context. Object descriptors contain owner, DACL, SACL; DACL grants/denies, while SACL can hold auditing/mandatory labels. Existing handles have granted access; not every operation reruns an identical open check.

For ordinary specific rights only, excluding owner/privilege exceptions, restricted-token checks, and object ACEs, set remaining=requested. In ACE order, matching deny intersecting remaining rejects. Matching allow removes granted bits. Empty remaining succeeds; reaching the end with bits left denies. This is not a universal union-all-allows-minus-all-denies algorithm.

DACL mask trace

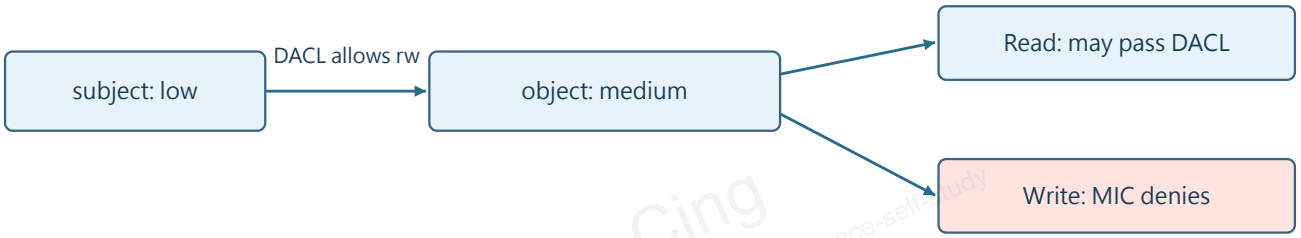
Request={r, w}; token U includes enabled Students. DACL=[deny(Students, w), allow(U, rw)]. Initially remaining={r, w}; the first intersects w and immediately denies. Without Students, skip it and allow U removes both bits, succeeding.

Deliberately placing allow(U, rw) first satisfies the request before reaching later deny. Deny does not override every allow regardless of order. Preferred canonical order places relevant explicit denies before allows; inheritance needs more ordering rules.

NULL DACL is absent and allows in this ordinary model. Empty DACL exists with no ACEs and grants nothing. Neither bypasses mandatory policy or special privilege rules.

MIC and UAC

One DACL grant; mandatory policy still distinguishes read/write



Assume NO_WRITE_UP only. Adding NO_READ_UP also denies the read branch.

Low-integrity subject, medium object, DACL allows r, w. With only NO_WRITE_UP, write fails while read can pass DACL/other checks. NO_READ_UP additionally denies read. Integrity primarily resists low-trust modification; it is not a full confidentiality-clearance model.

MIC and UAC (continued)

Windows 7 UAC can run an administrator's ordinary applications under a filtered medium token; controlled elevation obtains the corresponding higher token. A prompt neither creates rights the user never held nor identifies malicious programs.

Inheritance, copy/move, and propagation to existing objects depend on filesystem/API options. Inspect the final descriptor at creation. Parent ACL changes do not universally rewrite existing handles.

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Composing security arguments

Authentication binds credentials/keys to subjects; authorization limits operations; cryptographic schemes protect messages; freshness/receiver state prevent replay; transaction/recovery connects authorization to a recoverable effect. One layer cannot supply another's missing premise.

Valid MAC replay still needs denial. TLS ending at a compromised proxy does not guarantee end-to-end secrecy. MIC can reject DACL-authorized writes. NX does not prevent authorization-data corruption. State premises, failure consequences, and observable checks.

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Proofs, examples, and configuration evidence

Fermat/CRT proves RSA correctness for all messages; a modular-power table checks one. Keystream leakage follows an XOR identity. Bayes connects IDS denominators; TPR alone is not precision. Historical cases retain versions, endpoints, and omissions. One replay test, DACL trace, or scan supports that observation without replacing general transition/threat-model reasoning.

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References

Silberschatz, Galvin, and Gagne, *Operating System Concepts*,9th edition, Chapter 15, printed pages657–702. Cases, diagrams, and exercises are original and contain no operational attack payloads.

RFC5246, § 7.4.7.1,7.4.9,8.1, F.1.1.2 covers historical TLS1.2 RSA premaster, Finished, and forward secrecy. RFC7568, § 3 prohibited SSL3 in 2015. Microsoft DACL checks and MIC document masks and integrity conditions. Windows 7 retains its historical scope without later AppContainer features.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Chapter route

How can multiple guests remain isolated while preserving instruction semantics, device interfaces, and resource bounds?

Prerequisites are OS structure, traps, address translation, and I/O.

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Learning objectives

- Distinguish type 1 and type 2 hypervisors.
- Analyze trap-and-emulate and binary translation.
- Explain memory/I/O virtualization safety and costs.

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Overview

A system VM provides CPU, memory, device interfaces to a guest OS. A vCPU stores state; a pCPU executes it.

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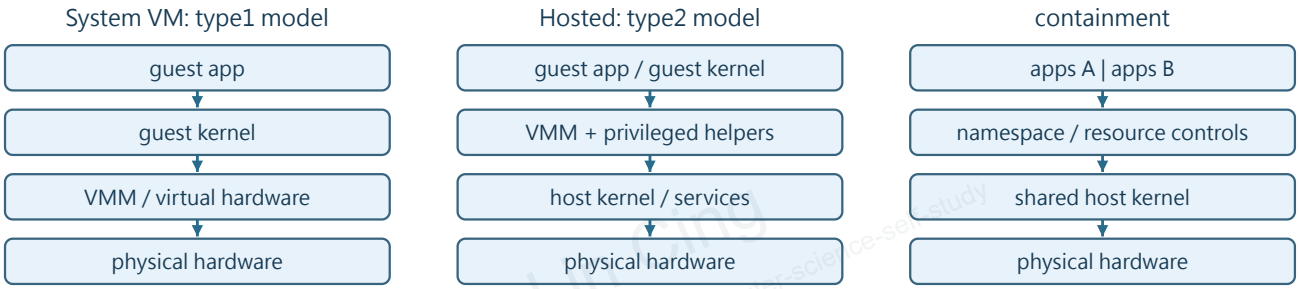
Host, guest, and VMM boundaries

Resource-control layers differ: VMs and shared-kernel containers



The VMM/hypervisor manages physical resources and implements virtual interfaces. Guest applications call their guest OS, which controls virtual CPUs/devices. A guest syscall need not immediately enter the host kernel.

Host, guest, and VMM boundaries (continued)

A vCPU retains guest registers, PC, state, and virtual interrupts until scheduled on a pCPU. Two VMs can both name physical page5 while second-stage mappings select different host frames. The VMM isolates shared CPUs, memory, devices, and management interfaces.

System VMs, process VMs such as JVM, cross-ISA emulators, and OS containment expose different abstractions. Identify the provided interface, kernel ownership, and physical-resource controller rather than relying on a type number.

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History

VM/370 minidisks partition shared resources. Fidelity, resource control, and efficiency are separate goals.

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Motivation and design goals

IBM VM/370 in the 1970s presented separate physical-disk regions as guest minidisks for multiple OSs on one mainframe. The abstraction preserves ranges/isolation rather than exposing unrestricted disk registers.

Fidelity preserves specified guest-visible behavior except explicit differences. Resource control prevents guests seizing VMM-controlled resources. Efficiency permits most safe work to execute directly. Neither goal promises a fixed slowdown or identical timing for every workload.

Reliable interception of sensitive instructions needing mediation is a common sufficient condition for traditional trap-and-emulate. Otherwise use translation, paravirtualization, or hardware assistance. Full software emulation can interpret an ISA without dedicated VM instructions; hardware changes technique/efficiency, not the logical possibility of emulation.

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Benefits and features

Consolidation still faces simultaneous peaks and shared failures. Snapshots, clones, and backups have different consistency/failure boundaries.

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Isolation and shared failure

VMs support OS development, multiversion testing, and low-utilization server consolidation. Hypervisor/device-emulator vulnerabilities or a shared failed host can still affect several VMs. CPU, memory, I/O quotas are needed against interference; separate addresses alone do not prevent slow neighbors.

An eight-core host reserves one for VMM. A/B/C peak at 3/3/2 cores, totaling8>available7. Average demand of one each cannot guarantee no queueing at simultaneous peaks. Overcommit depends on workloads/service goals and creates no compute capacity.

Shared networks/disks introduce trust boundaries. Management can clone, snapshot, read disk images, making its authority part of the TCB even when guests are isolated.

Suspend, snapshots, clones, and backups

Suspend retains resumable CPU/RAM/device state; a disk snapshot retains block versions at a point; a clone creates an independently runnable copy. Copying active disk files need not capture one consistent instant or matching RAM/disk state. Use shutdown, supported consistent snapshots, or guest application quiescence.

Crash consistency resembles recoverable power-loss disk state; application consistency additionally coordinates flush/transactions. External mail, payments, databases outside the snapshot are not undone by rollback. Clones must address duplicate IP/MAC, machine identities, and credentials.

Snapshots sharing the source's failure domain are not independent backups. Templates replicate both configurations and embedded vulnerabilities/credentials. Cloud/VDI APIs retain these isolation/external-state constraints.

Building blocks

Traps, translation, hypercalls, and hardware assistance have different premises. Shadow and two-stage translation require distinct mapping/permission/cost arguments.

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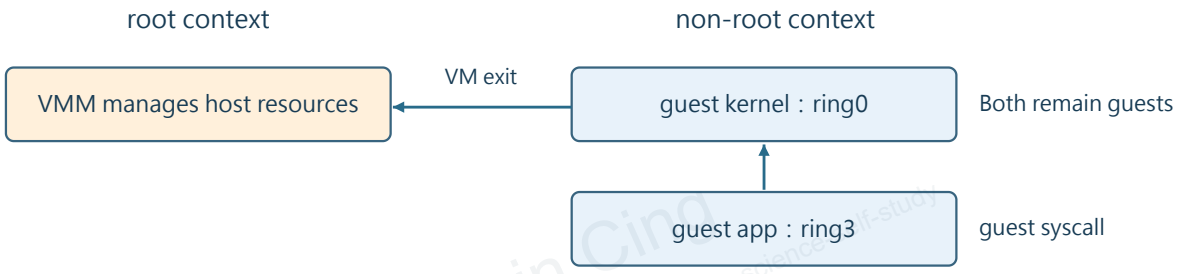
Virtual modes versus physical modes

User/kernel and root/non-root are separate privilege axes



Hardware-assisted model: non-root ring0 ≠ host root. Selected events exit; not every guest syscall.

In a traditional model without a separate VM mode, the VMM holds physical privilege while guest kernel/user execute with reduced physical rights. VMM retains virtual guest mode. A guest-kernel privileged operation traps, is checked against virtual authority, and is emulated. An illegal guest-user operation must produce a guest exception rather than gain authority by entering VMM.

Virtual modes versus physical modes (continued)

When a guest kernel disables interrupts, VMM clears vCPU.virtualIF and delays that guest's maskable interrupt injection. It must not permanently disable host interrupts serving other guests/devices. Re-enabling permits pending events according to the virtual architecture while retaining host control.

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Sensitive instructions that do not trap

An instruction may silently ignore some changes at reduced physical privilege, leaving a guest kernel's expected flags wrong without notifying VMM. Historical x86 POPF behavior under particular flags/privileges illustrates the gap. Lack of trap does not imply harmlessness.

Binary translation reads guest blocks, replaces mediated operations with safe equivalent VMM behavior, and caches executable translations. It must preserve indirect branches, self-modifying code, page-permission changes, and exception-visible state. Changed source code invalidates stale translations.

Paravirtualization instead changes a cooperative guest to use hypercalls. It can address similar problems with different guest ABI/modification requirements. Hardware assistance does not remove scheduling, device-emulation, and I/O costs.

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Hardware-assisted execution

Root/non-root or host/guest context adds an axis beyond user/kernel. Guest kernel can run non-root ring 0, app non-root ring 3, while VMM manages root context. VM entry loads guest state; configured events exit, save state/provide reason, and let VMM handle/resume.

Architecture/controls determine exits; not every instruction or syscall exits. Fidelity/control require correct intercepts, state retention, event injection. IOMMU and interrupt remapping isolate/route device activity separately from CPU virtualization. EPT alone does not constrain all DMA.

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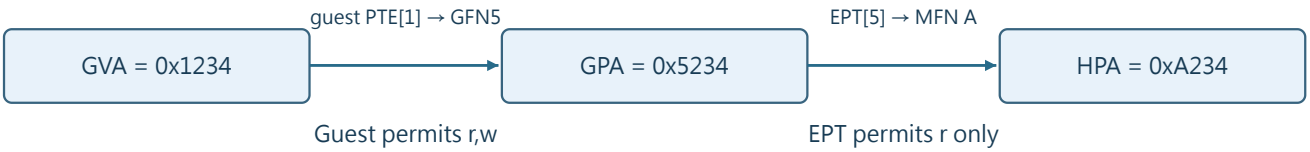
Shadow tables and hardware two-stage translation

Compose addresses and permissions at both stages



4096 B pages preserve offset 0x234 at both stages. Read passes; second stage denies write.

Shadow maps VPN1 → MFN A directly, requiring synchronized guest PTE updates and invalidation.

Guest tables map GVA→GPA and VMM maps GPA→HPA. Hardware two-stage EPT/NPT combines both during walking. A shadow table is a VMM-maintained GVA→HPA composition used by an ordinary MMU, requiring interception/tracking of guest table changes. Extra tables do not make them one algorithm.

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Shadow tables and hardware two-stage translation (continued)

With4096 B pages, GVA0x1234=(VPN1, offset 0x234); guestPTE[1]=(GFN5, rw) gives GPA0x5234. EPT[5]=(MFN0xA, r) gives HPA0xA234. Read needs permission at both; EPT denies write despite guest w. Guest-table denial follows guest fault semantics; second-stage violation uses VMM handling. They are not one identical fault.

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Why uncached nested walks are not twice as expensive

Assume m guest levels, n second-stage levels, no TLB/walk caches or large pages. Each guest PTE read requires n second-stage references plus its own read, totaling m(n+1). Translating the final data GPA adds n. Thus

$$T_{tables} = m(n+1) + n = (m+1)(n+1) - 1.$$

Add one final data access. For m=n=4,24 table reads plus data=25, not4+4. This is a cold model, not every load. TLB hits, walk caches, huge pages lower it. Shadows have a shorter hardware walk but VMM maintenance on guest updates.

Invalidate translation caches under architectural rules on mapping/permission changes. Correct tagging or flushing must prevent one VM's equal GVA from reusing another's mapping.

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VM types and implementations

Compare lifecycle, type 0/1/2, PV, process VMs, emulation, and containment by abstraction. Shared-ring counters, ownership, and publication order preserve state.

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Lifecycle and type responsibilities

Creation validates CPU, memory, disk, network needs and policy, allocates vCPUs, address spaces, devices, then loads the guest. Running VMs may pause, resume, snapshot, migrate. Deletion first stops devices/events before resource reclamation; deleting management metadata while DMA/network identity remains active is insufficient.

Type0 partitions via firmware/hardware with dedicated/shared resources. Type1 directly manages host resources, including VMMs in some general-purpose kernels. Type2 uses host process/device/memory services, often with privileged helpers/drivers and hardware assistance. Type2 does not mean no host awareness or no VT-x.

Types do not universally rank performance/functionality. Nesting depends on VMM, CPU, configuration, not solely type 0. Compare management scope, TCB, and resource paths.

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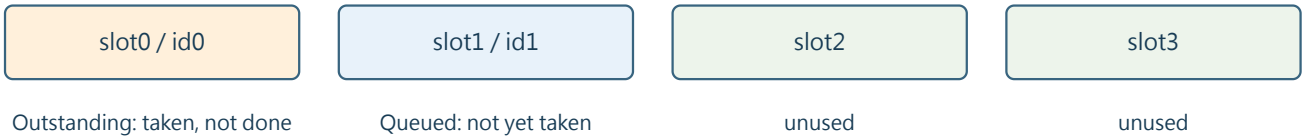
Paravirtual I/O and a four-slot ring

Four slots, ordered completion: backend consumption does not free a slot



Outstanding: taken, not done | Queued: not yet taken | unused | unused

req_prod=2 · req_cons=1 · rsp_prod=0 · rsp_cons=0 ; used=2 · free=4−2=2 。

Complete id0: rsp_prod→1. Frontend consumes response, rsp_cons→1; only then release the slot.

Historical reference is Xen SOSP2003 cooperation, hypercalls, shared rings. This simplified single-producer/consumer model is not every Xen ABI. Counters increase monotonically; slot=index mod4. With req_prod2, req_cons1, rsp_prod0, rsp_cons0, queued=1, outstanding=1, response=0, used=2, free=2.

Paravirtual I/O and a four-slot ring (continued)

Backend completing id0 advances rsp_prod1, making outstanding0, response1. Frontend consumes it, advances rsp_cons1, leaving used=req_prod−rsp_cons=1, free3. Backend taking a request does not permit descriptor reuse while completion is pending.

Ring synchronization and validation

Write descriptor before release-publishing req_prod. Consumer acquire-observes prod before reading descriptor. Write response before publishing rsp_prod; frontend consumes it before advancing rsp_cons to release the slot. Event-channel/doorbell notifications do not replace counter predicates. Recheck before sleeping against lost wakeups.

Maintain$0 \le req_prod - rsp_cons \le N$ and $rsp_cons \le rsp_prod \le req_cons \le req_prod$ in this ordered-completion model, keeping regions disjoint. Backend validates ID, operation, block bounds, length, and grants/ownership. Shared memory does not authorize arbitrary guest DMA into host. Out-of-order completion needs another ID/slot model.

Process VMs, emulation, and containment

JVM/.NET expose bytecode/runtime rather than a full guest OS. Cross-ISA emulation interprets/translates instructions/machine state to run incompatible binaries. Neither is classified solely by owning hardware ring 0.

Solaris10 zones/BSD jails share the host kernel while separating namespaces, resources, and security contexts. Different userland/network views do not create independent kernels. Shared-kernel failure differs from VM isolation. Solaris global zone manages non-global zones; services/resource controls need configuration rather than an independent kernel scheduler per zone.

Testing another kernel needs a system VM/emulator; separating same-kernel services can use containment; bytecode needs its runtime. Choose an appropriate interface rather than a universal winner.

Virtualization and OS components

Guest schedules threads; host schedules vCPUs. CPU runtime differs from wall time. Ballooning, COW, IOMMU, storage, and migration require explicit states.

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Two scheduling layers

Two pCPUs serve A 0/A 1 and B0/B1, ignoring host overhead. A runs0–10 ms, B10–20, A 20–30, B30–40. A thread needing20 ms CPU on A 0 completes at 30; B's counterpart at 40. Equal long-run halves do not make every20 ms budget have40 ms response.

Guest-ready threads wait while their vCPU lacks pCPU time. Descheduling a guest lock holder can waste another vCPU's spinning allocation. Separate virtual clock, wall clock, vCPU runtime. Expired virtual timers may await vCPU scheduling. Guest priorities alone cannot provide host-level real-time admission/isolation/scheduling guarantees.

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Balloon direction and capacity conservation

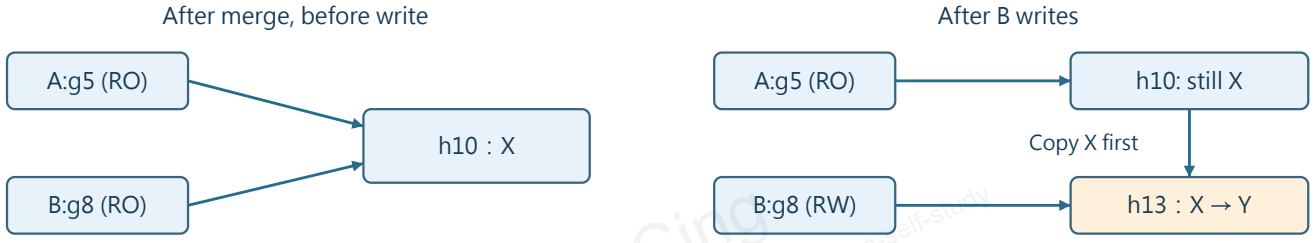
Use Waldspurger's ESX OSDI2002 design. A guest balloon driver allocates/reserves guest pages and reports PFNs; VMM can reclaim their host backing while guest policy chooses other pages to reclaim/swap. Inflation reduces workload-usable guest memory; it creates no RAM.

A 4096 MiB host reserves512, leaving3584 guest backing. A/B demand2304 each,4608 total not yet fully backed, exceeding availability by1024. Balloon A 256 and B768 to target A 2048+B1536=3584, plus VMM 512=4096. This conserves demand/allocation targets; it does not claim physical RAM once held4608 guest MiB simultaneously. Nominal sizes can remain while usable backing changes at a cost.

Balloon pages cannot remain workload-accessible. Protect unbacked mappings; erroneous guest access must fault/reback rather than reach frames reassigned elsewhere. Deflation restores safe backing first. Uncooperative/slow drivers can require VMM paging/limits; balloon success is not unconditional.

Sharing, COW, and double paging

Compare bytes before sharing; split host frames before writing



B write fault: allocate/copy → change B mapping and invalidate → allow write. A stays unchanged.

A:g5→h10 and B:g8→h12 both contain X. Hashes select candidates; full-byte comparison and synchronization precede merging into h10 with write removed and references tracked. B's write faults: copy X to h13, change B:g8→h13, invalidate old translation as required, then allow Y. A retains X. Never allow write before splitting.

Sharing, COW, and double paging (continued)

VMM swapping lacks guest victim knowledge. Guest can write to virtual swap while VMM also swaps the host frame holding its page, causing double paging/extra I/O. Ballooning lets guest choose victims but is not universally faster. Sharing scans cost CPU/synchronization and depend on tenant-isolation policy; cross-tenant sharing is not assumed always enabled.

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I/O paths and virtual networking

Emulation validates virtual-register operations before host-driver service. Paravirtual devices use cooperative queues. Passthrough lets a device serve a guest directly, still requiring IOMMU isolation, interrupt routing, and safe reset/ownership transfer.

If DMA0x5000 maps to host 0xA000, IOMMU must confine the permitted range. CPU EPT alone is not DMA confinement. Passthrough may require extra device-state support for snapshots/migration; RAM copying alone is insufficient.

Bridging joins an L2 network. NAT maps internal addresses/ports to external tuples. A 10.0.2.15:5000 and B10.0.2.16:5000 can use host:40001/40002 and share one public IP. Firewalls still define guest-to-guest/external channels.

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Storage mappings and snapshot chains

Guest LBA maps through an image/host filesystem to storage, or directly to a block volume; a virtual disk need not be one ordinary file. Thin provisioning postpones allocation without guaranteeing all logical capacities can be filled together.

Base={0:A,1:B,2:C}; after snapshot, active delta={1:B′}. Read1 finds B′ in delta, read 2 falls through to C. Selecting base restores B at 1 but does not retract sent mail. A shared base cannot be deleted while clones reference it. Backups need the required chain, metadata, and consistent point.

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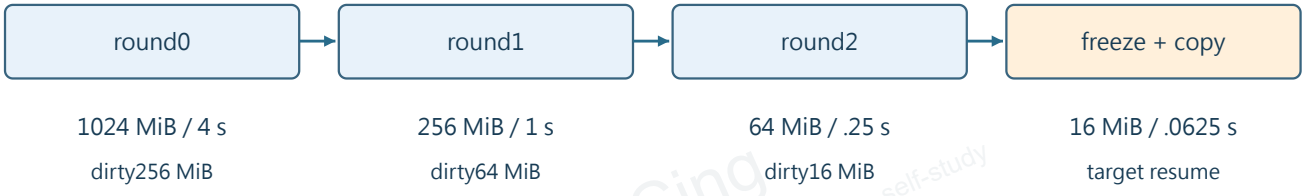
Pre-copy migration round by round

Normal pre-copy: total time differs from downtime



B=256 MiB/s, d=64 MiB/s. Total 1360 MiB / 5.3125 s; pause .0625 s, excluding control cost.

Assume compatible VMM/CPU, shared storage, no unmigratable device, effective copy B=256 MiB/s, distinct dirtied pages d=64 MiB/s, and ignore protocol/CPU-state cost. R0=1024 copies in 4 s, creating R1=256; next1 s creates R2=64; next0.25 s creates R3=16. Freeze and copy16 in 0.0625 s. Total1360 MiB over5.3125 s; pause0.0625 s.

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Pre-copy migration round by round (continued)

Generally $R(i+1)=(d/B)R(i)$, geometrically shrinking if $d/B<1$. At $d \geq B$ it does not shrink; cap rounds, reduce dirtying, extend pause, or change method. Repeated dirty pages, variable bandwidth, and finite RAM make reality richer than this explicit model.

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Stop-and-copy and one active owner

Freeze includes state-changing DMA/devices/events, not only user code. Save vCPU/device/dirty RAM and required network state. Source must stop externally executing the guest before ownership transfer; reclaim source after target operation is established.

If target starts but acknowledgment is lost, source restarting can create split brain. A safe protocol needs explicit commit/ownership and failure recovery. Waiting for acknowledgment alone is not a proof of all failure cases. This simplified normal path proves a consistent handoff, not transparent availability under arbitrary partitions.

Maintain external address/MAC/routing reachability. TCP tolerating a short pause does not make downtime zero. Shared storage is this model's premise; block migration exists elsewhere, so it does not prove all live migration excludes disk copying.

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Examples

Compare the book's Workstation architecture with Xen2003/ESX2002 research. Java SE 7 bytecode/runtime is a different abstraction from system VMs.

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Workstation, Xen, and ESX

The book's circa2013 hosted Workstation diagram routes guest app→guest kernel→virtual device/CPU→VMM/host services→hardware. Ordinary instructions can execute directly; I/O can use host services. It is an architecture example, not a release's complete implementation or measurement.

Xen SOSP2003 uses cooperative guests/hypercalls/shared rings. ESX OSDI2002 manages overcommit with ballooning/content sharing. Their periods/interfaces differ. Early Xen PV is not every later Xen mode; one paper benchmark does not establish a universal performance ratio.

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JVM verification and execution

In a Java SE 7-era abstraction, class loading/verification checks types, operand stacks, and legal control flow. Execution interprets or JIT-compiles; runtime supplies heap/GC/security. JVM has no separate general-purpose guest kernel and still uses host memory, threads, I/O.

For iconst_2;iconst_3;iadd;istore_1, stack is[]→[2]→[2,3]→[5]→[], local1=5. Starting empty with iadd lacks its two integers and can be rejected by verification before a hardware underflow. This does not prove business correctness.

JIT can use hotness/tiering rather than compile a whole method on first call. GC follows reachability/collector rules rather than deciding semantic future usefulness. Separate language specifications from implementation choices.

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Two required properties

Fidelity preserves specified guest-visible transitions; resource control prevents escaping CPU, address, device management. Traps, translation, hypercalls, hardware assistance implement these properties rather than prove themselves.

Permission intersection, ring publication/reuse, COW-before-write, and single-owner migration keep guest-visible and host-managed state consistent. A successful demonstration does not cover every transition.

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Denominators and boundaries

VCPU count is not pCPU capacity; nominal RAM is not current host backing; logical disk size is not allocated bytes; snapshots do not roll back the outside world; migration transfer time is not downtime. Identify the layer of every number. Exercises reconstruct modes, addresses, walk counts, ring counters, balloon capacity, COW, and migration recurrences.

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References

Silberschatz, Galvin, and Gagne,*Operating System Concepts*,9th edition, Chapter 16, printed pages711–738. Figures/cases are original; shadow and hardware two-stage translation remain separate.

Xen, SOSP2003, § 2–3 gives historical PV/hypercall/shared-I/O context. ESX ballooning and content sharing, OSDI2002 cover guest allocation/host reclamation, byte comparison, and COW. JVM SE 7 § 4.10 specifies verification types/stack rules. Capacity/migration calculations use explicit original assumptions.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Chapter route

How can services remain understandable and recoverable without shared memory, synchronized clocks, or failure-free nodes?

Prerequisites are network stacks, RPC, processes, and filesystem interfaces.

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Learning objectives

- Compare network and distributed operating systems.
- Analyze naming, routing, RPC, and failure semantics.
- Check DFS cache consistency and availability.

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Advantages

Sharing, speedup, and reliability require cost/failure assumptions. Nodes communicate by messages without a perfect common clock.

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Message passing and its limits

Each node has local memory/OS and exchanges messages. Local/remote are relative to the requester. Resource sharing, computation speedup, reliability, and communication motivate distribution without following automatically from it.

Of a 100 s job,20 s is serial,80 s divides across four nodes, and coordination costs 5 s. T4=20+80/4+5=45 s, speedup100/45≈2.222. Ignoring coordination, the infinite-node Amdahl bound is100/20=5. More nodes cannot remove serial work and can increase coordination cost.

Three independently failing interchangeable read replicas each have p=.01 over the interval. With reliable networking/metadata and any replica sufficient, unreadability is $p^3=10^{-6}$. Shared power/rack or required metadata-leader survival violates this independence model. Replica count is not a reliability proof.

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Network-based operating systems

Remote login, file transfer, and DFS expose different sharing semantics. Data, computation, and process migration move different state.

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Transparency and machine boundaries

Network OSs expose remote hosts/copies explicitly. Distributed OSs/middleware can unify naming, access, and migration. Transparency packages location/invocation and failure semantics; it does not remove physical latency/failure.

Remote login executes a remote session; transfer creates another file copy; DFS provides shared remote-file behavior. SSH/SFTP still need host authentication/user authorization; ordinary FTP is not inherently encrypted. Anonymous services need constrained namespaces/operations. Naming, identity mapping, network faults, and conflicting copies require rules.

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Moving data or computation

For1 GiB input,1 MiB result,100 MiB/s throughput, and20 ms RTT, moving all input costs at least1024/100+.020=10.260 s before local compute. Sending computation to a remote CPU taking.500 s and returning 1 MiB takes.500+.010+.020=.530 s, with negligible request. Include CPU queues, startup, serialization, and output size before generalizing.

Whole-file caching pays upfront for later local access. On-demand caching transfers requested blocks. Reading only1 MiB makes a whole1 GiB transfer waste1023 MiB; reading all blocks through unpipelined RTT-bound RPCs can instead favor bulk transfer.

Process migration moves/reconstructs PC, registers, memory, open files/sockets, and environment. Sending code for fresh remote execution is computation placement, not migration of running state. Cross-ISA/ABI execution needs compatibility rather than copying register bits blindly.

Network structure

LAN/WAN are not fixed throughput rankings. Separate serialization, propagation, queueing, and processing. BDP/window bounds govern in-flight data.

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Scope and latency components

LANs commonly cover smaller administrative/geographic areas; WANs connect sites. Capacity/latency depend on links, distance, queues, load. Switches, routers, and APs have different roles. Switched Ethernet is not universally one shared collision bus; a congested LAN can underperform a high-capacity WAN.

Latency≈L/R+propagation+queue+processing. A 1500 B packet on1 Gbit/s serializes in 12 µs. One-way5 ms propagation dominates. A tenfold rate increase reduces serialization to1.2 µs without dividing propagation by ten.

100 Mbit/s equals12.5 decimal MB/s before headers; MiB/s uses2^{20} B. Wireless nominal rates differ from goodput. These are explicit models rather than current product maxima.

Bandwidth-delay product

100 Mbit/s and40 ms RTT give BDP=4×10^{6} bits=500000 B≈488.28 KiB. A 64 KiB outstanding-data limit bounds throughput at 65536・8/.040=13.1072 Mbit/s despite100 Mbit/s capacity.

This assumes no loss, fixed RTT, and no other bottleneck. TCP congestion/receive windows, ACK behavior, slow start, retransmission further constrain results. Send buffers remove at most one bound, not congestion/server CPU. A 16 B RPC can be dominated by RTT rather than serialization, explaining the importance of placement/cache.

Communication structure

DNS names, IP addresses, ports, and stable object identity differ. Routing and connection semantics require explicit state/order assumptions.

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Names, addresses, ports, and objects

Store.lab.test is a domain name, possibly resolving to198.51.100.20 or IPv6. IP addresses are routable endpoint addresses, not users or permanent machine identities. Service identity also involves protocol/port/application naming. A TCP connection uses source/destination addresses/ports and protocol; a port is not a PID.

DNS uses resolver, queries, caches, and delegation rather than a server per label. Assume root delegates test and test delegates lab.test. On a cache miss, resolver follows known-root referrals to an authoritative A record. TTL bounds cache lifetime without immediate global invalidation.

At t0, an A record=.20 with TTL600 can remain cached at t100 with 500 s left. An authoritative change to.30 at t200 need not replace it until expiry at 600. Only successful new resolution obtains a checked current value. DNS integrity and endpoint authentication require corresponding mechanisms.

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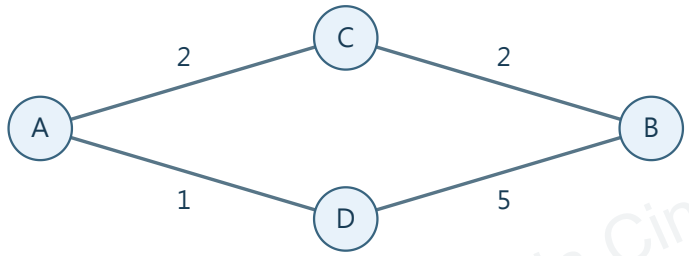
Routing on a fixed graph

One nonnegative weighted graph: every distance update



From A: D=1, C=2, B=∞

Settle D: B=1+5=6

Settle C: B=min(6,2+2)=4

Settle B: A → C → B, cost4

If C–B fails, recompute A→D→B, cost6. Dynamic protocols need detection/convergence time.

Edges are A–C2, C–B2, A–D1, D–B5. Dijkstra starts dD1, dC2, dB∞. Settling D gives dB6; settling C lowers it to4; settling B yields A→C→B, cost4. Losing C–B gives A→D→B, cost6 after recomputation.

Routing on a fixed graph (continued)

Static routing uses configured/admin rules; virtual circuits select session paths; dynamic protocols update metrics/reachability. Dynamic does not mean every packet chooses the emptiest link or converges without loops/drops. A path preserves order only with suitable link/queue behavior; transport supplies end-to-end stream ordering.

Control plane computes routes; data plane forwards from tables. SDN can separate them but still needs consistent updates, failure handling, and protected management.

Switching and connections

Frames, packets, and datagrams have layer-specific headers/limits and can vary in length. Application messages can span packets or share a TCP read. TCP reserves no dedicated physical circuit; applications can implement reliability over UDP.

Circuit switching reserves path resources; message switching stores/forwards whole messages; packet switching pipelines fragments with statistical multiplexing. Across two1 Mbit/s links, ignoring headers/propagation/queues, a 4000 B message takes 64 ms. Four1000 B packets each take8 ms/hop; pipeline completion is(2+4−1) • 8=40 ms.

With40 B per-packet headers, it becomes5 • 1040 • 8/10^6=41.6 ms. One40 B whole-message header gives64.64 ms. These are store-and-forward assumptions, not all cut-through/circuit networks.

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Communication protocols

OSI describes responsibilities, not fixed hardware assignments. Payload, headers, MTU, and wire byte-times use different denominators.

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Layer responsibilities

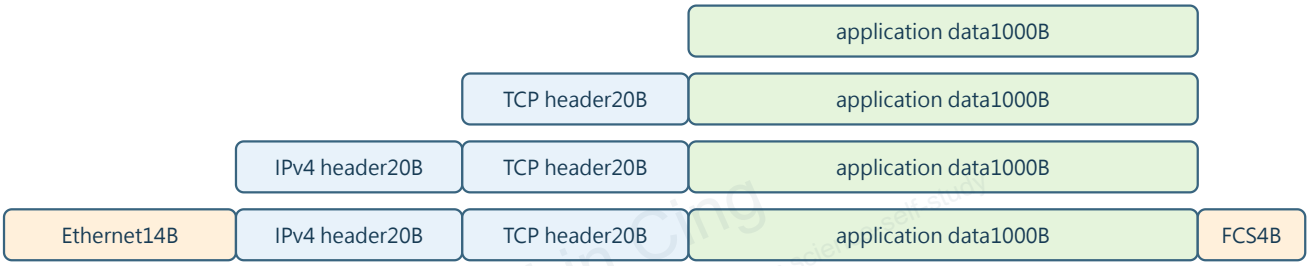
Physical carries signals/bits; link carries frames; network addresses/forwards; transport serves process endpoints; session manages dialogue; presentation encodes representations; application defines service operations. TCP/IP often places session/presentation in libraries/applications. Fewer named layers does not prove efficiency.

TCP offers ordered reliable bytes and flow/congestion control without message boundaries. UDP preserves datagrams without equivalent built-in delivery/order. IP is best effort. NIC offload can move checksums/segmentation into hardware, so no universal three-hardware/four-software split exists.

Serialization specifies lengths, byte order, encoding, versions. Raw C structs with padding/pointers are not portable protocols.

Encapsulation length accounting

Each layer adds headers/trailers; account for every length



Widths are not to scale. IP total1040 B; MAC frame1058 B. Preamble/SFD and IFG are separate.

1000 B application data plus20 B TCP gives1020 B segment;20 B IPv4 gives1040 B IP total. EthernetII14 B header+1040+4 FCS=1058 B. Add8 B preamble/SFD and specified12 byte-times IFG for 1078 wire byte-times.

Encapsulation length accounting (continued)

WithIPv4 MTU1500, this fits one frame. MSS subtracts TCP header from available IP payload;1500 is not all application data. VLAN/options/tunnels/security change accounting. TLS endpoints are not every forwarding router. A lower checksum detects accidental corruption rather than authenticating malicious changes with a secret key.

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TCP/IP example

ARP resolves the next hop; each hop rebuilds link addresses. IPv4 TTL/checksum change while IP endpoints remain in the no-NAT model.

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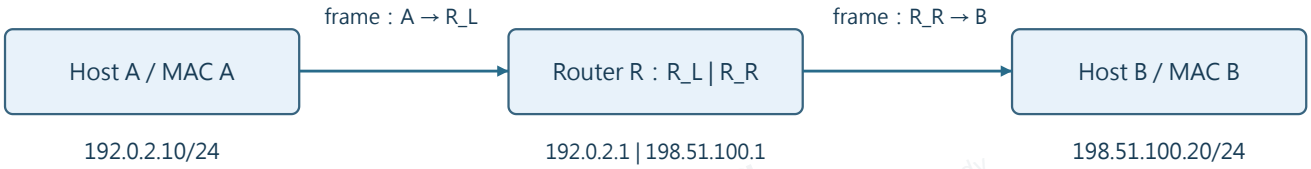
Two Ethernet LANs

Across LANs without NAT: each hop uses different MAC addresses



A ARPs next-hop 192.0.2.1, not remote B on the left LAN.

IP stays192.0.2.10 → 198.51.100.20. Router decrements TTL64→63; recompute checksum/FCS.

No NAT/tunnel. A=192.0.2.10/24, MAC A. Router left=192.0.2.1, MAC R_L; right=198.51.100.1, MAC R_R. B=198.51.100.20/24, MAC B. DNS gives B; /24 comparison selects next-hop192.0.2.1.

Two Ethernet LANs (continued)

Without cached R_L, A broadcasts ARP for 192.0.2.1 on its LAN. R answers. Frame1 has srcA, dstR_L, IP srcA, dstB. A does not ARP remote B directly. Router looks up route, decrementsTTL64→63, updates IPv4 checksum, resolves B on the right LAN, and sends frame2 srcR_R, dstB with unchanged A→B IP endpoints.

B decapsulates and demultiplexes by transport port/connection. Routers normally do not forward ARP broadcasts across L2 domains; ARP replies do not cryptographically prove identity.

Header changes and minimum frames

Each hop rebuilds Ethernet addresses/FCS. IPv4 TTL/checksum change; source/destination IP, TCP endpoints, payload remain under the stated exclusions. Saying only Ethernet changes is false. TTL reaching zero prevents normal forwarding; error reports can themselves be lost/filtered.

Untagged EthernetII has6 B destination,6 B source,2 B EtherType,46–1500 B data/pad,4 B FCS, minimum64 B from destination through FCS. Padding is not included in IP total length; preamble/SFD is outside that frame denominator.

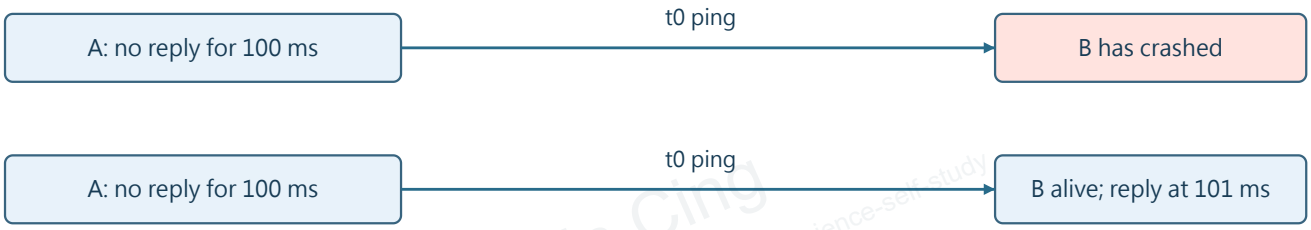
A 4 B TCP payload plus20+20 headers is44 B IP, requiring2 B pad. Frame14+46+4=64 B. Add8 B preamble/SFD and96-bit IFG=12 byte-times to84. At100 Mbit/s, it occupies6.72 µs. IFG is idle time, not12 bytes appended to IP; high-speed PHY tolerances are outside this model.

Robustness

Unbounded delays make timeout suspicion rather than proof. Fencing, transactional deduplication, and recovery handle different partial failures.

Failure-detector indistinguishability

Same local observation, different remote states



A has the same history through 100 ms. Without delay bounds, timeout does not prove crash.

In an asynchronous system without known delay bounds, A pings at 0 and sees no reply by100 ms. B may have crashed, partitioned, lost messages, or merely run slowly. Compare B crashed with B alive but replying at 101 ms. A's local history through100 is identical. Any algorithm declaring crash solely from it errs in the second execution.

Failure-detector indistinguishability (continued)

With heartbeat intervalH1 s, max send scheduling delays.2 s, and one-way network boundδ.1 s, the largest normal arrival gap is≤H+s+δ=1.3 s, assuming the previous arrival could be immediate. Timeout1.5 provides margin under these premises. Invalidating the delay bound invalidates the no-false-suspicion guarantee.

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New leaders need old-leader fencing

A writes shared storage. Partitioned B times out and declares itself leader while A still writes, creating split brain. Heartbeats/local declarations do not guarantee exclusion.

A correct coordinator issues monotonic epochs. Storage atomically installs currentEpoch43 before new B operates, and every write checks token==43. A's42 writes fail. This requires correct issuance/installation and every write path enforcing it; partitions cannot independently increment into legitimate ownership. Confirmed hardware/access isolation can also fence old nodes.

Recovering A synchronizes committed state, membership/epoch, and unfinished requests before becoming follower. Old local data does not restore leadership. IDs/generations protect against stale callbacks/messages after restart.

Lost RPC replies and deduplication

Counter 10 receives increment(id17), commits11, and loses reply. Naive retry yields12. TCP retransmission supplies connection-byte delivery, not cross-reconnection application deduplication.

In the explicit at-most-once model, a server transaction checks durable table[id]. Existing IDs return saved results; new IDs atomically persist counter++ and table[id]=result before replying. Retrying17 returns 11. Updating counter before recording ID permits duplicate increment after crash; recording ID first permits lost effect.

The proof requires shared transaction/recovery scope, unique IDs, and retention long enough for retries. Permanent disconnection can still leave the client uncertain. External email/payment outside the transaction needs another protocol.

Fault domains and failover capacity

Two active servers each handle100/s with total load160. Losing one leaves100 and at least60/s unsustainable. Replicas being able to take over does not preserve SLO without spare capacity, shedding, or degradation.

Stateless worker retries can still duplicate effects. Stateful services additionally need logs, recovery, replication, locks. RAID addresses selected disk faults, not cross-site availability or backups of logical corruption. Recover durable committed state, reconcile in-flight operations, then serve. Missing commit reply does not imply uncommitted work.

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Design issues

Transparency retains failure semantics; the narrowest resource limits scale. HDFS uses historical1.2.1 roles and explicit replica/reconstruction costs.

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Observable transparency and scaling

Location transparency hides placement;migration transparency preserves naming/operations after movement;replication transparency hides copies. Failure transparency cannot indefinitely hide a nonreturning operation. User mobility also needs identity, environment, and session consistency.

Ten workers at 100/s offer1000/s, but a one-core coordinator spending2 ms/request caps the system at 500/s. An eleventh worker does not lift it. Caching, batching, or partitioning can alter the bottleneck's per-request demand. Replication/rebuild traffic consumes capacity too.

Hadoop1.2.1 HDFS NameNode manages namespace/block locations;DataNodes store/transfer blocks;client data bypasses NameNode. Large blocks, streaming, and a simplified write model favor bulk data rather than full POSIX sharing or arbitrary low-latency transactions.

HDFS payload and flow

Custom64 MiB blocks and replication3 split130 MiB into64,64,2. Logical130 yields390 MiB replicated payload, excluding metadata, checksums, padding/allocation. The last2 MiB need not transfer a full64 MiB of payload but still needs block location metadata.

Client gets block/DataNode locations from NameNode, then reads DataNodes directly; writes can use a DataNode pipeline. Heartbeats/block reports detect missing replicas and trigger re-replication with delay/network/disk cost.

Secondary NameNode assists checkpointing;it is not an automatic hot standby. Fault-domain-aware placement helps data availability, while metadata and client reachability remain required.

Distributed file systems

Separate names, FIDs, locations, and cache validity. NFSv3 stable modes, COMMIT, and verifiers define restart handling.

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Names, stable identities, and locations

Location transparency hides physical placement;independence preserves names after movement. ServerA:/x exposes server location. Client mounts can hide it differently per client;global namespaces add common naming/management.

Path/course/a.txt→FID(volume7, file 12, generation 3), volume7→serverA. Moving to serverB changes only the second map. Reusing deleted file 12 must change generation or stale FIDs can identify the new file. Path→FID and volume→location caches have different invalidation conditions.

Replicated/cached location maps still need versions/coherence. Indirection does not eliminate updates. Diskless boot requires firmware/initial protocols to load kernel before DFS exists; an unloaded DFS cannot load itself.

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Complete cache-cost accounting

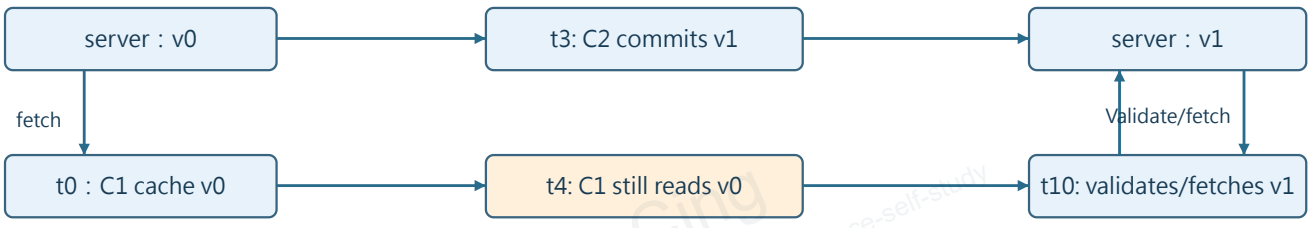
100 reads of 4 KiB at 5 ms remote cost total500 ms uncached. Ten misses at 5 plus90 hits at.1 cost59 ms and40 KiB network payload versus400 KiB. This specified workload does not prove a universal hit ratio.

Per-read 1 ms validation adds100, for 159 ms. Cache hits need not avoid network RTT. Larger cache units can improve read-ahead or waste bandwidth on a one-byte read and reduce resident-block counts at fixed capacity.

Memory cache is volatile;disk cache retains durable bytes but needs metadata recovery/freshness checks. Servers cache too;server RAM acceptance is not stable storage. Write-through/back specify propagation timing, while the next layer's return defines durability separately.

Cache consistency timelines

Server versions and client cache visibility have different timelines



Periodic checks permit staleness. Strong callback/lease models recall conflicting read rights before write completion.

Client-initiated validation:C1 reads/validatesv0 at 0;C2 completesv1 at 3;C1 validating next at 10 can still returnv0 at 4. This is not latest-completed-write visibility on each read. Open-time validation needs within-open semantics;every-access validation costs network and still needs coordination with the actual read.

Cache consistency timelines (continued)

Server-initiated coherence tracks cache/read/write rights and invalidates/recalls them, or disables cache on conflict. A strong model requires confirmation that conflicting read permissions are revoked/invalidated or their controlled leases expired before write completion. During partition, unlimited old-client reads cannot coexist with an immediate-visibility claim.

AFS/OpenAFS callbacks avoid checking every access, but do not rewrite application-owned bytes already read. Save/close and cache-manager refresh affect observations. Session, close-to-open, and stronger DFS guarantees are not synonyms.

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NFSv3 state and write stability

RFC1813 READ/WRITE carry filehandle, offset, count rather than relying on equivalent server OPEN sessions. Server caches/duplicate-request state and separate lock protocols can still exist. Authentication, permissions, and handle lifetime remain relevant.

WRITE requests/replies distinguish UNSTABLE, DATA_SYNC, FILE_SYNC. UNSTABLE success is not durable data. Clients requiring later COMMIT/verifier checks retain retransmittable bytes. Weak cache-consistency attributes assist change detection without proving global linearizability.

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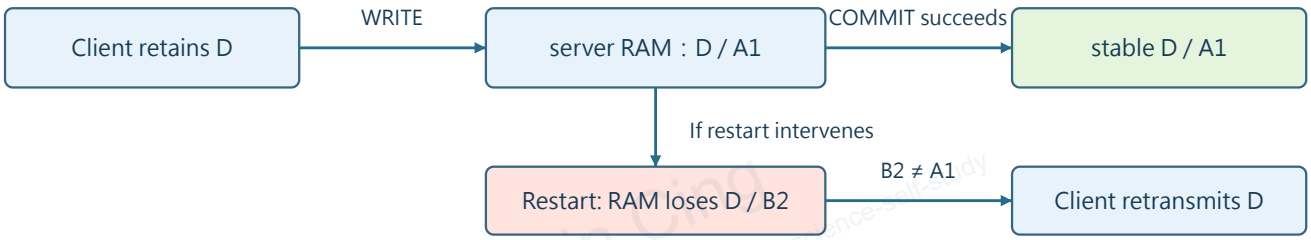
Restart after an unstable write

NFSv3: normal and restart paths after one successful WRITE



After UNSTABLE, retain retransmission data until a stable result with a matching verifier.

Client sends4 KiB D at offset 0, UNSTABLE. Server receives into RAM and replies UNSTABLE, verifierA1;client retains D. Normal COMMIT succeeds with A1, establishing protocol-stable storage and permitting release of that retransmission obligation.

Restart after an unstable write (continued)

If restart loses volatile D, new instance usesB2. A COMMIT/WRITE returningB2≠A 1 cannot certify the old data. Retransmit unconfirmed D and obtain the required stable result. COMMIT success without verifier comparison is insufficient.

This requires retained client bytes, correct server stability/verifier/range behavior, and does not give arbitrary concurrent writers application transactions. Durability, visibility, and RPC deduplication differ.

One remote operation across layers

Service name→DNS address→route→ARP→link/IP/transport→RPC ID→authorization/durable commit→client cache. Frame delivery does not prove RPC completion;TCP bytes do not prove one business execution;NFS UNSTABLE does not prove persistence.

Transparency offers a consistent interface inside stated boundaries without eliminating latency, partial failures, copies, or external effects. Insufficient observations require unknown/suspected rather than treating timeout as logical proof.

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Reconstructable states and arguments

Split serial/parallel/coordination cost. Track bits/bytes, headers, RTT, windows. Draw reliability fault domains. Link RPC IDs to atomic effects;separate DFS names, identities, locations, visibility, durability. General guarantees depend on transitions:unbounded-delay timeout cannot prove crash, fencing needs resource-side epoch enforcement, and deduplication needs sufficient retention. Exercises include counterexamples, normal paths, and restart states.

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References

Silberschatz, Galvin, and Gagne,*Operating System Concepts*,9th edition, Chapter 17, printed pages 741–774. Numerical cases, figures, and exercises are original.

RFC1034 covers DNS;RFC826 ARP;RFC1813, § 3.3.7,3.3.21 NFSv3 stable writes/COMMIT/verifiers. Addresses here are documentation examples. HDFS1.2.1 architecture and users guide define historical roles/checkpointing. OpenAFS concepts supplies callback context without assumed universal chunk defaults. Cisco frame/IPG reference distinguishes frame fields from idle gap;the numerical case fixes untagged100 Mbit/s conditions.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Chapter route

How does a UNIX-influenced kernel compose processes, resources, and modular interfaces?

Prerequisites are processes/threads, virtual memory, filesystems, and modules. This case fixes Linux v3.2 rather than describing unchecked current behavior.

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Learning objectives

- Distinguish historical design from present systems.
- Trace fork/exec, module loading, and page-cache paths.
- Analyze historical CFS/predecessor, reclaim, and filesystem boundaries.

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History

Kernel mechanisms differ from libc, ELF interpreters, shells, and distribution integration. Equal kernels do not guarantee every user binary runs. Examples fix ISA and configuration.

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Kernel, libraries, and distributions

Linux v3.2 provides tasks, VM, VFS, IPC, networking, and drivers. GNU tools, shell, libc, and package managers mostly run in user space. A distribution integrates kernel, tools, configuration, installation, and updates;its name does not specify every CONFIG option.

Two teaching machines share v3.2/x86. An ELF requests PT_INTERP=/lib/ld-linux.so.2. A provides it;B does not. Both find the executable, but B fails obtaining the interpreter. Kernel-version equality is insufficient. Static/dynamic linking, ISA, ABI, and library symbol versions impose different compatibility conditions.

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Kernel, libraries, and distributions (continued)

Component	Responsibility	Observable failure
Kernel image/CONFIG	Privileged syscalls/device support	Missing driver prevents device use
ELF interpreter/libc	Usually user-mode loading/API wrappers	Missing interpreter fails exec;missing symbol fails linking
Shell/utilities	User processes using services	Missing command does not remove fork/read
Distribution policy	Packages/default integration	Different permissions/startup behavior

From its1991 beginnings, Linux expanded to SMP and many architectures. Modules, fine-grained locks, and standard interfaces address hardware diversity, multitasking, and compatibility. Concrete claims still depend on versioned source/configuration.

Design principles

The modular monolithic kernel contains scheduling, memory, files, networking, and drivers. Modules ease deployment while sharing kernel authority/failure scope.

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Protection boundaries inside a modular kernel

Scheduler, memory, VFS, and most drivers share one kernel protection domain. Loading independently does not create isolated user servers. Ordinary function calls cross module boundaries without necessarily switching address spaces.

For one synchronous buffered read, syscall enters kernel, VFS calls filesystem, then block/driver code. User/kernel crossings change mode;internal calls do not. Waiting can schedule another task, adding a distinct context/address-space switch. No universal read switch count follows.

A bad driver pointer can corrupt kernel data. ABI/type/signature checks do not create another isolation domain. A user-space file service can use page protection but requires IPC, data-transfer, and server-failure protocols. Chapter 2 compares fixed paths;this chapter connects Linux objects.

UNIX/POSIX APIs specify observable fd/process/signal semantics, not a unique internal structure. Read can serve disk, pipe, socket with different seeking, blocking, and EOF contracts.

Kernel modules

Init/cleanup register drivers, filesystems, services. Loading resolves symbols, versions, dependencies. Signature enforcement is version/CONFIG-dependent, not universal v3.2. Unload must account for requests, references, and deferred work.

The book separates module management, user-mode loading tools, driver registration, and hardware-resource conflict resolution. Relocation connects imports to permitted exports;initialization/registration makes service callable;teardown reverses publication and waits for quiescence.

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Module loading as a resource transaction

Driver M imports A/B and registers D. Validate format/version, allocate/relocate, resolve exports, initialize, then publish. Finding A without B cannot expose a callable half-built D. If init allocates a buffer then registration fails, unwind successful acquisitions in reverse order.

Stage	Visibility	Resources	Failure action
Before resolution	Not generally callable	Load region	Reject validation failure
Resolve A/B	Not LIVE	Relocated code/data	Release on unresolved import
Init	Service being built	Queues, buffers, registration	Undo completed acquisitions
LIVE	New calls retain references	Registered service	Enforce per-call lifetime

M shares kernel privilege. Provenance/signature policy and runtime lifetime safety are different questions.

Zero references do not close entry points

This driver assigns one M reference per handle/request. Initially three handles plus two requests give5.

Stage	New references?	Handles	Requests	References	May free code?
LIVE	Yes	3	2	5	No
GOING, entries withdrawn	No	3	2	5	No
Handles closed	No	0	2	2	No
I/O finished/legally canceled	No	0	0	0	Callbacks may remain
All callbacks/workers quiescent	No	0	0	0	Yes

Checking zero before stopping entry lets another CPU retain and execute code about to be freed. State/registration protocol first prevents entry, then drains use. RCU callbacks need their grace periods;workqueues, timers, DMA need their own synchronous cancellation/quiescence. One RCU wait is not sufficient for every module.

Process management

Copy_process calls dup_task_struct to allocate child task/kernel stack immediately. Without CLONE_VM, copy_mm creates a child mm/page tables. COW defers copying private writable user-page contents, not task metadata.

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Four layers of fork state

In v3.2, copy_process/dup_task_struct allocates child task/stack, then initializes PID, credentials, scheduler, and other execution context. Ordinary fork uses copy_mm/dup_mm for independent mm_struct/page tables. Copy_page_range may map shared private frames write-protected. There is no wait-until-first-metadata-write allocation of task_struct.

Layer	Ordinary fork	First private write
Task/kernel stack	Separate PID, state, stack objects	Not deferred by data COW
Mm/page tables	Separate objects	Child updates its PTE
Private user contents	Shared frame, temporarily nonwritable	Faulting side copies needed page
Open-file descriptions	Existing descriptors can share objects	Offset sharing is unrelated to user-page COW

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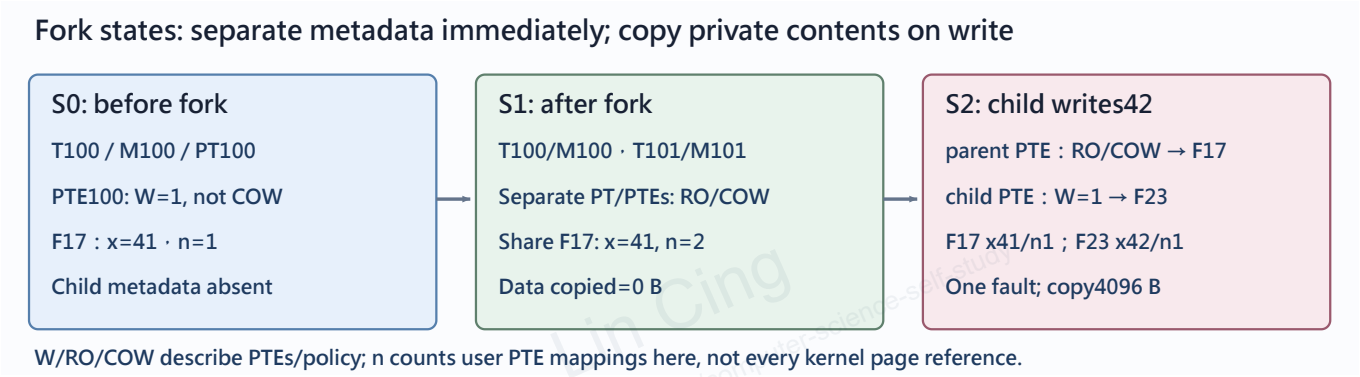
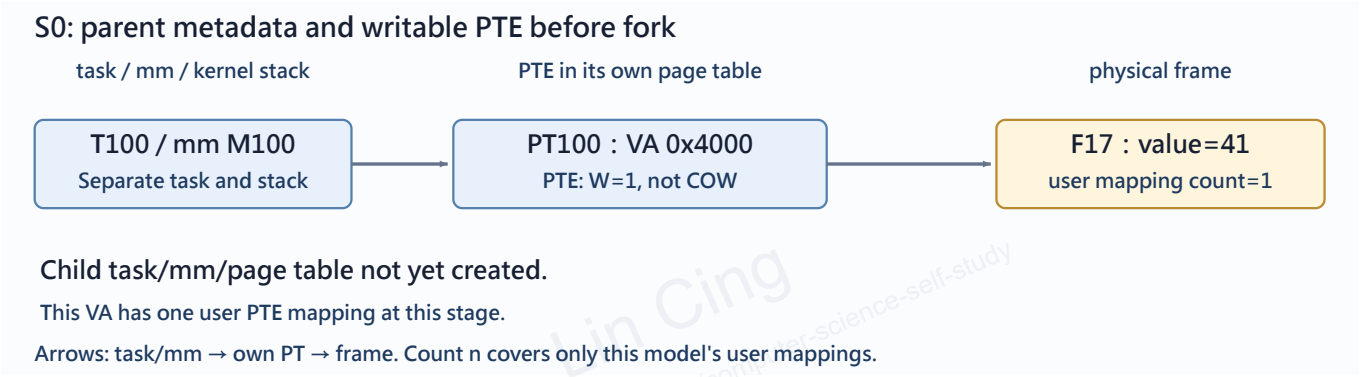
Four layers of fork state (continued)



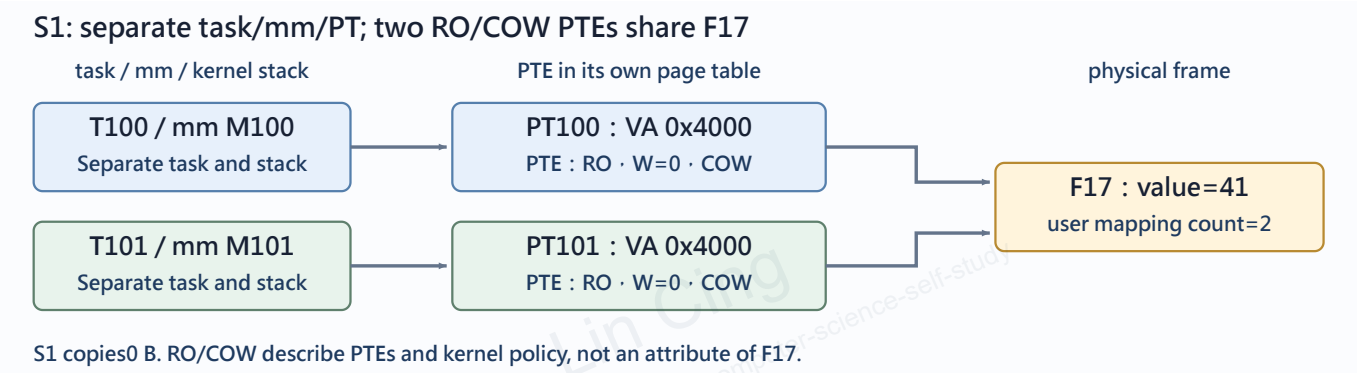
Clone parameterizes sharing. CLONE_VM shares mm;CLONE_FILES descriptor table;CLONE_FS filesystem context;CLONE_SIGHAND handler table. Thread-like sharing does not make ordinary fork share task metadata.

S0 before fork



T100/M100/PT100 maps VA0x4000 to frame17, x=41, W=1, user-PTE mapping count1. Child task does not exist. This is the baseline, not a preexisting COW state.

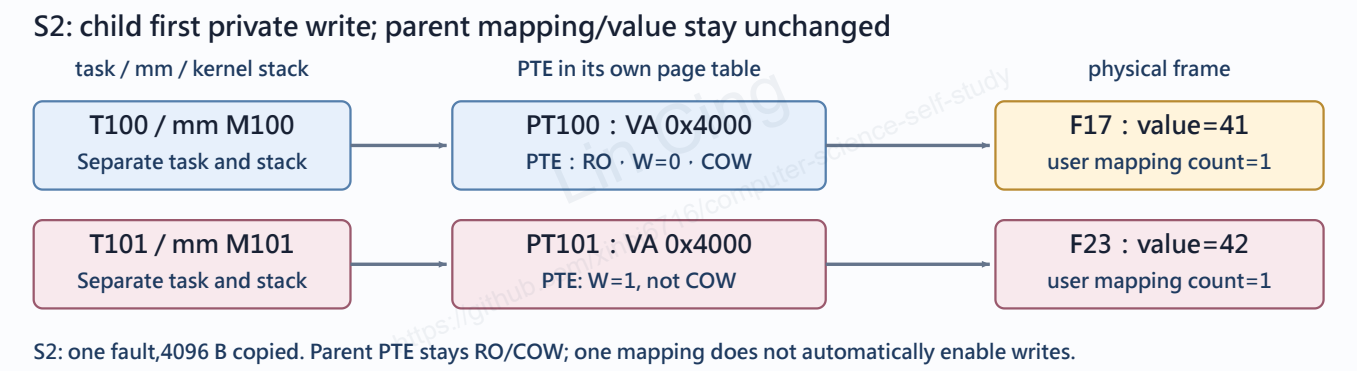
S1 separate metadata, shared frame



Child T101/M101/PT101 already has separate PID, state, kernel stack, mm, page table. Both PTEs mapF17 with writes disabled under COW policy. F17 remains41, mapping count2, data copied0 bytes.

S2 first child write

Assume one4 KiB anonymous private page, logically writable VMA, x at offset 0 ofVA0x4000, initial 41. Exclude huge pages, other mappings, DMA, concurrent writes. COW is a policy label, not an x86 hardware PTE bit named COW. Counts cover this model's user PTE mappings rather than all Linux page references.



S2 first child write (continued)

State	Parent	Child	Frame contents/mappings
S0	T100/M100/PT100:F17, W1	Absent	F17:x41, n1
S1	T100/M100/PT100:F17, W0, COW	T101/M101/PT101:F17, W0, COW	F17:x41, n2;copied0
S2 writes 42	Parent unchanged	T101/M101/PT101:F23, W1	F17:x41, n1;F23:x42, n1;one fault,4096 B copied

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Observable counterexamples and CLONE_VM

If child and parent task were one object, setting child PID 101 would change parent PID too. Correctly, child ready→blocked leaves parent ready. Shared data pages do not imply shared tasks.

Successful exec in this single-thread child preserves process PID while replacing its program image. Early failure before destroying the old image can return an error. After v3.2 flush_old_exec crosses the point of no return, some later failures terminate by signal rather than restore the old image.

CLONE_VM is a different explicit option:two tasks share mm/tables. With writableF17, child writes 42, parent sees42 without this private-COW fault. Ordinary fork instead yields parent41/child42. These finite models expose differences rather than claim live Linux tests;real verification needs corresponding source, fault traces, and frame/PTE counters.

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Clone flags are constrained

Mode	Task/TID	TGID	Mm	Fd table	Existing descriptions
Fork	New	New group	New, private pages may COW	Copied	Shared references
CLONE_VM only	New	Can be a new group	Shared	May be copied	References can share
VM/SIGHAND/THREAD	New	Shared group	Shared	Depends on CLONE_FILES	By table entries

V3.2 rejects THREAD without SIGHAND, and SIGHAND without VM. VM/FS/FILES/SIGHAND alone omit pthread-like thread-group/TGID/signal lifecycle semantics. ParentTID100/TGID 100 can fork101/101 or create thread102/100. Both have independent tasks/stacks despite shared mm.

Scheduling

Fair-class CFS weights real runtime into vruntime. RT classes use separate priorities. Local preemption/IRQ controls are not SMP mutual exclusion;configuration/shared contexts determine locking.

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Real time converted to virtual runtime

For v3.2 fair class, greater weight means slower virtual accumulation and more CPU share. With nice0 baseline1024,

$$\Delta v = \Delta t \frac{1024}{w}.$$

Continuously runnableA, nice0, w1024 andB, nice5, w335 share an ideal6 ms target period in ratios1024/1359 and335/1359, about4.521/1.479 ms. Both virtual increments are about4.521 ms. This weighted calculation does not promise exact per-period segments;tick, wakeup, minimum granularity, and other classes change traces.

The fair rbtree's handling of a running entity follows scheduling paths;not every task is always in one tree. RT FIFO/RR can outrank fair and do not compare the same vruntime. Historical predecessors explain design pressures without becoming v3.2's scheduler. Chapter 6's fixed HRTICK/periodic-tick Gantts provide intermediate selections.

Local reentry versus SMP exclusion

A process context and same-CPU IRQ handler both acquire L. If the process holds L without masking that IRQ, the handler interrupts and spins forever because the holder cannot resume. Use the appropriate irq-save protocol and restore prior state rather than unconditionally enable IRQs.

Mechanism	Prevents	Does not prevent
Local preemption disable	Scheduler task switch here	IRQ, other CPUs
Local ordinary IRQ disable	Corresponding local handlers	Other CPUs, NMI
Atomic spinlock	Cooperating contexts on same lock	Nonparticipants, invalid DMA
Irq-save spinlock	Same-lock SMP races and specified local IRQ reentry	Mechanisms outside the protocol

Local reentry versus SMP exclusion (continued)

Do not acquire a blocking mutex in nonsleepable paths. Keep spinlock regions short. NMI/softirq/configuration variations need separate analysis rather than one universal priority diagram.

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Memory management

Virtual memory includes anonymous pages, page cache, slab, and reclaim with context-specific allocation constraints. Dirty/cache visibility is not durable storage.

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Zones under a fixed x86-32 configuration

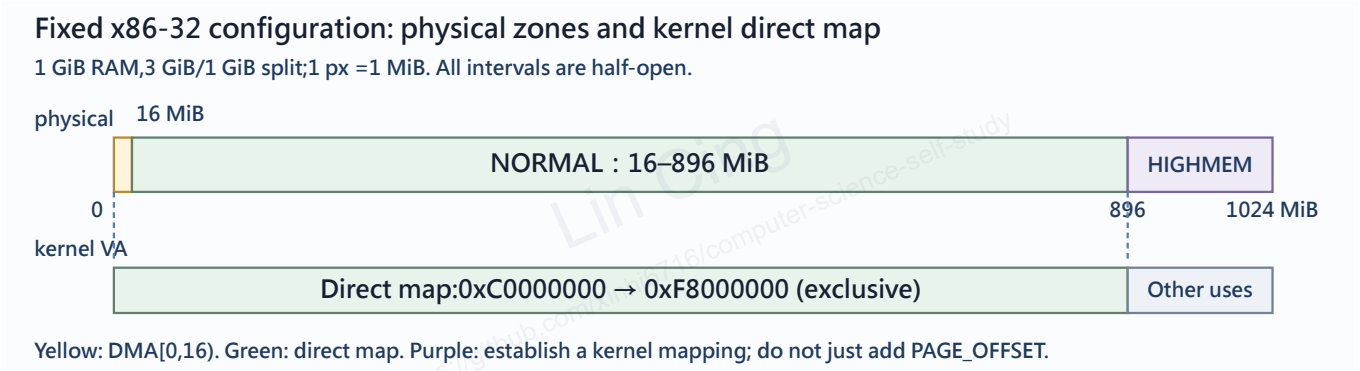
Fix v3.2, x86-32,4 KiB pages,3 GiB user/1 GiB kernel split,896 MiB permanent direct-map coverage, and1 GiB RAM. These are configuration-specific coverage intervals, not all-free memory.

Physical range	Zone	Constraint
[0,16 MiB)	DMA	Selected legacy ISA limits
[16,896 MiB)	NORMAL	Ordinary permanent direct map
[896 MiB,1 GiB)	HIGHMEM	Requires suitable temporary kernel mapping

PA0x01002000 plus PAGE_OFFSET0xC0000000 gives kernelVA0xC1002000 here. HighmemPA0x39000000 cannot simply be declaredVA0xF9000000, whose region has other uses. Establish a valid mapping. User PTEs can still map highmem:VPN0x40000→PFN0x39000 mapsVA0x40000123 toPA0x39000123.

Zones under a fixed x86-32 configuration (continued)

The physical allocator follows zone/order constraints;buddy allocates power-of-two contiguous pages, slab smaller objects. Device masks, IOMMU, bounce buffers, and DMA API determine DMA addresses;not all DMA is limited to16 MiB.



ELF bytes and zero-fill

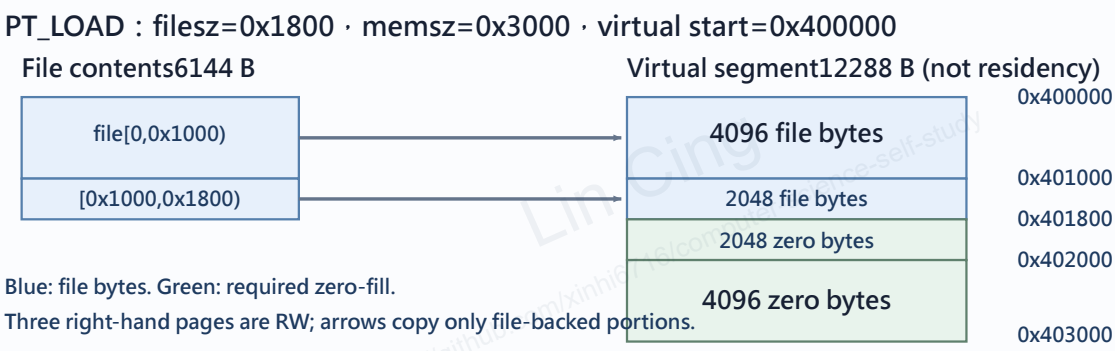
This separate example uses an NX-capable x86-64 ELF permission model, not the preceding x86-32 layout. PT_LOAD has p_offset0, p_vaddr0x400000, p_filesz0x1800, p_memsz0x3000, RW. Memory beyond filesz must read zero rather than later file garbage.

Virtual page	Contents	Permission
[0x400000,0x401000)	File[0,0x1000)	RW
[0x401000,0x402000)	File[0x1000,0x1800), then 0x800 zero bytes	RW
[0x402000,0x403000)	0x1000 zero bytes	RW

File contents6144 B plus zero tail6144 equal12288 virtual B. Creating a VMA does not make every page resident. Separate RX segments describe code;private writes use COW, connecting fork and fault handling.

ELF bytes and zero-fill (continued)

PT_INTERP selects a user-mode dynamic interpreter. Kernel maps program/interpreter, builds initial stack, and enters it;dynamic linker resolves DT_NEEDED/symbols. Kernel need not resolve every function call, and libraries need not always map at one VA.



Reclaim preserves backing and live references

Track mapping, backing, dirty state, and reclaim eligibility separately. Reuse cannot occur while valid CPU/device references still require a frame. Clean file-cache data is reconstructable only from valid backing. Dirty data may be the only latest copy;complete the required writeback before marking the covered state clean. Remove unsafe mappings/users under the reclaim protocol and release necessary references before reuse. A cached page can remain referenced while no application currently executes on it. Writeback cleanliness alone is not a full filesystem durability transaction.

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File systems

VFS connects inode, dentry, file, superblock objects into one namespace. Filesystem-specific journaling/COW, not the interface alone, establishes crash consistency.

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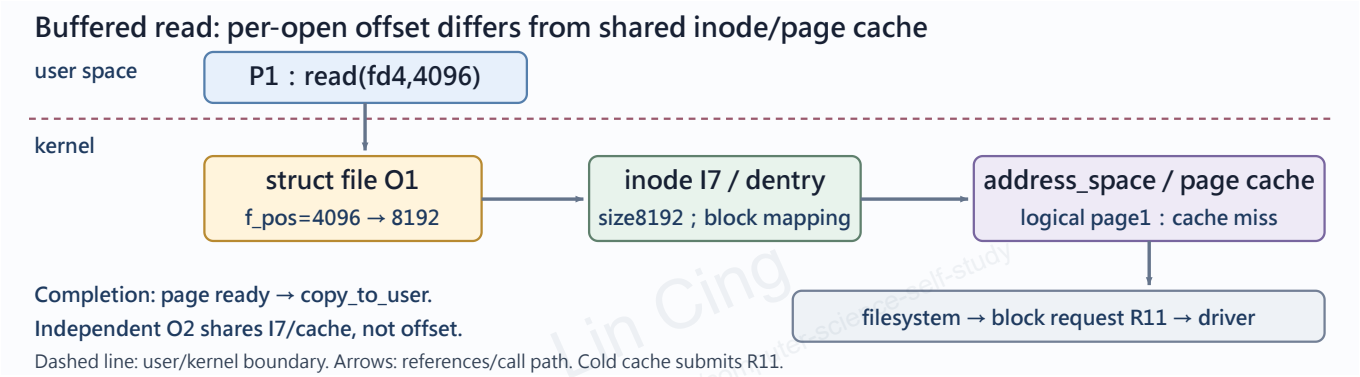
Four VFS objects in one buffered read

Fix v3.2, ordinary ext3 buffered read, size 8192, page 4096, openedP1 fd4→struct file O1 withf_pos4096, dentry/inodeI7 references. I7.address_space manages shared file pages;superblock identifies the filesystem instance.

Object	State/references	Incorrect assumption
Superblock	Filesystem instance/operations	Contains per-open offset
Dentry	Parent/name→inode, possibly negative	One unique name per file
Inode	Identity, mode, size, mapping, address_space	One global current offset
Struct file O1	F_pos, flags, operations, path	Every fd necessarily has its own offset

Read4096 retains O1, validates operation/user buffer, and dispatches file ops. A miss on logical page 1 submits mapped blocks, waits for page readiness, copies to user, and setsO1.f_pos8192. Independent O2 can use I7's page cache;dup shares O1. Cache hits save block I/O without eliminating lookup/copy costs.

Four VFS objects in one buffered read (continued)



Ext3 block groups and procfs boundaries

Ext3 organizes inodes, data, bitmaps, block groups for locality. Nearby allocation is a policy target, not an invariant that every file stays contiguous. Growth/fragmentation can move later blocks while mappings remain correct.

Under the teaching data=ordered model, a 6000 B write dirties two page-cache pages. Data precedes metadata commit to avoid publishing pointers to unready contents. Write success differs from fsync, and metadata atomicity does not make all in-place user writes atomic. Data=writeback and data=journal have different contracts.

Procfs /proc/123/status formats live kernel state rather than necessarily reading persistent blocks. ReusedPID123 can later denote another process;monitoring needs start time/other identity evidence. Separate proc reads do not form a simultaneous system snapshot. A fixed16-bit PID or always-PID-in-upper-inode-bits rule is not assumed for v3.2.

Input and output

Block, char, network drivers manage requests, IRQs, DMA, completion through different paths. They share privileged failure scope;DMA memory cannot be unmapped early.

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Three driver paths and terminal states

Block devices offer block/sector addressing through filesystem/request layers. Character drivers define file operations without universal seeking. Network devices use sockets/protocols/packets rather than universally seekable device files.

Fix one cold-page synchronous read, resident user buffer, no read-ahead, requestR11 over kernel bufferF40. Queue→in-flight device ownership→terminal success/error. Publish results and release mapping according to completion/lifetime. Real layers can merge/split;one page/request is this input model.

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Three driver paths and terminal states (continued)

Outcome	Hardware evidence	Terminal notifications	Application	Reuse F40
Success	I/O ends, ownership returns	One	4096 B	After copying/references end
EIO	Request stopped/error-completed	One	Error with no prior bytes	After DMA/references end
Timeout, hardware may continue	Stop unproved	Cannot claim success/safe release	Cancellation/reset protocol	Not yet

An error completion still completes;excluding it from completion counts strands waiters. Timeout is not hardware cessation. Late DMA after free corrupts the next owner's data.

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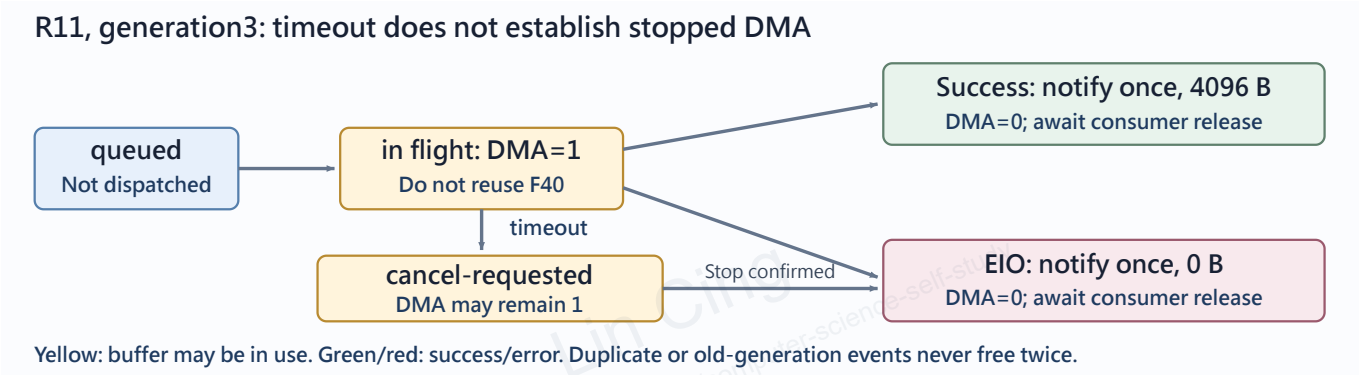
Three driver paths and terminal states (continued)



Duplicate completion and reset

Submission identity is(id11, generation 3). First valid terminal event atomically changes in-flight→terminal and notifies once. Duplicates of that generation are ignored/logged without another notify/free. Reusingid11 withgeneration4 rejects old3 events.

Generation protects software identity, not DMA quiescence. Cancel/reset must establish hardware no longer uses old addresses before synchronization/unmap/reuse. Device-specific late-IRQ/error protocols matter beyond a timeout number.

Synchronous read finishes copying before return. Async user buffers remain valid through the completion contract. Queue state, ownership, and notification count belong in one consistent model.

Interprocess communication

Pipes, shared memory, queues, signals, sockets have different lifetimes, permissions, and blocking. Shared memory needs synchronization;streams need framing;datagrams preserve messages but can lose, truncate, or reorder.

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Pipe EOF requires every writer to close

Parent fd3→read descriptionR, fd4→write descriptionW. Fork copies fd tables while retaining R/W objects. A parent consumer and child producer must close unused endpoints.

Event	Parent writer	Child writer	Buffered bytes	Read result
After fork	1	1	0	Block or EAGAIN, not EOF
Child writesABC/closes	1	0	3	ABC available
Consumer drains	1	0	0	Still no EOF
Parent closes last writer	0	0	0	Later read returns 0

Streams preserve byte order, but multiwriter atomicity depends on PIPE_BUF, size, and mode rather than every arbitrary write being indivisible. Shared memory supplies neither queueing nor exclusion;choose locks, semaphores, or an atomic protocol.

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Signals, wait queues, and atomic semop

Blocked standard signals can coalesce repeated instances into one pending event, so they are not reliable counters. Real-time signals can queue and sigqueue can carry limited payload. Handlers remain constrained by async-signal-safety.

A waiter that tests done=false before enqueue/sleep can miss a producer's intervening done=true+wake. Establish a wakeable waiting state first, then recheck. If true, do not sleep. Recheck after waking because another waiter can consume the resource. Wake makes runnable rather than immediately executing or permanently satisfying the condition.

System V semop atomically batches operations within a semaphore set. WithS0=1, S1=0, the batchS0-=1, S1-=1 must not permanently decrementS0 while waiting forS1. Publish no partial effect before all can complete. Permissions, limits, deletion, and errors remain part of the contract.

Network structure

Sockets, protocols, routing, interface drivers, and hardware compose the historical path. Flow control, checksums, fragmentation, queues, interrupt mitigation affect distinct properties.

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Application and wire byte denominators

Fix IPv4/TCP, no options, MTU1500, no TSO/GSO. Send accepts6000 B. MSS=1500−20−20=1460;wire TCP payloads1460,1460,1460,1460,160 sum6000. Five IP/TCP headers add200, L3 bytes6200, excluding Ethernet, ACK, retransmission.

Sk_buff, TCP send queue, routing, and driver retain different states. Offload can split one large sk_buff into many wire packets, so counts differ. Send acceptance proves neither peer acknowledgment nor application processing nor disk durability.

For10.1.2.3 with routes/0,10.0.0.0/8,10.1.0.0/16,10.1.2.0/24, the longest match is/24. FIB structures, caches, hooks vary by version/CONFIG;longest-prefix semantics do not imply one universal hash layout.

Queueing, retransmission, and causal evidence

In a fixed trace relative to enqueue0, dispatch2 ms, transmit completion3, ACK13. Queue wait2, dispatch-to-transmit1, post-transmit ACK wait10. These claims follow linked timestamps.

Aggregate queue length20, retransmissions3, device errors1 cannot prove queueing caused a long tail. Loss, reordering, or unrelated device faults may explain counts. Link the same request/flow/window's enqueue, dispatch, completion, drop, ACK evidence, or use a controlled intervention before making a causal claim.

TCP sequence handling restores ordered delivered bytes despite IP loss/reordering. UDP preserves datagram boundaries without delivery guarantees. A TCP upper layer does not make every driver/IP queue lossless or ordered.

Security

Authentication identifies a subject;authorization checks credentials, object, operation. PAM and kernel enforcement differ. Capabilities can override DAC, while execute/saved IDs/retained handles impose additional boundaries.

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Authentication and authorization

PAM composes user-space authentication, account, session policies;kernel resource checks still use credentials. Salt separates stored values without adding password entropy. KDF cost/access protection differ. Old eight-character schemes are not current recommendations.

Credentials include real, effective, saved, filesystem IDs, groups, capabilities. Ordinary file DAC selects owner/group/other then requested bits;ACL, LSM, capability, mount conditions can change the decision. Successful open can create a continuing handle not revoked by later pathname-mode changes.

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Root and regular-file DAC

Fix v3.2, owner1000, mode 0400, rw mount, no ACL/LSM denial, separate read/write/execute requests. Effective capabilities belong to the corresponding user namespace;exclude other policy.

Caller	Read	Write	Execute	Rule
Fsuid1000, no DAC capability	Allow	Deny	Deny	Owner bits100
Fsuid2000, no capability	Deny	Deny	Deny	Other000
Fsuid0, CAP_DAC_OVERRIDE	Allow	Allow	Deny	Override read/write;regular execute needs some x bit
Fsuid0, no relevant capability	Deny	Deny	Deny	Numeric UID alone adds no missing capability
Fsuid2000, CAP_DAC_READ_SEARCH only	Allow	Deny	Deny	Supplies regular-file read only

Root and regular-file DAC (continued)

Generic_permission first performs ordinary checking, then tests CAP_DAC_OVERRIDE on failure;execute also requires nonzero S_IXUGO. Mode 0401 supplies an x bit, allowing that DAC override, while noexec, format, and other policies still apply. Thus mode 0400 does not force capable-root write denial, and root does not bypass every security layer.

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Temporary versus permanent privilege drop

A setuid-root program hasreal1000, effective0, saved0. Seteuid(1000) yields1000/1000/0 and can later restore effective0 under saved-ID rules. It is temporary.

Permanent dropping additionally handles saved IDs, groups, capability sets, and privileged handles. Changing euid alone does not recall opened descriptors. Securebits, user namespaces, and required resources affect implementation;this ID trace is not a complete sandbox recipe.

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One successful read across mechanisms

Continue fd4→O1→I7, offset 4096, read 4096, cold cache, resident user buffer. Retain fd/validate→cache lookup→filesystem map→R11 block I/O→device completion→ready page→copy_to_user→O1.offset 8192→return4096. Waiting can switch tasks, unlike an ordinary internal function call.

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Errors, partial progress, and lifetimes

If R11 terminates EIO before any copy, read returns error without advancing offset. Terminal notification is still exactly one, and buffer reuse waits for stopped DMA/released references. With earlier copied bytes, an API may return that successful prefix and expose errors later;not every failed read has zero progress. This fixed example explicitly excludes prior bytes.

Fork sharing pages does not share tasks;independent opens sharing inode do not share offsets;send acceptance is not remote receipt;dirty-cache writes are not durable. Identity, authority, lifetime, and completion contracts preserve these distinctions.

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References

- *Operating System Concepts*,9th edition, Chapter 18, printed pages 781–824; § 18.6 p801 Fig18.3 covers x86-32 zones; § 18.8 p816 Fig18.8 covers I/O responsibilities.
- Linux v3.2 fork.c defines copy_process, dup_task_struct, copy_mm, dup_mm, and clone constraints;memory.c defines page copying/write faults;sched.h defines flags.
- Fork API describes modern memory semantics;versioned implementation remains v3.2 source.
- V3.2 namei.c supplies DAC/execute rules;binfmt_elf.c supplies PT_INTERP and post-flush_old_exec failure paths.
- Signal semantics distinguishes standard/realtime queueing and payload;historical implementation uses corresponding source.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Chapter route

How does layered Windows architecture organize objects, processes, I/O, and services across user/kernel boundaries?

Prerequisites are OS structure, protection domains, and file/I/O mechanisms.

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Learning objectives

- Identify user/kernel-mode components.
- Analyze object-manager handles and granted access.
- Separate Windows 7 history from contemporary implementations.

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History

Fix Windows 7/NT architecture. APIs, user services, and kernel components have different authority/failure boundaries. Historical subsystems are not universal current support.

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One NT-family case

The case distinguishes user subsystems/services from kernel-mode executive, kernel, HAL, drivers. It does not substitute DOS-based Windows, Mach, or later Windows additions.

Term	Meaning	Does not imply
Win32 API	Application interfaces for processes, objects, I/O	Every API makes a syscall
NT kernel/executive	Trusted scheduling, memory, object, I/O	Every component has its own address space
Compatibility	Runs existing programs in specified environment	Every ISA, bitness, legacy subsystem supported
Portability	Concentrates hardware dependencies	Driver binaries run unchanged across CPUs

OS/2, Win16, POSIX labels in historical diagrams retain their period/version scope. They do not make every Windows 7 installation run every old binary. Subsequent cases specify execution, configuration, failure assumptions.

ReadFile across trusted components

Library preparation enters a controlled kernel interface that checks handle/buffer. I/O manager, filesystem, cache, memory manager select a hit or driver path. Device waits can block the thread before completion returns a result.

An API name alone specifies neither blocking, copy count, IRP presence, nor context-switch count. Cached/fast paths differ. Record open flags, object type, cache state, and completion mode first. Layering includes many kernel-mode components rather than a mandatory user-server hop or independent failure domain per component.

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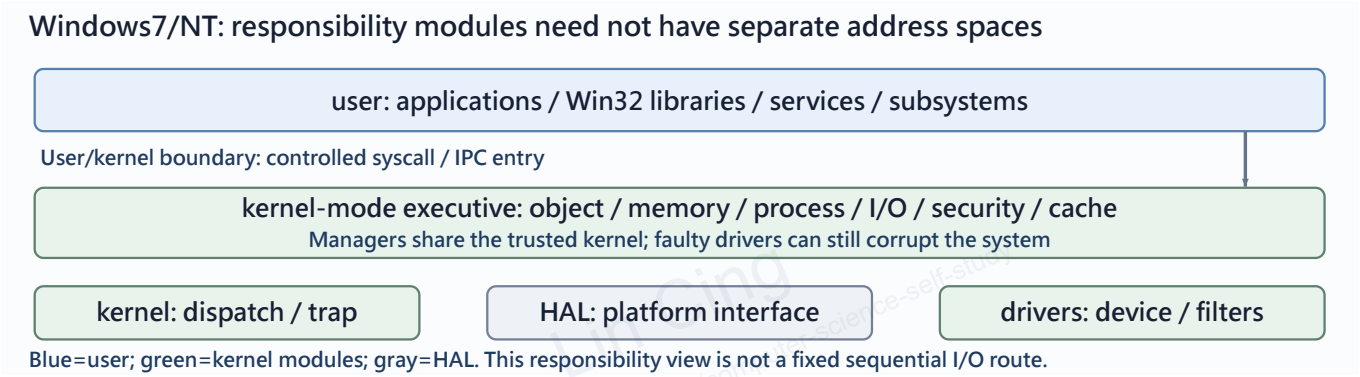
ReadFile across trusted components (continued)



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Design principles

Connect security, reliability, portability, and other goals to enforcement. Unicode lengths and PnP/power lifetimes expose concrete boundary cases.

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Goals and mechanisms

Goal	Mechanism	Not guaranteed by its name
Security	Tokens, descriptors, granted handle access	Bug-free kernel/drivers
Reliability	Boundaries, references, error recovery	Shared-kernel fault containment
Performance	Caches, async I/O, scheduling	Uniformly low latency
Extensibility	Object/driver interfaces	Every extension is safe
Portability	HAL, architecture-specific layers	Cost-free porting
Internationalization	Unicode, locale data	Characters equal bytes
Power/dynamic devices	Power states, PnP lifetime	Removal without outstanding I/O

HAL abstracts selected platform, interrupt, timer details without removing every memory-model/driver responsibility. SMP support does not make shared updates race-free.

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Unicode length units

Win32 W interfaces use UTF-16 code units;A interfaces perform code-page conversion. Neither means UTF-8 bytes or Unicode scalar counts.

A, U+1F600, B encodes0041 D83D DE00 0042:four16-bit units, three scalars, eight data bytes, plus two-byte NUL. A four-byte buffer is insufficient. Capacity in wchar_t units must not receive a byte count;byte length must not receive scalar count. Unicode APIs also do not guarantee canonical filename normalization.

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PnP/power depend on ownership

The lifecycle model is working→stop-pending→stopped→removed, not the complete PnP IRP list. Stop-pending blocks new submission, drains/cancels existing work with confirmed completion, then releases device context, DMA maps, buffers.

Freeing context before an outstanding IRQ/request ends causes use-after-free. Requested cancellation does not prove DMA stopped. Removal can finish requests with errors rather than guaranteed success. No resource can pass to an incompatible owner while execution/device references can still touch it.

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System components

Dispatcher, object, memory, I/O, security managers maintain different state. Thread priority, IRQL, handle authority, and buffer ownership are not interchangeable.

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Kernel, executive, HAL, and services

Component	Responsibility	State
Kernel dispatcher	Select/dispatch threads, synchronization/trap foundations	Ready queues, thread states
Executive managers	Objects, memory, process, I/O, security, cache	Handles, PTE/PFN, IRPs
HAL	Platform, interrupt, timer abstraction	Routing/control
Drivers	Devices, filters, filesystems	Device objects, queues, buffers
User services/subsystems	Login, background services, application environments	User address spaces, IPC

Kernel/executive are responsibility labels, both privileged. A thread entering kernel uses its kernel stack without creating a separately scheduled second thread.

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Thread priority differs from IRQL

Dispatcher chooses the highest-priority eligible ready thread:variable 1–15, realtime 16–31, idle special 0. Variable boosts decay no lower than base and do not enter realtime range. Conditions affect boost/quantum;the later q2 is not universal.

IRQL governs current-processor interrupt/kernel activity and permitted operations, not another0–31 thread priority.

Common x86/x64 IRQL	Execution	Constraint
PASSIVE_LEVEL0	Threads/workers	Waiting still follows API rules
APC_LEVEL1	Selected kernel APC work	Specific APC/call constraints
DISPATCH_LEVEL2	Ordinary DPC/spinlock regions	No nonzero blocking wait/pageable access fault
Device DIRQL	ISR/device synchronization	Shorter, more restricted work

Raising one CPU's IRQL does not exclude other CPUs. Realtime thread priority does not guarantee hard deadlines. Driver priority rules define these constraints.

A fixed priority trace

One CPU, no boosts/decay, equal-priority q2, higher priority immediately preempts. A preempted task keeps residual slice and returns to its priority queue's front. P1/P2 priority8, ready0, bursts4/3. P3 priority10, ready1, burst2.

Interval	Reason	Result
0–1 P1	First equal-priority arrival	Preempted with burst3/slice1
1–3 P3	Higher priority10	Complete
3–4 P1	Queue front, residual slice	Burst2, quantum expires, tail
4–6 P2	Priority8 queue front	Burst1, tail
6–8 P1	Queue front	Complete
8–9 P2	Only ready task	Complete

CompletionsP1/P2/P3=8/9/3, total CPU 9. This isolates queue/quantum rules;actual boosts, quanta, processors require their own calculation.

Dispatcher objects are not universal queues

Manual-reset events remain signaled until ResetEvent and can release many waiters. Auto-reset satisfies one wait then resets. Repeated SetEvent on an already signaled event does not accumulate independent tokens. Counted resources need semaphores or explicit queues.

A mutex has an owner. WAIT_ABANDONED signals abnormal prior-owner termination and requires checking protected-data consistency;acquisition does not repair it. Mutexes alone do not prevent multi-lock ordering deadlocks.

This model exposes semaphore count0…limit. At0, waiters sleep;release can hand off a permit. Do not confuse it with Chapter 5's alternative internal signed counter.

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ISR, DPC, APC, and worker contracts

ISR checks/acknowledges source and performs minimum work. Ordinary DPC defers work but remains DISPATCH_LEVEL, so it cannot perform arbitrary blocking filesystem I/O. Sleepable/pageable work goes to an appropriate PASSIVE worker with retained context references.

An APC targets a thread. Ordinary user APCs generally run during its alertable wait, not arbitrary user instructions. Sleep differs from SleepEx(TRUE). Special kernel APC/startup paths have additional rules outside this summary.

ISR→DPC→worker retains context before queueing and releases after worker completion. Freeing before the queued worker uses it is invalid. IRQL and ownership jointly determine legal work.

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Handles are process-table indexes

A's0x40→F, grantedR while B's0x40 can name eventE. Integers alone do not transfer authority. Inheritance, DuplicateHandle, or named reopening provide explicit authorized sharing.

Step	A table	B table	F lifetime
Initial	0x40→F, R	Empty	A retains
SAME_ACCESS duplicate	Unchanged	0x88→F, R	Both retain
A closes	Empty	Still0x88	Retained
B closes, internal I/O remains	Empty	Empty	Still retained
Last internal reference ends	Empty	Empty	Cleanup permitted by type rules

SAME_ACCESS preserves rights. Other desired-access behavior depends on object/API authority, not a universal subset rule. Source/target process handles need rights such as PROCESS_DUP_HANDLE. DuplicateHandle.

Tokens and ordered DACLs

Choose the effective token, including impersonation, with SIDs, groups, privileges. Descriptor contains owner, DACL, SACL;DACL grants/denies while SACL principally audits.

This simplified model uses enabled SIDs, ordinary ACEs, specific rights, excluding privilege, integrity, owner/object-ACE exceptions. AbstractR1/W2 are not actual Win32 masks. RequestRW starts remaining3.

ACE	Match	Remaining/result
ALLOW Alice R	Yes	RemoveR, leaveW
DENY Staff W	Alice inStaff	IntersectsW, deny/stop
ALLOW Staff RW	Not reached	Cannot repair denial

Moving the final allow before deny can satisfy all remaining bits first. Order matters;canonical ACE ordering is preferred. NULL DACL imposes no DACL restriction;empty DACL grants nothing in this ordinary model. Ordered access checks.

Reservation, commitment, and residency

Fix4 KiB pages,64 KiB allocation granularity, private base0x100000. Reserve64 KiB of process VA, then commit pages 0x101000/0x102000 with 8 KiB charge, not necessarily physical frames.

Step	Reserved VA	Private committed	Resident here
Reserve	64 KiB	0	0
Commit two pages	64 KiB	8 KiB	0
Touch0x101123	64 KiB	8 KiB	4 KiB zero-fill
Decommit first, no other reference	64 KiB	4 KiB	0
Release region	0	0	0

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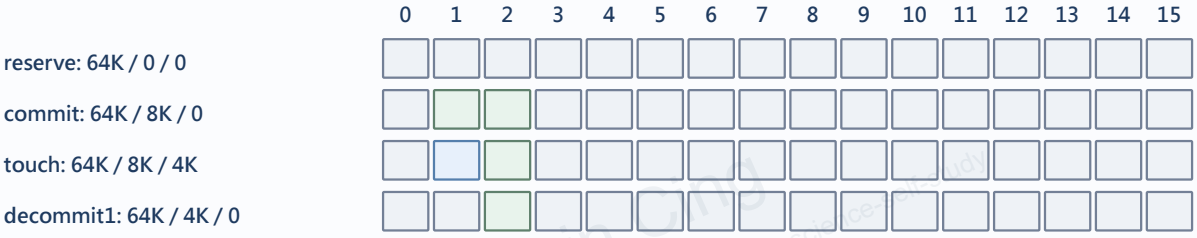
Reservation, commitment, and residency (continued)

Sixteen 4 KiB pages in 64 KiB: distinguish reserve, commit, and resident

Gray=reserved, uncommitted; green=committed, nonresident; blue=resident and committed.



Rows show reserved VA / committed private / resident. Releasing the entire region makes all three zero.

Reserve/commit can fail;commit follows system backing commitments rather than immediately zero-writing pagefile pages. Equal VA numbers in different processes do not compete for one shared virtual address. VirtualAlloc.

PTE, PFN, and working-set trimming

PTEs describe virtual mappings/software states;PFN database describes frames, references, lists. A legal nonresident page can be demand-zero, section/file/pagefile-backed, or recoverable from RAM. Not every fault reads storage.

VA0x401123 has VPN0x401, offset 0x123. A validPTE→PFN0x2A gives PA0x2A123. Writing RO without legal COW is an access violation, not permission to allocate around protection.

Trimmed clean backed pages can enter standby rather than vanish. Modified pages need writeback before reuse;age alone cannot free pinned/I/O-held frames. Standby recovery can be a soft fault;missing RAM backing may require storage. Windows 7 reference-history/working-set heuristics are neither future-aware OPT nor exact LRU. Track frames, PTEs, pending writes together.

Processes, sections, and ALPC authority

CreateProcess includes an initial thread;another CreateThread makes two. Process owns address space, handles, token;threads own registers, user/kernel stacks, execution state. Closing a process handle does not terminate it. Exited-thread objects can remain observable/waitable through retained handles.

Sections map file/pagefile backing at different process VAs with independent rights and shared-data synchronization. ALPC/ports package client/server requests and authority. Shared sections save copies without removing lengths/ownership checks.

A maps offset 0 at 0x100000, B at 0x500000. A+16 and B+16 can share bytes, but literal pointer0x100010 is not B's pointer. Transfer section-relative offset 16 and length.

IRP dispatch down, completion up

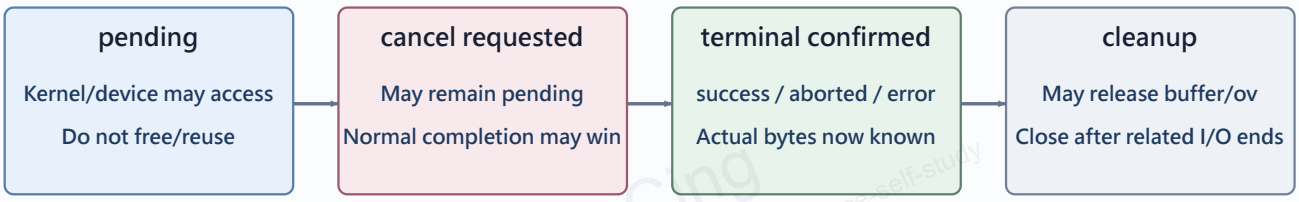
I/O manager can send IRPs through filesystem/filter/device stacks with per-stack state. Lower completion returns status/actual bytes through completion paths. Cached/fast operations need not repeat the same full chain.

Time	Ownership/state	Allowed action
Submit	Valid buffer, unique operation state	Do not reuse while pending
Pending	Driver/device can access	Cancel requests work;does not free
Hardware/cancel completion	Status/bytes established	Notify under completion rules
Caller confirms terminal	Success/aborted/error known	Release buffer/OVERLAPPED

CancelIoEx races normal completion. Success means cancellation requested;ERROR_NOT_FOUND can mean no matching pending operation. Collect the original result regardless. Cancellation contract.

IRP dispatch down, completion up (continued)

Cancel requested ≠ I/O complete: retain buffer/OVERLAPPED until terminal



This path requests cancellation. Pending may complete normally; cancel success is not the operation result.
Use unique ov/buffer per outstanding operation. Win7 needs a count argument; final bytes come from completion.

Cache/memory managers and reads

Cached reads find file ranges in filesystem/cache views. Resident data copies to caller. A view fault invokes paging read through storage to fill a frame, then satisfies the original read. Paging/cached requests are related but not two application deliveries.

For4096 B, a hit copies 4096 to caller. A miss adds4096 B device→cache DMA before that copy. CPU copy cost, DMA bytes, storage reads differ;this is not two CPU copies. Lazy writing batches dirty data while delaying durability. Filesystem/storage log/data flush ordering establishes persistence rather than hit/write success alone.

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PnP, power, registry, and boot dependencies

PnP coordinates discovery, driver stacks, removal;power manager coordinates power states. Drain hardware-touching requests before releasing resources. Driver objects, device objects, and physical presence/usability have different lifetimes.

Registry hives, keys, values store configuration, possibly volatile state, rather than ordinary file paths. Changes take effect under service reload/restart/notification rules, not instantaneously in every process.

Boot progresses through firmware/boot manager/OS loader, kernel/HAL/boot drivers, kernel initialization, sessions/services, logon. BIOS/UEFI differ;no universal MBR file sequence is assumed. Page tables/storage drivers must exist before dependent memory/file services.

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Terminal services

Sessions/desktops do not create a VM per user. Namespace scope, object authorization, and process-local handles differ.

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Session, window station, desktop, and process

Terminal services/fast user switching preserve interactive sessions, desktop/input views, and processes. Windows 7 service/session arrangements must not be confused with older versions.

Resource	Independence across sessions	Reason
User address space	Per-process mappings	Session numbers do not replace MMU
Window stations/desktops, some names	Session-related scope	Object rights/namespace still matter
Kernel, drivers, RAM	Shared	Common faults/resource limits
NTFS/network services	Can be shared	Credentials/object policy

The abstraction does not prove unrestricted simultaneous interactive connections;product, licensing, deployment limits are separate.

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Local/Global scope is not authorization

S1/S2 Local\Ready events can be distinct session objects. Authorized Global\Ready access can reach one global object while retaining DACL/create/open rights. Global never means bypass checks.

Handle0x80 in different processes can denote unrelated objects. Equal names require matching namespace and authority. Session ID is not trusted user identity. Switching the foreground user need not terminate background-session processes or release their locks, CPU, memory.

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File system

NTFS uses MFT attributes and VCN/LCN mappings. Metadata recovery, user-data durability, and snapshots preserve different invariants.

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Volumes, MFT, attributes, extents

Windows 7 supports multiple filesystems;not every volume is NTFS. Clusters allocate storage;LCNs are volume-relative. Disk offsets additionally include partition/volume start.

MFT records contain attributes. Small data can be resident in its record;nonresident data uses VCN→LCN extents. Nonresident here does not mean absent from RAM. In a 1024 B record with 700 already used,324 remains. Data 300 fits this budget;512 does not. Names/attributes/format change the threshold;300 is not universal.

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One complete VCN-to-LCN example

Cluster4096 B, volume starts at disk byte1048576. Half-open extents are:

Start VCN	Length	Start LCN	Mapping
0	4	100	VCN0–3→LCN100–103
4	3	200	VCN4–6→LCN200–202
7	2	None	VCN7/8 are holes, logical zeros
9	2	190	VCN9/10→LCN190/191

File byte20603=5・4096+123 selects extent2. LCN=200+(5−4)=201. Volume byte201・4096+123=823419;disk byte=1048576+823419=1871995. A hole returns zeros without treating None asLCN0.

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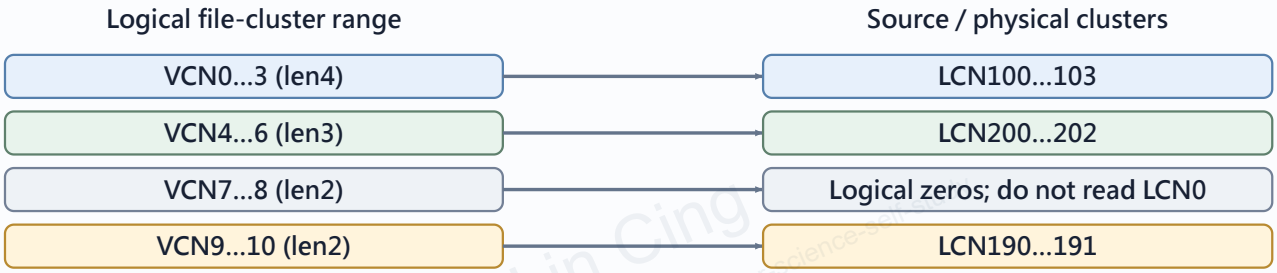
One complete VCN-to-LCN example (continued)

One NTFS file: VCN is file-relative, LCN volume-relative; gray=hole



20603 → VCN5/123 → LCN201 → volume 823419 → disk 1871995; cluster 4096 B, volume base 1048576.

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File references, indexes, and metadata files

The book's NTFS reference combines48-bit MFT record and16-bit sequence. Reuse changes sequence to reject mismatched stale references. Finite16-bit sequences do not prove everlasting uniqueness against wrap.

Directory indexes sort keys/references. Balanced B+ paths have logarithmic height, but insertion/splitting can reorganize nodes. Small indexes fit root/record;larger ones allocate index blocks. Cached directory sizes/times need not reread current contents on every enumeration.

MFT, bitmap, log, volume, and other metadata use special files and common cache/I/O services. Cluster bitmap, extent, size updates still need coordinated consistency rather than independent successful writes.

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Metadata WAL before home updates

This single-transaction undo/redo model is not complete NTFS log encoding or concurrent recovery. Complete records/commit are recognizable;flush is trustworthy. Steal permits precommit home writes;no-force permits postcommit home lag.

Old metadata:size 4096, extents[100], bitmap200=0 free. New:size 8192, extents[100,200], bit1 allocated. Persist each old/new log before its home update. Durable commit establishes metadata transaction success.

Crash prefix	Home may contain	Recovery
No complete log	Old	No unlogged home mutation allowed
Full log, no commit	Mixed	Undo old values to complete old
Full log, durable commit	Old/mixed/new	Redo new values to complete new

Choose a consistent metadata group rather than arbitrary size/bitmap fragments. Actual NTFS also manages LSNs, concurrency, checkpoints beyond this table.

Recovery invariant and checkpoint scope

Without commit, every permitted changed home field has durable old data for undo. With commit, all required new values are durable for redo. Assigning old/new values is idempotent, so repeated recovery chooses the same state. Logs must remain available under the model's failure assumptions.

Checkpoint does not mean every log older than a time threshold can be discarded. Active-transaction undo and dirty-metadata redo remain until safe truncation conditions hold.

Metadata consistency is not user-data transaction atomicity. Cached write success or NTFS logging does not atomically update two business files. Applications need their own flush/publication/transaction protocol.

ACLs, encryption, and repair

NTFS uses Windows descriptors/DACLs, not POSIX owner/group/other masks. Traverse-bypass privilege and explicit FILE_TRAVERSE checks are different contexts;Windows does not universally ignore path protection.

EFS encrypts file contents;BitLocker protects volume/storage under its keys/unlock conditions. ACLs alone do not protect unencrypted media read elsewhere. Encryption supplies neither backups nor application transactions.

Bad-sector remapping supplies new locations, not unreadable lost contents without a valid mirror/parity/source. RAID/spares, metadata recovery, and data repair guarantee different things.

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Sparse allocation, compression, USN, and snapshots

The example spans11 logical clusters, including two holes. Nine data clusters allocate36864 B without compression while logical size can be45056 B. Hole writes later require allocation. This is not a process VA reservation.

Compression units and content/configuration determine compression results, not a fixed ratio. USN tracks changes for consumers/indexes;it differs from recovery logging and can wrap. A history gap requires reconstruction rather than treating it as an eternal backup.

VSS provides a point-in-time storage view. Application updates across related fields can require writer coordination/quiescence for application consistency.

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COW snapshots retain the old version

LiveX41, Y80 creates snapshotS. Before first updateX42, retain oldX41 forS, then publish new live. After liveX43, S still sees41 rather than 42. UnchangedY can share old80.

Time	Live X	S X	Y
After snapshot	41	41	Shared80
First write	42	41	80
Second write	43	41	80

Reclaim old backing only after every retaining snapshot/reader releases it. Shared-storage failure can destroy live and snapshot together, so it is not independent backup media.

Networking

Winsock, NDIS, WSK, and redirectors serve different layers. Remote timeout, cache, and message modes retain explicit failure contracts.

Interfaces and protocol responsibilities

User Winsock offers sockets;NDIS connects adapter/protocol driver models. TDI is historical;Microsoft directs new Vista-and-later kernel network modules toward WSK. It is not Windows 7's only new-driver route. These responsibilities do not determine a fixed process-switch count per packet.

Mechanism	Role	Does not supply
TCP/IP	Addressing, packets, reliable bytes	Application framing/exactly-once effects
SMB redirector/server	Remote file operations	Identical local/remote failures
Named pipe	Local/remote pipe semantics	Interchangeable byte/message modes
RPC	Request/reply call abstraction	Timeout proves nonexecution
COM	Object interfaces/marshalling	Every call crosses machines

Specify versions, authentication, encryption, errors separately. HTTP/WebDAV/VPN names do not replace settings or imply contemporary security guidance.

UNC paths and partial failures

Opening `\\server\share\report` selects a redirector encoding network requests. Server applies its credentials/filesystem policy and replies status/bytes. Client, network, server, storage can fail independently.

Server writes but reply is lost. Client timeout does not prove no write;retrying append can duplicate. Protocol request identity, deduplication/retry, and durability rules must handle it. TCP reliability alone is not business exactly-once.

Similar APIs can have different share/offline-cache visibility and error timing. Open success does not promise an immortal connection.

Named-pipe messages and directory identities

A 500 B message read into200 B can return200 plusERROR_MORE_DATA;read the remaining300 of that same message. FALSE need not mean zero-byte fatal failure. This does not apply to TCP framing.

Domains integrate shared identity/management. Active Directory combines directory/authentication-related services rather than merely an encrypted database. Equal display names can have different SIDs;servers check effective credentials/token/object policy.

Offline client caching creates reconnection versions/conflicts. Cached success is not remote durable commit. Sharing follows the contracts in Chapters11/17.

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Programmer interface

Processes, threads, fibers, UMS, pools have different scheduling/lifetimes. Overlapped submission, cancellation, and terminal completion are distinct.

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Scheduling units and pools

CreateProcess includes an initial thread and separately observable/waitable process/thread handles. Additional threads share address space/resources but have separate PC, stack, TLS. Handle names do not replace references.

Fibers switch in user code within a thread. A blocking syscall on that thread does not let ordinary fiber switching make the same kernel thread run another fiber simultaneously. Windows 7 UMS coordinates user scheduling with managed kernel state under platform/API constraints;it does not make all blocking free or every API fiber-compatible.

Pools reuse workers/queues while retaining work-data lifetimes. One-worker jobA waiting for queued jobB in the same pool can starve/deadlock it.

Sharing handles and file positions

Inheritance requires handle inheritability and CreateProcess conditions/handle list. DuplicateHandle requires source/target process authority. Named opening checks the selected namespace/policy again.

DUPLICATE_SAME_ACCESS retains granted mask;copying read-only to a child creates no write right. Different requested access follows object/API rules, not arbitrary amplification or a universal subset theorem.

Duplicated handles to one file object can share position;independent CreateFile can have separate positions. Seek/read races remain. Overlapped operations carry their own offsets/state.

Three ReadFile starting outcomes

Use regular files opened FILE_FLAG_OVERLAPPED, independent OVERLAPPED and valid buffer per outstanding operation, and a manual-reset event. Windows 7 requires non-NULL lpNumberOfBytesRead, so provide a DWORD;pending final bytes come from completion.

Initial result	Meaning	Lifetime
TRUE	Operation completed;inspect actual bytes	Process final result
FALSE, ERROR_IO_PENDING	Submitted, not ordinary failure	Retain buffer, ov, event
FALSE, other error	Follow file/pipe/device contract	Establish whether work remains

Confirm pending work with an appropriate mechanism such as GetOverlappedResult. Do not assume later GetOverlappedResultEx exists in Windows 7.

Six bytes, four bytes, then EOF

FileABCDEFGHIJ has10 B. Each request 6 uses an explicit offset. Initial completion can be immediate or pending;final results follow:

Operation	Offset	Request	Final bytes/data	Next step
R0	0	6	6, ABCDEF	Reuse state only after terminal
R1	6	6	4, GHIJ	Total prefix10
R2	10	6	EOF,0	Stop rather than spin

Regular-file EOF can appear as successful zero or ERROR_HANDLE_EOF depending on synchronous/overlapped context;handle the applicable result. Pipes/devices differ. Cancellation can still finish normally;collect the actual outcome.

Virtual memory, heaps, mappings, and TLS

VirtualAlloc manages reserve/commit/protection. Heap managers subdivide regions;HeapFree need not release the whole region. Sections/views map file/pagefile backing at process-local VAs with compatible offset/size/file bounds.

TLS provides per-thread slots/logical ownership rather than MMU isolation. Another thread with a pointer can access the same process memory. Objects outliving threads need ownership/synchronization.

Commit, working set, and heap counts differ. Reserve64 KiB, commit8 KiB, resident4 KiB can hold simultaneously;one memory number cannot describe all three.

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Complete cancellation and cleanup

While kernel/device may access an operation, retain buffer, OVERLAPPED, event, context. For pendingR, request CancelIoEx(h,&ov), collect terminal through the original completion path, handle normal/aborted/other error and actual bytes, then release data/ov. Close shared handle/event only after all related operations terminate.

Normal completion may win;successful abort can yield ERROR_OPERATION_ABORTED. ERROR_NOT_FOUND from cancellation does not replace original outcome. Never free merely because cancel returned or reuse one ov for simultaneousR1/R2. IOCP/other notification modes have their own delivery rules;follow the selected contract.

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States of one read

Layer	Track	Earlier success can still leave
Authority	Local handle, granted mask, effective token	NoW or later network failure
Virtual memory	Buffer VA, commit, residency, rights	Reserved but uncommitted memory
I/O	IRP/operation, pending, actual bytes	Submission without completion
Cache/filesystem	Offset, VCN/LCN, metadata	Cache write without user durability
Device/network	DMA, connection, storage contract	Requested cancel without stopped hardware

IRQL constrains present kernel activity;thread priority selects ready execution. Confusing them invalidates ISR/DPC waiting and lifetime reasoning.

Reconstructing the examples

The fixed q2 trace gives completions8/9/3 from residual slices/queues. One region can reserve64 KiB, commit8, and reside4. File byte20603→VCN5/123→LCN201→volume823419→disk1871995, with holes separate. Reads at 0/6/10 finish6/4/EOF;pending, cancel request, terminal differ.

These instantiate general invariants:operations need granted rights;live I/O forbids premature free;extent, size, bitmap agree;one CPU's IRQL does not exclude other CPUs/shared-memory users.

References

Silberschatz, Galvin, and Gagne,*Operating System Concepts*,9th edition, Chapter 19, printed pages 829–885. Original diagrams/numerical cases retain Windows 7 scope;they do not claim complete scheduler heuristics, NTFS log encoding, or performance measurements.

Microsoft IRQL rules,VirtualAlloc,DuplicateHandle,file rights,DACL order, andNULL/empty DACLs support the applicable basic contracts. Later enclave options are outside scope.

ReadFile,GetOverlappedResult, andCancelIoEx define submission/results/cancellation, including Windows 7's non-NULL count pointer. WSK supplies the historical TDI transition boundary.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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Chapter route

Which computing needs and constraints drove OS evolution, and how do historical designs relate to today's abstractions?

Prerequisites are OS fundamentals, process/memory/file abstractions, and architecture history.

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Learning objectives

- Compare historical systems by requirements/constraints.
- Distinguish inherited concepts from version-specific mechanisms.
- Avoid equating historical products with current systems.

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Feature migration

Compare constraints, mechanisms, and new contracts. Similar names do not establish identical address units, execution agents, or protection boundaries.

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Concepts migrate without identical implementations

The chapter follows constraint→mechanism→contract, not a strict chronology. Batch sequencing reduces operator waits;spooling decouples device producers/consumers;time sharing adds interaction/CPU reclamation;virtual memory separates information identity from residence.

Constraint	Mechanism	New obligations
Slow manual job changes	Resident monitor	Detect termination/protect monitor
Slow cards/printing	Offline I/O, spool	Queues, capacity, errors/retries
Long exclusive occupancy	Time sharing	Timers, isolation, response, fairness
Expensive fast memory	Paging/backing	Restartable faults, mapping, replacement

Multics, UNIX, and smaller successors share multiuser, hierarchical-file, and sharing concepts. Direct code descent requires source/design evidence beyond conceptual similarity.

Four comparison dimensions

A process can mean static worker, dynamically created process, or a Mach task's thread. Paging can mean relocation without demand loading. Identify:

1. Whether execution agents are dynamic and dedicated per job or reused.
2. Byte/word addressing, page size, and table purpose.
3. Software-layer, privilege, address-space, or machine boundaries.
4. Ownership of faults, timeouts, buffers, and retry/release conditions.

Different answers prohibit transferring modern API/performance promises solely by name. Numerical demonstrations specify rules rather than claim historical measurements.

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Early systems

Batch/monitors reduce manual waits;spooling overlaps device work. One printer remains bounded by its service demand.

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From manual operation to resident monitors

Compilation, assembly, loading, running required media changes/termination observation. Batching groups similar needs;monitor control records choose next steps. Controllers/drivers still perform device operations.

Three jobsF, C, F each need2 CPU minutes. Each compiler setup costs 3, including initial setup. Original order has9 setup+6 CPU=15. ReorderingF, F, C has6+6=12, assuming reorderability and excluding response/dependencies.

Protect the monitor against user overwrite or halt. Control-record/data separation needs syntax, length, and escaping rules rather than assuming data rarely contains control characters.

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Offline I/O, buffers, spools, and multiprogramming

Offline equipment converts cards to tape before mainframe processing, saving host waiting but adding media-batch/movement latency. Buffers absorb production/consumption mismatch until finite capacity backpressures. Disk spools retain independent jobs/data/queue metadata for overlap, unlike imagining random simultaneous access to both tape ends. Multiprogramming separately runs another ready job while one waits, and requires scheduling/isolation.

For three jobs each CPU 2 then print4, with one CPU/printer, CPU runsJ1 0–2, J2 2–4, J3 4–6. Printer runsJ1 2–6, J2 6–10, J3 10–14. Total14, CPU utilization6/14, printer 12/14. Serial execution takes 18. Spooling does not halve the printer's12 units of work.

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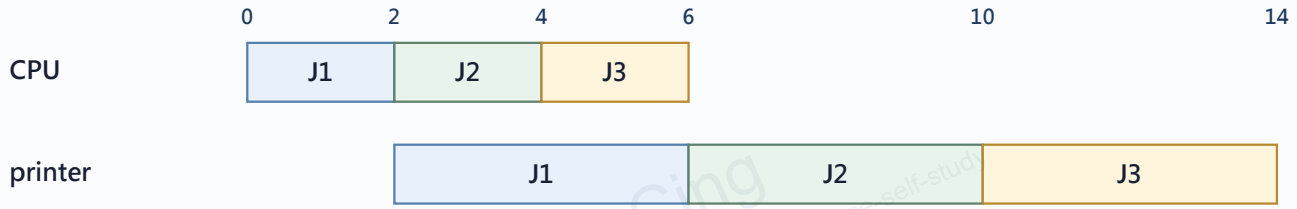
Offline I/O, buffers, spools, and multiprogramming (continued)

Three jobs: CPU 2 each, one printer 4 each; total 14

One color per job. CPU and printer each serialize; different devices overlap.



J2/J3 output can queue. Printer work remains 12. CPU busy 6/14, printer 12/14; serial total 18.

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Atlas

One-level storage separates information addresses from core/drum residence. Learning replacement includes the all-t=0 fallback.

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Information identity and automatic backing

Atlas combines fast core/drum behind a one-level store with 512-word pages and associative mappings. The book's16 Kiwords core has32 frames, not16 KiB. A four-frame teaching model retains512 words/page.

Virtual word address3597=7・512+13 maps page 7→frame 2 to core word address1037. If absent, preserve restartable execution, load into an available frame, update mapping, and retry. One-level store does not make drum as fast/directly accessible as core. Transfers, replacement, and live-frame ownership remain necessary.

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Three learning-replacement branches

Kilburn1962 defines t as time since last use and T as the preceding inactive interval, sampled by executed instructions rather than wall time. Choose first any t>T+1;otherwise among t≠0 choose maximalT−t;otherwise all t=0, choose maximalT. This example breaks ties by table order.

Case	(t, T) entries	Branch/victim
A	(6,3),(2,5),(0,9)	First:6>4, first page
B	(2,4),(1,6),(0,8)	Second:nonzero scores2/5, second
C	(0,2),(0,5),(0,3)	Third:largestT5, second

Past intervals predict future use rather than supply OPT knowledge. A predicted-distant page used next can add a fault. Branch3 makes valid all-recently-used states selectable. Original paper.

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XDS-940

Paging can relocate without demand paging. Reentrant code can share while writable state remains private.

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Relocation is not demand loading

Historical virtual space16 Kiwords, physical64 Kiwords, page 2 Kiwords give eight virtual pages/process and32 frames. Register-held tables scatter resident pages;whole-image swapping changes resident processes.

P virtual page 3→frame 11, offset 17 gives VA6161 and PA11・2048+17=22545 words. No byte conversion or unspecified word width is assumed. A legal missing-page demand fault is additional behavior absent from this relocation-only case. Atlas automatic demand loading and XDS relocation differ despite both having tables.

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Shared code and controlled execution

Two processes each use eight pages. Sharing two safely read-only reentrant code pages needs2+6・2=14 frames instead of 16. Shared instruction pages do not privatize a secretly shared writable global buffer;separate code and mutable state.

Converting XDS-930 to940 added user/monitor distinction, privileged I/O/halt, and controlled system calls. Dynamic subprocesses/shared memory additionally require lifecycle, synchronization, termination cleanup. These protections work together.

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THE

Layers and explicit synchronization decompose reasoning. A software layer is not automatically a hardware protection domain.

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Lower-layer contracts before upper-layer reasoning

Dijkstra's THE is batch multiprogramming by cooperating sequential processes, not primarily terminal interaction.

Level	Responsibility/hidden details
0	Processor allocation/timer;sequential-process abstraction
1	Segment controller/backing;information-segment identities
2	Console-message interpreter;physical console allocation
3	I/O buffering/logical communication units
4	User programs
5	Operator, not an implemented software level

Layers describe dependency rather than separate address spaces or mandatory traps. Segments identify information;core/drum pages hold it. EWD196.

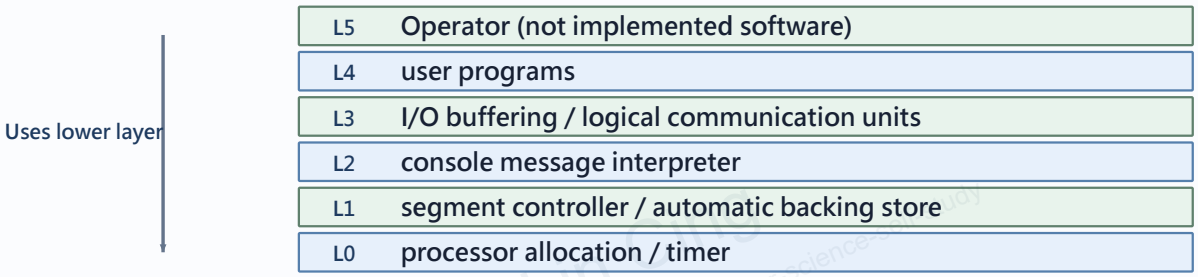
Induction over layer contracts

Assume layer i guaranteesCi fromC0…C(i−1), with stated interference constraints, and does not corrupt lower hidden state. Hardware/processor control establishesC0;induction establishes each successive contract. Layering alone proves nothing without those obligations.

Console messages can reside in segments and use level 1 retrieval. If level 1 instead needs level 2 to fetch its own pages, the dependency cycle invalidates bottom-up induction. Synchronization cannot assume relative speed:consume only produced segments, and give a printer only one incompatible owner. Waiting changes progress, not correctness based on "usually slower."

Induction over layer contracts (continued)

THE: higher levels use lower abstractions, not separate address spaces



Induction: Ci uses/preserves C0…C(i−1). Cycles or uncontrolled shared state invalidate the argument.

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RC4000

A nucleus supplies primitives while upper processes choose policy. Finite buffers require unique ownership and explicit return conditions.

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Policy above the nucleus

RC4000 supports constructing different systems rather than hardwiring one batch/time-sharing policy. Nucleus supplies process control, protection, messages. The book's fixed eight-word messages use a shared pool and per-process queues. Send-message/wait-message and send-answer/wait-answer form request/reply. Primitive atomicity protects internal transitions, not an entire server transaction.

Drivers translate registers/interrupts into process-like message interfaces. Devices do not thereby become ordinary processors executing arbitrary user code.

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Conservation with two buffers

Pool={b0, b1};retain one buffer through request/reply completion. This is an ownership model, not instruction-level original-machine emulation.

Step	Free	Server queued/processing	Awaiting client reply
Initial	b0, b1	Empty	Empty
C1 sends	b1	b0 queued	Empty
C2 sends	Empty	b0, b1 queued	Empty
S receives/answers b0	Empty	b1 queued	b0
C1 consumes/returns	b0	b1 queued	Empty

Each buffer has exactly one state;free+queued+processing+reply=2. Third submission must wait/fail as specified rather than allocateb0 again. S waiting for C1 while C1 waits for S still deadlocks. Small kernels/atomic send do not eliminate circular wait.

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CTSS

Interaction and swapping shape feedback. Specify initial level, quantum growth, and promotion separately.

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Short interaction versus long CPU bursts

MIT CTSS demonstrates interactive time sharing compatible with existing batch programs. Moving images between scarce memory/backing makes excessively short slices costly. Compatibility concerns preserving that environment, not every future API. The7090/7094 memory counts36-bit words, not equal KiB.

Exponential feedback demotes work exhausting a full slice and gives longer lower-priority slices, while short interactions gain earlier service. I/O promotion, aging, and starting level depend on the chosen version/model.

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Complete exponential-feedback trace

Fix q_i=2^i, start0, one burst11, no competitors/promotions;demote only when a full slice expires before completion.

Dispatch	Level	Allowed	Used	Total CPU	Remaining
1	0	1	1	1	10
2	1	2	2	3	8
3	2	4	4	7	4
4	3	8	4	11	0

Full slices throughk sum2^(k+1)−1 by the geometric series. With swap overhead.5/dispatch, four dispatches cost2, total 13. RR q1 needs11 dispatches, overhead5.5, total 16.5. This compares the fixed long-burst/cost model rather than every interactive workload.

Multics

Segments, files, paging, and selective sharing combine with ring enforcement. Rings and software layers provide different guarantees.

A computer utility with controlled sharing

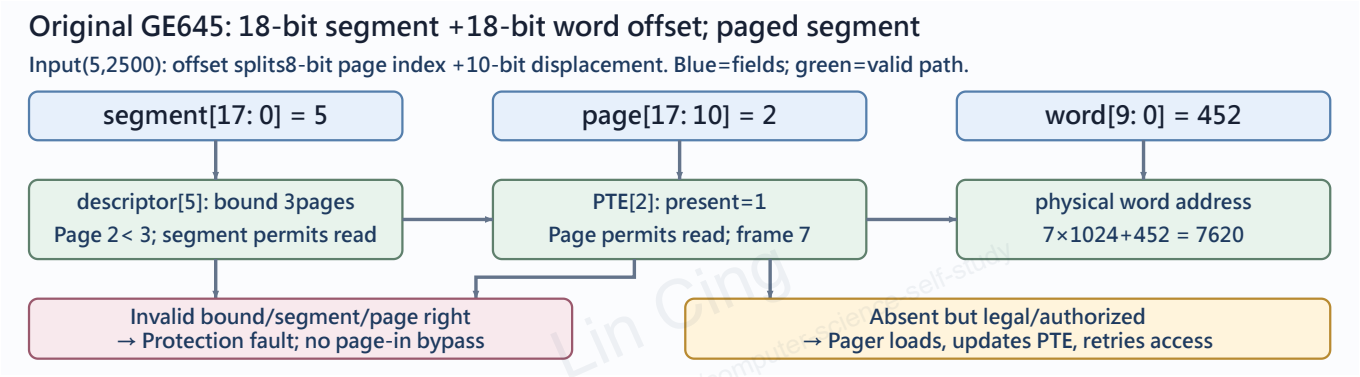
Multics targets remote terminals, continuous service, growing capacity, and selective sharing. Hierarchical files, segments, paging, rings jointly support those goals rather than one vague early-large-OS label.

Fix original GE645 hardware:18-bit segment number and18-bit word offset, each0…2^{18}−1. A 1024-word page uses10 offset bits, leaving8 page-index bits. Descriptor bounds can be smaller than field capacity. Original design.

Segment5, offset 2500=2 • 1024+452:first inspect descriptor, bound, rights, then segment page 2. Frame7 gives core word address7620. Segment identity is independent of physical frame.

Here segment 5 is paged, bound 3 pages, and segment/page permit read. Page 2<3 and presentPTE2→frame 7 allow access. Invalid bound/permission is a protection fault that page-in cannot bypass. Only an authorized absent page invokes pager, mapping update, retry.

A computer utility with controlled sharing (continued)



Rings versus layers

Rings enforce privilege/entry rules;THE layers organize abstraction/dependency. Modules need not each own a ring, and drawing layers proves no protection boundary.

Resolving a segment name still requires object access lists, execution context, entry rules. Shared code/private writable state need mapping/protection contracts. The developers'1972 retrospective describes integration exposing unexpected path costs and iterative redesign. Useful concepts do not prove the first interface partition efficient;smaller successors need not retain every mechanism. Retrospective.

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IBM OS/360

Family compatibility does not erase MFT/MVT differences. Later virtual memory and CP/CMS add distinct abstractions.

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Compatibility and allocation variants

System/360 sought common software/tools across capacities. OS/360 MFT uses fixed regions;MVT variable regions. One variant does not specify the entire family.

A 256 KiB example has four64 KiB regions and jobs50,50,70. Two50 jobs waste28 KiB internally. Remaining128 total cannot fit70 into one64 region. This is the chosen configuration, not all possible OS/360 layouts.

Variable holes50,30,40 total 120 but lack a contiguous70, an external-fragmentation example. Moving/compacting requires relocation support/cost;load-time-bound addresses cannot be arbitrarily changed.

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Later virtual memory and CP/CMS

System/370-era OS/VS/MVS uses new VM hardware rather than original360 paging semantics. CP/67 provides VMs;CMS is a single-user system within them. CP supplies multi-CMS isolation rather than CMS itself becoming a shared multiuser kernel.

Two guests can both see address1000 while VMM maps their memory views to different host resources. Mediate guest privilege;guest supervisor status does not authorize arbitrary host writes. Family compatibility, relocation, VM isolation, and interaction are distinct. JCL describes batch resources/steps;option count proves neither correctness nor a fixed performance ratio.

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TOPS-20

TENEX/TOPS-20 combines command assistance and memory support. Syntax usability does not replace object authorization.

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Interaction and virtual memory

In the book's history, BBN added paging to PDP-10-related hardware and developed TENEX. DEC later acquired rights and formed DECSYSTEM-20/TOPS-20. Command completion/help reduces syntax-memory burden;virtual memory manages addresses/residence. Shell suggestions do not authorize page-table changes, and paging does not infer desired commands.

A machine-like private execution view with many simultaneous users still needs CPU scheduling, terminal I/O, file rights, reclamation. Faster CPUs alone do not provide interaction.

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Usability and security complement one another

In an explicit toy COPY source destination parser, valid syntax creates a request. Execution still checks source read and destination-parent/file write contracts. Autocomplete success does not grant access.

Checking destination then later writing its pathname can race name replacement, as in Chapter 11 TOCTOU. Protected references, clear errors, and understandable interaction must work together. This toy illustrates responsibility, not literal TOPS-20 syntax.

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CP/M and MS-DOS

Small-platform compatibility differs from modern protected kernels. A shared-address-space convention does not confine untrusted code.

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Resource limits and isolation

CP/M supplies early microcomputer commands/program/device services. MS-DOS follows an IBM-PC single-user/compatibility path. Their protection differs from later multiuser systems;they are not merely smaller NT/Linux.

Counts such as64 KiB/640 KiB must distinguish CPU addressability, conventional layout, OS reservations, and extensions. A common program region is not every version's absolute physical limit.

Without enforced memory isolation, programs can overwrite system data despite APIs instructing them to use services. Hostile programs need not follow conventions. Hardware privilege plus trusted mediation is required.

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A shared-space counterexample

Toy RAM[0,1024) has monitor[0,256), job[256,1024), without hardware bounds. Store100 corrupts monitor even if correct libraries never do so. This is not an actual DOS layout.

Protected base256, limit768 validates0≤a<768 then translates256+a. Addresses−1/768 fail;0/767 reach256/1023. Users cannot modify base/limit. This proves range confinement, not progress against infinite loops, which needs a timer, or shared-file policy. One hardware feature supplies one boundary rather than a full modern OS.

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Macintosh and Windows

GUI design does not determine preemption, memory protection, or driver boundaries. Cooperative/preemptive progress rests on different premises.

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Desktop interaction versus kernel guarantees

Early Macintosh/Windows made desktop interaction central;later generations added fuller memory, multitasking, networking. Specify generation. Windows 7 here is NT, not early DOS-dependent Windows.

Windows, mouse, menus are interaction models;preemption, address protection, compatibility, drivers are separate. Two windows prove neither parallel processes nor isolated address spaces. A blocked UI thread can freeze a GUI while kernel scheduling continues. Trace event, queue, handler, and awaited service.

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Cooperative and preemptive progress

On one CPU, A loops forever without yield while B is ready. If switching requires yield/block, B may never run. With serviceable timer and a scheduler giving both slices, preemption runsB at a slice boundary.

If A instead loops unbounded in a nonpreemptible kernel lock region, user slices alone cannot guarantee progress. Include kernel/IRQ bounds. This toy compares protocols without classifying every desktop generation identically. Even a scheduledB may wait for A's global resource;preemption does not eliminate deadlock/priority inversion.

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Mach

Mach2.x and3 place BSD differently. Separate task/thread/port authority from copy/mapping ownership.

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Versioned abstractions

Mach builds on Accent-related research with tasks/address spaces, threads, IPC, VM. A task owns address space/port rights;threads execute and share their task's memory.

Model	BSD placement	Boundary
Mach2.x/2.5	Much remains in kernel	File service need not cross a user server
Mach3 microkernel	UNIX services in user servers	Controlled IPC across address spaces

Kernel-checkable port rights authorize messages;equal integers do not transfer rights across tasks. Shared memory also exists, so not all exchange requires byte-copy messages.

A fixed IPC cost and isolation model

ClientC sends32 B to user serverS, which replies4096 B. Each direction copies user→kernel→other user twice, without shared-page/out-of-line optimization.

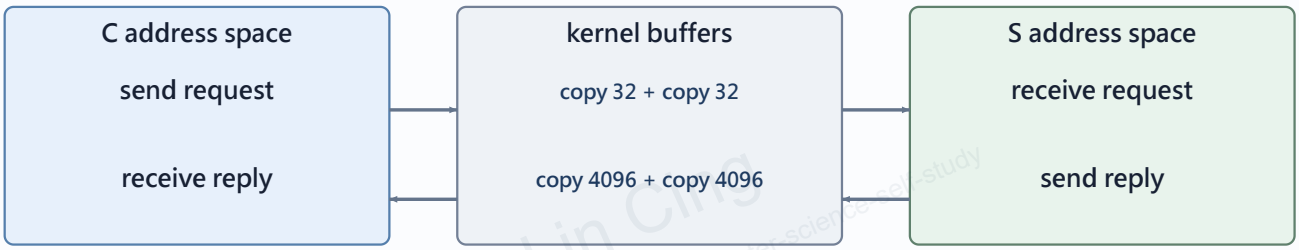
Path	Mode transitions	Address-space changes	Copy bytes
C user→K→S user	2	1	64
S user→K→C user	2	1	8192
Total	4	2	8256

Address protection confines direct corruption from server crash, but client still needs errors, timeouts, reconstruction. Smaller kernel does not prove universal availability. IPC/VM integration can avoid copies while adding mapping, rights, page lifetime, COW costs rather than becoming free.

A fixed IPC cost and isolation model (continued)

Mach3 copy model: 32 B request +4096 B reply; 8256 B copied

Blue=client C; green=server S; gray=kernel. Two copies/direction, no shared-page optimization.



Controlled kernel transfer: 4 mode transitions,2 address-space changes. Check port rights and buffer lifetime.

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Other systems and unequal processor roles

The book also discusses Burroughs MCP's systems language, segmentation, multiprocessing, and CDC6600 coordination among processing activities. Multiple processors need not be symmetric peers. Identify roles, memory authority, I/O responsibility before applying an SMP scheduler.

High-level languages improve expression, type/module checking, maintenance, while synchronization, ABI, hardware control still need correctness. Assembly does not logically imply failure. Language choice changes inspectable conditions rather than replacing contracts.

Compatibility ecosystems, hardware costs, deployment organizations also affect survival. No single feature here explains commercial success or supplies current-market evidence.

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Mechanisms across cases

Problem	Case	Required conditions
CPU/slow-I/O overlap	Monitors/spools	Queues, buffers, ownership, overlappable work
Nonresident information	Atlas	Mapping, restartable fault, total victim selection
Scattered/shared code	XDS-940	Relocation, RO code, private mutable state
Decomposed correctness	THE	Acyclic dependencies, layer contracts
Policy above nucleus	RC4000	Message ownership, pool limits, wait relationships
Interaction/long-burst tradeoff	CTSS	Feedback, promotion, swap-cost rules
Selective sharing/naming	Multics	Separate segment, page, protection
Service isolation	Mach3	IPC rights, address spaces, failure handling

General proofs beyond finite demonstrations

A trace exposes missing t=0 branches, word/byte confusion, or duplicate buffer allocation. General claims require rules covering all legal states.

Atlas selects a victim from every nonempty finite frame set:if branch 1 applies, use it;otherwise if any nonzero t exists, a finite maximum score exists;otherwise branch 3 chooses maximalT. This proves total selection, not minimum faults.

Buffer conservation holds initially and under each atomic ownership transition. Appearing simultaneously free and awaiting reply violates that invariant. Layer induction requiresCi to depend only on lower contracts while preserving them;cycles/uncontrolled hidden-state mutation invalidate it.

These prove total selection, resource safety, and compositional correctness, respectively. None alone establishes an entirely reliable OS.

References

- Silberschatz, Galvin, and Gagne,*Operating System Concepts*,9th edition, Chapter 20, printed pages 887–904. Historical cases retain their configurations. Numerical models, questions, SVGs are original;word counts are not byte counts.
- Kilburn1962 one-level store, pp231–232 supplies the three learning branches, including all-t=0. Historical intervals predict rather than implementOPT.
- Dijkstra EWD196 defines THE levels/information segments versus storage pages without implying independent address spaces.
- GE645 design specifies18-bit segment and18-bit word offset, split8+10 for 1024-word pages, not later6180 pointer formats.
- Multics first seven years,1972 describes utility goals/iteration rather than current markets.
- Tom Van Vleck on CTSS supplies7094/36-bit/compatibility context. Fixed q_i=2^i cases are not historical per-tick reconstructions.

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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